Maths is nothing but a plus, minus, times and divide.

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Algebra

Simplify expressions using the laws of rational exponents

As always, I am going to use examples to ease into the concept.

Example

$2^3 \times 2^4$ versus $2^{\frac{1}{3}} \times 2^{\frac{1}{4}}$

Solution

$2^3 \times 2^4$ versus $2^{\frac{1}{3}} \times 2^{\frac{1}{4}}$

Apply the law of exponents
$a^m \times a^n = a^{m+n}$

$= 2^{3+4}$ $= 2^{\frac{1}{3}+\frac{1}{4}}$

$= 2^7$ $= 2^{\frac{7}{12}}$

As you know, we apply the same exponential laws to the rational exponents $\left(2^{\frac{1}{3}} \times 2^{\frac{1}{4}}\right)$.

Example: Simplify the following exponents.

$2^3 \div 2^4$ versus $2^{\frac{1}{3}} \div 2^{\frac{1}{4}}$

Solution

$2^3 \div 2^4$ versus $2^{\frac{1}{3}} \div 2^{\frac{1}{4}}$

Apply the law of exponents
$a^m \div a^n = a^{m-n}$

$= 2^{3-4}$ $= 2^{\frac{1}{3}-\frac{1}{4}}$

$= 2^{-1}$ $= 2^{\frac{1}{12}}$

Apply the law $a^{-n} = \frac{1}{a^n}$

$= \frac{1}{2}$

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Such exponents are called **rational exponents**. A rational exponent is in the form;

$x^{\frac{a}{b}}$, where $a > 0 \; and \; b > 0$

You may write the rational exponent $x^{\frac{a}{b}}$ in the form $\sqrt[b]{x^a}$, it is the same thing; it's just that it's written in a different form. See such examples below.

$$4^{\frac{1}{2}} = \sqrt[2]{4^1} = \sqrt[2]{4} = 2$$

$$27^{\frac{1}{3}} = \sqrt[3]{27^1} = \sqrt[3]{27} = 3$$

$$8^{\frac{2}{3}} = \sqrt[3]{8^2} = \sqrt[3]{64} = 4$$

We will do more practice questions to improve your understanding.

Example: Simplify without using a calculator.

a) $\frac{2^3}{2^4}$ vs $\frac{2^{\frac{1}{3}}}{2^{\frac{1}{4}}}$ b) $\frac{2^4}{2^3}$ vs $\frac{2^{\frac{1}{4}}}{2^{\frac{1}{3}}}$ c) $4^{\frac{1}{2}}$ d) $9^{\frac{1}{2}}$

Solution

a) $\frac{2^3}{2^4}$ vs $\frac{2^{\frac{1}{3}}}{2^{\frac{1}{4}}}$

Apply the law of exponents $\frac{a^m}{a^n} = a^{m-n}$

$= 2^{3-4}$ vs $= 2^{\frac{1}{3}-\frac{1}{4}}$

$= 2^{-1}$ Apply the law $a^{-n} = \frac{1}{a^n}$ $= 2^{\frac{1}{12}}$

$= \frac{1}{2}$

b)

$$\frac{2^4}{2^3}$$

VS

$$\frac{2^{\frac{1}{4}}}{2^{\frac{1}{3}}}$$

Apply the law of exponents
$\frac{a^m}{a^n} = a^{m-n}$

$$= 2^{4-3}$$

$$= 2^1 = 2$$

$$= 2^{\frac{1}{4}-\frac{1}{3}}$$

$$= 2^{-\frac{1}{12}} = \frac{1}{2^{\frac{1}{12}}}$$

Apply the law $a^{-n} = \frac{1}{a^n}$

c) $4^{\frac{1}{2}}$

Firstly, turn 4 into its exponential form. 4 is 2^2 in its exponential form.

$$= 2^{2\times\frac{1}{2}} = 2^{\frac{2}{2}} = 2^{\frac{1}{1}} = 2^1 = 2$$

d) $9^{\frac{1}{2}}$

Firstly, turn 4 into its exponential form. 9 is 3^2 in its exponential form.

$$= 3^{2\times\frac{1}{2}} = 3^{\frac{2}{2}} = 3^{\frac{1}{1}} = 3^1 = 3$$

Example: Simplify without using a calculator.

a) $\frac{3^3 \times 3^2}{2^4 \times 2^2}$ b) $\frac{a^{\frac{3}{4}} \times a^{\frac{1}{4}}}{b^{\frac{2}{3}} \times b^{\frac{1}{3}}}$

Solution

a) $\frac{3^3 \times 3^2}{2^4 \times 2^2}$

On both the numerator and denominator, apply the law of exponents; $a^m \times a^n = a^{m+n}$.

$$= \frac{3^{3+2}}{2^{4+2}} = \frac{3^5}{2^6}$$

b) $\frac{a^{\frac{3}{4}} \times a^{\frac{1}{4}}}{b^{\frac{2}{3}} \times b^{\frac{1}{3}}}$

On both the numerator and denominator, apply the law of exponents $a^m \times a^n = a^{m+n}$

$$= \frac{a^{\frac{3}{4}+\frac{1}{4}}}{b^{\frac{2}{3}+\frac{1}{3}}} = \frac{a^{\frac{4}{4}}}{b^{\frac{3}{3}}} = \frac{a^1}{b^1} = \frac{a}{b}$$

Example: Simplify without using a calculator.

a) $4^{\frac{1}{2}} \times 2^3$ b) $\frac{27^{\frac{1}{3}}}{3^{\frac{1}{2}}}$ c) $\frac{8^{\frac{1}{3}}}{2}$

Solution

a) $4^{\frac{1}{2}} \times 2^3$, turn 4 into its exponential form, 4 is 2^2 in its exponential form.

$$= 2^{2\times\frac{1}{2}} \times 2^3 = 2^{\frac{2}{2}} \times 2^3 = 2^1 \times 2^3 = 2^{1+3} = 2^4 = 16$$

b) $\frac{27^{\frac{1}{3}}}{3^{\frac{1}{2}}}$, turn 27 into its exponential form, 27 is 3^3 in its exponential form.

$$= \frac{3^{3\times\frac{1}{2}}}{3^{\frac{1}{2}}} = \frac{3^{\frac{3}{2}}}{3^{\frac{1}{2}}} = 3^{\frac{3}{2}-\frac{1}{2}} = 3^1 = 3$$

c) $\frac{8^{\frac{1}{3}}}{2}$, turn 8 into its exponential form, 8 is 2^3 in its exponential form.

$$= \frac{2^{3\times\frac{1}{3}}}{2} = \frac{2^{\frac{3}{3}}}{2} = \frac{2^1}{2} = \frac{2}{2} = 1$$

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Equations and Inequalities

Linear Equations

Fill in the empty box.

$\square + 1 = 2$

So you are looking for number that you add to 1 to get 2. That number is clearly 1. We call this process solving an equation.

$\square + 1 = 2$

You might know this as "solving the x".

You may replace this box with an x, and have;

$\square + 1 = 2$

Think of the "x" as an empty box you need to fill, it is just the same thing.

So, when a question comes in the form $x + 1 = 2$, and says solve for x, they basically asking you to fill in the empty box in
$\square + 1 = 2$.

So whatever you put in the empty box is the value of "x". However, there will be a shortcut method you will be using in this chapter.

Example: Solve for x in the equation below: $x + 1 = 2$

The aim is to remain with "x" only on one side, so we move all the numbers to the other side to leave *"x"* on its own. In mathematics, when a number is moved from one side to another, the sign in front it changes. Let's follow two quick steps to solve the equation $x + 1 = 2$

Step 1: Move all the numbers on one side to leave "x" alone.

$x + 1 = 2$

$x = 2 - 1$

Step 2: Do the calculations.

$x = 1$

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The answer $x = 1$ means that if we go back to the equation $x + 1 = 2$ and replace x with 1, the equation will be true.

$x + 1 = 2$

$1 + 1 = 2$

$2 = 2$

Replace the x with 1 to get the equation 1 + 1 = 2

Let's do more examples.

Example: Solve for x in the equation below: $x - 1 = 2$

The aim is to remain with "x" only on one side, so we move all the numbers to the other side to leave "x" on its own. In mathematics, when a number is moved from one side to another, the sign in front it changes. Let's follow two quick steps to solve the equation $x - 1 = 2$

Step 1: Move all the numbers on one side to leave "x" alone.

$x - 1 = 2$

$x = 2 + 1$

The -1 on the left of the equation moved to the right and then changed to +1.

Step 2: Do the calculations.

$x = 3$

The answer $x = 3$ means that if we go back to the equation $x - 1 = 2$ and replace x with 3, the equation will be true.

$x - 1 = 2$

$3 - 1 = 2$

$2 = 2$

Replace the x with 3 to get the equation 3 - 1 = 2

Such simple "solve for x" equations come in various basic forms.

Form 1

$$x + 2 - 3 = 6 + 5 + 2x$$

As shown above, an equation may come with an x and various numbers on both sides of the equation. Your aim here is to put the x's on one side and the numbers on the other side of the equation.

Form 2

$$2(x+1)+2=5$$

An equation may come with brackets as shown, which you would have to expand and simplify.

Form 3

$$2x=4$$

The x on one side may have a number in front of it. We will learn how to solve such an equation a bit later.

We now get into the examples of each of the forms, starting with form 1.

Form1 examples

Example: Solve for x in the following equation:

$2x+4=5+3+x$

Solution

$2x+4=5+3+x$

Your aim here is to put the x's on one side and the numbers on the other side of the equation.

$2x-x=5+3-4$

$2x-1x=5+3-4$

For 2x – 1x, just deal with the numbers first: 2 – 1 = 1. Then your answer for 2x – 1x = 1x = x

$x=4$

To check if your answer is correct, replace the x with 4 in the original equation.

$2x+4=5+3+x$

$2(4)+4=5+3+(4)$

Replace the x with 4

$8+4=8+4$

$12=12$. The sides are therefore equal, this means the value of x = 4 is correct.

Example: Solve for x in the following equation:

$-2x + 4 = 5 - 3x$

Solution

$-2x + 4 = 5 - 3x$

Your aim here is to put the x's on one side and the numbers on the other side of the equation.

$-2x + 3x = 5 - 4$

For -2x + 3x, just deal with the numbers first: -2 + 3 = 1. Then your answer for -2x + 3x = 1x = x

$x = 1$

To check if your answer is correct, replace the x with 1 in the original equation.

$-2(1) + 4 = 5 - 3(1)$

Replace the x with 1

$-2 + 4 = 5 - 3$

$2 = 2$

Form 2 Examples

Example: Solve for x in the following equation:

$2(x + 1) + 2 = 5 + x$

Solution

Start by multiplying out the brackets.

$(2 \times x) + (2 \times 1) + 2 = 5 + x$

$2x + 2 + 2 = 5 + x$

$2x + 4 = 5 + x$

$2x - x = 5 - 4$

$x = 1$

To check if your answer is correct, replace the *x* with 1 in the original equation.

$2(1 + 1) + 2 = 5 + 1$

$2(2) + 2 = 5 + 1$

$4 + 2 = 6$

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$6 = 6$

Form 3 examples

Example: Solve for x in the following equation.

$2x = 4$

Solution

Remember that the aim is to remain with x "alone" on one side. We know that $2x$ is the same as $2 \times x$. So you may rewrite the equation as $2 \times x = 4$, it's the same thing.

Again, we also know $2 \times x$ that is the same as $x \times 2$. So we may rewrite the equation as $x \times 2 = 4$

Now remember when we dealt with plus and minus in front of a number, we change the sign of the number when we move it to the other side. We also do something similar with × and ÷.

$$x \times 2 = 4$$

In this case, we move the $\times 2$ to the other side, and when it moves, the × changes to ÷ as shown below.

$x \times 2 = 4$

$x = 4 \div 2$

$x = 2$, that's the answer.

To check if your answer is correct, replace the x with 2 in the original equation.

$2x = 4$

$2(2) = 4$

$4 = 4$, the sides are equal, so our answer is correct.

Example: Solve for x in the following equation.

$3x = 9$

Solution

Remember that the aim is to remain with x "alone" on one side. We know that $3x$ is the same as $3 \times x$. So you may rewrite the equation as $3 \times x = 9$, it's the same thing.

Again, we also know $3 \times x$ that is the same as $x \times 3$. So we may rewrite the equation as $x \times 3 = 9$

Now remember when we dealt with plus and minus in front of a number, we change the sign of the number when we move it to the other side. We also do something similar with × and ÷.

$$x \times 3 = 9$$

In this case, we move the $\times 3$ to the other side, and when it moves, the × changes to ÷ as shown below.

$x \times 3 = 9$

$x = 9 \div 3$

$x = 3$, that's the answer.

To check if your answer is correct, replace the x with 3 in the original equation.

$3x = 9$

$3(3) = 9$

$9 = 9$, the sides are equal, so our answer is correct.

Example: Solve for x in the following equation.

$\frac{x}{2} = 4$

Solution

Remember that the aim is to remain with x "alone" on one side. We know that $\frac{x}{2}$ is the same as $x \div 2$.

So we may rewrite the equation as $x \div 2 = 4$

Now remember when we dealt with plus and minus in front of a number, we change the sign of the number when we move it to the other side. We also do something similar with × and ÷.

$x \div 2 = 4$

In this case, we move the $\div 2$ to the other side, and when it moves, the ÷ changes to × as shown below.

$x \div 2 = 4$

$x = 4 \times 2$

$x = 8$, that's the answer.

To check if your answer is correct, replace the x with 8 in the original equation.

$\frac{x}{2} = 4$

$\frac{\mathbf{8}}{2} = 4$

$4 = 4$, the sides are equal, so our answer is correct.

Example: Solve for x in each of the two equations below.

$2x = 4$ vs $\frac{x}{2} = 4$

Solution

$2x = 4$

$2 \times x = 4$

$x \times 2 = 4$

$x = 4 \div 2$

$x = 2$

vs

$\frac{x}{2} = 4$

$x \div 2 = 4$

$x = 4 \times 2$

Note that you will also have different combinations of the forms. In order to be comfortable with such forms, you need to do a lot of practice questions.

Question: Solve for x in each of the following equations.

a) $3x + 2 = 6 - x$

b) $5x - 6 = 4 + 3x$

c) $3(x + 2) = 5x$

d) $\frac{1}{2}x = 4$

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e) $\frac{3}{2}x = 4$

f) $\frac{3}{2}x = 4x + 1$

h) $\frac{3}{2}x = 4x - 1$

i) $\frac{3}{2}x + 5 = 4x - 1$

j) $\frac{3}{2}x + 3 = \frac{4}{5}x + 7$

Solution

a) $3x + 2 = 6 - x$

$3x + x = 6 - 2$

$4x = 4$

$4 \times x = 4$

$x \times 4 = 4$

$x = 4 \div 4$

$x = 1$

b) $5x - 6 = 4 + 3x$

$5x - 3x = 4 + 6$

$2x = 10$

$2 \times x = 10$

$x \times 2 = 10$

$x = 10 \div 2$

$x = 5$

c) $3(x + 2) = 5x$

$3x + 6 = 5x$

$3x - 5x = -6$

$-2x = -6$

$-2 \times x = -6$

$x \times -2 = -6$

$x = -6 \div -2$

$x = 3$

d) $\frac{1}{2}x = 4$

$\frac{1}{2} \times x = 4$

$x \times \frac{1}{2} = 4$

$x = 4 \div \frac{1}{2}$

$x = 4 \times \frac{2}{1}$

$x = 8$

e) $\frac{3}{2}x = 4$

$\frac{3}{2} \times x = 4$

$x \times \frac{3}{2} = 4$

$x = 4 \div \frac{3}{2}$

$x = 4 \times \frac{2}{3}$

$x = \frac{8}{3}$

f) $\frac{3}{2}x = 4x + 1$

$\frac{3}{2}x - 4x = 1$

$-\frac{5}{2}x = 1$

$-\frac{5}{2} \times x = 1$

$x \times -\frac{5}{2} = 1$

$x = 1 \div -\frac{5}{2}$

$x = 1 \times -\frac{2}{5}$

$x = -\frac{2}{5}$

h) $\frac{3}{2}x = 4x - 1$

$\frac{3}{2}x - 4x = -1$

$-\frac{5}{2}x = -1$

$-\frac{5}{2} \times x = -1$

$x \times -\frac{5}{2} = -1$

$x = -1 \div -\frac{5}{2}$

$x = -1 \times -\frac{2}{5}$

$x = \frac{2}{5}$

i) $\frac{3}{2}x + 5 = 4x - 1$

$\frac{3}{2}x - 4x = -1 - 5$

$$-\frac{5}{2}x = -6$$

$$-\frac{5}{2} \times x = -6$$

$$x \times -\frac{5}{2} = -6$$

$$x = -6 \div -\frac{5}{2}$$

$$x = -6 \times -\frac{2}{5}$$

$$x = \frac{12}{5}$$

j) $\frac{3}{2}x + 3 = \frac{4}{5}x + 7$

$$\frac{3}{2}x - \frac{4}{5}x = 7 - 3$$

$$\frac{3}{2}x - \frac{4}{5}x = 7 - 3$$

$$\frac{7}{10}x = 4$$

$$\frac{7}{10} \times x = 4$$

$$x \times \frac{7}{10} = 4$$

$$x = 4 \div \frac{7}{10}$$

$$x = 4 \times \frac{10}{7}$$

$$x = \frac{40}{7}$$

Quadratic Equations

Fill in the empty box: $\square^2 = 4$

The question is basically asking: "What number can you put in the empty box such that you get an answer of 4 when you put a power of 2 on it?"

The number we are looking for is 2.

$\mathbf{2}^2 = 4$. As you can see, **2** to the power of 2 is 4. Let's look at our question again: $\square^2 = 4$

We may replace the box with an x and have: $x^2 = 4$

Think of the x as an empty box that you have to fill, it is just the same thing. Such an equation where the highest power of x is 2 is called a quadratic equation. Quadratic equations exist in various basic forms. In this section, we have devised three different forms.

Form 1

$x^2 = 4$

The equation comes with an x^2 and one or more numbers. One other example is $x^2 - 1 = 8$, which you simplify by taking the -1 to the other side to get $x^2 = 8 + 1$,which in turn gives $x^2 = 9$

I will soon show you how to solve such an equation.

Form 2

$x^2 + x = 0$

Form 2 is made up of x^2, x and a "0". The equation may come rearranged in many different ways. For example, the above equation may be written as $x^2 = -x$, if you take the x to the left-hand side, the equation becomes $x^2 + x = 0$. All you have to do is move everything on one side so that you may see the form of the equation.

Form 3

$x^2 + x - 6 = 0$

Form 3 is made up of x^2, x and a number that's "not" a "0". In this case, the number is -6. For example, the above equation may be written as $x^2 + x = 6$, if you take the 6 to the left-hand side, the equation becomes $x^2 + x - 6 = 0$. All you have to do is move everything on one side so that you may see the form of the equation.

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If you come across any other form different to the above three forms, simplify it and you will end with one of the three above forms. We will now learn how to solve each of the forms.

Form 1 Examples

Example: Solve for x in the equation $x^2 = 4$

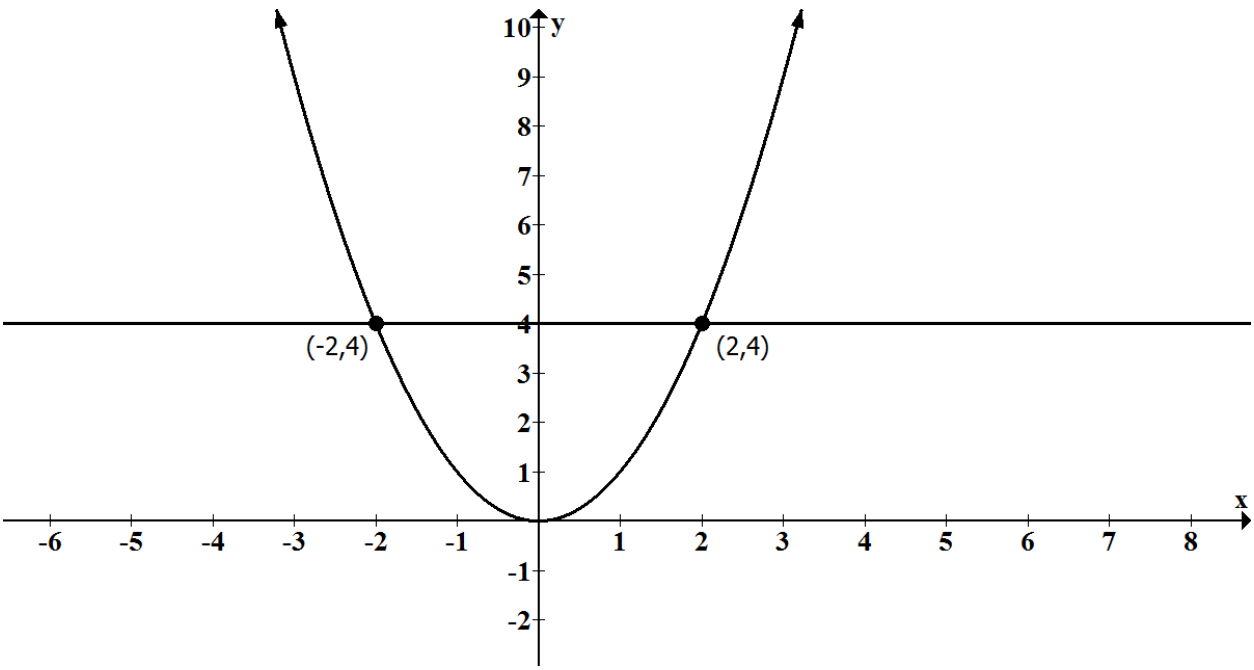
There are two basic methods we can use to solve such a form, the graphical method and algebraic method.

Graphical method

Let's plot the graph of x^2. We start with the table of values and then plot it.

x	-3	-2	-1	0	1	2	3
y	9	4	1	0	1	4	9

Below is the graph of x^2 and the graph of 4.



Remember, the equation $x^2 = 4$ is basically asking you for the points where the graph of x^2 meets the graph of 4. As you can see from the diagram, the two graphs meet at points A and B. Point A is *(-2 ; 4)* and point B is *(2 ; 4).* These are the points where the two graphs have the same y-values, the same y-values means they are equal. So when an equation comes as $x^2 = 4$, it is just asking you to give the x-coordinates of the points where the graphs of x^2 and 4 have the same y-values. Such x-values are -2 and 2. So are our answers are

$x = -2 \; and \; x = 2$

We can check if our answers are correct by substituting them in the original equation:

$x^2 = 4$, substitute -2 first.

$(-2)^2 = 4$

$4 = 4$, so -2 is correct.

Now substitute 2.

$x^2 = 4$

$(2)^2 = 4$

$4 = 4$, so 2 is also correct.

As mentioned, the method we just used is called the graphical method. There is shorter method called the algebraic method. We are going to redo the question using the algebraic method and see if we get the same answer.

Algebraic method

Solve for x in the following equation: $x^2 = 4$

Remember that the aim is to remain with x alone on one side and not x^2. So we must find a way to get rid of the "power 2" on top of the x.

$x^2 = 4$

Step 1: Move the 4 to the left-hand side.

$x^2 - 4 = 0$

Step 2: Use the quadratic formula to find the values of x.

$x^2 - 4 = 0$ can be rewritten to $x^2 + 0x - 4 = 0$

Use the quadratic formula to find the values of x:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

We have $x^2 + 0x - 4 = 0$, so $a = 1, b = 0\ and\ c = -4$. Substitute these values into the quadratic formula.

$$x = \frac{-0 \pm \sqrt{0^2 - 4 \times 1 \times -4}}{2(1)}$$

$x = \frac{-0+\sqrt{0^2-4\times1\times-4}}{2(1)}$ or $x = \frac{-0-\sqrt{0^2-4\times1\times-4}}{2(1)}$

Use your calculator.

$x = 2$ or $x = -2$

The algebraic method has given us similar answers to those from the graphical method. Attempt the following examples.

Example: Solve for x using both methods

a) $x^2 = 9$

b) $x^2 = 16$

Solution

a)

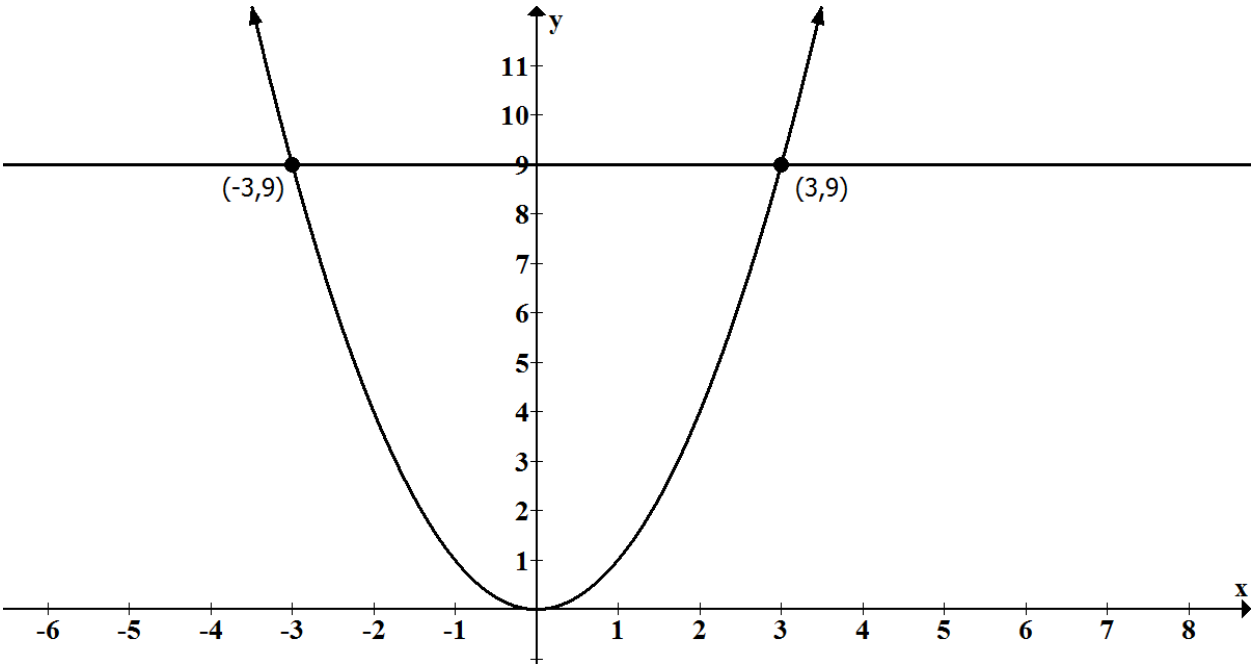
Graphical Method

$x^2 = 9$

Let's plot the graph of x^2. We start with the table of values and then plot it.

x	-3	-2	-1	0	1	2	3
y	9	4	1	0	1	4	9

Below is the graph of x^2 and the graph of 9.



Remember, the equation $x^2 = 9$ is basically asking you for the points where the graph of x^2 meets the graph of 9. As you can see from the diagram, the two graphs meet at points A and B. Point A is (-3 ; 9) and point B is (3 ; 9). These are the points where the two graphs have the same y-values, the same y-values means they are equal. So when an equation comes as $x^2 = 9$, it is just asking you to give the x-coordinates of the points where the graphs of x^2 and 9 have the same y-values. Such x-values are -3 and 3. So are our answers are

$x = -3 \; or \; x = 3$

Algebraic Method

$x^2 = 9$

Step 1: Move the 9 to the left-hand side.

$x^2 - 9 = 0$

Step 2: Use the quadratic formula to find the values of x.

$x^2 - 9 = 0$ can be rewritten to $x^2 + 0x - 9 = 0$

Use the quadratic formula to find the values of x:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

We have $x^2 + 0x - 9 = 0$, so $a = 1, b = 0\ and\ c = -9$. Substitute these values into the quadratic formula.

$$x = \frac{-0 \pm \sqrt{0^2 - 4 \times 1 \times -9}}{2(1)}$$

$x = \frac{-0+\sqrt{0^2-4\times1\times-9}}{2(1)}$ or $x = \frac{-0-\sqrt{0^2-4\times1\times-9}}{2(1)}$

Use your calculator.

$x = 3$ or $x = -3$

The algebraic method has given us similar answers to those from the graphical method.

b)

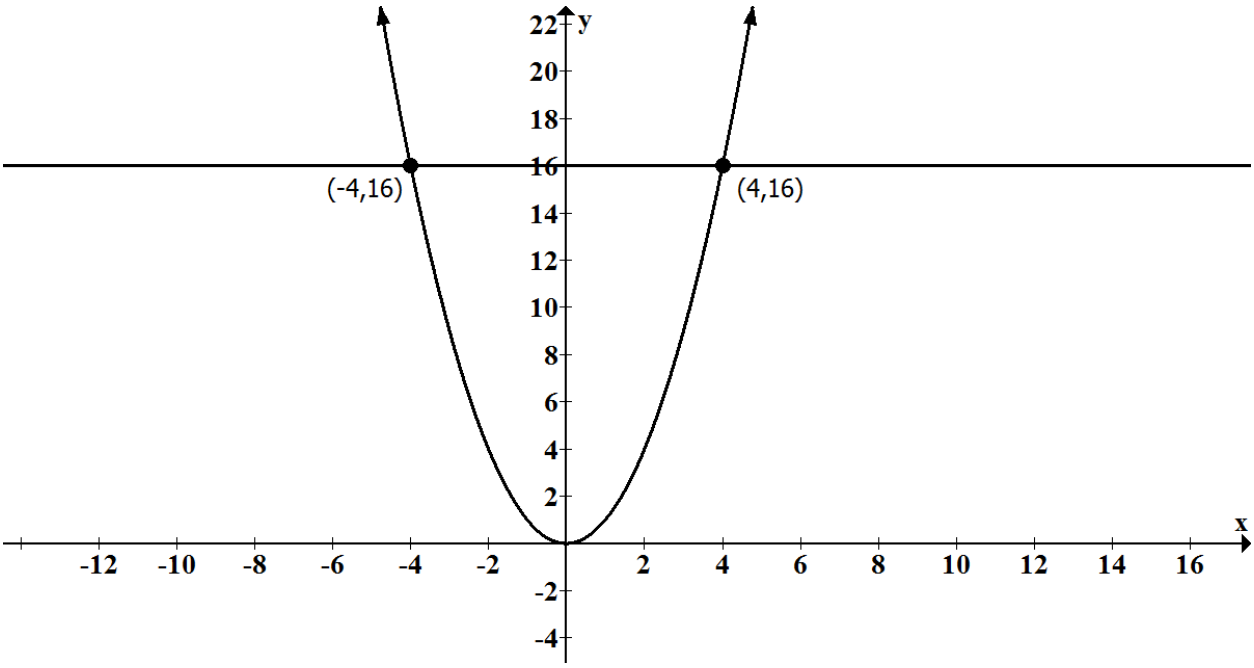
Graphical Method

$x^2 = 16$

Let's plot the graph of x^2. We start with the table of values and then plot it.

x	-3	-2	-1	0	1	2	3
y	9	4	1	0	1	4	9

Below is the graph of x^2 and the graph of 16.



Remember, the equation $x^2 = 16$ is basically asking you for the points where the graph of x^2 meets the graph of 16. As you can see from the diagram, the two graphs meet at points A and B. Point A is (-4 ; 16) and point B is (4 ; 16). These are the points where the two graphs have the same y-values, the same y-values means they are equal. So when an equation comes as $x^2 = 16$, it is just asking you to give the x-coordinates of the points where the graphs of x^2 and 16 have the same y-values. Such x-values are -4 and 4. So are our answers are

$x = -4 \; or \; x = 4$

Algebraic Method

$x^2 = 16$

Step 1: Move the 16 to the left-hand side.

$x^2 - 16 = 0$

Step 2: Use the quadratic formula to find the values of x.

$x^2 - 16 = 0$ can be rewritten to $x^2 + 0x - 16 = 0$

Use the quadratic formula to find the values of x:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

We have $x^2 + 0x - 16 = 0$, so $a = 1, b = 0 \; and \; c = -16$. Substitute these values into the quadratic formula.

$$x = \frac{-0 \pm \sqrt{0^2 - 4 \times 1 \times -16}}{2(1)}$$

$x = \frac{-0+\sqrt{0^2-4\times1\times-16}}{2(1)}$ or $x = \frac{-0-\sqrt{0^2-4\times1\times-16}}{2(1)}$

Use your calculator.

$x = 4$ or $x = -4$

Form 2 examples

Remember that form 1 is usually in a form like $x^2 + x = 0$

Just like form 1, there are two basic methods we can use, the graphical method and the algebraic method. We will start with the graphical method.

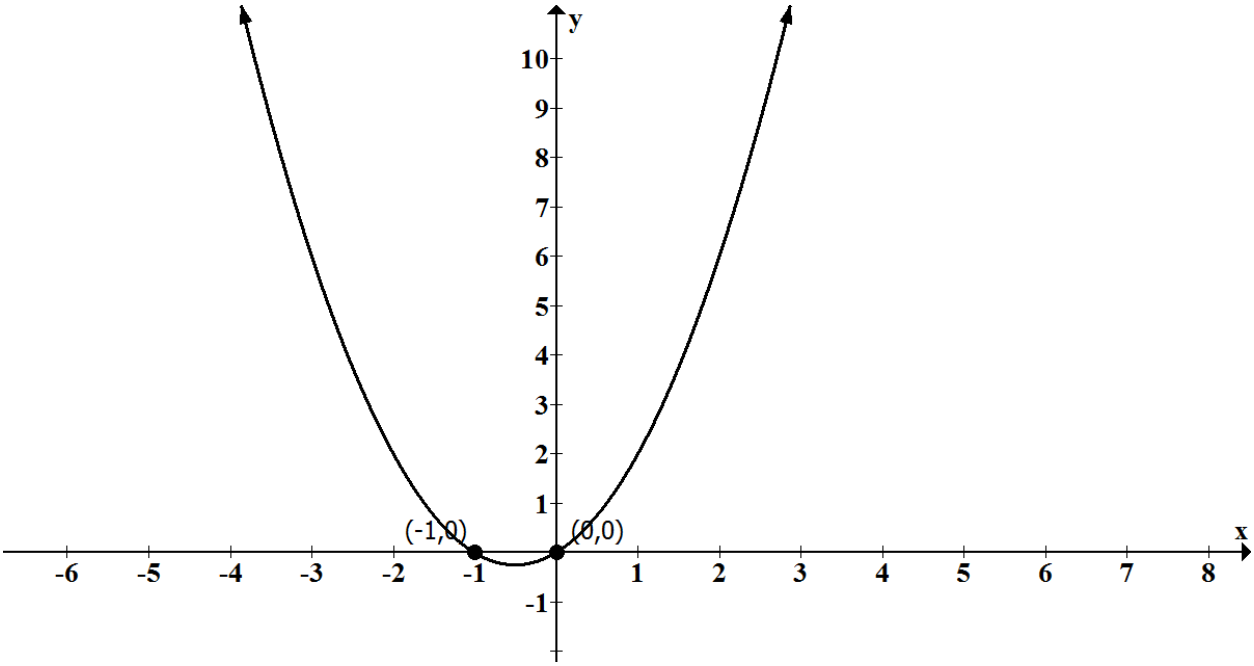
Graphical method

$x^2 + x = 0$

Let's plot the graph of $x^2 + x$. We start with the table of values and then plot it.

x	-3	-2	-1	0	1	2	3
y	6	1	-2	-3	-2	1	6

Below is the graph of $x^2 + x$ and the graph of 0, which is just the x-axis.



Remember, the equation $x^2 + x = 0$ is basically asking you for the points where the graph of x^2 meets the graph of 0 (x-axis). As you can see from the diagram, the two graphs meet at points A and B. Point A is (-1 ; 0) and point B is (0 ; 0). These are the points where the two graphs have the same y-values, the same y-values means they are equal. So when an equation comes as $x^2 + x = 0$, it is just asking you to give the x-coordinates of the points where the graphs of $x^2 + x$ and 0 have the same y-values. Such x-values are -1 and 0. So are our answers are

$x = -1 \; or \; x = 0$

Algebraic Method

$x^2 + x = 0$

Factor out x.

$x(x + 1) = 0$

$x = 0$ or $x + 1 = 0$

$x = 0$ or $x = -1$, gives the same answers as in the graphical method.

Question

a) $x^2 - x = 0$

b) $x^2 - 2x = 0$

c) $2x^2 - x = 0$

d) $4x^2 - 2x = 0$

e) $x^2 - \frac{2}{3}x = 0$

Solution

a) From now on, we are going to use the algebraic method because it is quicker.

Algebraic Method

$x^2 - x = 0$

Factor out x.

$x(x - 1) = 0$

$x = 0$ or $x - 1 = 0$

$x = 0$ or $x = 1$

b)

$x^2 - 2x = 0$

Factor out x.

$x(x - 2) = 0$

$x = 0$ or $x - 2 = 0$

$x = 0$ or $x = 2$

c)

$2x^2 - x = 0$

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Factor out x.

$x(2x-1)=0$

$x=0$ or $2x-1=0$

$x=0$ or $2x=1$

$x=0$ or $2\times x=1$

$x\times 2=1$

$x=1\div 2$

$x=0$ or $x=\frac{1}{2}$

d)

$4x^2-2x=0$

Factor common factor *2x*.

$2x(2x-1)=0$

$2x=0$ or $2x-1=0$

$x=0$ or $2\times x=1$

$x\times 2=1$

$x=1\div 2$

$x=0$ or $x=\frac{1}{2}$

e)

$x^2-\frac{2}{3}x=0$

Factor out x.

$x(x-\frac{2}{3})=0$

$x=0$ or $x-\frac{2}{3}=0$

$x=0$ or $x=\frac{2}{3}$

Under exam conditions, it is best to use the quadratic method. In general, any quadratic equation can be rearranged into the form $ax^2 + bx + c = 0$. Once in this form, you can then use the algebraic method to solve the equation. The main thing here is to ensure that the quadratic equation is in the above form.

Examples: Solve for x.

a) $2x^2 + 5x + 3 = 0$

b) $x^2 + 5x + 6 = 0$

c) $2x^2 + 3x + 1 = 0$

d) $x^2 + 6x = -3$

Solution

a) $2x^2 + 5x + 3 = 0$

The equation is in the form $ax^2 + bx + c = 0$, so we can use the quadratic formula to find the values of x:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

We have $2x^2 + 5x + 3 = 0$, so $a = 2, b = 5\ and\ c = 3$. Substitute these values into the quadratic formula.

$$x = \frac{-5 \pm \sqrt{5^2 - 4 \times 2 \times 3}}{2(2)}$$

$$x = \frac{-5 + \sqrt{5^2 - 4 \times 2 \times 3}}{2(2)} \text{ or } \quad x = \frac{-5 - \sqrt{5^2 - 4 \times 2 \times 3}}{2(2)}$$

Use your calculator.

$$x = -1 \text{ or } x = -\frac{3}{2}$$

b) $x^2 + 5x + 6 = 0$

The equation is in the form $ax^2 + bx + c = 0$, so we can use the quadratic formula to find the values of x:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

We have $1x^2 + 5x + 6 = 0$, so $a = 1, b = 5\ and\ c = 6$. Substitute these values into the quadratic formula.

$x = \frac{-5 \pm \sqrt{5^2 - 4 \times 1 \times 6}}{2(1)}$

$x = \frac{-5 + \sqrt{5^2 - 4 \times 1 \times 6}}{2(1)}$ or $x = \frac{-5 - \sqrt{5^2 - 4 \times 1 \times 6}}{2(1)}$

$x = -2$ or $x = -3$

c) $2x^2 + 3x + 1 = 0$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

We have $2x^2 + 3x + 1 = 0$, so $a = 2, b = 3\ and\ c = 1$. Substitute these values into the quadratic formula.

$x = \frac{-3 \pm \sqrt{3^2 - 4 \times 2 \times 1}}{2(2)}$

$x = \frac{-3 + \sqrt{3^2 - 4 \times 2 \times 1}}{2(2)}$ or $x = \frac{-3 - \sqrt{3^2 - 4 \times 2 \times 1}}{2(2)}$

$x = -\frac{1}{2}$ or $x = -1$

d) $x^2 + 6x = -3$

You should firstly rearrange the equation to the form $x^2 + bx + c = 0$. We do so by moving the -3 to the left-hand side.

$x^2 + 6x + 3 = 0$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

We have $1x^2 + 6x + 3 = 0$, so $a = 1, b = 6\ and\ c = 3$. Substitute these values into the quadratic formula.

$$x = \frac{-6 \pm \sqrt{6^2 - 4 \times 1 \times 3}}{2(1)}$$

$x = \frac{-6 + \sqrt{6^2 - 4 \times 1 \times 3}}{2(1)}$ or $x = \frac{-6 - \sqrt{6^2 - 4 \times 1 \times 3}}{2(1)}$

$x = -3 + \sqrt{6}$ or $x = -3 - \sqrt{6}$, $x = -0.55$ or $x = -5.45$

Exponential Equations

Consider the following equation.

$2^{\square} = 2^3$

What number must we put in the empty box so that the two sides are equal? Well, the number is obviously 3.

$2^{\mathbf{3}} = 2^3$

However, such a question is asked differently in mathematics. The above question could've been asked as follows:

Solve for *x* in the equation $2^x = 2^3$. This question is similar to $2^{\square} = 2^3$, we just replaced the empty box with an *"x".*

$2^x = 2^3$. The answer is clearly 3, so we say that *x = 3*. Attempt the following questions to check if you have understood the idea so far.

Solve for x

a) $3^x = 3^{\frac{1}{2}}$ b) $2^x = 2^3$ c) $2^x = 8$ d) $2^{2x} = 2^3$

Solution

a) $3^x = 3^{\frac{1}{2}}$

$\boxed{\textit{If } \boldsymbol{a^x = a^y}\textit{, then } \boldsymbol{x = y}}$

When apply the above rule, our answer is $x = \frac{1}{2}$

b) $2^x = 2^3$

$\boxed{\textit{If } \boldsymbol{a^x = a^y}\textit{, then } \boldsymbol{x = y}}$

When apply the above rule, our answer is $x = 3$

c) $2^x = 8$, put 8 into its exponential form, $8 = 2^3$

$2^x = 2^3$ $\boxed{\textit{If } \boldsymbol{a^x = a^y}\textit{, then } \boldsymbol{x = y}}$

When apply the above rule, our answer is $x = 3$

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d) $2^{2x} = 2^3$

If $\boldsymbol{a^x = a^y}$ ***, then*** $\boldsymbol{x = y}$

When apply the above rule, our answer is $2x = 3$

$2x = 3$, to get rid of the "2" in front of "*x*", divide both sides by 2.

$$\frac{2x}{2} = \frac{3}{2}$$

$$\frac{\cancel{2}x}{\cancel{2}} = \frac{3}{2}$$

$$x = \frac{3}{2}$$

Question

a) $2^{x+2} = 2^5$ b) $2^{x+2} = 2^{5x+3}$ c) $5^{2x+5} = 25^{3x}$

d) $\left(\frac{1}{2}\right)^{x+2} = 2^{5x+3}$ e) $2^x = \frac{1}{2}$ f) $3^{x+1} = \left(\frac{1}{3}\right)^{5x}$

Solution

a) $2^{x+2} = 2^5$

$x + 2 = 5$

$x = 5 - 2$

$x = 3$

b) $2^{x+2} = 2^{5x+3}$

$x + 2 = 5x + 3$

, keep the *x's* on the left-hand side, and the numbers on the right-hand side.

$x - 5x = 3 - 2$

$-4x = 1$

, to get rid of the "-4" in front of "*x*", divide both sides by -4.

$$\frac{-4x}{-4} = \frac{1}{-4}$$

$$\frac{\cancel{-4}x}{\cancel{-4}} = \frac{1}{-4}$$

$$x = -\frac{1}{4}$$

c) $5^{2x+5} = 25^{3x}$

$5^{2x+5} = (5^2)^{3x}$

$5^{2x+5} = 5^{2\times 3x}$

$5^{2x+5} = 5^{6x}$

$2x + 5 = 6x$

, keep the x's on the left-hand side, and the numbers on the right-hand side.

$2x - 6x = -5$

$-4x = -5$

, to get rid of the "-4" in front of "*x*", divide both sides by -4.

$$\frac{-4x}{-4} = \frac{-5}{-4}$$

$$\frac{\cancel{-4}x}{\cancel{-4}} = \frac{-5}{-4}$$

$x = \frac{5}{4}$

d) $\left(\frac{1}{2}\right)^{x+2} = 2^{5x+3}$

We first change the $\frac{1}{2}$ to 2^{-1}. We know this from the rule: $\frac{1}{a} = a^{-1}$

$(2^{-1})^{x+2} = 2^{5x+3}$

$2^{-1(x+2)} = 2^{5x+3}$

$2^{-1x-2} = 2^{5x+3}$

If $\boldsymbol{a^x = a^y}$ ***, then*** $\boldsymbol{x = y}$

$-1x - 2 = 5x + 3$

$-1x - 5x = 3 + 2$

$-6x = 5$

, to get rid of the "-6" in front of "*x*", divide both sides by -6.

$\frac{-6x}{-6} = \frac{5}{-6}$

$\frac{\cancel{-6}x}{\cancel{-6}} = \frac{5}{-6}$

$x = -\frac{5}{6}$

e) $2^x = \frac{1}{2}$

$2^x = 2^{-1}$

$x = -1$

f) $3^{x+1} = \left(\frac{1}{3}\right)^{5x}$

We first change the $\frac{1}{3}$ to 3^{-1}. We know this from the rule: $\frac{1}{a} = a^{-1}$

$3^{x+1} = (3^{-1})^{5x}$

$3^{x+1} = 3^{-5x}$

$x + 1 = -5x$

If $\boldsymbol{a^x = a^y}$ ***, then*** $\boldsymbol{x = y}$

$x + 5x = -1$

$-6x = -1$

, to get rid of the "-6" in front of "*x*", divide both sides by -6.

$\frac{-6x}{-6} = \frac{-1}{-6}$

$\frac{\cancel{-6}x}{\cancel{-6}} = \frac{1}{-6}$

$x = -\frac{1}{6}$

Linear Inequalities

Attempt the example. Solve for x in the following inequality.

$$x + 2 > 3$$

You solve this in a similar way to the equations. The aim is to remain with x alone on one side, so move the 2 to the right-hand side.

$x + 2 > 3$

$x > 3 - 2$

$x > 1$, that's the answer.

Question

a) $x + 2 > 10$

b) $x - 5 < 3$

Solution

a) $x + 2 > 10$

$x > 10 - 2$

$x > 8$

b) $x - 5 < 3$

$x < 3 + 5$

$x < 8$

Check out the example below.

$2x > 4$

$2 \times x > 4$

$x \times 2 > 4$

$x > 4 \div 2$

$x > 2$

The only difference between inequalities and equations is the treatment of the "negative" when we multiply or divide. For example, solving the inequality $-2x > 4$.

$-2x > 4$

$-2 \times x > 4$

$x \times -2 > 4$

At this point, we know that the next step is to move the -2 to the right-hand side. When a negative number moves from one side of an inequality to the other side, such that it divides or multiplies the other side, the direction of the "inequality" changes.

$x \times -2 > 4$

$x < 4 \div -2$

$x < -2$

Another example

$\frac{x}{-2} > 4$

$x \div -2 > 4$

$x < 4 \times -2$

$x < -8$

Question: Solve the inequalities

a) $x \times -2 > 4$ vs $x - 2 > 4$

b) $x \div -2 > 4$ vs $x - 2 > 4$

Solution

a) $x \times -2 > 4$ vs $x - 2 > 4$

$x < 4 \div -2$ $x > 4 + 2$

$x < -2$ $x > 6$

This was just to show you that the direction only changes when the moving negative has a "×" or "÷" in front it.

b)

$x \div -2 > 4$	vs	$x - 2 > 4$
$x < 4 \times -2$		$x > 4 + 2$
$x < -8$		$x > 6$

This was just to show you that the direction only changes when the moving negative has a "×" or "÷" in front it.

Note that when the moving number is positive, the direction of the sign doesn't change. Check out the example below.

$2x > 4$

$2 \times x > 4$

$x \times 2 > 4$

$x > 4 \div 2$

$x > 2$

The direction doesn't change because the moving number (2) is positive.

Note that we treat "≥" and "≤" in exactly the same way we do ">" and "<".

Question

a) $x + 5 < 8$

b) $3x + 3 > 12$

c) $3x + 3 \geq 12$

d) $-x + 7 \leq 5x + 31$

Solution

a) $x + 5 < 8$

$x < 8 - 5$

$x < 3$

b) $3x + 3 > 12$

$3x > 12 - 3$

$3x > 9$

$3 \times x > 9$

$x \times 3 > 9$

$x > 9 \div 3$

$x > 3$

c) $3x + 3 \geq 12$

$3x \geq 12 - 3$

$3x \geq 9$

$3 \times x \geq 9$

$x \times 3 \geq 9$

$x \geq 9 \div 3$

$x \geq 3$

d) $-x + 7 \leq 5x + 31$

Firstly, group the x's on one and the numbers on the other side.

$-x - 5x \leq 31 - 7$

$-6x \leq 24$

$-6 \times x \leq 24$

$x \times -6 \leq 24$

$x \geq 24 \div -6$

$x \geq 4$

Sometimes you may get two inequalities in "one".

For example, solve for x in the following inequality.

$1 < x + 2 < 3$

Remember, the aim is to have x left "alone". In this case, we must somehow get rid of the $+2$ next to the x.

$1 < x + 2 < 3$

The only way to get rid of the $+2$ is to put -2 next to it, so that we have $+2 - 2$, which is 0. But as you know, what you do on the left-hand side you also do on the right-hand side, so you would also have to put -2 on the other sides too.

$1 - 2 < x + 2 - 2 < 3 - 2$

$-1 < x + 0 < 1$

$-1 < x < 1$

Try $2 < x + 3 < 5$

To get rid of the $+3$ next to x, put -3. You also do the same for the other sides.

$2 - 3 < x + 3 - 3 < 5 - 3$

$-1 < x + 0 < 2$

$-1 < x < 2$

Try $2 < x - 3 < 5$

To get rid of the -3 next to x, put $+3$. The number you put must have the opposite sign. You also do the same for the other sides.

$2 + 3 < x - 3 + 3 < 5 + 3$

$5 < x + 0 < 8$

$5 < x < 8$

Try $1 < x - 0.5 < 4$

To get rid of the -0.5 next to x, put $+0.5$. The number you put must have the opposite sign. You also do the same for the other sides.

$1 + 0.5 < x - 0.5 + 0.5 < 4 + 0.5$

$1.5 < x + 0 < 4.5$

$1.5 < x < 4.5$

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Note that the same method applies to "≥" and "≤".

There's also multiplication and division involved in such inequalities. For example, $4 < 2x < 6$

Remember that the aim is to be left with "x" on its own and not "$2x$". In this case, we should find a way to get rid of the "2" in front of the "x".

$$4 < 2x < 6$$

As with equations, you divide $2x$ by 2 to get rid of the 2 in front of x.

$4 < 2x < 6$

$\frac{4}{2} < \frac{2x}{2} < \frac{6}{2}$

Remember that what you do on the left-hand side you also do on the right-hand side. So we divide all the sides by **2**.

$\frac{4}{2} < \frac{\cancel{2}x}{\cancel{2}} < \frac{6}{2}$

$2 < 1x < 3$

$2 < x < 3$

Try doing $4 < -2x < 6$

$4 < -2x < 6$

$\frac{4}{-2} < \frac{-2x}{-2} < \frac{6}{-2}$

$\frac{4}{-2} < \frac{\cancel{-2}x}{\cancel{-2}} < \frac{6}{-2}$

$-2 < 1x < -3$

$-2 < x < -3$

Example: Solve for x in the inequality $3 < \frac{x}{2} < 5$

Solution

$$3 < \frac{x}{2} < 5$$

Note that this one involves division instead of multiplication. The aim is to get rid of the "2" that "divides" the x. To get rid of the 2, multiply $\frac{x}{2}$ by 2. We also do the same for the 3 and 5.

$$3 < \frac{x}{2} < 5$$

$$3 \times 2 < \frac{x}{2} \times 2 < 5 \times 2$$

$$6 < \frac{x}{2} \times 2 < 10$$

$$6 < \frac{2x}{2} < 10$$

$$6 < \frac{2x}{2} < 10$$

$$6 < x < 10$$

Example: Solve for x in the inequality $3 < \frac{x}{4} < 5$

Solution

$$3 < \frac{x}{4} < 5$$

Note that this one involves division instead of multiplication. The aim is to get rid of the "4" that "divides" the x. To get rid of the 4, multiply $\frac{x}{4}$ by 4. We also do the same for the 3 and 5.

$$3 < \frac{x}{4} < 5$$

$$3 \times 4 < \frac{x}{4} \times 4 < 5 \times 4$$

$$12 < \frac{x}{4} \times 4 < 20$$

$$12 < \frac{4x}{4} < 20$$

$$12 < \frac{4x}{4} < 20$$

$$12 < x < 20$$

Example: Solve for x in the inequality $3 < \frac{3}{2}x < 5$

Solution

$$3 < \frac{3}{2}x < 5$$

$$3 < \frac{3}{2} \times x < 5$$

$$3 < x \times \frac{3}{2} < 5$$

The aim is to get rid of the $\frac{3}{2}$ in front of the x. To get rid of the $\frac{3}{2}$, multiply $x \times \frac{3}{2}$ by $\frac{2}{3}$. We also do the same for the 3 and 5.

$$3 \times \frac{2}{3} < x \times \frac{3}{2} \times \frac{2}{3} < 5 \times \frac{2}{3}$$

$$2 < x \times 1 < \frac{10}{3}$$

$$2 < x < \frac{10}{3}$$

We multiplied by $\frac{2}{3}$ because it cancels out $\frac{3}{2}$. The whole idea is to multiply with the "opposite" of $\frac{3}{2}$. Apply the same idea in the next example.

Example: Solve for x in the inequality $3 < \frac{4}{5}x < 5$

Solution

$$3 < \frac{4}{5}x < 5$$

$$3 < \frac{4}{5} \times x < 5$$

$$3 < x \times \frac{4}{5} < 5$$

The aim is to get rid of the $\frac{4}{5}$ in front of the x. To get rid of the $\frac{4}{5}$, multiply $x \times \frac{4}{5}$ by $\frac{5}{4}$. We also do the same for the 3 and 5.

$$3 \times \frac{5}{4} < x \times \frac{4}{5} \times \frac{5}{4} < 5 \times \frac{5}{4}$$

$$\frac{15}{4} < x \times 1 < \frac{25}{4}$$

$$\frac{15}{4} < x < \frac{25}{4}$$

Questions

a) $\frac{1}{2} < x + 1 < 3$
b) $-2 < 2x \leq 4$
c) $3 < -2x \leq 6$
d) $-1 < \frac{x}{6} < 6$
e) $-1 < 2x + 3 < 6$

Solution

a) $\frac{1}{2} < x + 1 < 3$

$$\frac{1}{2} - 1 < x + 1 - 1 < 3 - 1$$

$$-\frac{1}{2} < x + 0 < 2$$

$$-\frac{1}{2} < x < 2$$

b) $-2 < 2x \leq 4$

$$\frac{-2}{2} < \frac{2x}{2} \leq \frac{4}{2}$$

$$\frac{-2}{2} < \frac{\cancel{2}x}{\cancel{2}} < \frac{4}{2}$$

$-1 < x < 2$

c) $3 < -2x \leq 6$

$\frac{3}{-2} < \frac{-2x}{-2} \leq \frac{6}{-2}$

$-\frac{3}{2} < \frac{-\cancel{2}x}{-\cancel{2}} < \frac{6}{-2}$

$-\frac{3}{2} > 1x > -3$

$-3 < x < -\frac{3}{2}$

d) $-1 < \frac{x}{6} < 6$

$-1 < x \div 6 < 6$

$-1 \times 6 < x < 6 \times 6$

$-6 < x < 36$

e) $-1 < 2x + 3 < 6$

$-1 - 3 < 2x + 3 - 3 < 6 - 3$

$-4 < 2x + 0 < 3$

$-4 < 2x < 3$

$\frac{-4}{2} < \frac{2x}{2} \leq \frac{3}{2}$

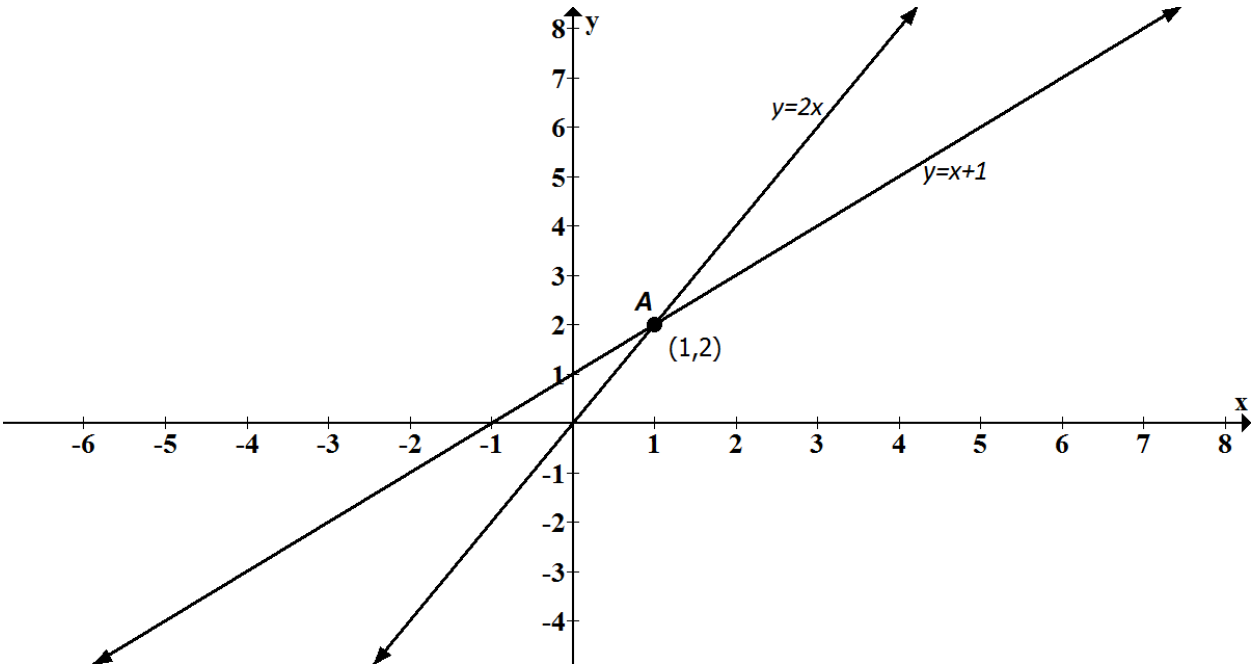
$\frac{-4}{2} < \frac{\cancel{2}x}{\cancel{2}} < \frac{3}{2}$

$-2 < 1x < \frac{3}{2}$

$-2 < x < \frac{3}{2}$

Simultaneous equations

This is a very easy section. Consider the two graphs; $y = x + 1$ and $y = 2x$. Look at the graphs and write the coordinates of the point at which the two graphs meet.



The coordinates are $(1; 2)$. So the *x-coordinate* is $x = 1$ and *y-coordinate* is $y = 2$.

With simultaneous equations, they ask you to find the coordinates of where the graphs meet but ask it differently. The question would go like this;

Solve for x and y simultaneously if $y - x = 1$ and $y = 2x$.

We are going to use a so-called "Substitution method".

Substitute one of the equations into the other.

We have $y - x = 1$ and $y = 2x$

Since $y = 2x$, substitute it into the first equation on y.

$2x - x = 1$

$1x = 1$

$x = 1$

We've found the x-value. We now have to find the y-value. We do so by substituting the $x = 1$ into the second equation.

$y = 2x$

$y = 2(1)$

$y = 2$, We have found the y-value too.

Therefore, we have $x = 1$ and $y = 2$.

This means that the graphs of the $y - x = 1$ and $y = 2x$ meet the point $(1; 2)$

We got the same answer when we used the graphical method. That's how we use the substitution method. Improve your understanding by attempting the following examples.

Example: Solve for x and y simultaneously if $y - x = 3$ and $y = 4x$

Substitute one of the equations into the other.

We have $y - x = 3$ and $y = 4x$

Since $y = 4x$, substitute it into the first equation on y.

$4x - x = 1$

$3x = 1$

$x = \frac{1}{3}$

We've found the x-value. We now have to find the y-value. We do so by substituting the $x = \frac{1}{3}$ into the second equation.

$y = 2x$

$y = 2 \times \frac{1}{3}$

$y = \frac{2}{3}$, We have found the y-value too.

Therefore, we have
$x = \frac{1}{3}$ and $y = \frac{2}{3}$

This means that the graphs of the $y - x = 3$ and $y = 4x$ meet the point $\left(\frac{1}{3}; \frac{2}{3}\right)$

Functions

Linear Function

In this section we are going to learn more about the behavior of the linear function, reflection about the axes, shift, y-cut, domain and range.

Reflection through the y-axis

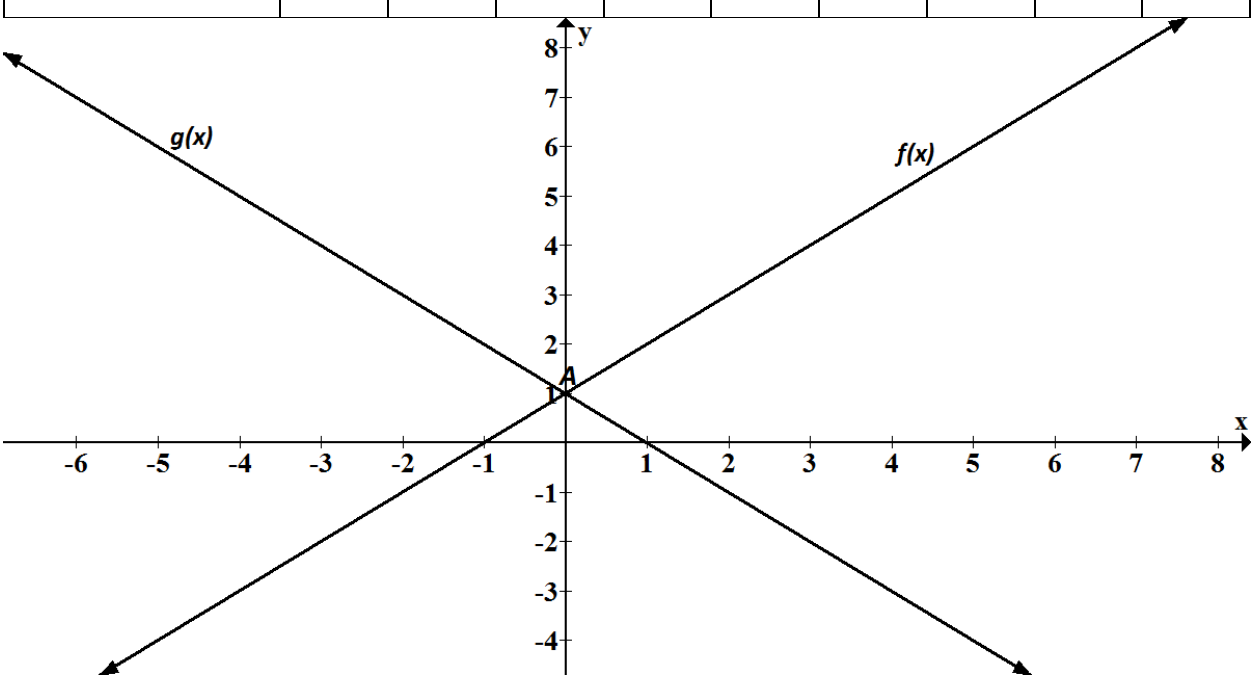
Compare the graph $y = x + 1 \;\; vs \;\; y_1 = -x + 1$

Let's name $y = x + 1$ function $f(x)$, so $f(x) = x + 1$,

Let's name $y_1 = -x + 1$ function $g(x)$, so $g(x) = -x + 1$,

Below we fill in the table of values for both functions and draw their graphs as well.

x	-4	-3	-2	-1	0	1	2	3	4
$f(x) = x + 1$	-3	-2	-1	0	1	2	3	4	5
$g(x) = -x + 1$	5	4	3	2	1	0	-1	-2	-3



Look at the part of $f(x)$ in the first quadrant until point *A* (in yellow). Look at the part of $g(x)$ in the second quadrant until point A. Notice that the part of $g(x)$ in the second quadrant is a mirror reflection of the part of $f(x)$ in the first quadrant, through the *y-axis*.

Again, look at the part of $f(x)$ in the third quadrant until point A. Look at the part of $g(x)$ in the fourth quadrant until point A, notice that the part of $g(x)$ in the fourth quadrant is a mirror reflection of the part of $f(x)$ in the third quadrant.

Therefore we can safely conclude that the function $g(x) = -x + 1$ is mirror reflection of the function $f(x) = x + 1$ through the *y-axis* (the *y-axis is the mirror)*.

Notice that we could've concluded that without having to draw the graph. We see that by comparing the functions.

$f(x) = x + 1 \qquad vs \qquad g(x) = -x + 1$

$f(x) = x + 1$ is our original function; to get $g(x)$, we just place the "$-$" in front of the "x" in $f(x) = x + 1$. See that below.

$$x + 1 \rightarrow -x + 1$$

$$f(x) \rightarrow g(x)$$

Thus every time we place a "$-$" in front of the "x" in a linear function, just know we are basically reflecting it through the *y-axis*.

Example

Consider the function $f(x) = x + 2$

Let's put a "$-$" in front of the x to get $g(x)$

$x + 2 \rightarrow -x + 2$

So our $g(x)$ is $g(x) = -x + 2$

Now let's check if this is true by drawing the graphs of $f(x) = x + 2 \quad vs \quad g(x) = x + 2$

x	-4	-3	-2	-1	0	1	2	3	4
$f(x) = x + 2$	-2	-1	0	1	2	3	4	5	6
$g(x) = -x + 2$	6	5	4	3	2	1	0	-1	-2

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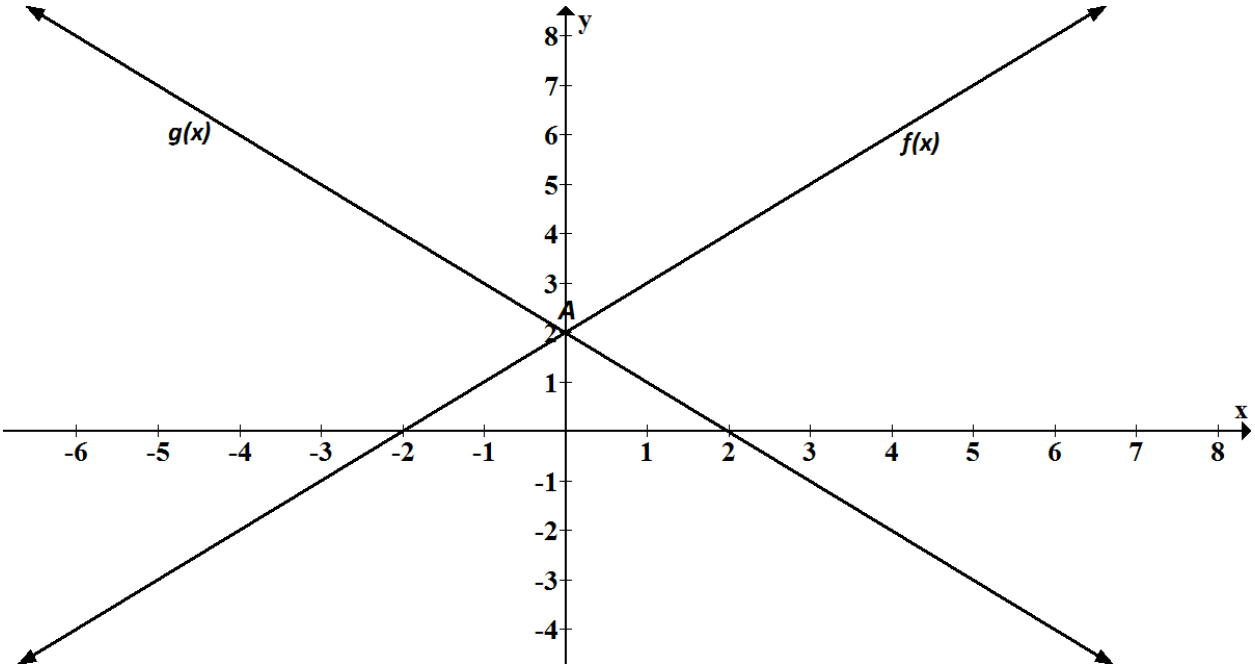


We can clearly see that the $g(x) = -x + 2$ is a mirror reflection of $f(x) = x + 2$ through the *y-axis.* The part of $g(x)$ in the second quadrant is a mirror reflection of the part of $f(x)$ in the first quadrant. The part of $g(x)$ in the fourth quadrant is a mirror reflection of the part of $f(x)$ in the third quadrant, that's why we say $g(x)$ is a mirror reflection of $f(x)$ (the *y-axis is the mirror)*. So this proves us right. Every time we put a "$-$" in front of "x" in a linear function we are basically reflecting through the *y-axis*.

Example

Consider the function $f(x) = x + 3$

Let's put a "$-$" in front of the x to get $g(x)$

$x + 3 \rightarrow -x + 3$

Our $g(x)$ is $g(x) = -x + 3$.

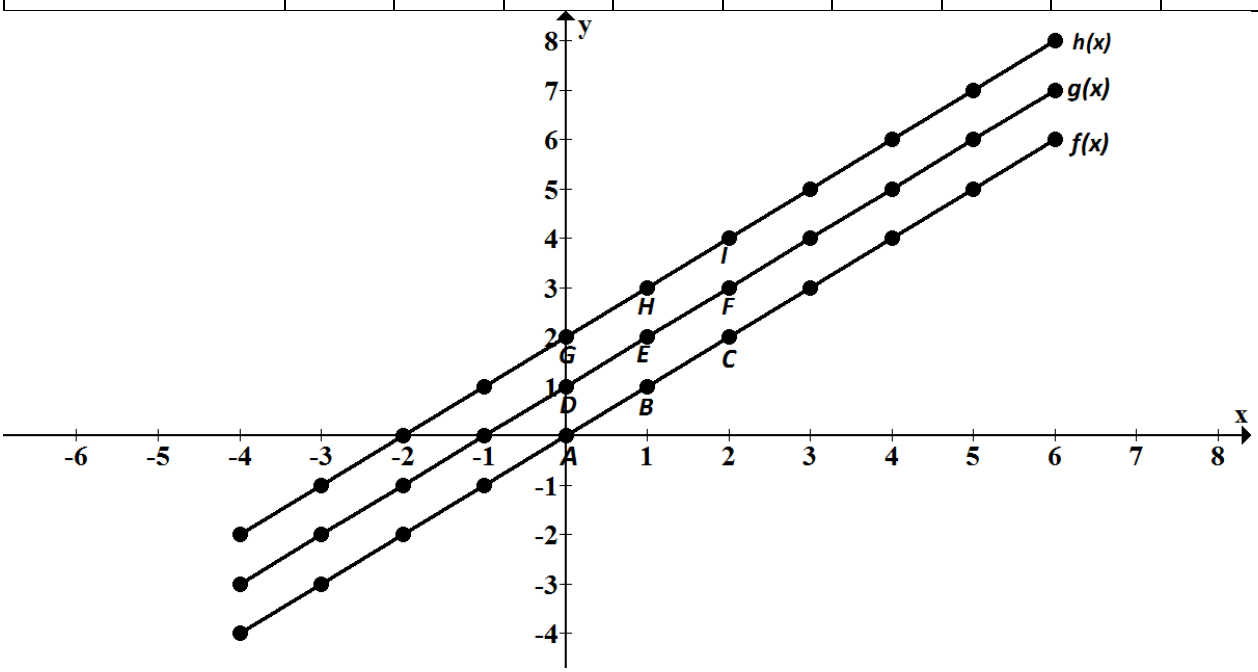
We can clearly see that the $g(x) = -x + 3$ is a mirror reflection of $f(x) = x + 3$ through the *y-axis.*

<u>The y-cut</u>

Compare the graph $f(x) = x$, $g(x) = x + 1$, $h(x) = x + 2$

Below we fill in the table of values for each of the functions and draw the representation graph.

x	-4	-3	-2	-1	0	1	2	3	4
$f(x) = x$	-4	-3	-2	-1	0	1	2	3	4
$g(x) = x + 1$	-3	-2	-1	0	1	2	3	4	5
$h(x) = x + 2$	-2	-1	0	1	2	3	4	5	6



Notice that $f(x)$ crosses the *y-axis* at 0, $g(x)$ crosses the *y-axis* at 1 and $h(x)$ crosses the y-axis at 2. These are called the *y-cuts* or the *y-intercepts*.

What do you notice on each of the functions below?

$f(x) = x = x + 0$

$g(x) = x + 1$

$h(x) = x + 2$

The number at the end represents the ***y-cut***.

In general, a linear function exists in the form $y = mx + c$, where is c is the *y-cut*.

Question

Write down the *y-cut (y-intercept)* of each of the following functions.

a) $f(x) = x + 6$
b) $g(x) = x - 8$

c) $k(x) = -x + \frac{1}{2}$

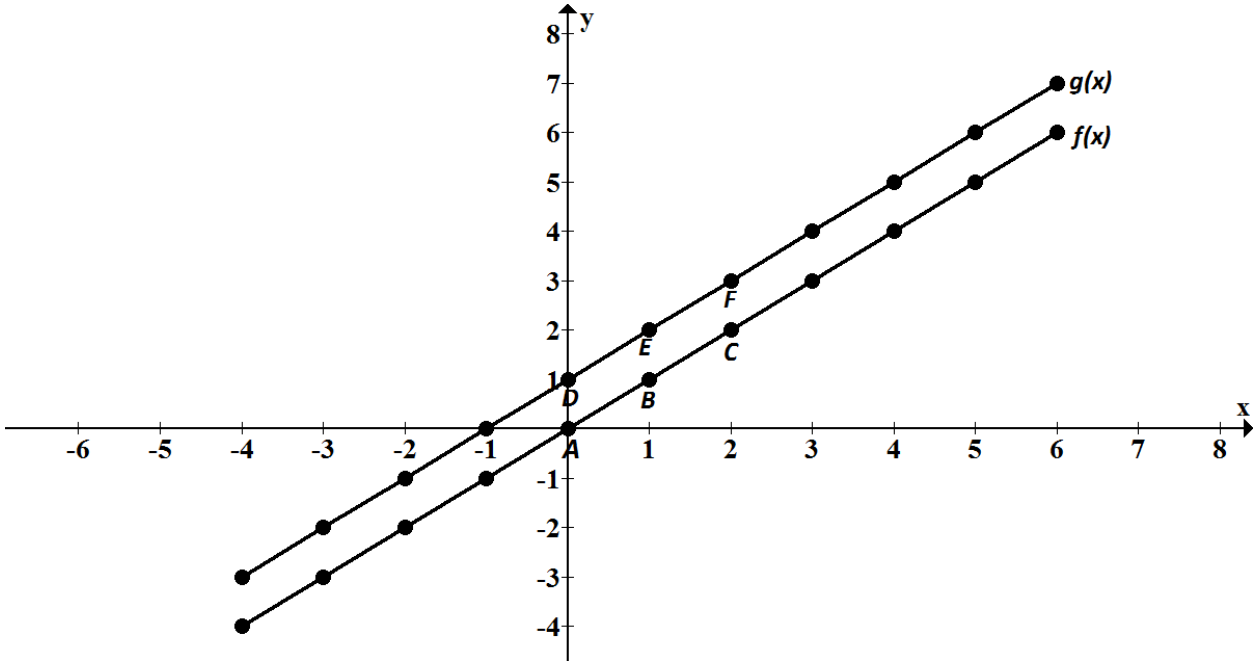
Solution

a) 8
b) -8
c) $\frac{1}{2}$

Translation (movement of the graph)

Consider the functions $f(x) = x$, $g(x) = x + 1$

x	-4	-3	-2	-1	0	1	2	3	4
$f(x) = x$	-4	-3	-2	-1	0	1	2	3	4
$g(x) = x + 1$	-3	-2	-1	0	1	2	3	4	5



If you were to walk from point A to point D, you'll have to walk 1 unit upwards. In fact, if you were to walk from any point on $f(x)$ to the corresponding point on $g(x)$, you'd have to walk 1 unit upwards. To see this clearer, let's compare these corresponding points, let's take points A, *B, C* from $f(x)$ and points *D, E and F* from $g(x)$.

$f(x)$: *A(0 ; 0)* *B(1 ; 1)* *C(2 ; 2)*

$g(x)$: *D(0 ; 1)* *E(1 ; 2)* *F(2 ; 3)*

Firstly, notice that when you move from point A to point D, the *x-coordinate* **does not** change, it remains 0, only the *y-coordinate* changes.

A(0;0)

D(0;1)

The *y-coordinate* changes from 0 to 1. So we basically added 1

Secondly, notice that when you move from point B to point E, the *x-coordinate* **does not** change, it remains 1, only the *y-coordinate* changes.

B(1;1)

E(1;2)

The *y-coordinate* changes from 1 to 2. So we basically added 1 to 1

Lastly, notice that when you move from point C to point F, the *x-coordinate* **does not** change, it remains 2, only the *y-coordinate* changes.

C(2;2)

F(2;3)

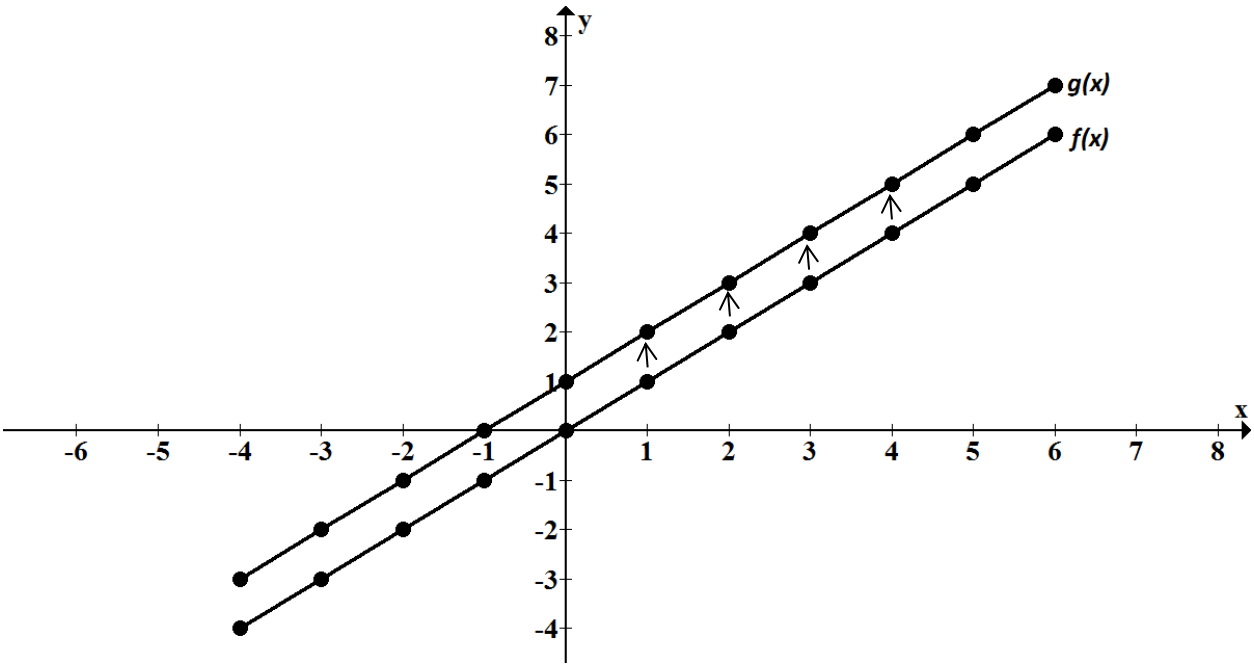
The *y-coordinate* changes from 2 to 3. So we basically added 1 to 2

From the points in $f(x) = x$, we added 1 to the *y-coordinate* to get the point on $g(x) = x + 1$. In other words, to get the graph on $g(x) = x + 1$, we simply move the graph of $f(x) = x$ 1 unit upwards. We can even see that from their functions below.

$f(x) = x \rightarrow g(x) = x + 1$

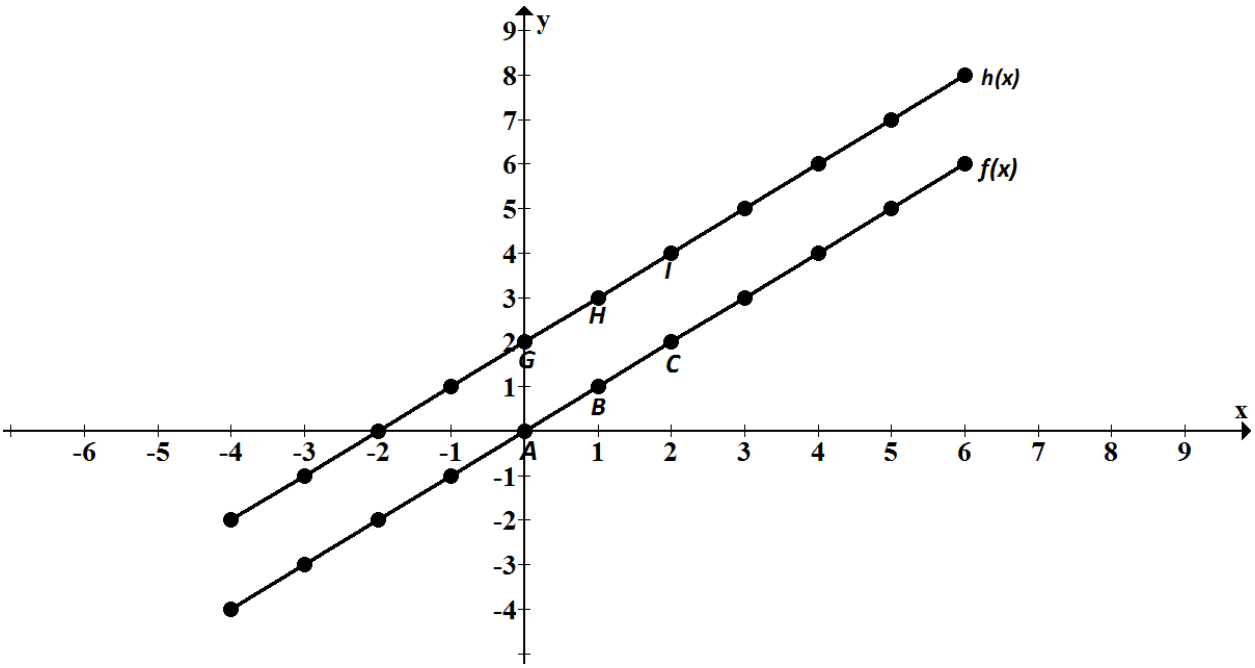
We just added 1 to x to get $x + 1$.

Mathematicians call this translation, the movement of $f(x)$ 1 unit upwards to get $g(x)$.



Now let's see how we can also translate $f(x) = x$ to $h(x) = x + 2$

x	-4	-3	-2	-1	0	1	2	3	4
$f(x) = x$	-4	-3	-2	-1	0	1	2	3	4
$h(x) = x + 2$	-2	-1	0	1	2	3	4	5	6



$f(x)$: A(0 ; 0) B(1 : 1) C(2 : 2)

$h(x)$: G(0 : 2) H(1 : 3) I(2 : 4)

Firstly, notice that when you move from point A to point G, the *x-coordinate* **does not** change, it remains 0, only the *y-coordinate* changes.

A(0;0)

G(0;2)

The *y-coordinate* changes from 0 to 2.
So we basically added 2 to 0

Secondly, notice that when you move from point B to point H, the *x-coordinate* **does not** change, it remains 1, only the *y-coordinate* changes.

B(1;1)

H(1;3)

The *y-coordinate* changes from 1 to 3.
So we basically added 2 to 1

Lastly, notice that when you move from point C to point I, the *x-coordinate* **does not** change, it remains 2, only the y-*coordinate* changes.

C(2;2)

I(2;4)

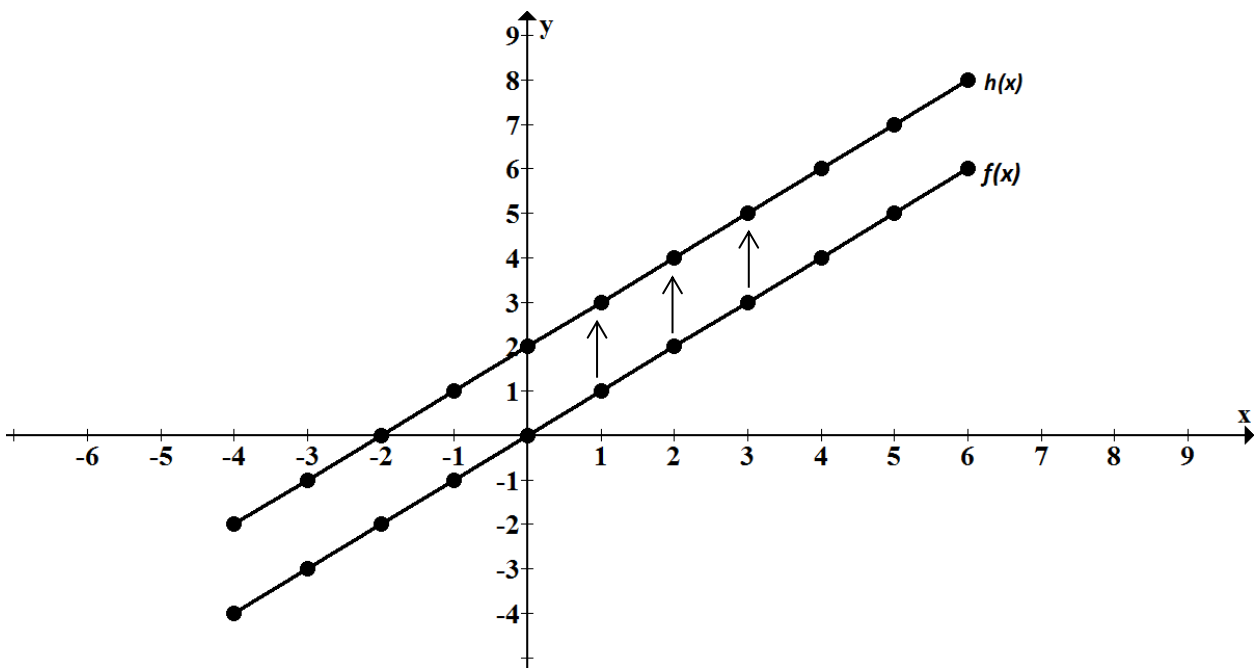
The *y-coordinate* changes from 2 to 4.
So we basically added 2 to 2

From the points on $f(x) = x$, we added 2 to the y-coordinate to get to get the point $h(x) = x + 2$

In other words, to get the graph of $h(x) = x + 2$, we simply move the graph of $f(x) = x \quad 2$ units upwards. We can also see this from the functions below.

$f(x) = x \rightarrow h(x) = x + 2$

We just added 1 to x to get $x + 2$.

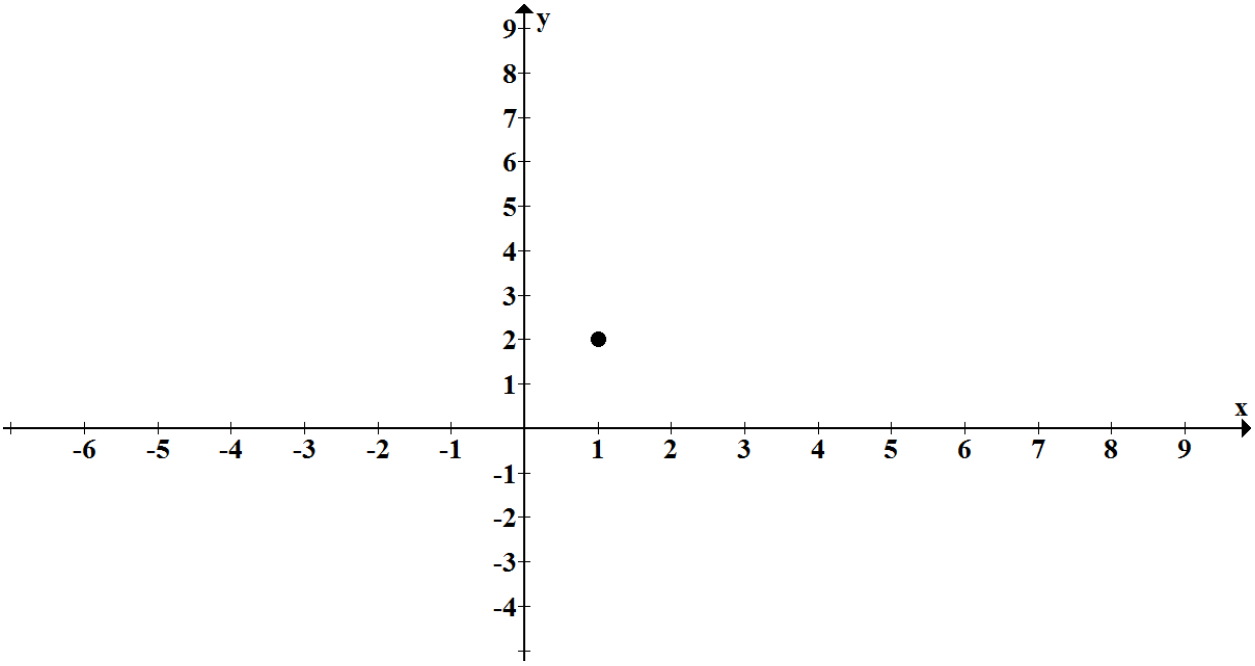


Domain

In the section, we are going to learn about the so called ***domain*** of a function. For now, I am going to give you a "loose" definition of ***domain***.

A domain is a set of all the *x-values* of a function. Remember, this is a loose definition, just to ease you into the understanding of it. I am also going to use a number of examples as a build-up to the real definition of Domain.

Example 1



As you can see, the point has coordinates *(1 ; 2).*

The *x-value* is 1 and the y-value is 2;

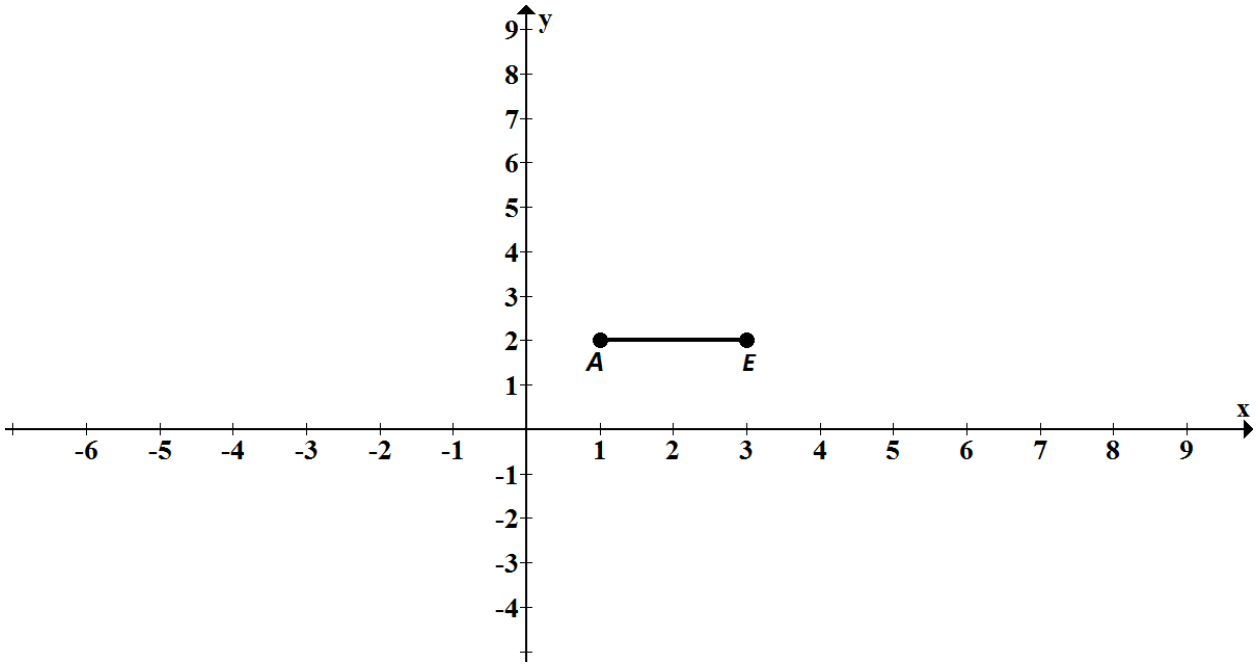
$$(x; y)$$
$$\downarrow \quad \downarrow$$
$$(1; 2)$$

Remember that we said that the domain is the set of all the *x-values* of a function, so for this point, there is only *one x-value*. Thus the domain of the point is 1.

Let's do another example.

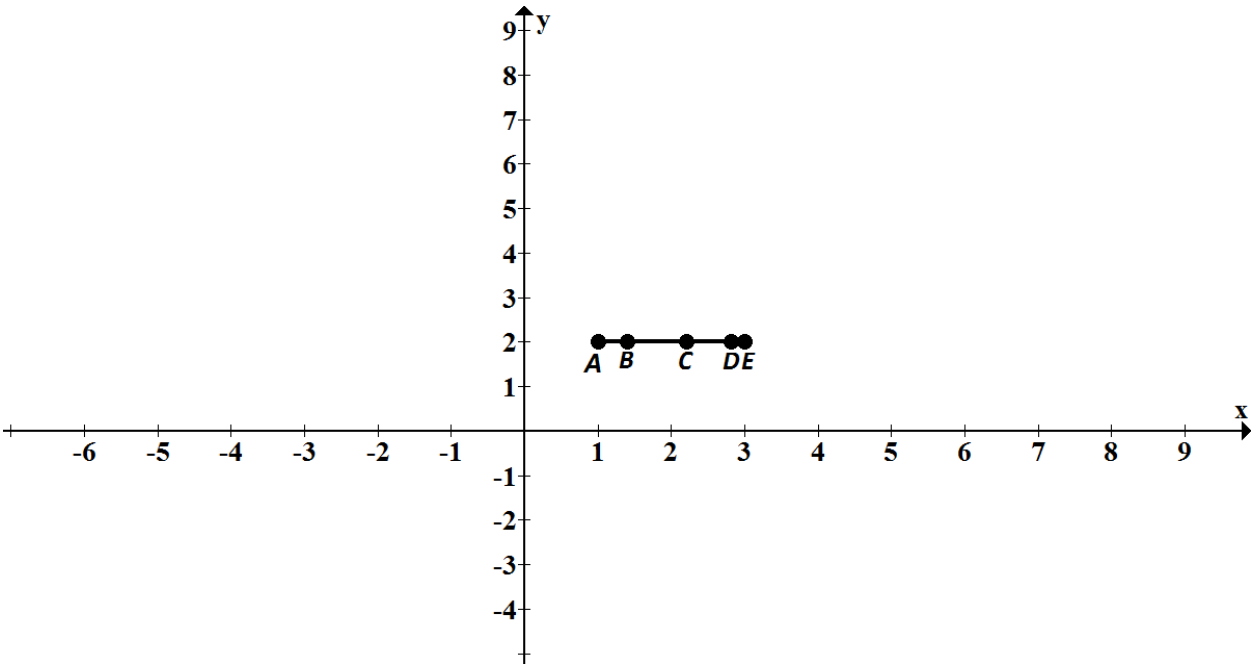
Example 2

Consider the short line on the Cartesian plane below.



The short line spans from point A to point E. Point *A* has coordinates *(1;2)* and point E has coordinates *(3;2).* So what is the domain of the line?

Well, the domain is the set of all the *x-values* of a function, so for this line, it is the x-value of point *A*, point *E* and all the other points in between them. Yes we also include the *x-values* of all the points between A and E. See the short line diagram below



Consider the points *A,B,C,D,E* on our short line *A(1:2),*

B(1,4 ; 2), C(2;2), D(2,9999 ; 2,) E(3;2).

Would it be correct to say our domain is

$x = \{1,\ 1.4\ ,\ 2.2\ ,\ 2.99999\ ,\ 3\}$?

Well that's very incorrect, very Wrong!

I mean what about the *x-values* of the points between A and B? What about the *x-values* of the points between B and C, between C and D, between D and E? All of these *x-values* must be included in the domain.

So to make things easy, we use the *x-values* of the starting point and the ending point of the line to generate the domain. In this case we use the *x-values* of *A(**1** : 2)* and *E(**3** :2)* (starting point and end point). We therefore say that the domain of the short line is all the x-values from ***1*** to ***3***. This includes all the fractions and decimals between ***1*** and ***3***.

Therefore the domain is; all x-values from ***1*** to ***3***.

There is a mathematical way of writing the statement all the "all x-values from ***1*** to ***3***":

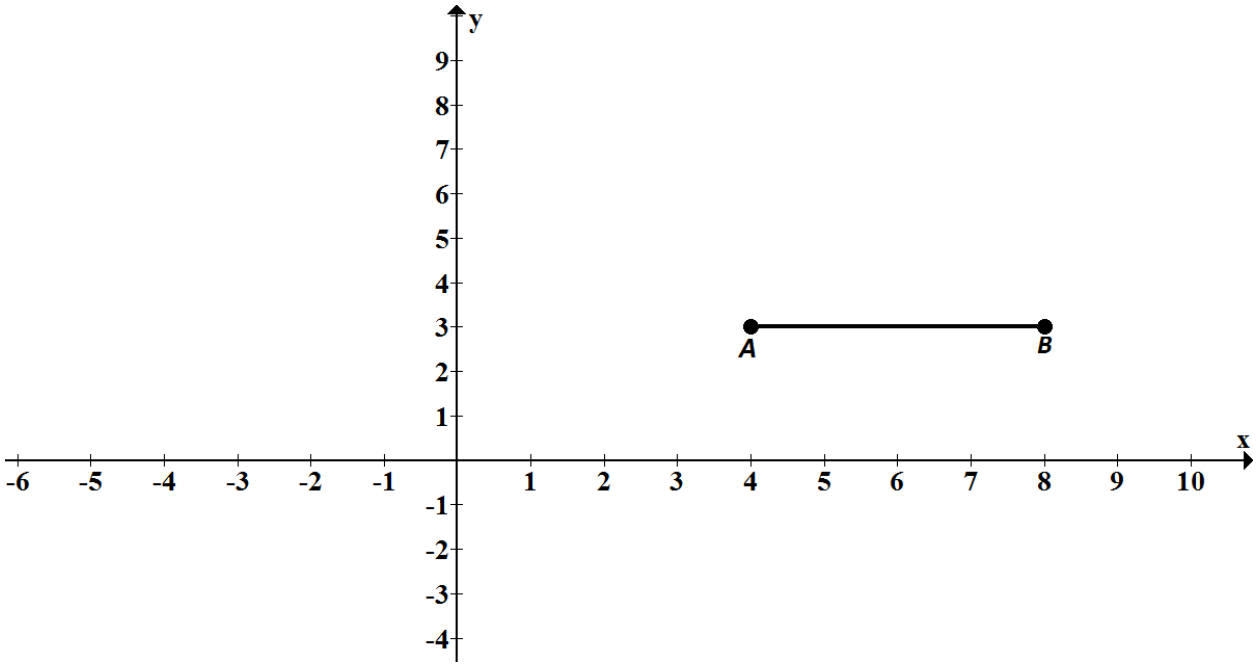
[1 ; 3]

This means all values from 1 to 3 , including 1 and 3 too.

Do more examples to improve your understanding.

Example 3

Consider the line on the Cartesian plane below.

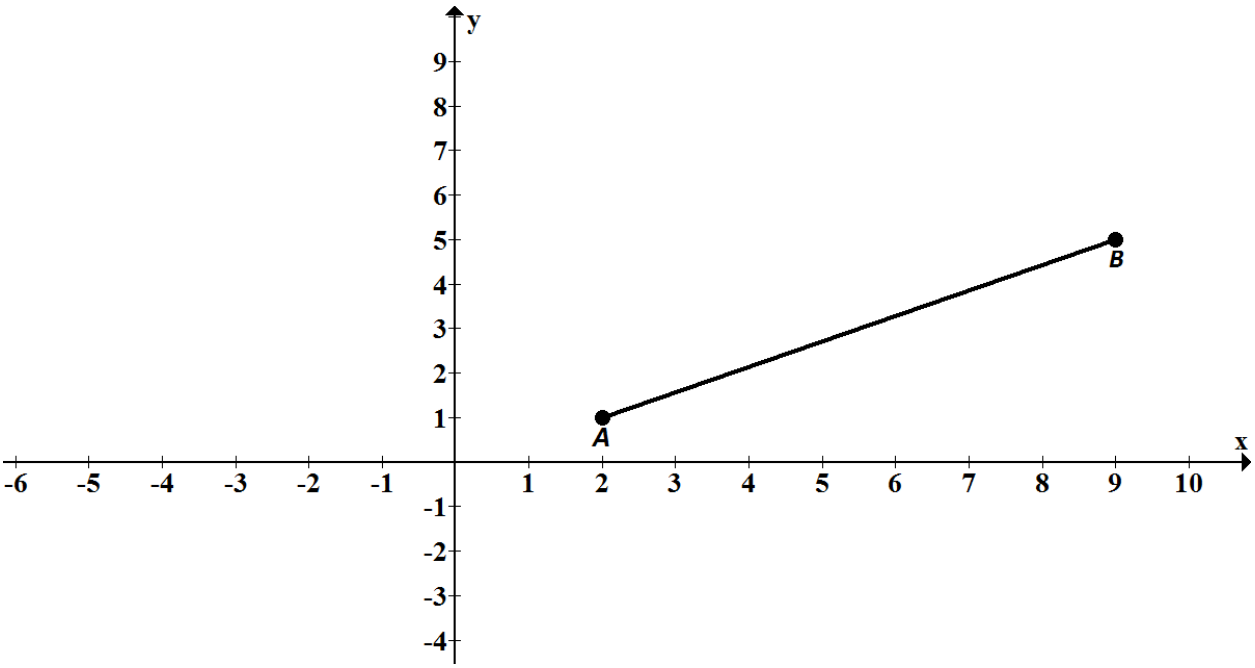


As you can see, the coordinates of A are *(**4**;3)* and the coordinates of B are *(**8**;3)*.

What is the domain of the line?

Well, like we did in the previous example, we use the *x-values* of the starting point and the ending point on the line. In this case, we use the x-values of *A(**4**;3)* and *B(**8**;3).* Therefore, the domain is all the *x-values* from ***4*** to ***8***, including 4 and 8. Mathematically we write: [4 ; 8].

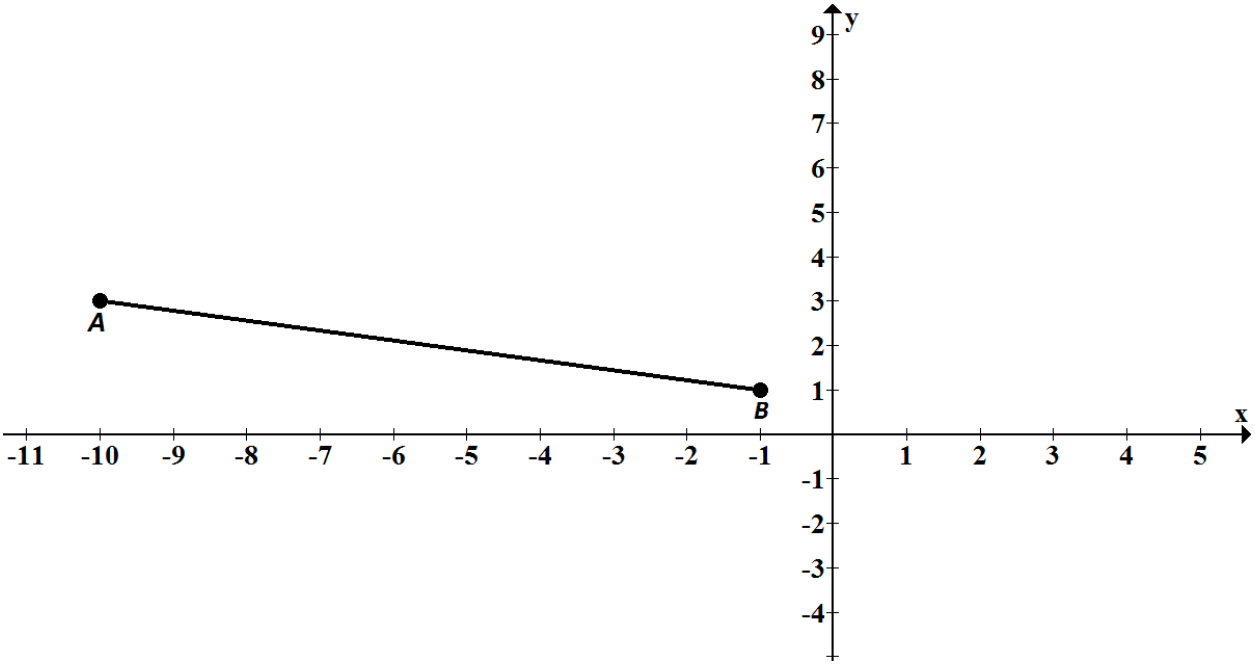
Example 4



As you can see, the coordinates of A are *(**2**;1)* and the coordinates of B are *(**9**;5)*. What is the domain of the line?

We use the x-values of the starting point and the ending point. In this case, we use the x-values of *A(**2**;1)* and *B(**9**;5).* Therefore the domain is all the x-values from 2 to 9, including 2 and 9. Mathematically we write: [2 ; 9].

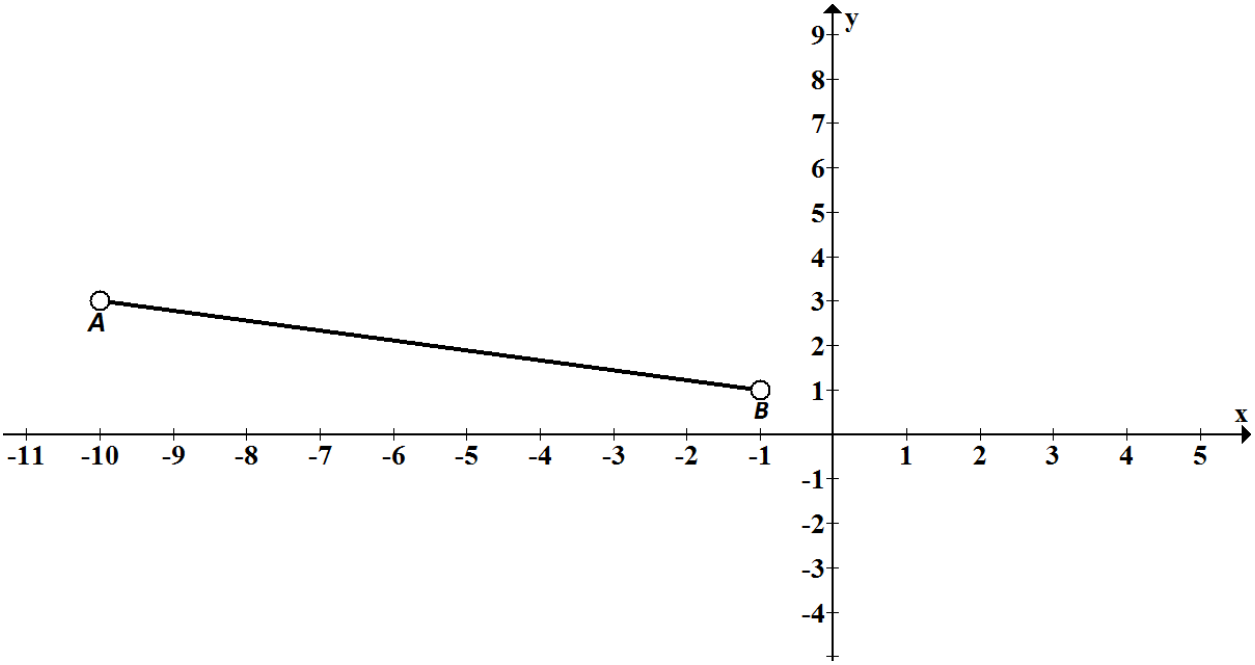
Example 5



The coordinates of A are *(-10:3)* and the coordinates of B are *(-1;1).* What is the domain of the line?

A(-10;3) and B(-1:1). Therefore the domain is all the x-values from -10 to -1, including, -10 and 1. Mathematically we write: *[-10 : -1].*

Example 6



Be careful with the function above. The open dots mean that the coordinates *(-10;1)* and *(-1:1)* are not part of the function. This function is slightly different to the ones in the previous example; the dots for example 5 are closed so *(-10;1) and (-1;1)* are part of the function.

So what is the domain of the function above?

The starting point has coordinates ***(-10**;1)* and the ending point has coordinates ***(-1**;1).* Therefore, the domain is all the *x-values* from ***-10*** to ***1***, but excluding ***-10*** and ***1***, this is because of the open dots at (-10;1) and (-1;1). So basically the domain is all the x-values in between -10 and 1.

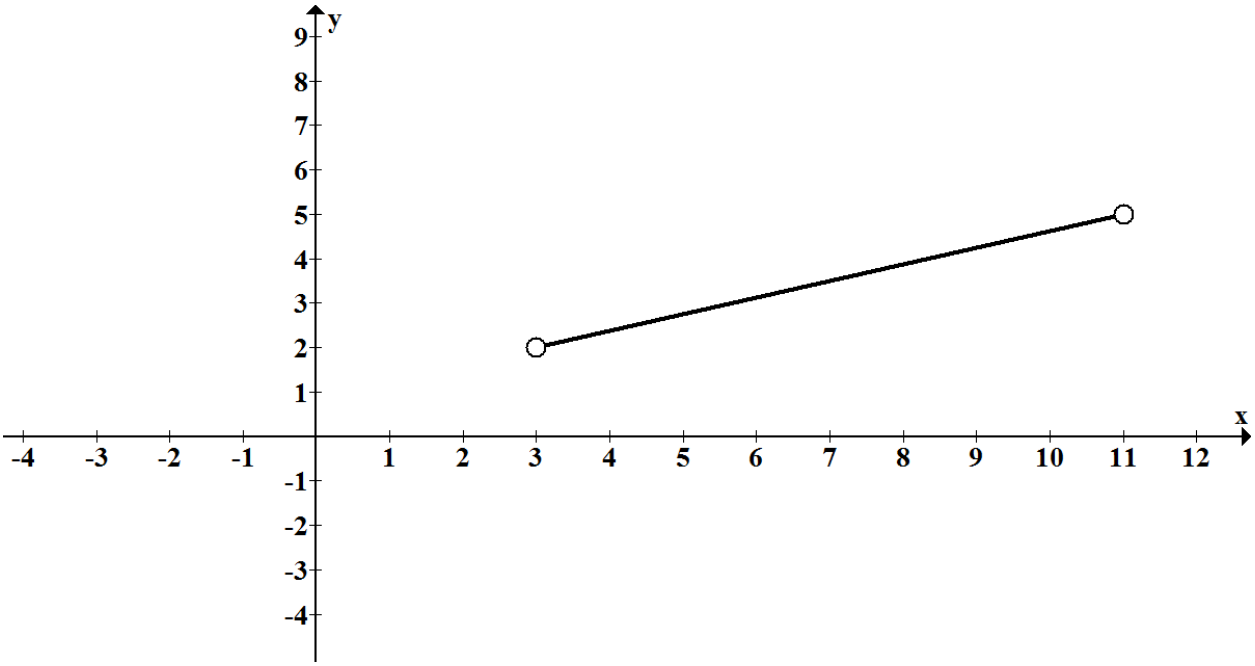
Mathematically we write: *(-10 ; 1)*

Notice the curvy brackets, () , they are different to the square brackets [] from the previous examples. This is because *-10* and *1* are excluded.

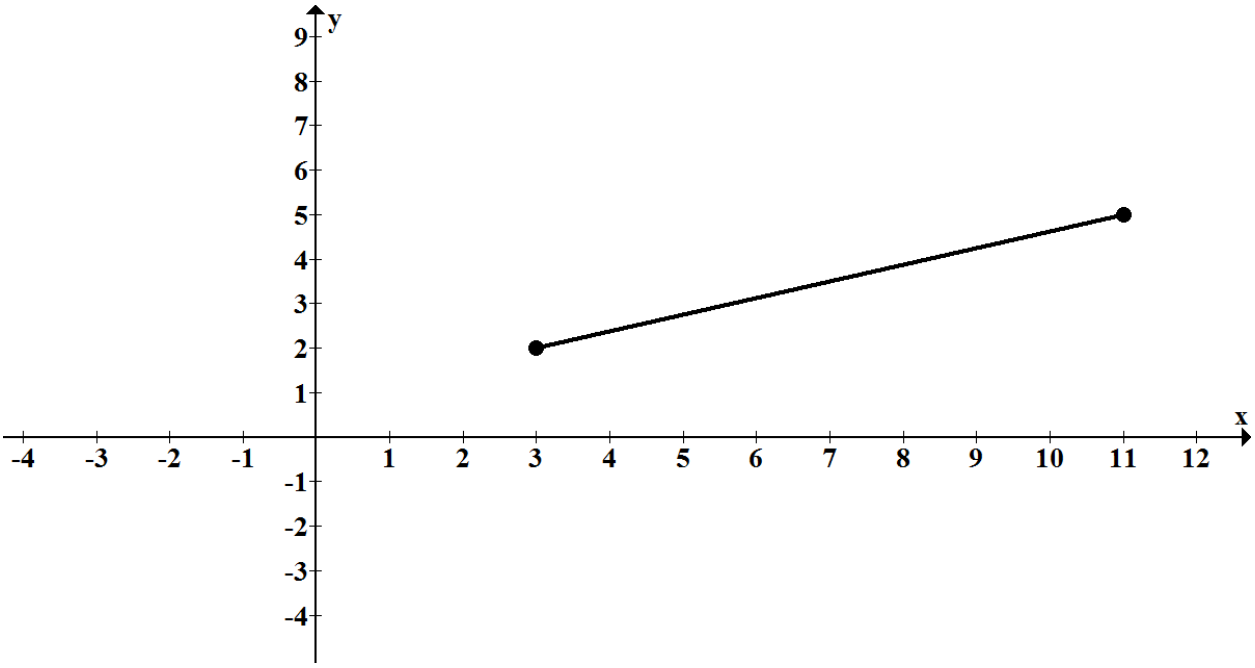
Question

Write down the domain of each of the following functions

a)


y
x
9
8
7
6
5
4
3
2
1
-1
-2
-3
-4
-4 -3 -2 -1 1 2 3 4 5 6 7 8 9 10 11 12

b)


y
x
9
8
7
6
5
4
3
2
1
-1
-2
-3
-4
-4 -3 -2 -1 1 2 3 4 5 6 7 8 9 10 11 12

Solution

a) The coordinates of the starting point are *(3;2*) and *(11;5)* for the ending point. However, there are open dots on the coordinates. So the domain is all *x-values* from 3 to 11, ***excluding*** 3 and 11. That is, all the *x-values* between 3 and 11. Mathematically we write: *(3 ; 11)* Take note of the curvy brackets.

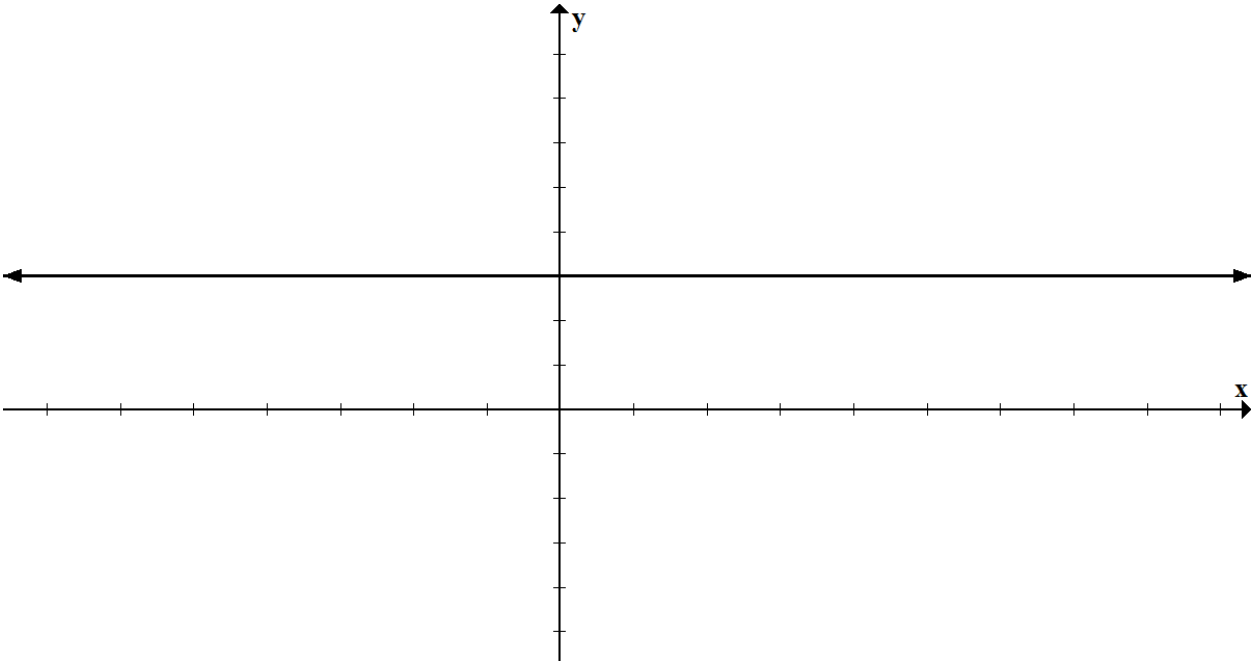
b) The coordinate of the starting point are *(3;2)* and *(11;5)* for the ending point. Note that the dots are closed, so the domain is all the x-values from 3 to 11, *including* 3 and 11. Mathematically we write : *[3 ; 11]* proper brackets. Take note the square brackets, it indicates that 3 and 11 are also included.

We can finally give an official definition of Domain.

A ***Domain*** is a set of all x-values for which a function is defined.

Let's do two more examples on domain. These two examples are ***super important***, so please don't skip them.

Example 8



What is the domain of the graph?

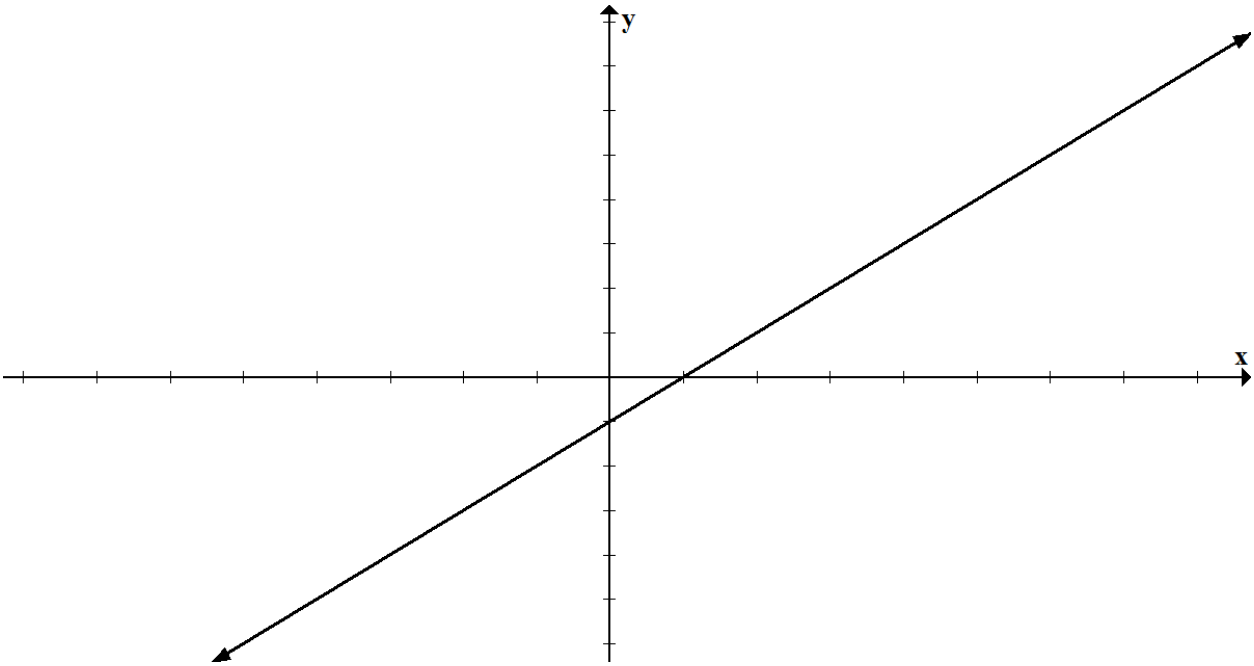
Remember that in order to find a domain, we need to have a starting point and an ending point, but it seems like we don't quite have them here. The arrow pointing to the **left** end indicates that the graph goes in that direction forever, that is, it goes to the negative infinity. So basically the starting point is negative infinity $-\infty$.

Similarly, the arrow pointing to the **right** end indicates that the graph goes in that direction forever, that is, it goes to positive infinity ∞. So basically the ending point is positive infinity: ∞. We use curvy brackets every time we deal with infinity, so our domain is $(-\infty; \infty)$.

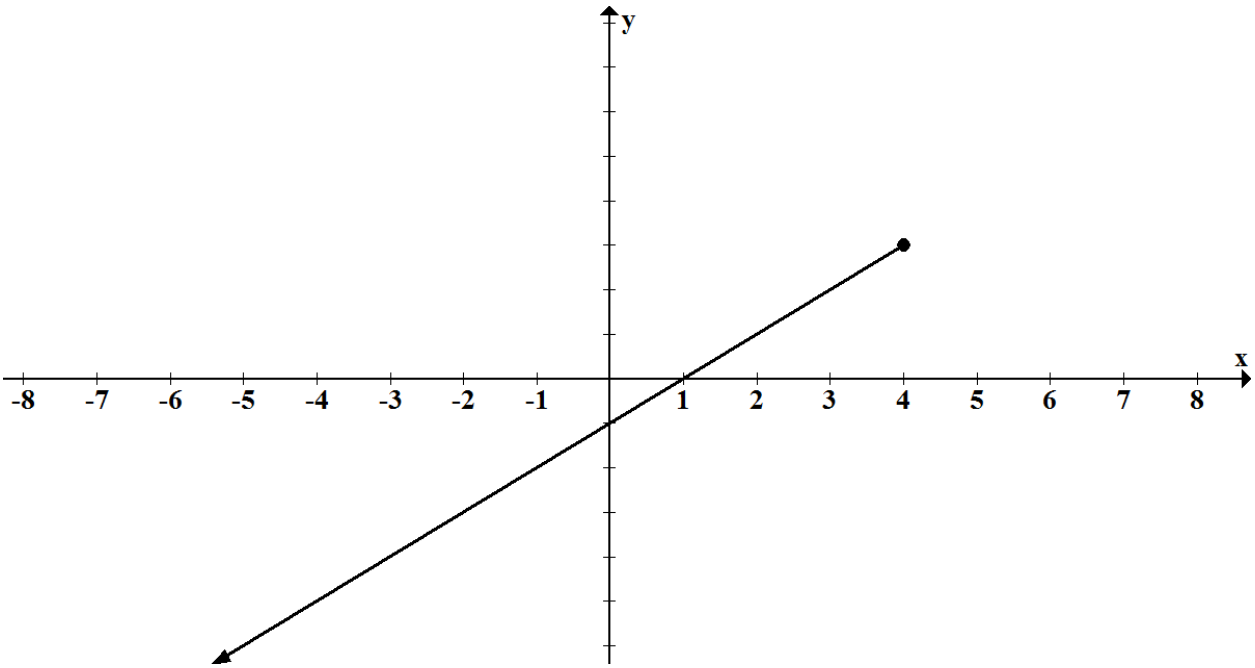
Question

Write down the domain for each of the following functions.

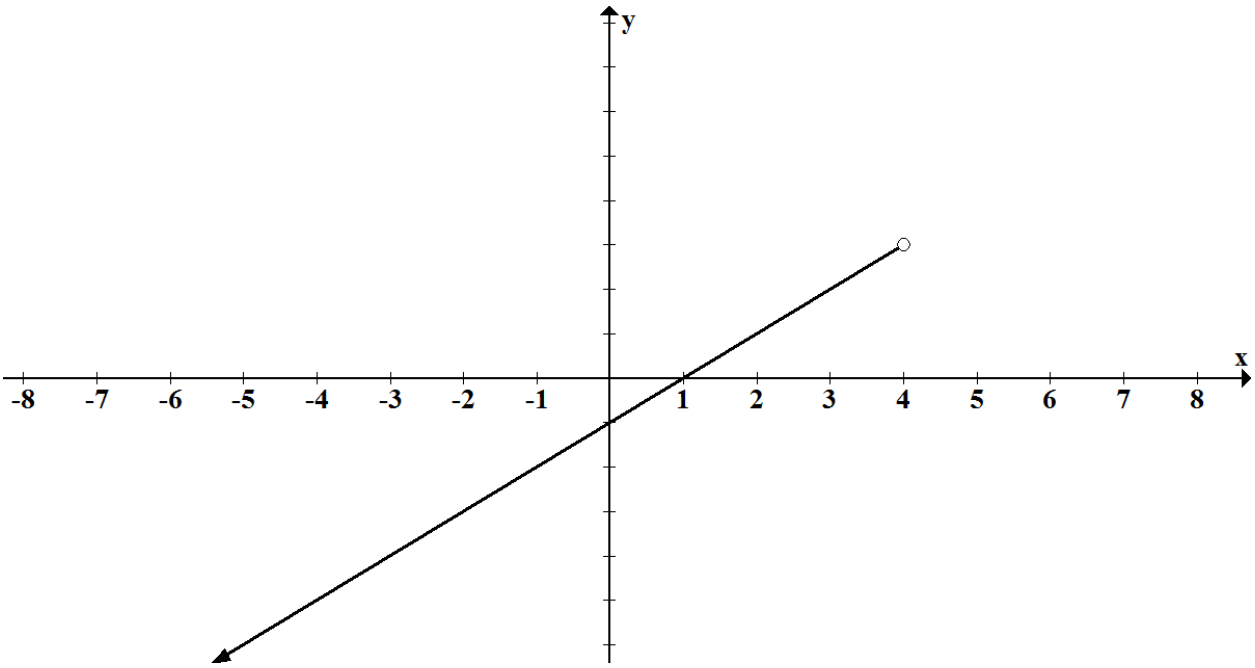
a)



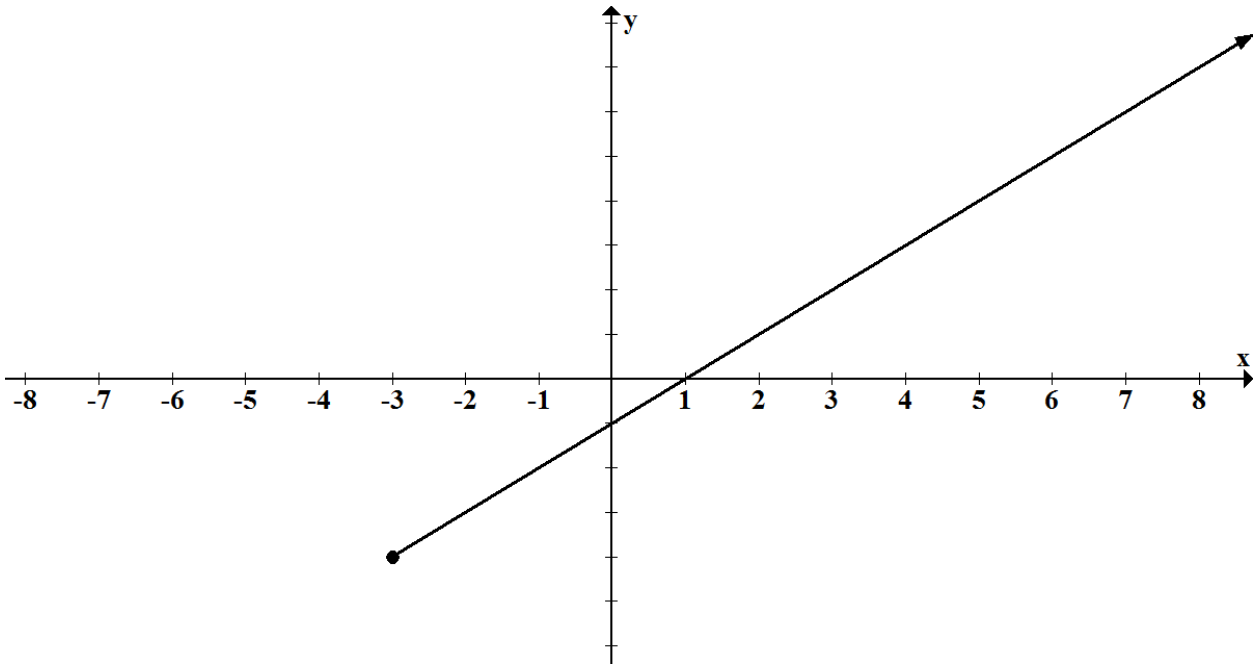
b)



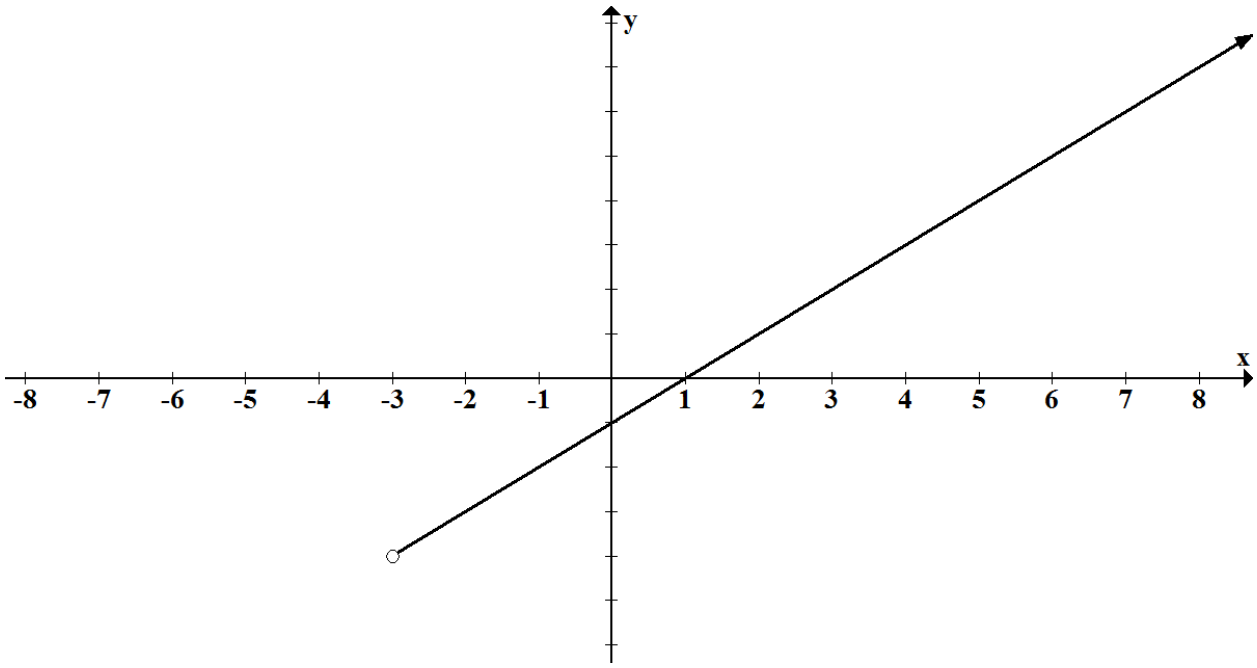
c)



d)



e)



Solution

a) $(-\infty; \infty)$

b) $(-\infty; 4]$, domain includes *4*

c) $(-\infty; 4)$, domain doesn't include *4*

d) $[-3;\ \infty)$, domain includes *-3*

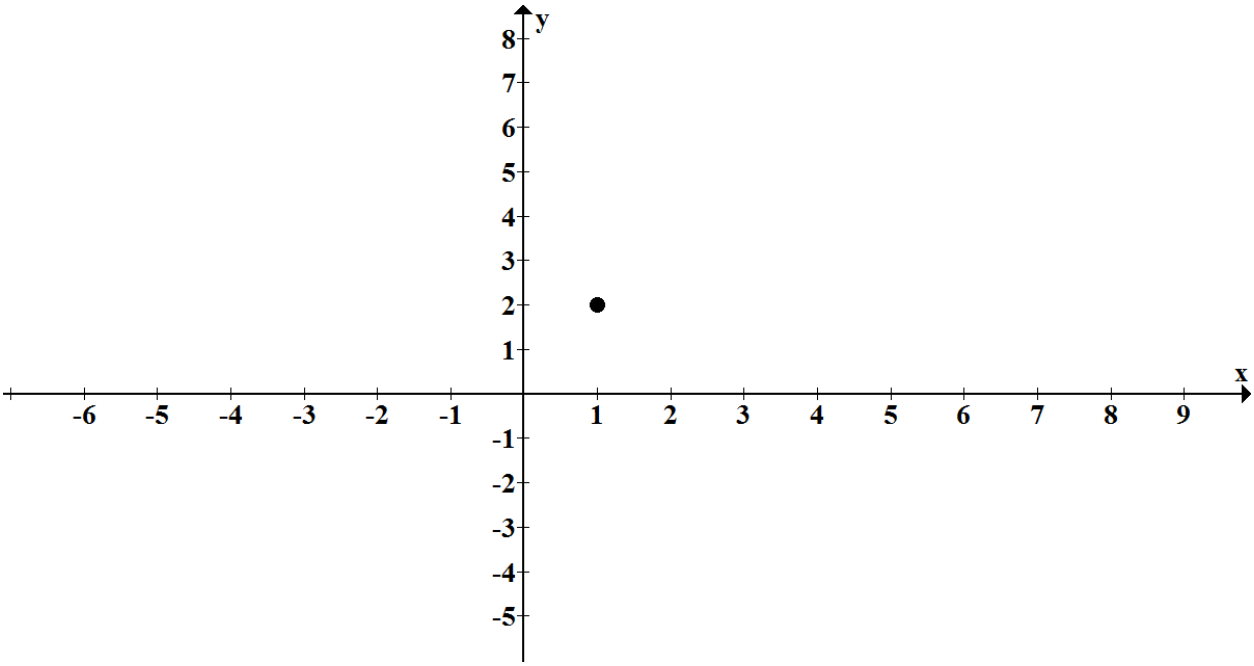
e) $(-3;\ \infty)$, domain doesn't include *-3*

Range

In this section we are going to learn the so called ***Range*** of a function. For now, I am going to give you a loose definition of a range. A ***Range*** is a set of all the *y-values* of a function. Remember that this is a loose definition just to ease you into the understanding of it. I am going to use a number of examples in the build up to the "real" definition of range.

Example 1

Consider the point in the Cartesian plain below.



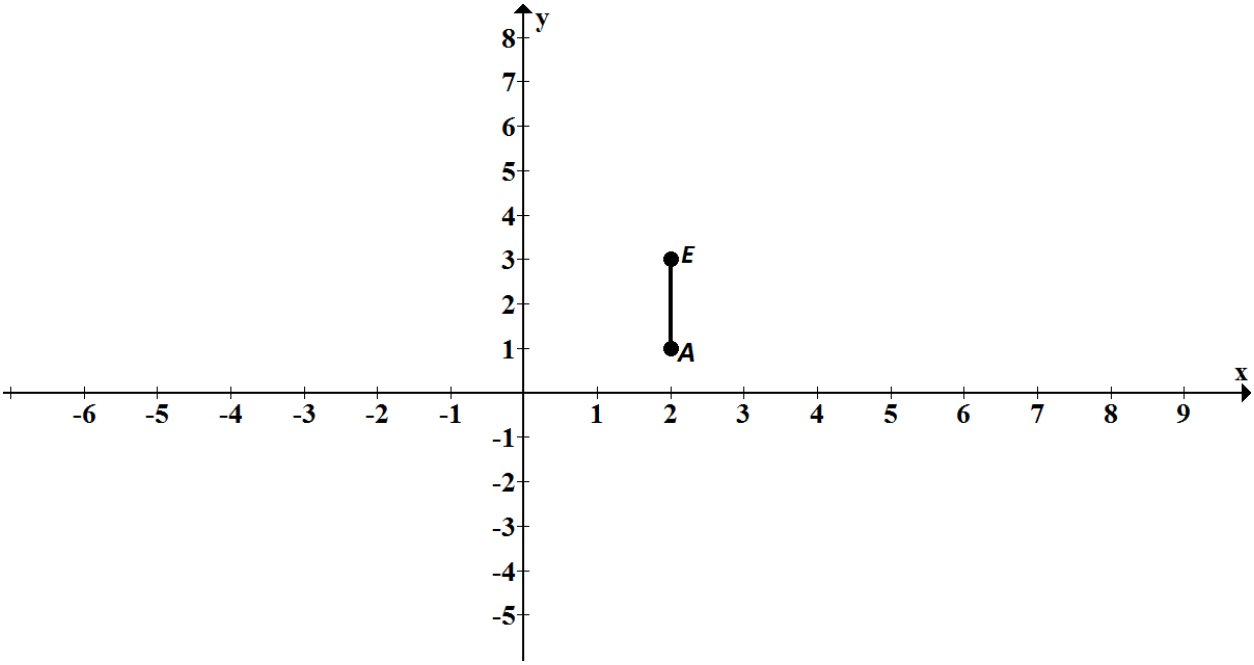
As you can see the point has coordinates *(2;1).* The x-value is 2 and the y-value is 1:

$$(x; y)$$
$$\downarrow \quad \downarrow$$
$$(2; 1)$$

Remember we said the range of a function is a set of all the y-values. Thus the range of the point is 1. Let's do another example.

Example 2

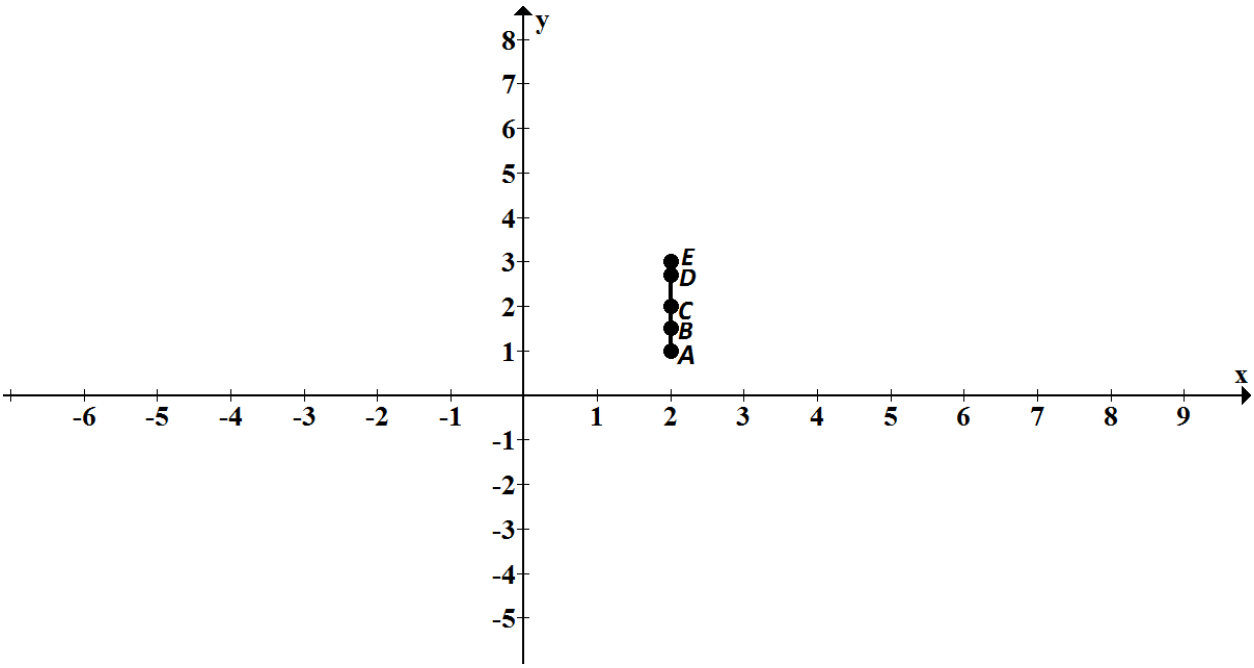
Consider the line on the Cartesian plane below.



The line spans from point A to E. Point A has coordinates *(2;1)* and point E has coordinates *(2;3)*.

So what is the range?

Well, the range is a set of all of all the y-values of point A, point E and all the other points between them. Yes, we also include the y-values of all the points between A and E. See the diagram below.



Consider the points *A,B,C,D and E*.

*A(2;**1**) B(2 ; **1.4**) C(2;**2**) D(2; **2,99999**) E(2;**3**)*

Would it be correct to say that our range is;

y = $\{1\ ,\ 1.4\ ,\ 2\ ,\ 2.9999\ ,\ 3\}$? That's very incorrect, very wrong! I mean what about the y-values of the points between B and C, between C and D, between D and E?

All of the y-values must be included in the range. So to make things easy, we use the *y-value* of the starting point and the ending point of the line to generate the range. In this case, we use the *y-value* of *A(2;**1**) and E(2;**3**).*

We say that the range of the line is all the y-values from *1* to *3*. This includes all the numbers, fractions and decimals between 1 and 3.

The range is therefore all values from 1 to 3, A mathematical way of writing this is : $[1; 3]$. The ***Range*** has a similar setup as the domain, the only difference is that the range is based on the y-values, while the domain is based on the *x-values*.

Quadratic (parabolic) Functions

In this section, we are going to learn more about the behavior of the ***Quadratic Function***, that is, the **reflection** about the axes, **shifting**, ***y-cut***, **domain** and **range**.

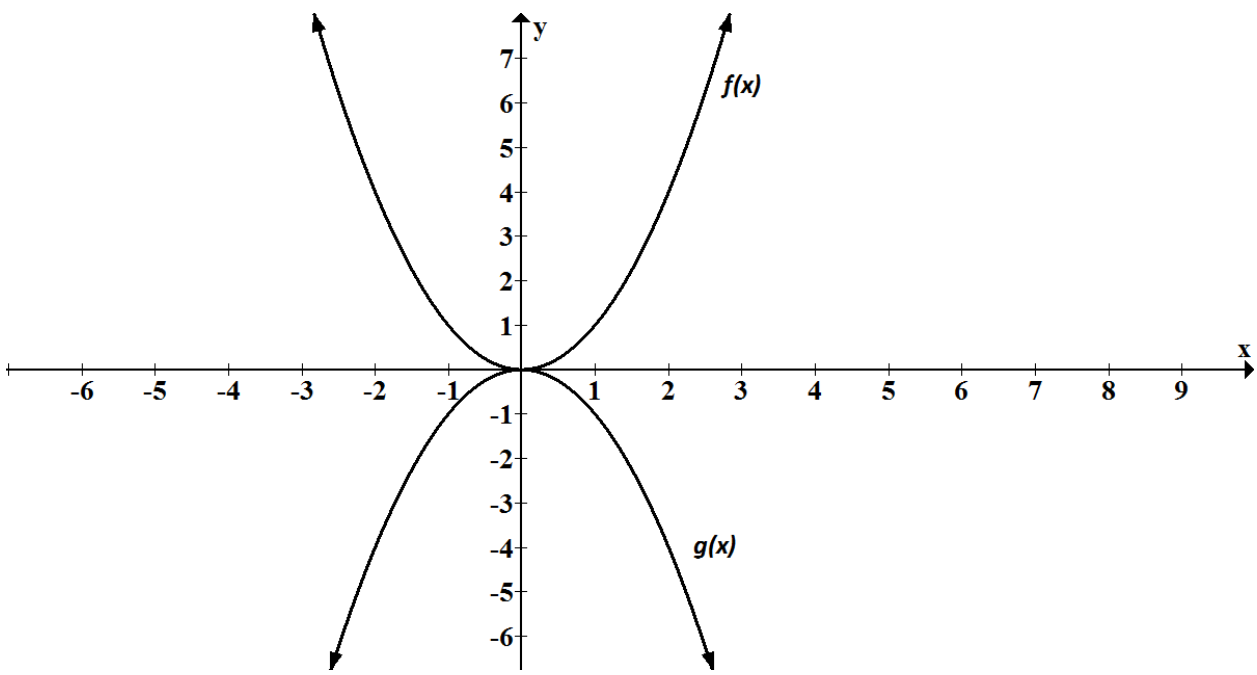
Reflection through the x-axis

Compare the function $y = x^2$ vs $y_1 = -x^2$

Let's name $y = x^2$ function $f(x)$ and name $y_1 = -x^2$ function $g(x)$.

Below we fill in the table of values for both functions and draw their respective graphs.

x	-4	-3	-2	-1	0	1	2	3	4
$f(x) = x^2$	16	9	4	1	0	1	4	9	16
$g(x) = -x^2$	-16	-9	-4	-1	0	-1	-4	-9	-16



Do notice the *g(x)* graph is a mirror replication of the *f(x)* graph through the *x-axis*. We can therefore safely conclude that the function $g(x) = -x^2$ is a mirror reflection of the function $f(x) = x^2$ through the *x-axis* (the *x-axis* is the mirror).

Notice that we could've concluded the above without having to draw the graph. We see that by comparing the functions.

$f(x) = x^2 \qquad vs \qquad g(x) = -x^2$

$f(x) = x^2$ is our original function; to get $g(x) = -x^2$, we just place the "$-$" in front of the "$whole\ function$" $f(x) = x^2$. See that below.

$$x^2 \rightarrow -(x^2) = -x^2$$

$$f(x) \rightarrow g(x)$$

Thus every time we place a "$-$" in front of the "$whole\ quadratic\ function$", just know we are basically reflecting it through the *x-axis*.

Let's do an example.

Example

Consider the function $f(x) = x^2 + 1$

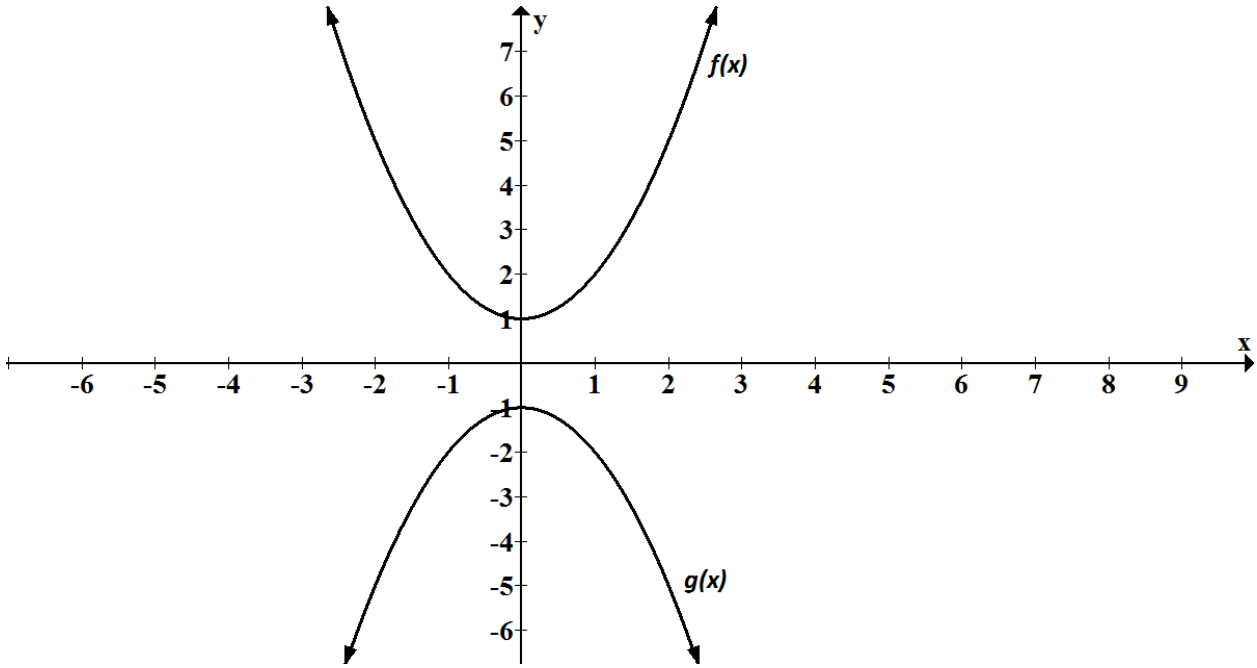
Let's put a " $-$ " in front of the entire $x^2 + 1$ to get $g(x)$, a reflection through the *x-axis*.

$x^2 + 1 \rightarrow -(x^2 + 1) = -x^2 - 1$

So our $g(x)$ is $g(x) = -x^2 - 1$

$g(x) = -x^2 - 1$ is a mirror reflection of $f(x) = x^2 + 1$ through the *x-axis*.

Now let's check if the above is true by drawing the graphs of $f(x)$ and $g(x)$.



We can see clearly that $g(x)$ is a mirror reflection of $f(x)$ through the x-axis.

So this proves us right. Every time we place a " $-$ " in front of an entire quadratic function, we are practically reflecting it through the x-axis.

Example

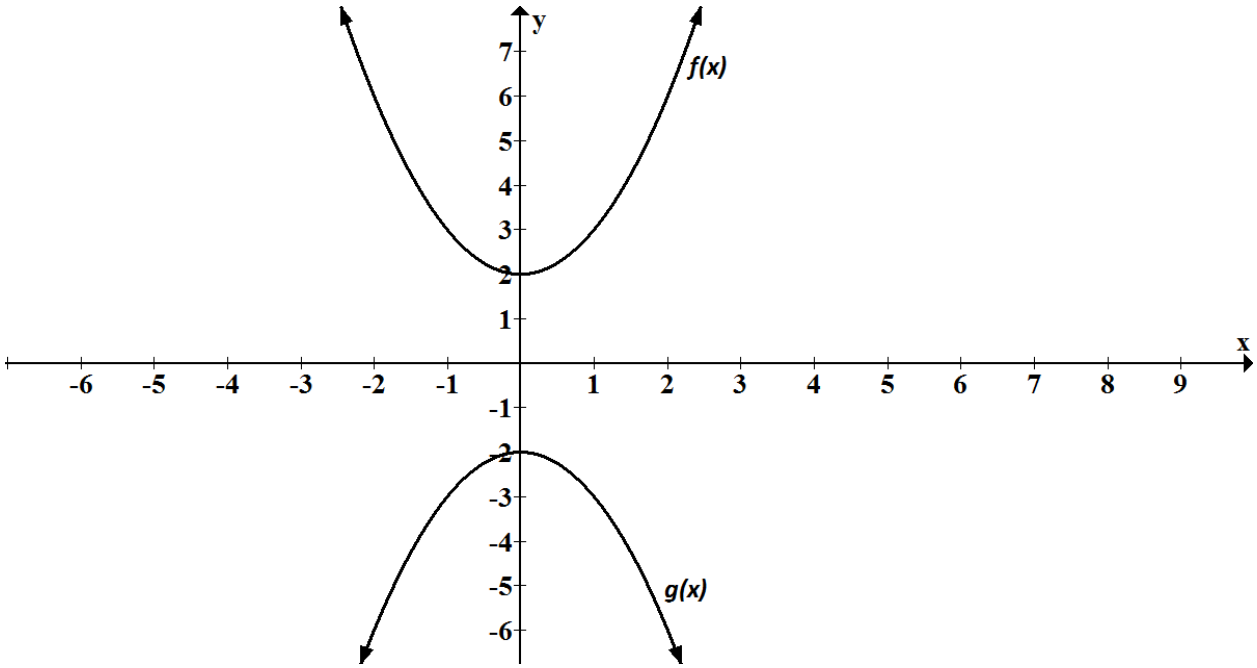
Consider the function $f(x) = x^2 + 2$

Let's put a " $-$ " in front of the entire $x^2 + 2$ to get $g(x)$, a reflection through the *x-axis*.

$x^2 + 2 \rightarrow -(x^2 + 2) = -x^2 - 2$

So our $g(x)$ is $g(x) = -x^2 - 2$

$g(x) = -x^2 - 2$ is a mirror reflection of $f(x) = x^2 + 2$ through the *x-axis*.



We can clearly see that $g(x)$ is a mirror reflection of $f(x)$.

Y-Cut

Compare the graph $f(x) = x^2$, $g(x) = x^2 + 1$, $h(x) = x^2 + 2$

Below we fill in the table of values for each of the functions and draw the respective graphs.

x	-4	-3	-2	-1	0	1	2	3	4
$f(x) = x^2$	16	9	4	1	0	1	4	9	16
$g(x) = x^2 + 1$	17	10	5	2	1	2	5	10	17
$h(x) = x^2 + 2$	18	11	6	3	2	3	6	11	18

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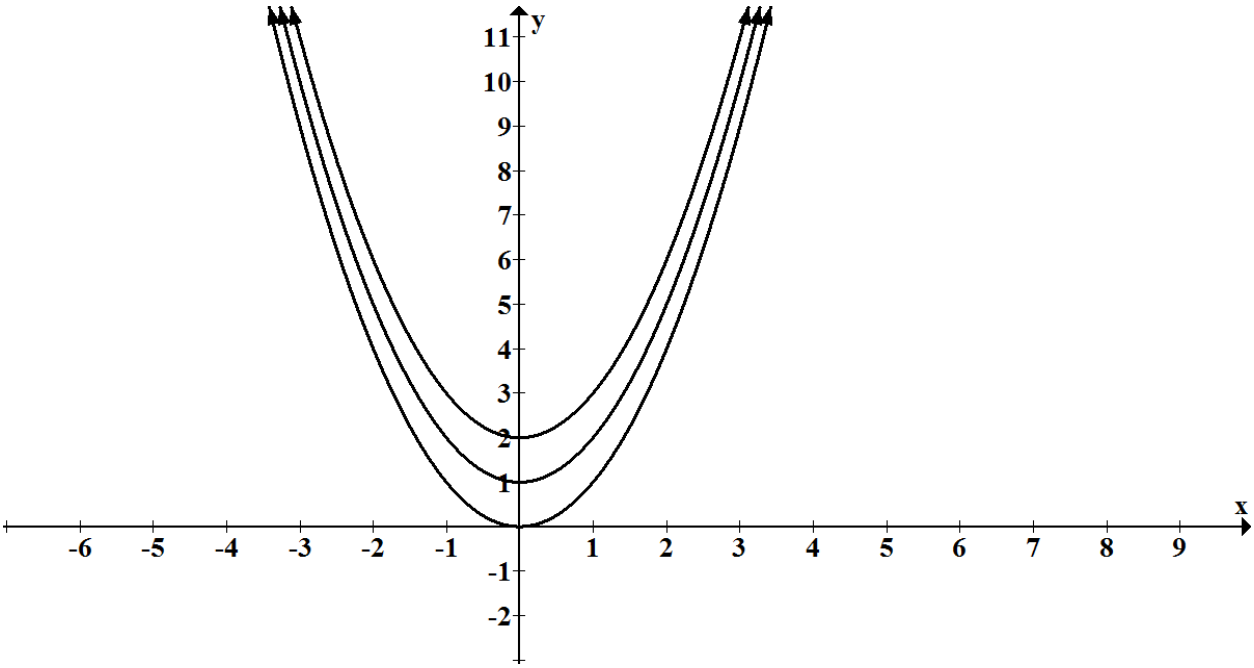


Notice that $f(x)$ crosses the y-axis at 0.

$g(x)$ crosses the y-axis at 1.

$h(x)$crosses the y-axis at 2.

These are called the *y-cuts or y-intercepts*. What do you notice about the function below?

$f(x) = x^2 = x^2 + 0$

$g(x) = x^2 + 1$

$h(x) = x^2 + 2$

The number at the end represents the ***y-cut***.

In general, a quadratic function exists in the form $y = ax^2 + bx + c$, where c is the *y-cut*.

Question

Write down the *y-cut (y-intercept)* of each of the following function

a) $x^2 + 6$
b) $x^2 + 8$
c) $-x^2 + \frac{1}{2}$
d) $x^2 + 4x + 5$
e) $x^2 - 6x + 9$

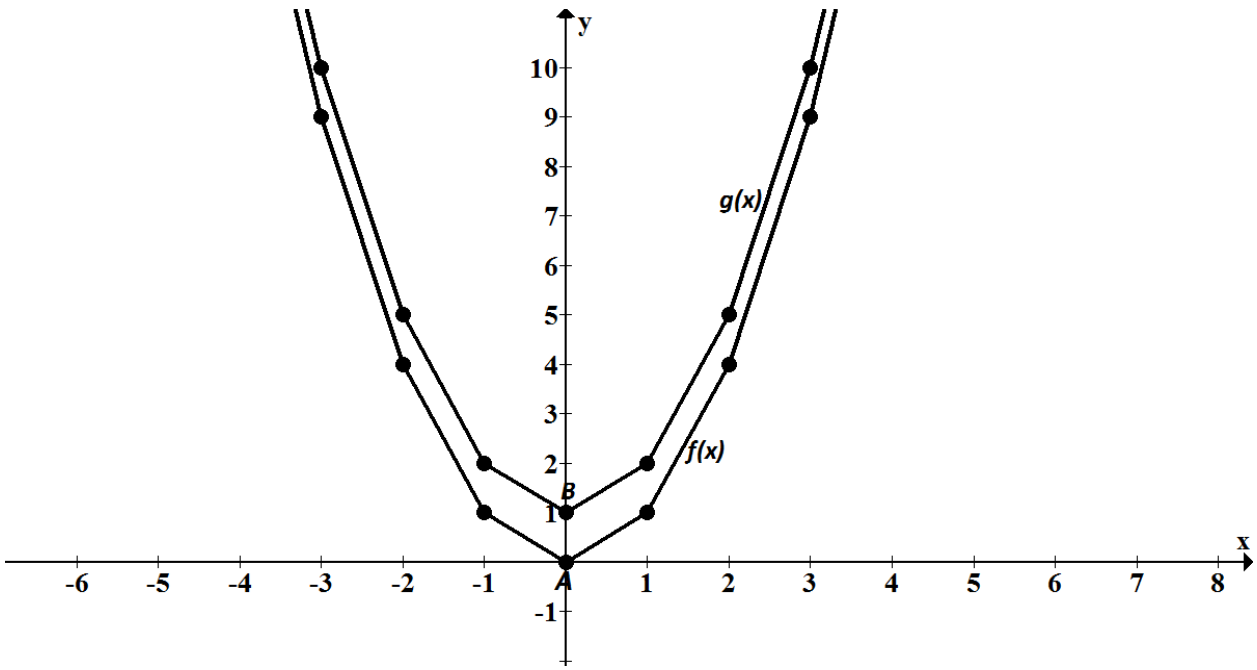
Solution

a) 6
b) -6
c) $\frac{1}{2}$
d) 5
e) 9

VERTICAL TRANSLATION (MOVING GRAPH UP AND DOWN)

Consider function $f(x) = x^2$ and $g(x) = x^2 + 1$. Below is the table of values for the functions and their respective graphs.

x	-4	-3	-2	-1	0	1	2	3	4
$f(x) = x^2$	16	9	4	1	0	1	4	9	16
$g(x) = x^2 + 1$	17	10	5	2	1	2	5	10	17



If you were to move from point A to Point B, you would have to walk 1 unit up. In fact, if you were to walk from any point on *f(x)* to the corresponding point on *g(x),* you would have to walk 1 unit upward. To see this clearly, let's compare the corresponding points on the table of values.

x	-4	-3	-2	-1	0	1	2	3	4
$f(x) = x^2$	16	9	4	1	0	1	4	9	16
$g(x) = x^2 + 1$	17	10	5	2	1	2	5	10	17

The value $f(x)$ here is 16 and 17 for $g(x)$. So you only move 1 unit from 16 to 17.

In fact, the value for $g(x)$ is 1 unit higher than that of $f(x)$ everywhere.

This is true for all corresponding points.

From the points on $f(x) = x^2$, we add 1 to the y-coordinates to get the corresponding points on $g(x) = x^2 + 1$.

In other words, to get the graph of $g(x) = x^2 + 1$, we simply move the graph of $f(x) = x^2$ 1 unit upwards.

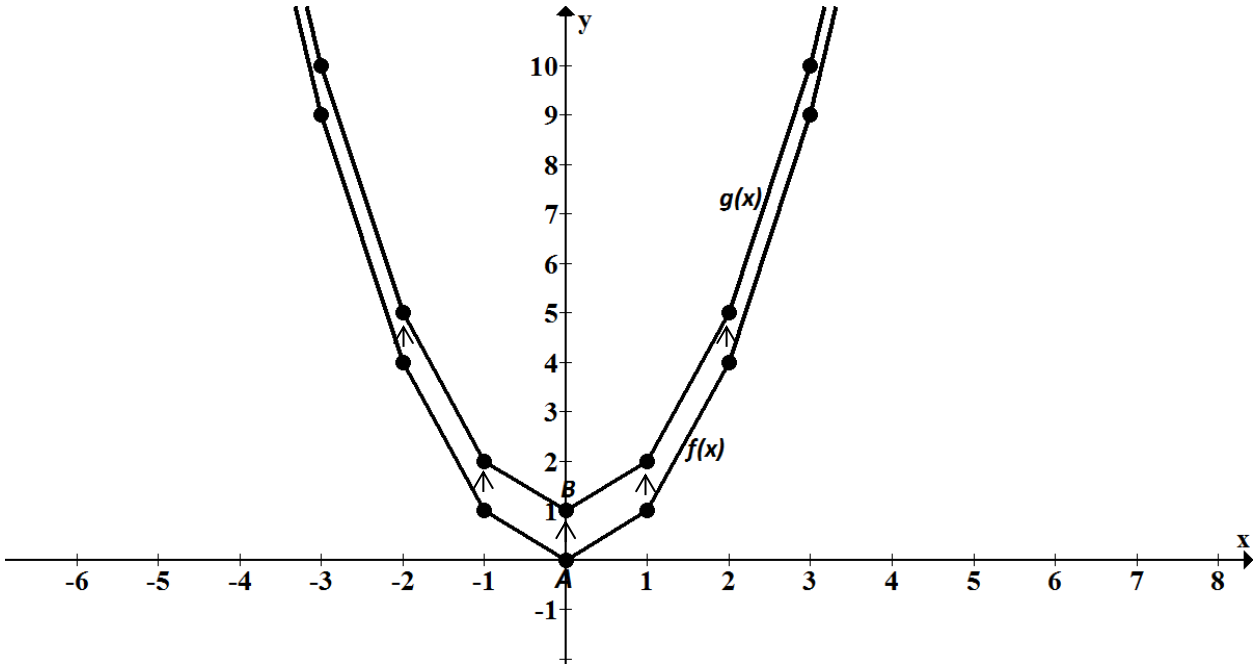
We can even see this from their respective functions.

$$f(x) \rightarrow g(x)$$

$$x^2 \rightarrow x^2 + 1$$

We just added 1 to x^2 to get $x^2 + 1$

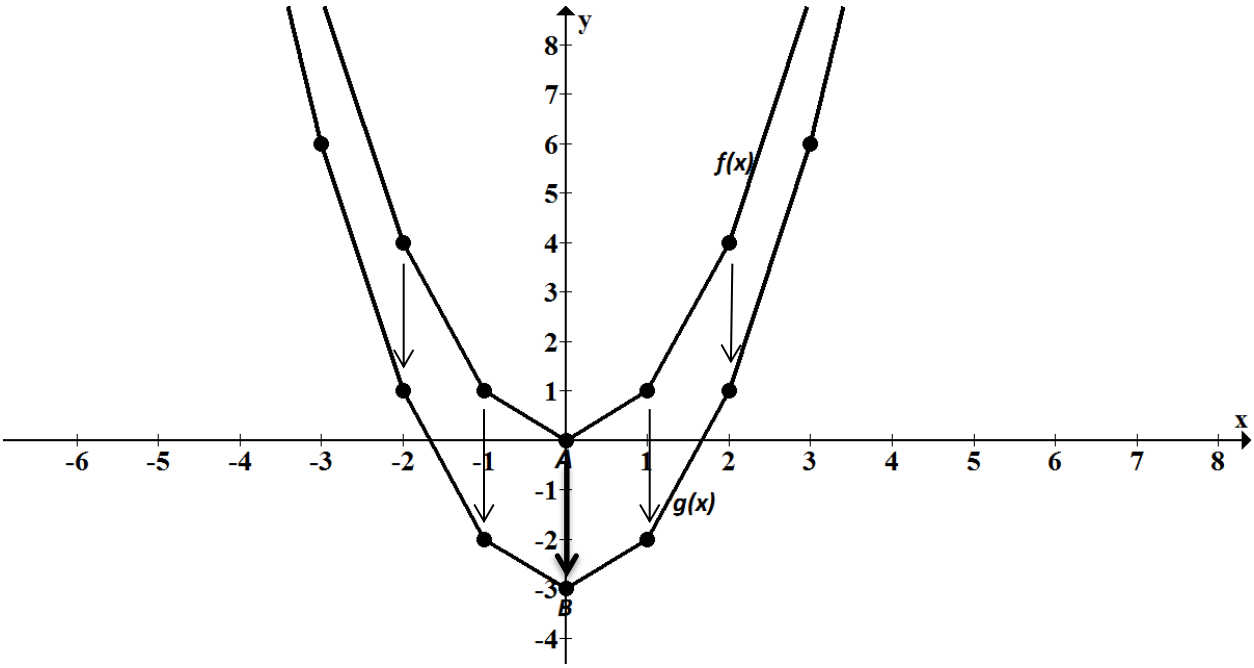
Mathematicians call this Translation. The movement of *f(x)* 1 unit upwards, to get *g(x)*.



Example

$f(x) = x^2$ to $g(x) = x^2 - 3$

x	-4	-3	-2	-1	0	1	2	3	4
$f(x) = x^2$	16	9	4	1	0	1	4	9	16
$g(x) = x^2 - 3$	13	6	1	-2	-3	-2	1	6	13



From the graph, we see that the movement from point *A* on *f(x)* to point *B* on *g(x)* is 3 movements ***downwards***.

We can even see this from their respective functions.

$$f(x) \rightarrow g(x)$$

$$x^2 \rightarrow x^2 - 3$$

We *"minused"* 3 from *f(x)* to get *g(x).*

Thus a "minus" on the whole function in translation is a downward movement of the function.

Question

The graph $f(x) = x^2$ is moved to get the graph of $g(x) = x^2 + 5$. How was $f(x)$ moved to get $g(x)$?

Solution

$f(x)$ was moved 5 units *UPWARDS* to get $g(x)$.

Question

The graph $f(x) = x^2$ is moved to get the graph of $g(x) = x^2 - 5$. How was $f(x)$ moved to get $g(x)$?

Solution

$f(x)$ was moved 5 units *DOWNWARDS* to get $g(x)$.

Question

The graph $f(x) = x^2 + 2x + 5$ is moved to get the graph of $g(x) = x^2 + 2x + 6$. How was $f(x)$ moved to get $g(x)$?

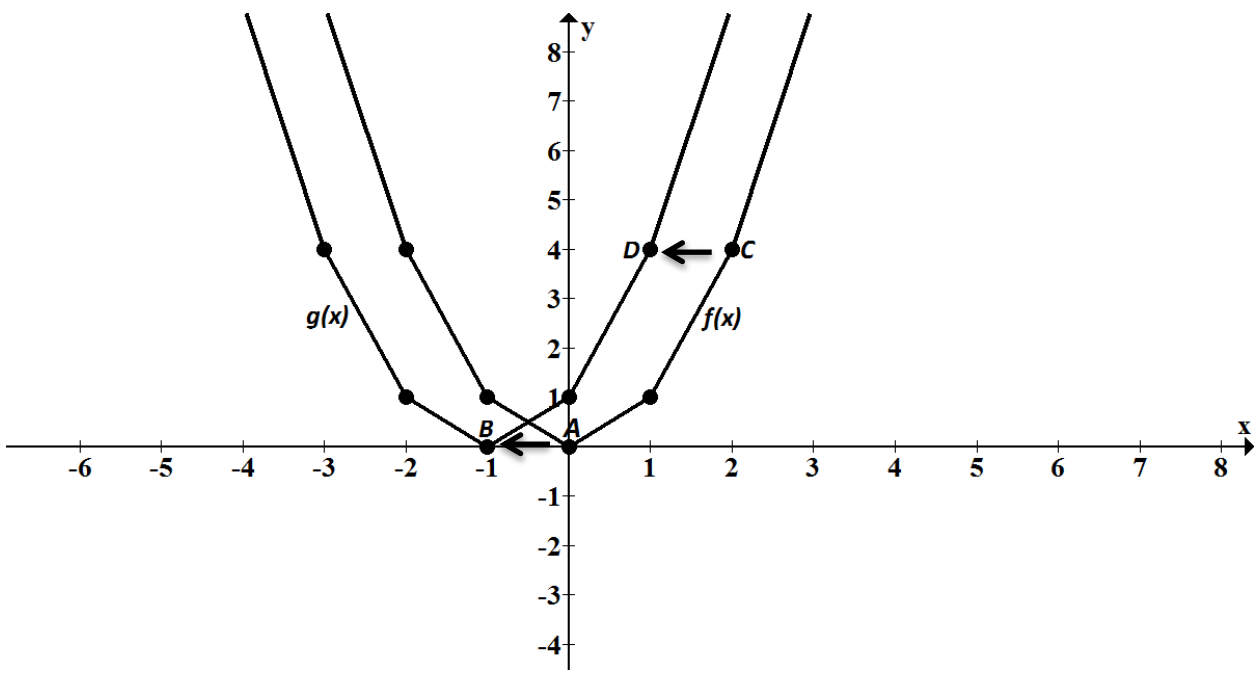
Solution

$f(x)$ was moved 1 unit *DOWNWARDS* to get $g(x)$.

Horizontal Translation (moving the graph left or right)

Consider the functions $f(x) = x^2$ and $g(x) = (x+1)^2$. Below is the table of values of the functions and their respective graphs.

x	-4	-3	-2	-1	0	1	2	3	4
$f(x) = x^2$	16	9	4	1	0	1	4	9	16
$g(x) = (x+1)^2$	9	4	1	0	1	4	9	16	25



If you were to move from point *A* to point *B f(x)* to *g(x)*, you'd have to walk 1 unit to the ***LEFT***.

In fact, if you walk from any point on *f(x)* to the corresponding point on *g(x)*, you'd have to walk 1 unit to the ***LEFT***. Check the movement from point C on *f(x)* to D on *g(x)*, it's also 1 unit to the left, the pattern is the same for the other points on those two graphs.

Therefore, to get the graph of $g(x) = (x+1)^2$, we simply shift the graph of $f(x) = x^2$ 1 unit to the left.

That's the ***EFFECT OF ADDING 1*** to the "x".

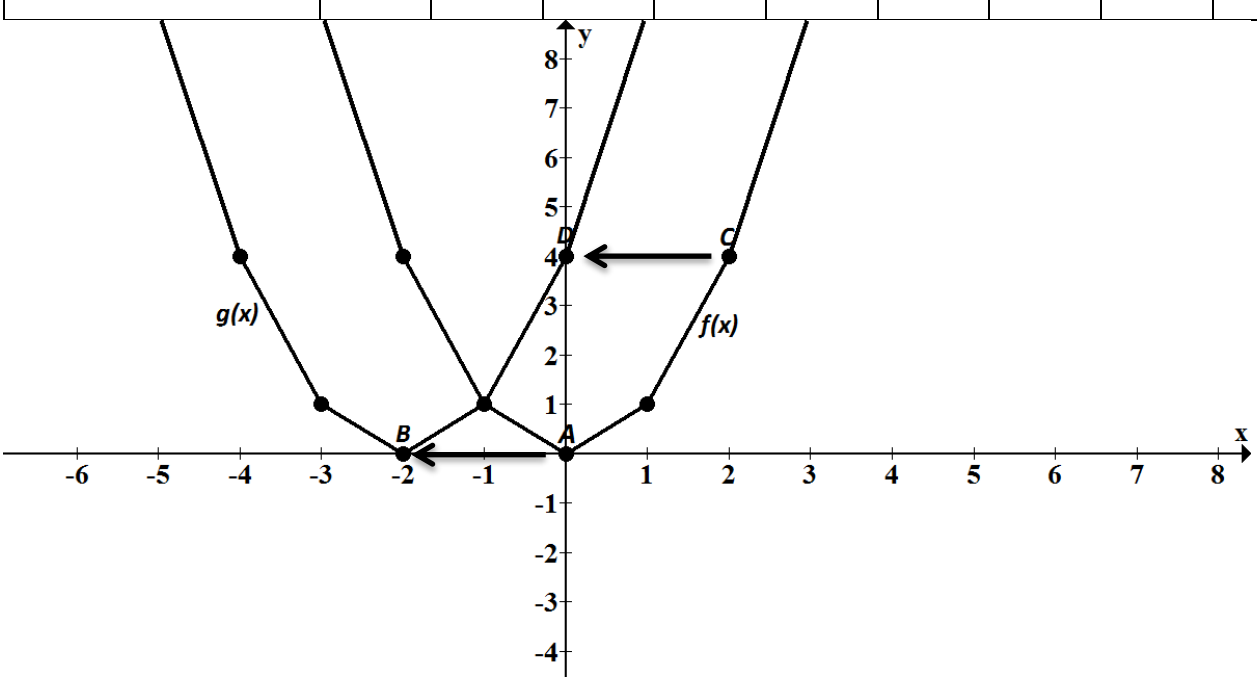
Let's do more examples.

Example

Consider the function $f(x) = x^2$ and $g(x) = (x+2)^2$

Below is the table of values and graph of the representative graphs.

x	-4	-3	-2	-1	0	1	2	3	4
$f(x) = x^2$	16	9	4	1	0	1	4	9	16
$g(x) = (x+2)^2$	4	1	0	1	4	9	16	25	36



In fact, if you walk from any point on *f(x)* to the corresponding point on *g(x)*, you'd have to walk *2* units to the ***LEFT***. Check the movement from point C on *f(x)* to D on *g(x)*, it's also 2 units to the left, the pattern is the same for the other points on those two graphs.

Therefore, to get the graph of $g(x) = (x+2)^2$, we simply shift the graph of $f(x) = x^2$ *2* units to the left.

That's the ***EFFECT OF ADDING 2*** to the "x".

In general, when we add a positive number that way on a quadratic function, it is basically a shift to the left, while subtraction means you are shifting the quadratic function to the ***right***.

Question

The graph of $f(x) = x^2$ is moved to get the function $g(x) = (x-3)^2$. How was *f(x)* moved to get *g(x)*?

Solution

f(x) was moved 3 units to the ***left*** to get *g(x).*

Question

The graph of $f(x) = x^2 + 2x + 4$ is moved to get $g(x) = (x+3)^2 + 2(x+3) + 4$. How was *f(x)* moved to get *g(x)*?

Solution

$f(x)$ was moved 3 units to the left to get $g(x)$. Be careful! When a function moves 3 units to the left, we put $+3$ next to every "x" on the function.

Question

The graph of $f(x) = x^2 + 2x + 4$ is moved to get $g(x) = (x-3)^2 + 2(x-3) + 4$. How was *f(x)* moved to get *g(x)*?

Solution

$f(x)$ was moved 3 units to the **right** to get $g(x)$. Be careful! When a function moves 3 units to the **right**, we put -3 next to every "x" on the function.

You should be careful when you are dealing with translation. It is very easy to confuse horizontal translation with vertical translation.

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Consider two different translations on the function $f(x) = x^2$

1. $f(x) = x^2 \rightarrow g(x) = (x+1)^2$

versus

2. $f(x) = x^2 \rightarrow g(x) = x^2 + 1$

The **first translation** is a shift of $f(x) = x^2$ 1 unit to the **left** since the $+1$ is included inside the square.

The **second translation** is a shift of $f(x) = x^2$ 1 unit **upwards** since the $+1$ is included at the **end** and **not** inside the square.

That's the difference between the vertical movement (sideways) and vertical movement (up or down).

Question

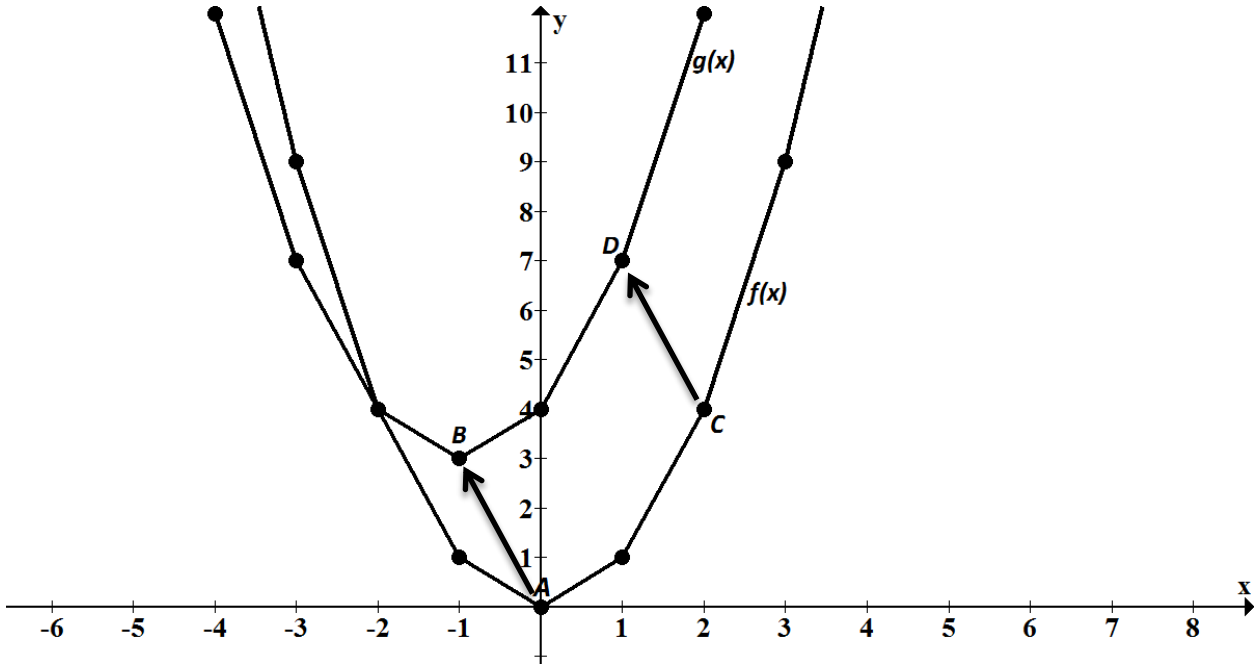
The graph of $f(x) = x^2$ is moved to get the graph of $g(x) = (x+1)^2 + 3$. How was *f(x)* moved to get *g(x)*?

Solution

Like I said above, the $+1$ next to "x" indicates a horizontal movement, that is, moving *f(x)* 1 unit to the **left**. The $+3$ at the end indicates vertical movement, that is, *f(x)* was also moved 3 units **upwards**.

Let's draw the graph of $f(x) = x^2$ and $g(x) = (x+1)^2 + 3$ to check if our answer is correct.

x	-4	-3	-2	-1	0	1	2	3	4
$f(x) = x^2$	16	9	4	1	0	1	4	9	16
$g(x) = (x+1)^2 + 3$	4	1	0	1	4	9	16	25	36



Focus on point A and point B, to move from point A to point B, we first move 1 unit to the **left**, and then 3 units **upwards** to get point. That proves the answer correct.

The same happens when we move from point C to point D. We move *C* 1 unit to the **left** and 3 units **upwards**. The same applies to all the other corresponding points on the graphs.

Question

The graph of $f(x) = x^2 + 4x + 5$ is moved to get the graph of $g(x) = (x+1)^2 + 4(x+1) + 5 - 2$. How was *f(x)* moved to get *g(x)*?

Solution

The $+1$ next to "$x's$" indicates a horizontal movement, that is, moving *f(x)* 1 unit to the **left**. The -2 at the end indicates vertical movement, that is, *f(x)* was also moved 2 units **downwards** (downwards because it is negative).

Turning Point of the Quadratic Function

The general form of a quadratic function exists in two forms, the ***Turning Point form*** and the ***Standard form***. I am going to use examples in the build-up of these terms.

Consider the function $f(x) = (x+2)^2 + 1$. This is the turning point *f(x).*

Consider the same function $f(x) = x^2 + 4x + 5$. This is the standard form of *f(x)*.

$f(x) = (x+2)^2 + 1$ $\downarrow$ Turning Point Form

$f(x) = x^2 + 4x + 5$ $\downarrow$ Standard Form

We may find one form using the other form. We can use the turning point form to get the standard form and the other way around.

Here's how we use the turning point form to get the standard form.

$(x+2)^2 + 1 = x^2 + 4x + 5$

You just simplify it.

$(x+2)^2 + 1$

$(x+2)(x+2) + 1$

$x^2 + 2x + 2x + 4 + 1$

$x^2 + 4x + 5$

That's how you get the standard form, you simplify the turning point form.

Now here's how to use the standard form to get the turning point form.

We use a method called "completing the square".

$x^2 + 4x + 5$

<u>Step 1</u>

Divide everything by the coefficient of x^2, that is, the number in front of x^2.

$x^2 + 4x + 5 = 1x^2 + 4x + 5$

The number in front of x^2 is 1, so divide everything by 1.

$\frac{1x^2}{1} + \frac{4x}{1} + \frac{5}{1} = x^2 + 4x + 5.$

<u>Step 2</u>

$x^2 + \mathbf{4}x + 5$

Take the coefficient of x and divide it by 2. The coefficient of x is $\mathbf{4}$, we divide this by 2.

$\frac{\mathbf{4}}{2} = \mathbf{2}$

<u>Step 3:</u> This step consists of two sub-steps a and b.

a) Take the value from step 2 and square it: $(\mathbf{2})^2 = \mathbf{4}$

Keep this value ($\mathbf{4}$), you are going to use it at the end.

b) Add the value from step 2 (the $\mathbf{2}$) to x and square both of them to get $(x + \mathbf{2})^2$. Then put the $+5$ from the original standard form to get $(x + \mathbf{2})^2 + 5$.

Finally, subtract the value we kept from step 3a (the $\mathbf{4}$) to get $(x + \mathbf{2})^2 + 5 - \mathbf{4}$.

That gives $(x + \mathbf{2})^2 + 5 - \mathbf{4} = (x + 2)^2 + 1$

That's how we get the turning point from the standard form.

Do some examples.

Example

$f(x) = x^2 + 10x + 6$ is in the standard form. Determine the turning point form of $f(x)$.

Solution

We use a method called "completing the square".

$x^2 + 10x + 6$

<u>Step 1</u>

Divide everything by the coefficient of x^2, that is, the number in front of x^2.

$x^2 + 10x + 6 = 1x^2 + 10x + 6$

The number in front of x^2 is 1, so divide everything by 1.

$\frac{1x^2}{1} + \frac{10x}{1} + \frac{6}{1} = x^2 + 10x + 6.$

<u>Step 2</u>

$x^2 + \mathbf{10}x + 6$

Take the coefficient of x and divide it by 2. The coefficient of x is $\mathbf{10}$, we divide this by 2.

$\frac{\mathbf{10}}{2} = \mathbf{5}$

<u>Step 3:</u> This step consists of two sub-steps a and b.

a) Take the value from step 2 and square it: $(\mathbf{5})^2 = \mathbf{25}$

Keep this value ($\mathbf{25}$), you are going to use it at the end.

b) Add the value from step 2 (the **5**) to x and square both of them to get $(x + \mathbf{5})^2$. Then put the $+6$ from the original standard form to get $(x + \mathbf{5})^2 + 6$.

Finally, subtract the value we kept from step 3a (the $\mathbf{25}$) to get $(x + \mathbf{5})^2 + 6 - \mathbf{25}$.

That gives $(x + \mathbf{5})^2 + 6 - \mathbf{25} = (x + 5)^2 - 19$

$(x + 5)^2 - 19$ is the turning point form of $x^2 + 10x + 6$

If by any chance you doubt your answer, you may simplify it and see if it gives the standard form you started with.

$= (x + 5)^2 - 19$

$= (x + 5)(x + 5) - 19$

$= x^2 + 5x + 5x + 25 - 19$

$= x^2 + 10x + 6$

The turning point form has taken us back to our standard form, this proves that our answer is correct.

Example

$f(x) = 2x^2 + 4x + 12$ is in the standard form. Determine the turning point form of $f(x)$.

Solution

We use a method called "completing the square".

$2x^2 + 4x + 12$

<u>Step 1</u>

Divide everything by the coefficient of x^2, that is, the number in front of x^2.

$2x^2 + 4x + 12$

The number in front of x^2 is 2, so divide everything by 2.

$\frac{2x^2}{2} + \frac{4x}{2} + \frac{12}{2} = x^2 + 2x + 6.$

<u>Step 2</u>

$x^2 + \mathbf{2}x + 6$

Take the coefficient of x and divide it by 2. The coefficient of x is $\mathbf{2}$, we divide this by 2.

$\frac{\mathbf{2}}{2} = \mathbf{1}$

<u>Step 3:</u> This step consists of two sub-steps a and b.

a) Take the value from step 2 and square it: $(\mathbf{1})^2 = \mathbf{1}$

Keep this value ($\mathbf{1}$), you are going to use it at the end.

b) Add the value from step 2 (the **1**) to x and square both of them to get $(x + \mathbf{1})^2$. Then put the $+6$ from the original standard form to get $(x + \mathbf{1})^2 + 6$.

Subtract the value we kept from step 3a (the $\mathbf{1}$) to get $(x + \mathbf{1})^2 + 6 - \mathbf{1}$

$(x + 1)^2 + 5$

Note that the coefficient of x^2 wasn't 1 in the beginning, it was 2, for that reason, we multiply everything by 2 at the end. This is the case every time the coefficient of x^2 is **Not** 1.

$2[(x + 1)^2 + 5]$

$2 \times (x + 1)^2 + 2 \times 5$

$2(x + 1)^2 + 10.$ That's our answer.

You may check your answer again.

$= 2(x^2 + 2x + 1) + 10$

$= 2x^2 + (2 \times 2x) + (2 \times 1) + 10$

$= 2x^2 + 4x + 2 + 10$

$= 2x^2 + 4x + 12$

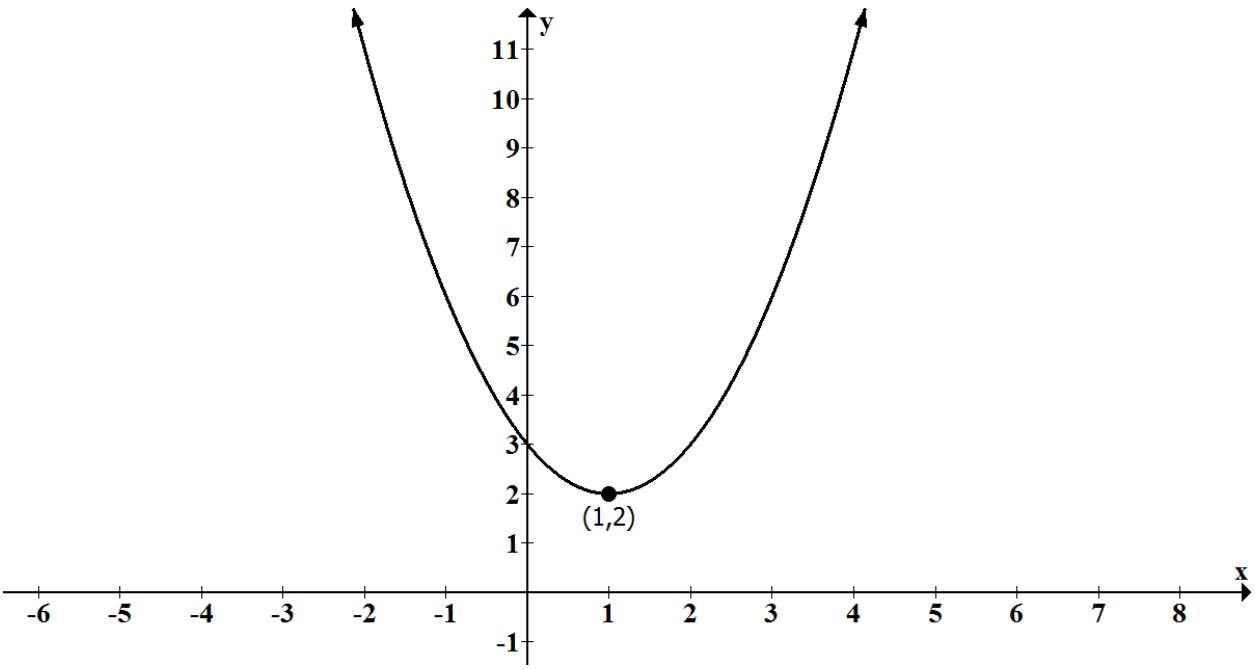
The turning point form has taken us back to our standard form, this proves that our answer is correct.

The Turning-Point using turning-point form

The turning-point on the graph of a ***quadratic function*** is simply the point at which the graph turns. Below are the examples of a turning-point.

Example

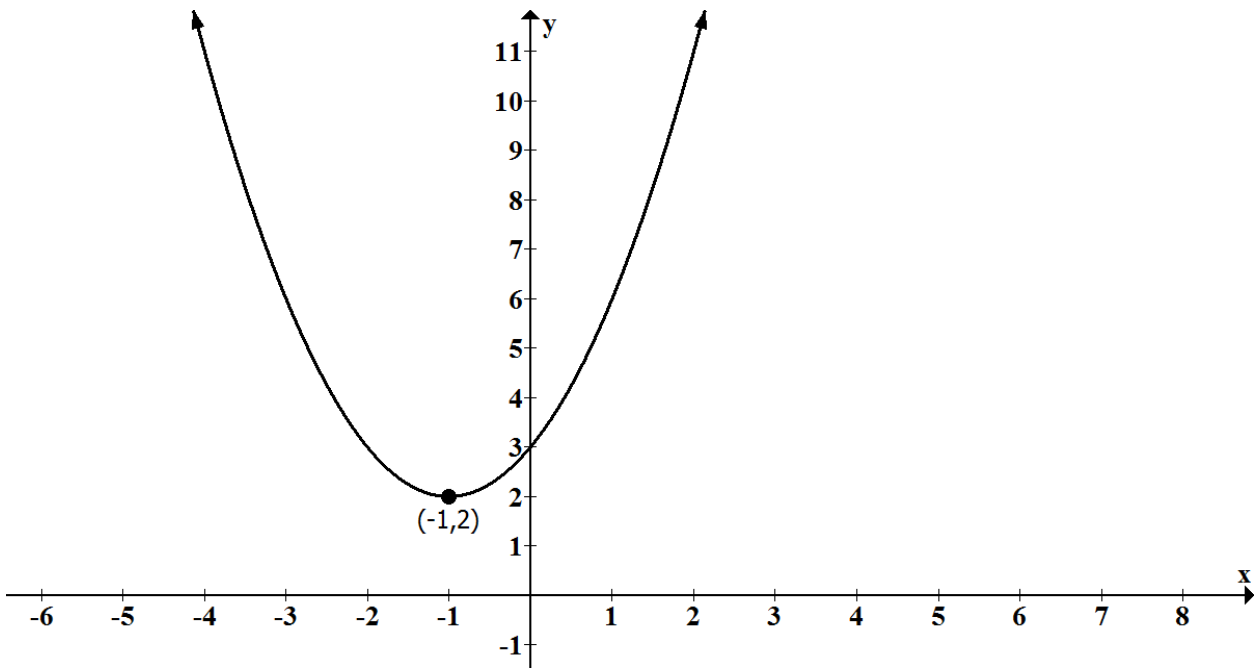
$f(x) = (x-1)^2 + 2$. Look for the turning-point on the graph below.



As you can see, the turning-point of this graph is at *(1;2)*

Example

$g(x) = (x+1)^2 + 2$. Look for the turning-point on the graph below.



As you can see, the turning-point of this graph is at *(-1;2)*

What do you notice about the coordinates of the turning-points and the functions in the two examples above?

For $f(x) = (x+1)^2 + 2$, the TP is *(-1,2).*

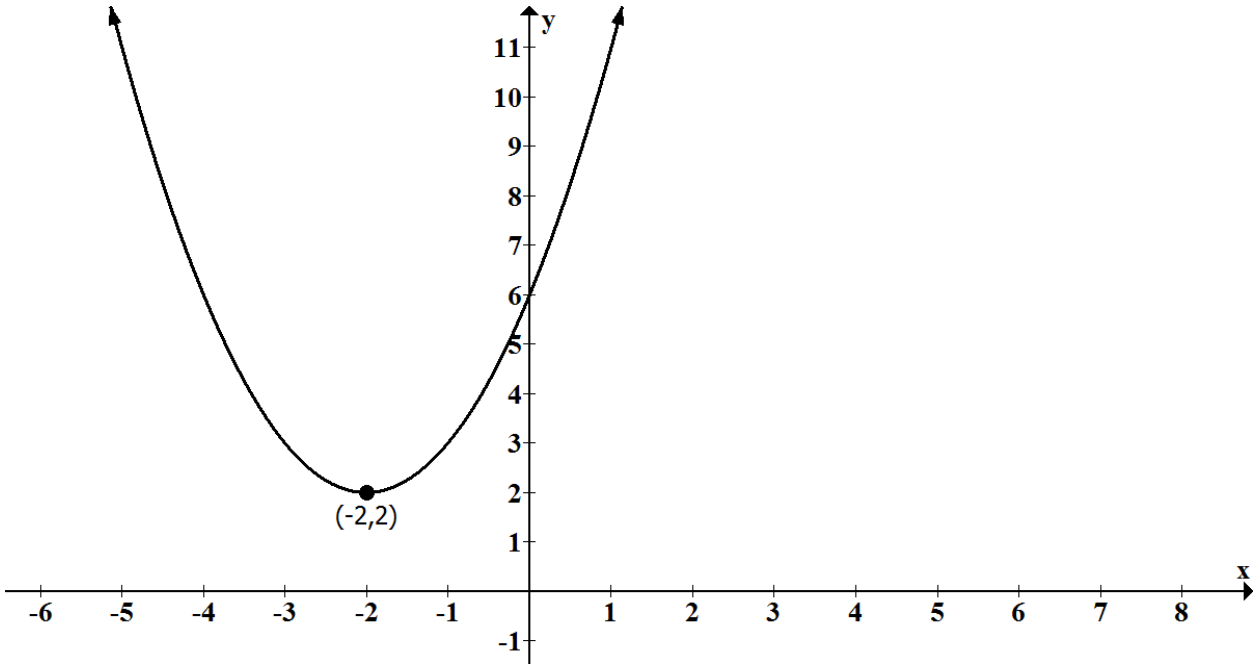
For $g(x) = (x-1)^2 + 2$, the TP is *(1,2).*

Therefore, we can safely conclude that the "opposite" of the number next to x is the x-coordinate of the TP, while the number at the end gives the y-coordinate of the TP. This is one of the advantages of having a quadratic function in the TP form.

Let's do more examples.

Example

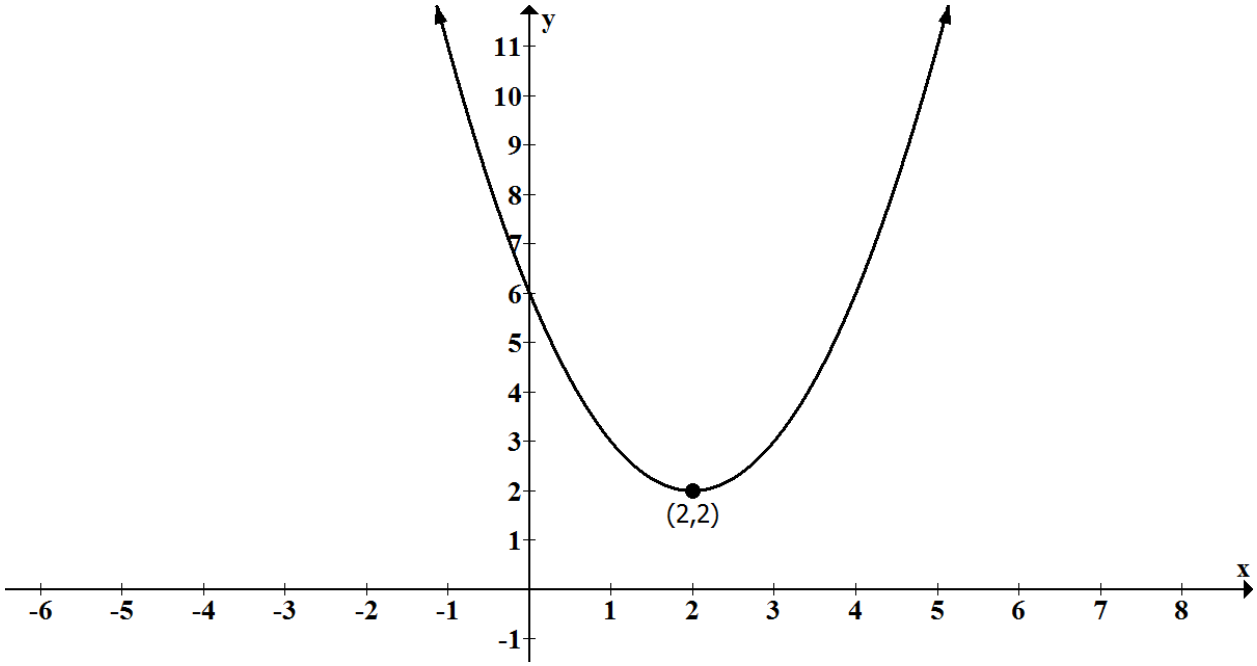
$f(x) = (x+2)^2 + 2$. Look for the turning-point on the graph below.



As you can see, the turning-point of this graph is at *(-2;2)*

Example

$g(x) = (x - 2)^2 + 2$. Look for the turning-point on the graph below.



As you can see, the turning-point of this graph is at *(-2;2)*

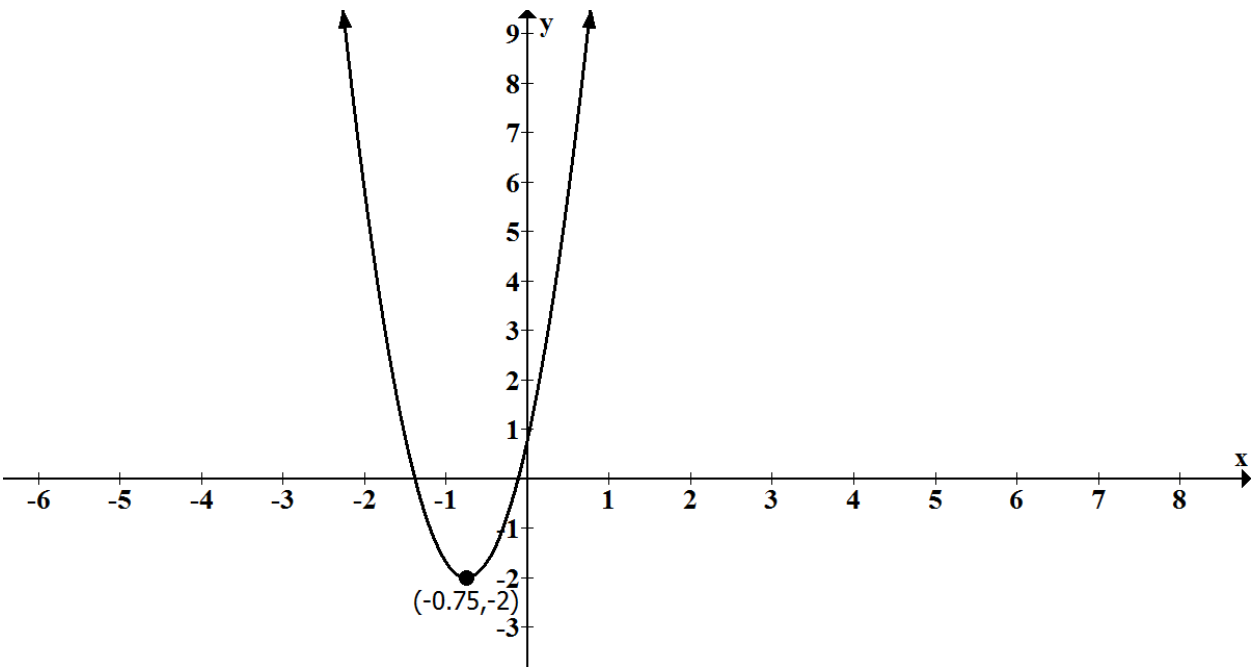
The same pattern continues.

For $f(x) = (x+2)^2 + 2$, the TP is *(-2,2).*

For $g(x) = (x-2)^2 + 2$, the TP is *(2,2).*

Example

$f(x) = 5\left(x + \frac{3}{4}\right)^2 - 2$. Look for the turning-point on the graph below.



As you can see, the turning-point of this graph is at $(-\frac{3}{4}; -2)$.

That's how we get the TP of a quadratic function when we have the turning-point form. In the next section, we learn how to find the turning-point using the standard form.

The Turning-Point using the Standard Form

The standard form of a quadratic function is

$f(x) = ax^2 + bx + c$

There is a formula for the x-coordinate of the Turning Point if the function is in standard form.

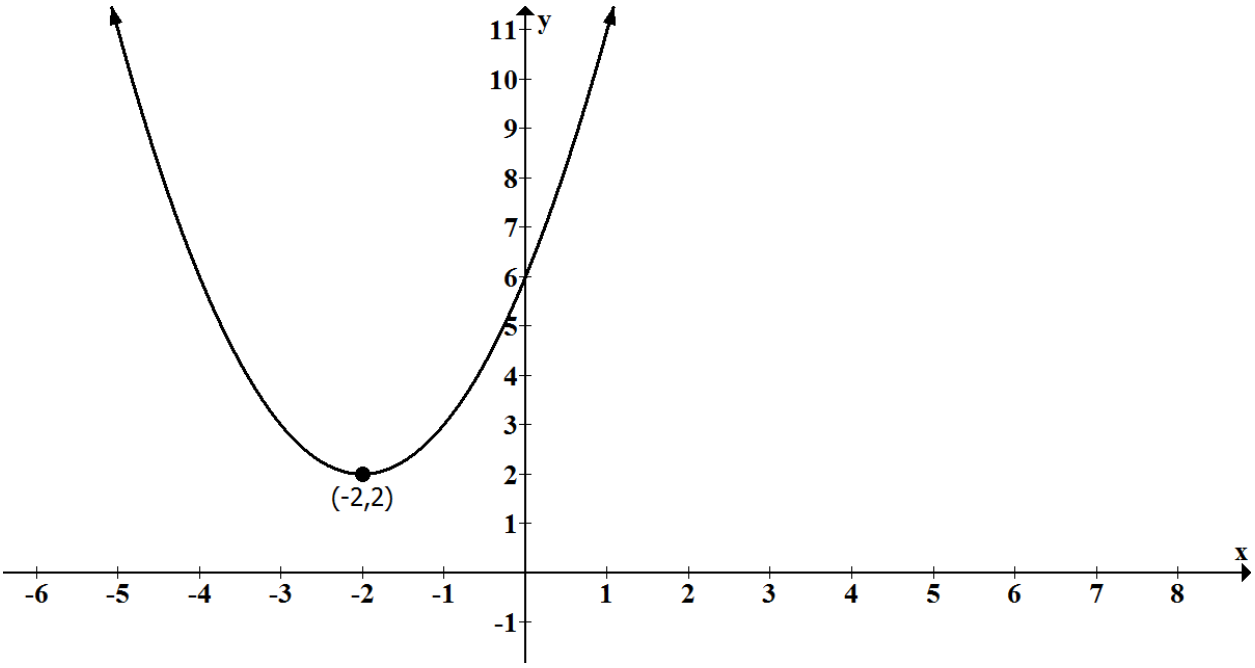
There is also a formula for the y-coordinate of the TP if the function is in the standard form.

$$x = -\frac{b}{2a} \qquad\qquad y = \frac{4ac - b^2}{4a}$$

Let's do a few examples to see how this actually works.

Example

$f(x) = x^2 + 4x + 6$. Look for the turning-point on the graph below.



As you can see, the turning-point of this graph is at *(-2;2).*

Now, let's use the formulas to see if they give us *(-2;2).*

$f(x) = x^2 + 4x + 6$, where $a = 1, b = 4, c = 6$

x-coordinate:

$x = -\frac{b}{2a}$

Substitute

$$x = -\frac{4}{2.1} = -\frac{4}{2} = -2$$

$x = -2$

So the x-coordinate of the TP is *-2*

y-coordinate:

$y = \frac{4ac-b^2}{4a}$

Remember that $a = 1, b = 4, c = 6$

Substitute

$$y = \frac{4.1.6 - (4)^2}{4.1}$$

$$y = \frac{24 - 16}{4}$$

$$y = \frac{8}{4} = 2$$

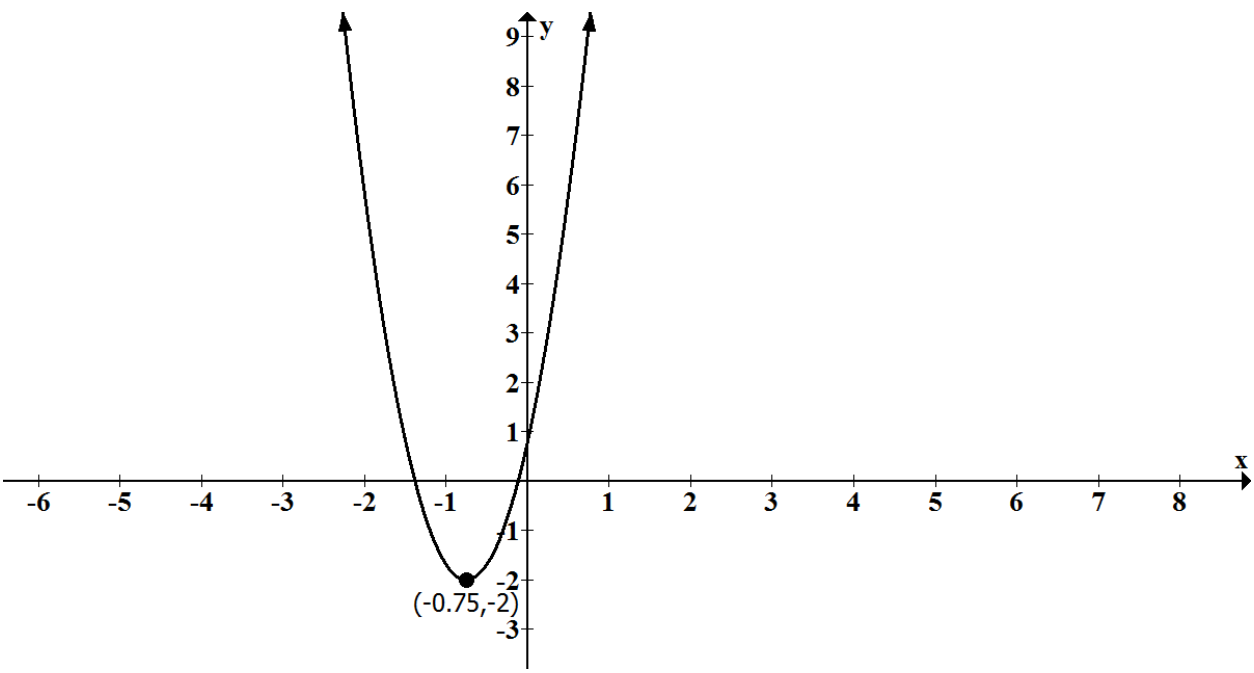
$$y = 2$$

So the y-coordinate of the TP is *2*

We have found that the coordinates of the TP using the formulas are *(-2;2),* which proves us right.

Example

$f(x) = 5x^2 + \frac{15}{2}x + \frac{13}{16}$. Look for the turning-point on the graph below.



As you can see, the turning-point of this graph is at $(-\frac{3}{4}; -2)$.

Now, let's use the formulas to see if they give us $(-\frac{3}{4}; -2)$.

$f(x) = 5x^2 + \frac{15}{2}x + \frac{13}{16}$ where $a = 5, b = \frac{15}{2}, c = \frac{13}{16}$

x-coordinate:

$x = -\frac{b}{2a}$

Substitute

$$x = -\frac{\left(\frac{15}{2}\right)}{2.5}$$

Make sure the $\left(\frac{15}{2}\right)$ is in brackets as shown above.

$$x = -\frac{\left(\frac{15}{2}\right)}{10}$$

$$x = -\frac{3}{4}$$

So the x-coordinate of the TP is $-\frac{3}{4}$

y-coordinate:

$y = \frac{4ac-b^2}{4a}$

Remember that $a = 5, b = \frac{15}{2}, c = \frac{13}{16}$

Substitute

$$y = \frac{4.5.\frac{13}{16} - \left(\frac{15}{2}\right)^2}{4.5}$$

$$y = \frac{\frac{65}{4} - \left(\frac{15}{2}\right)^2}{20}$$

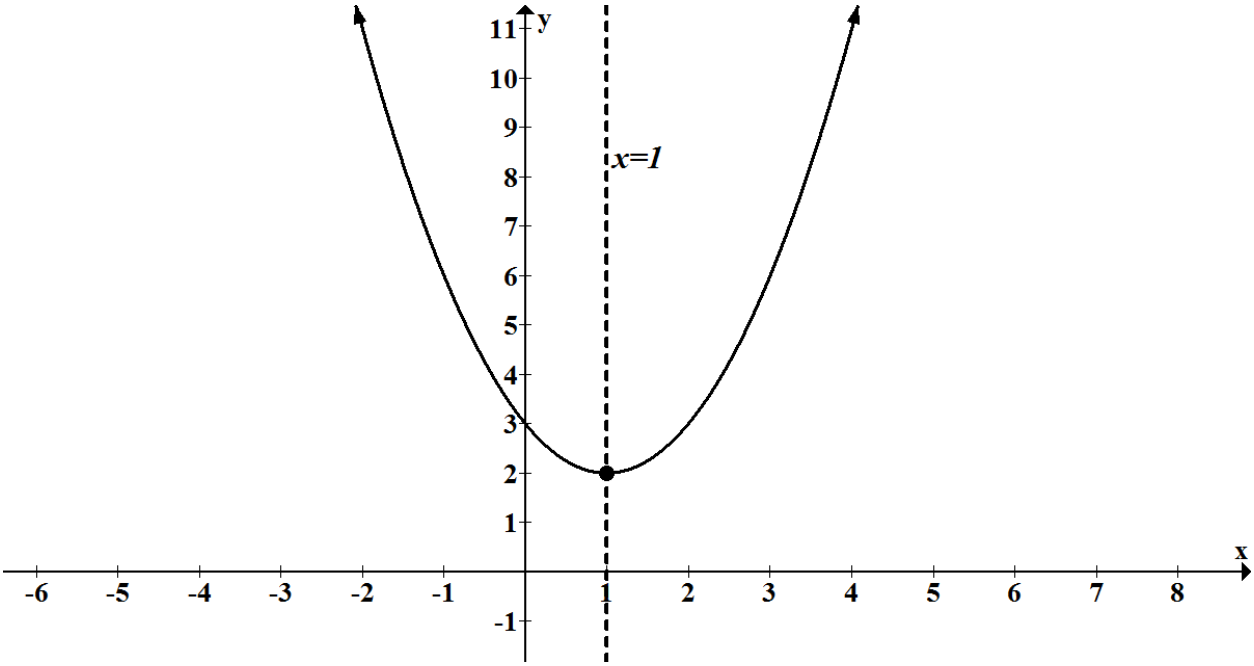
$y = -2$

So the y-coordinate of the TP is *-2*

We have found that the coordinates of the TP using the formulas are $\left(-\frac{3}{4}; -2\right)$, which proves us right.

Line of Symmetry

This short section is on the so called the *line of symmetry* of a quadratic function. Below is the graph of the function $f(x) = (x-1)^2 + 2$. The dotted line represents the line of symmetry. It is the line that basically cuts through the middle of the graph.



As you can see from the graph above, our line of symmetry is $x = 1$. In fact, the line of symmetry passes through the Turning-Point of the function, therefore it equals the x-coordinate of TP. We know from the function $f(x) = (x-1)^2 + 2$ that the TP is *(1,2).* We can then see that the line of symmetry is basically the *x-coordinate* of the TP.

Question

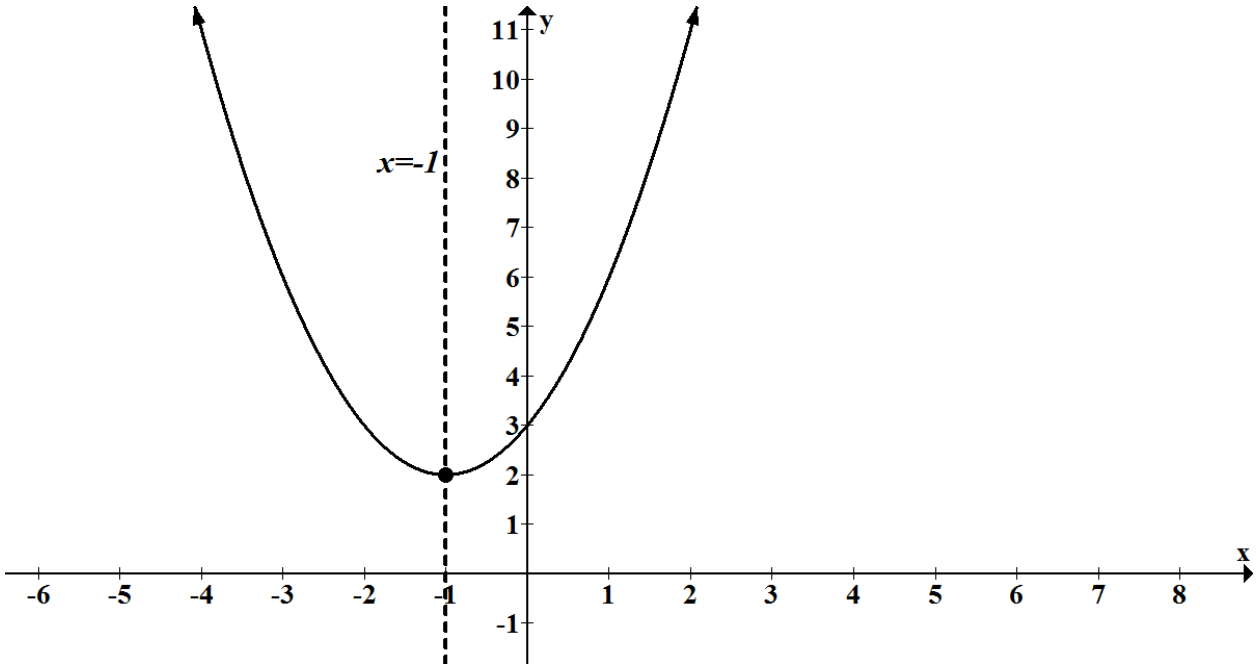
What is the line of symmetry of the function?

$f(x) = (x+1)^2 + 2$?

Solution

We know that the line of symmetry is the x-coordinate of the turning-point. From the function, we can tell the turning-point is $(-1; 2)$. Therefore, the line of symmetry is $x = -1$.

You may draw the graph of $f(x) = (x+1)^2 + 2$ to check if you're answer is correct.


y
x
x=-1
11
10
9
8
7
6
5
4
3
2
1
-1
-6
-5
-4
-3
-2
-1
1
2
3
4
5
6
7
8

Hyperbolic Function

Consider the function $y = \frac{2}{x}$

Question: Fill in the table of values below and draw the graph using the points.

x	-4	-3	-2	-1	0	1	2	3	4
y									

Solution

Our function here is $y = \frac{2}{x}$. We start by substituting -4 into the function.

$y = \frac{2}{-4}$. Punch this into your calculator to get the decimal answer.

$y = -0.5$

We repeat the same with the other values of x.

For $x = -3, y = \frac{2}{-3} = -0.6666 \ldots . = -0.67$

For $x = -2, y = \frac{2}{-2} = -1$

For $x = -1, y = \frac{2}{-1} = -2$

For $x = 0, y = \frac{2}{0}$, **Undefined**

For $x = 1, y = \frac{2}{1} = 2,$

For $x = 2, y = \frac{2}{2} = 1$

For $x = 3, y = \frac{2}{3} = 0.6666 \ldots = 0.67$

For $x = 4, y = \frac{2}{4} = 0.5$

Now let's fill the table.

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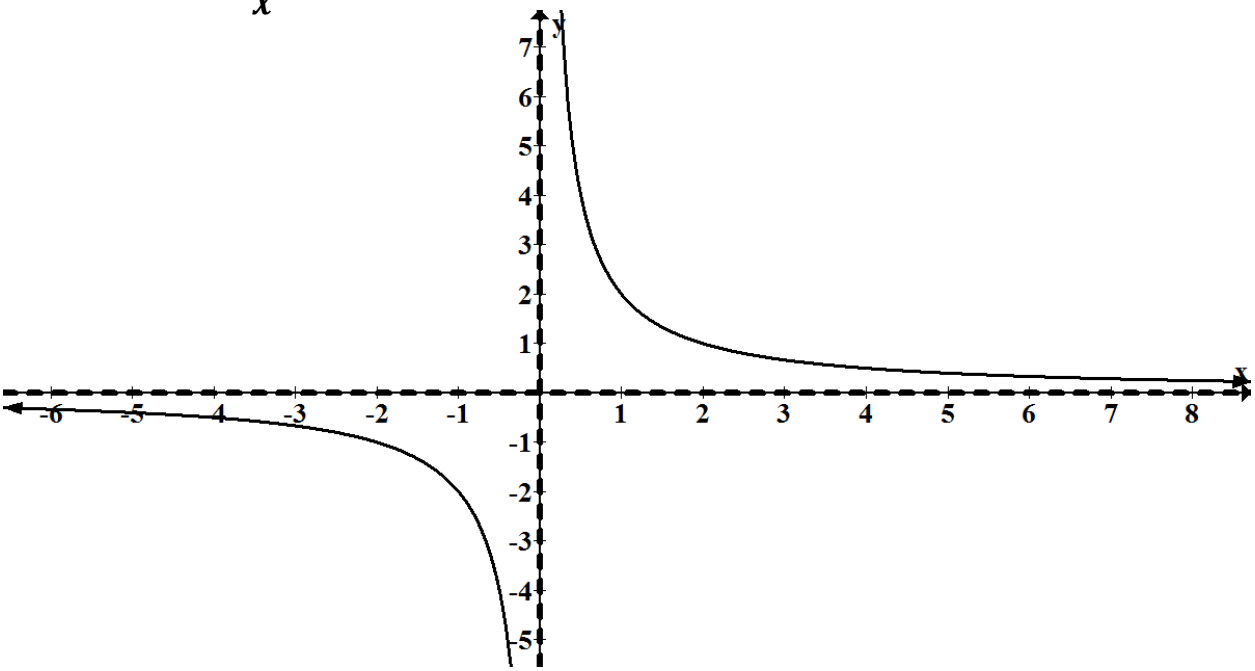
x	-4	-3	-2	-1	0	1	2	3	4
y	-0.5	-0.67	-1	-2	error	2	1	0.67	0.5

We nov $y = \frac{2}{x}$ d draw the graph.



Let's make a quick analysis on the graph. There are two important points to make.

- The graph has two parts, one in the 1st quadrant and the other in the 3rd quadrant.
- Notice that both parts get close to the axes but will never touch.

Question: Fill in the table of values below and draw the graph using the points.

Consider the function $y = \frac{-2}{x}$

x	-4	-3	-2	-1	0	1	2	3	4
y									

Solution

Our function here is $y = \frac{-2}{x}$. We start by substituting -4 into the function.

$y = \frac{-2}{-4}$. Punch this into your calculator to get the decimal answer.

$y = 0.5$

We repeat the same with the other values of x.

For $x = -3, y = \frac{-2}{-3} = 0.6666 \ldots . = 0.67$

For $x = -2, y = \frac{-2}{-2} = 1$

For $x = -1, y = \frac{-2}{-1} = 2$

For $x = 0, y = \frac{-2}{0}$, **Undefined**

For $x = 1, y = \frac{-2}{1} = -2,$

For $x = 2, y = \frac{-2}{2} = -1$

For $x = 3, y = \frac{-2}{3} = -0.6666 \ldots = -0.67$

For $x = 4, y = \frac{-2}{4} = -0.5$

Now let's fill the table.

x	-4	-3	-2	-1	0	1	2	3	4
y	0.5	0.67	1	2	error	-2	-1	-0.67	-0.5

We now plot the points and draw the graph.

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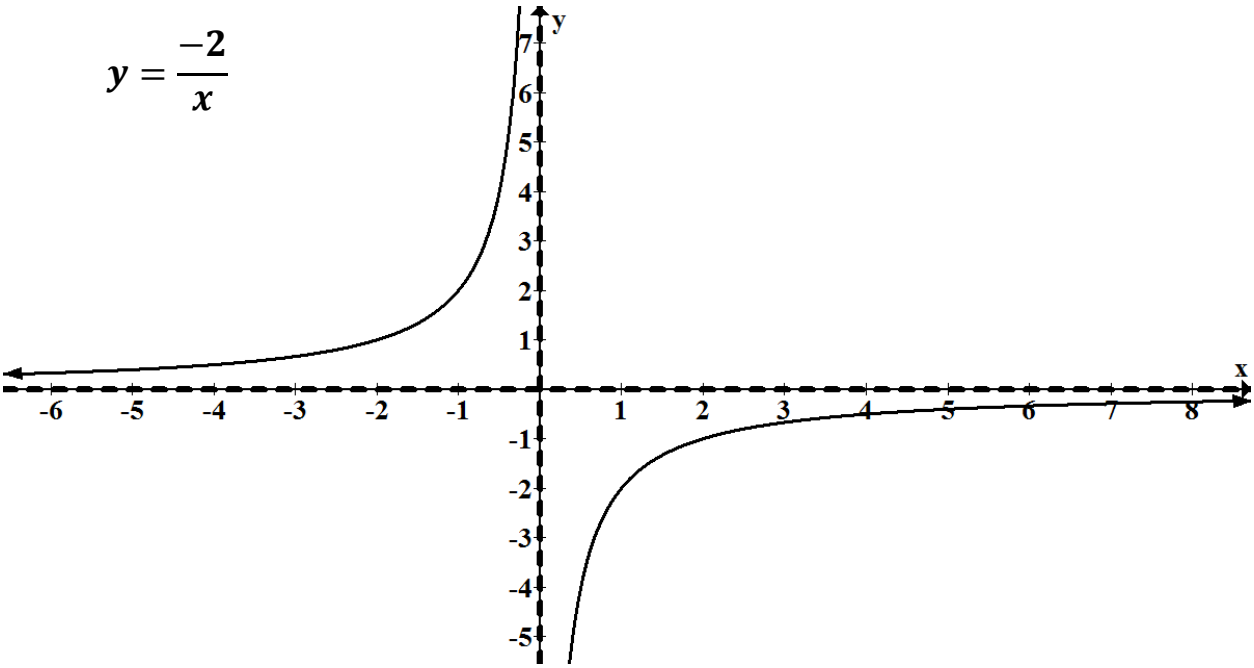
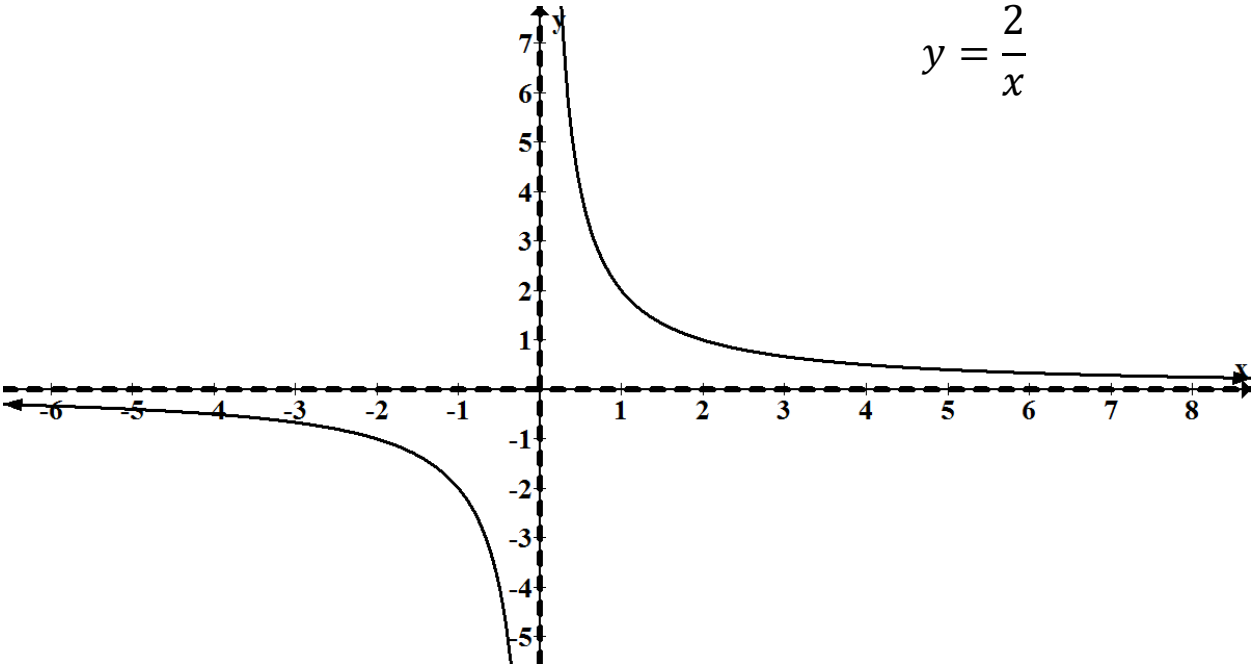


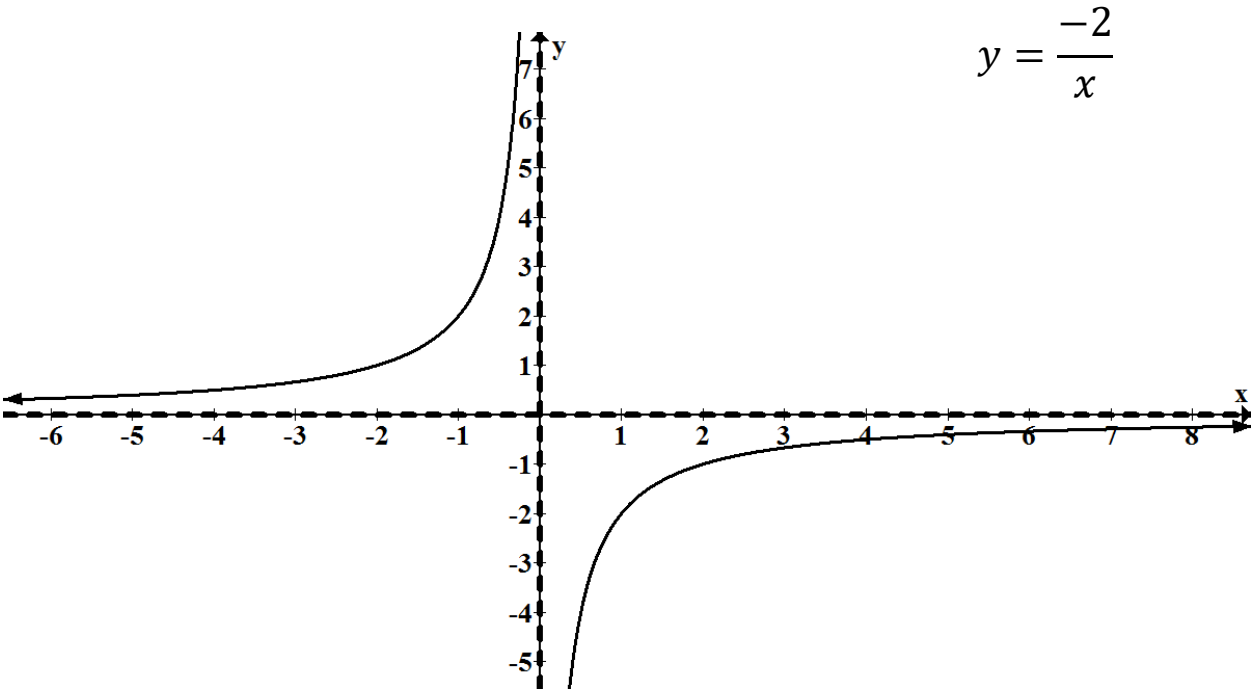
Again, we make a quick analysis on the graph. There are two important points to make.

- The graph has two parts, one in the 2nd quadrant and the other in the 4th quadrant.
- Notice that both parts get close to the axes but will never touch.

Do you notice the difference between the two graphs?



Versus



As you can see, the one with a "positive" 2 at the top has its parts in the 1st and 3rd quadrants, while the one with a negative "-2" at the top has its parts in the 2nd and 4th quadrants. That's the effect of having a negative number at the top.

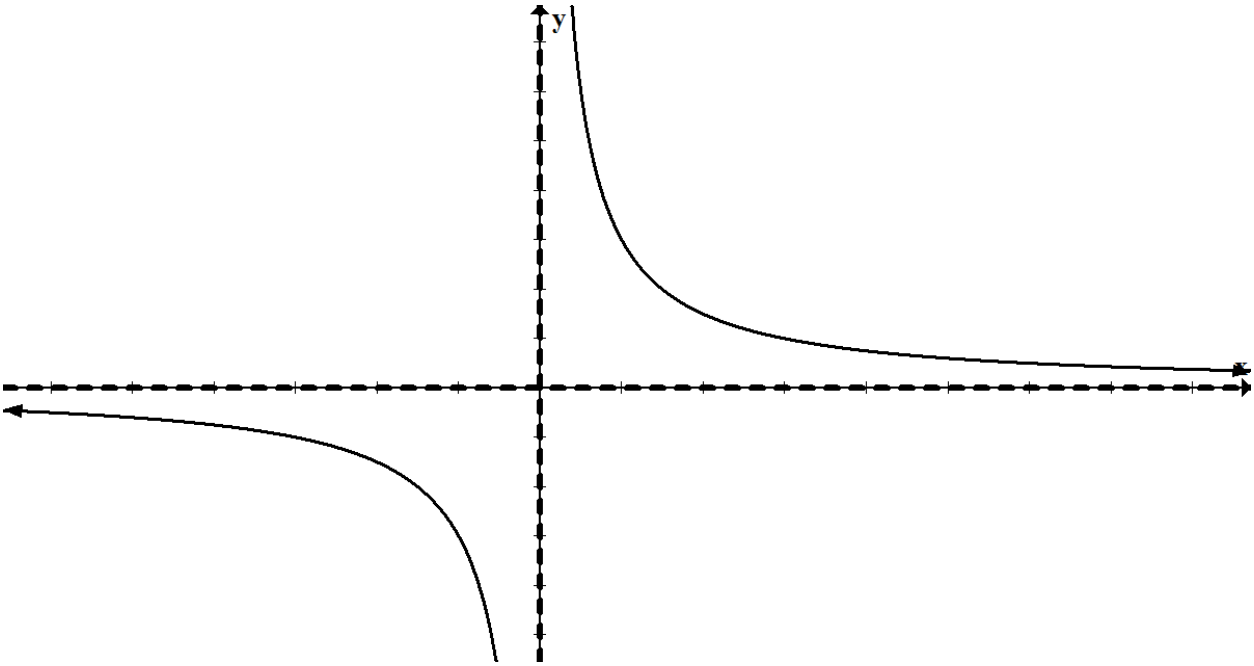
Question

a) Draw a rough sketch of the function $y = \frac{3}{x}$

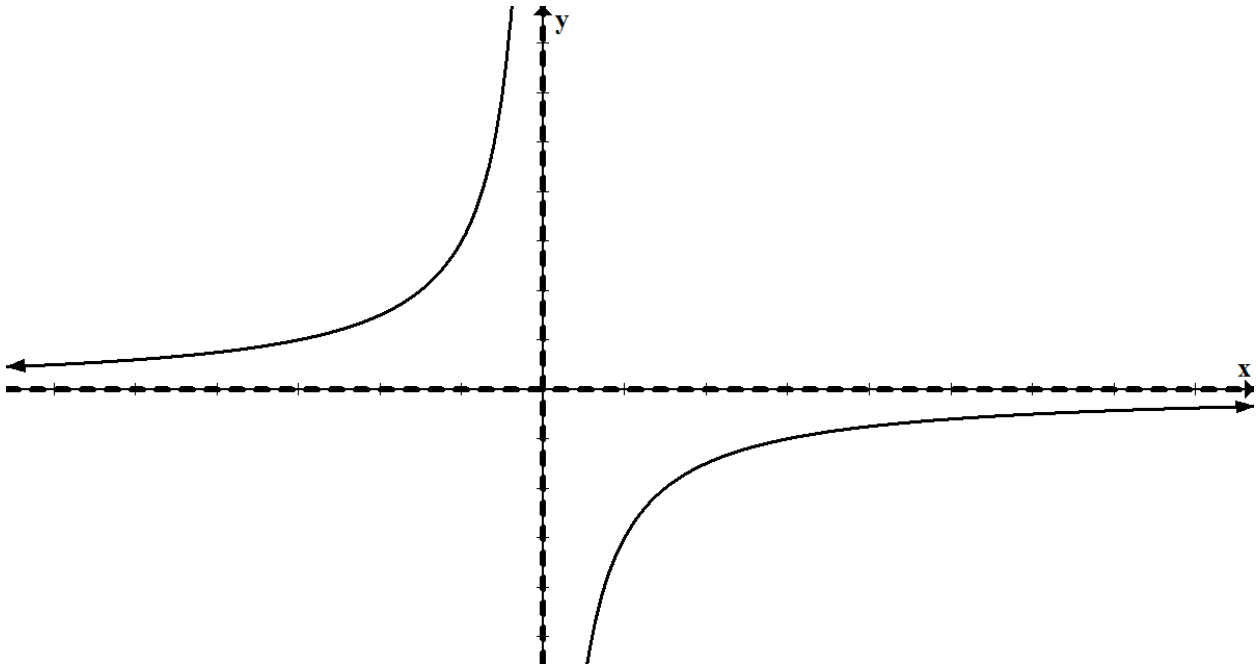
b) Draw a rough sketch of the function $y = \frac{-3}{x}$

Solution

a) Remember, when the number at the top is positive, the graph has its parts in the 1st and 3rd quadrants. So the graph would be something like this:



b) Remember, when the number at the top is positive, the graph has its parts in the 2nd and 4th quadrants. So the graph would be something like this:



In order to understand the hyperbolic function even better, we are going to relook at the function $y = \frac{2}{x}$

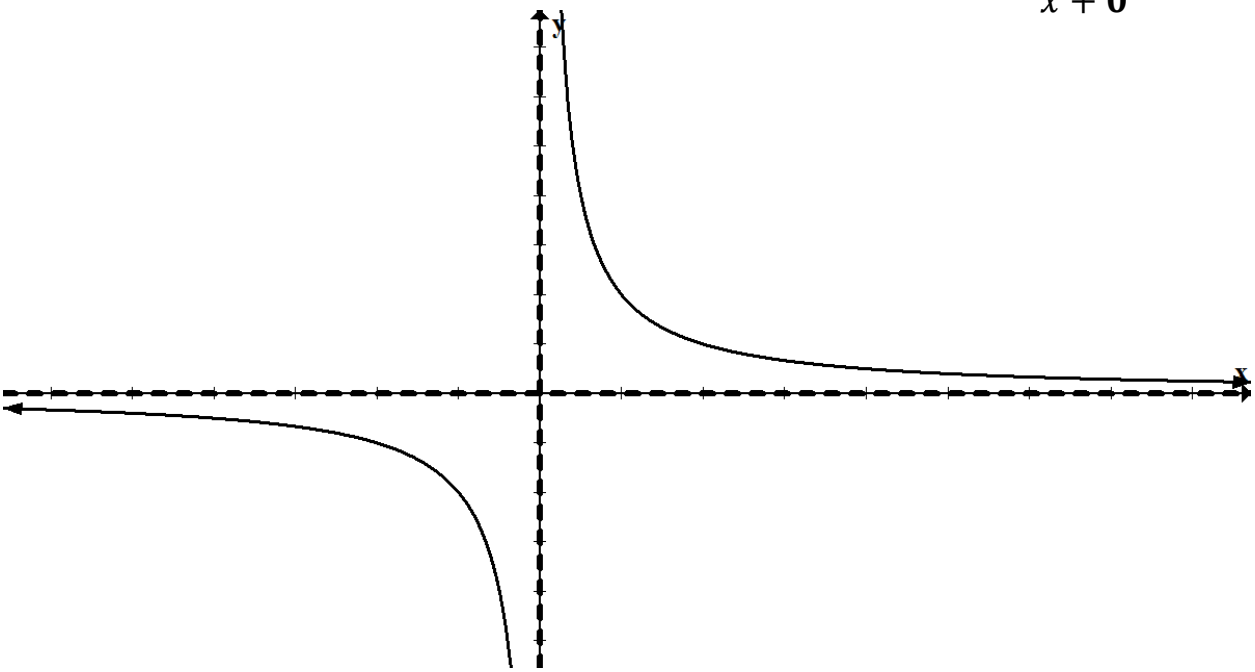
$y = \frac{2}{x}$, let's play around with it.

$y = \frac{2}{x} + \mathbf{0}$, this is just the same thing, let's add another 0 next to "x"

$y = \frac{2}{x+\mathbf{0}} + \mathbf{0}$, remember, this is still $= \frac{2}{x}$, we just added 0's, which doesn't have any effect

Draw the graph $y = \frac{2}{x+\mathbf{0}} + \mathbf{0}$, that's just the graph of $y = \frac{2}{x}$

$$y = \frac{2}{x + \mathbf{0}} + \mathbf{0}$$



We now compare the above with the function with another function: $y_1 = \frac{2}{x+\mathbf{1}} + \mathbf{3}$

$y_1 = \frac{2}{x+\mathbf{1}} + \mathbf{3}$ versus $y = \frac{2}{x+\mathbf{0}} + \mathbf{0}$

The difference is that y_1 has a "1" next to "x" and a "3" at the end. We fill the table of values and draw the graph of y_1. We will them check how the two functions differ.

x	-4	-3	-2	-1	0	1	2	3	4
y	2.3	2	1	error	5	4	3.7	3.5	3.4

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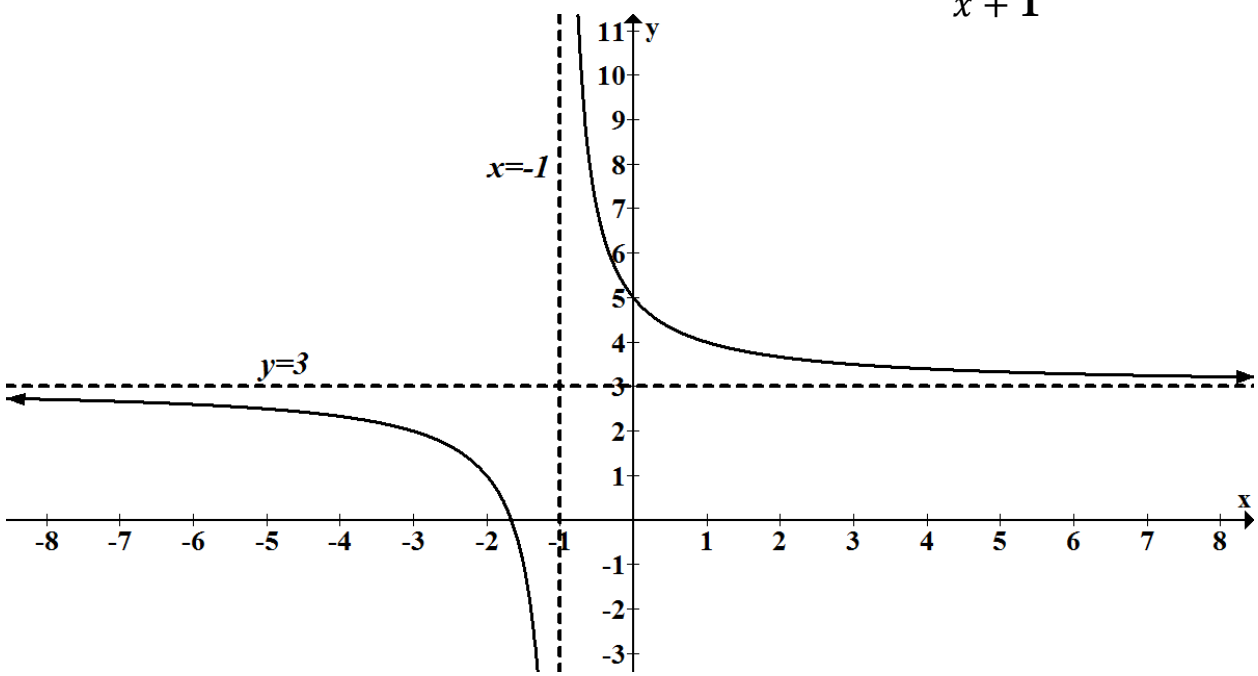
$$y = \frac{2}{x+1} + 3$$



Do you see the difference between the two graphs? Focus on the dotted lines. These are the lines to which the graphs get very close but never touch. For $y_1 = \frac{2}{x+1} + 3$, the vertical line is ***x = -1***, and the horizontal line is ***y = 3***. These numbers are also shown on the function itself.

$$y_1 = \frac{2}{x+1} + 3$$

x = -1 is the vertical line. Note that it has an opposite sign compared to the "+1" on the function

y = 3, the horizontal line.

In mathematics, we call these lines "***asymptotes***". For $y = \frac{2}{x+0} + 0$, the vertical line is ***x = 0***, and the horizontal line is ***y = 0***. These numbers are also shown on the function itself.

$$y_1 = \frac{2}{x+0} + 0$$

x = 0 is the vertical line

y = 0, the horizontal line.

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These are the lines to which the graphs get close but never cross. Let's summarize what we have just learnt.

The asymptote of $y = \frac{2}{x+\mathbf{0}} + \mathbf{0}$ are ***x = 0*** and ***y = 0***

The asymptote of $y_1 = \frac{2}{x+\mathbf{1}} + \mathbf{3}$ are ***x = -1*** and ***y = 3***

Therefore, we can safely conclude that the number next to "x" gives the vertical asymptote while the number at the end gives the horizontal asymptote.

In general, the hyperbolic function is in the form:

$$y = \frac{a}{x + \boldsymbol{p}} + \boldsymbol{q}$$

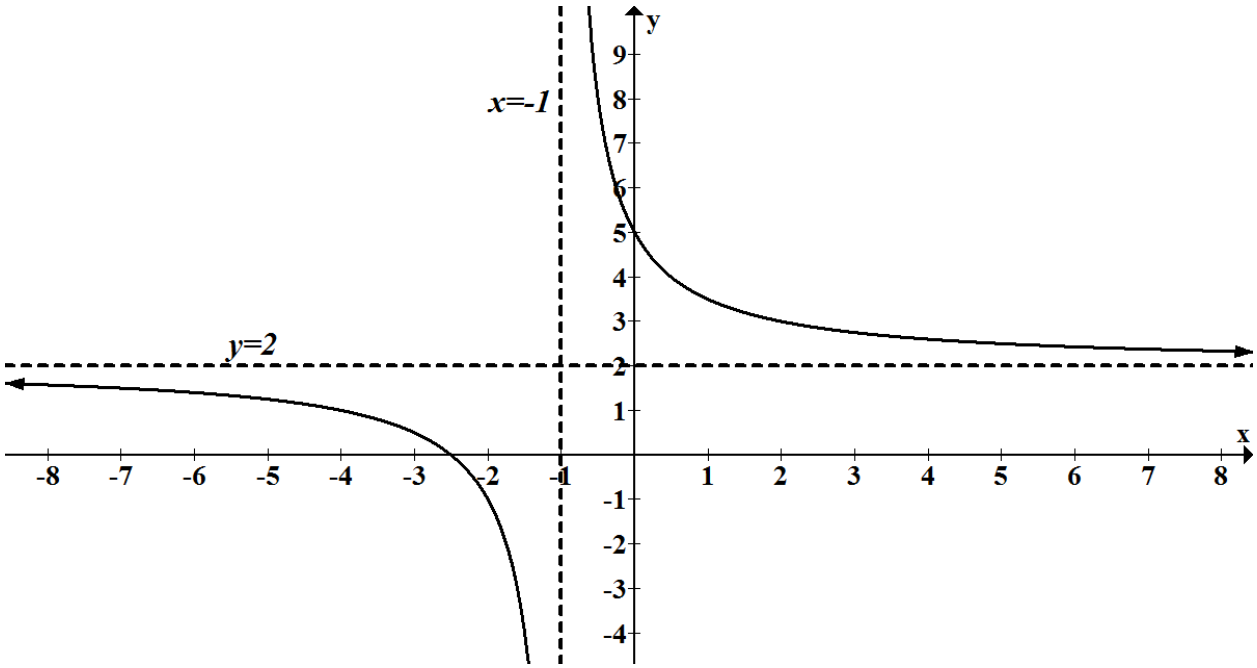
where the "opposite" of ***p*** gives the vertical asymptote and the value of ***q*** is the horizontal asymptote.

Below are some more examples of a hyperbolic function and their respective asymptotes.

Example:

$$y = \frac{3}{x + \mathbf{1}} + \mathbf{2}$$

p = 1 **q = 2**

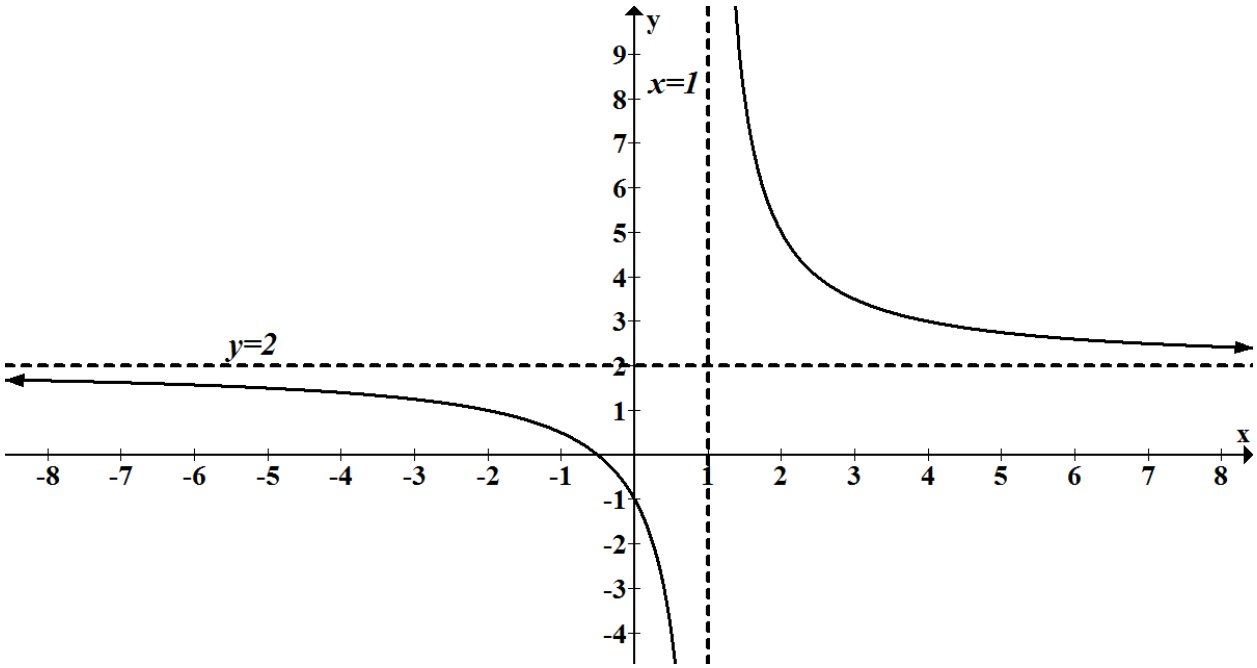


Remember, the vertical asymptote is the opposite of "p", so in this case, our vertical asymptote is x = -1 instead of "1". The horizontal asymptote is just the value of "q", thus our horizontal asymptote is y = 2. We can clearly see these asymptotes on the graph.

Example:

$$y = \frac{3}{x - \mathbf{1}} + \mathbf{2}$$

p = -1 **q = 2**

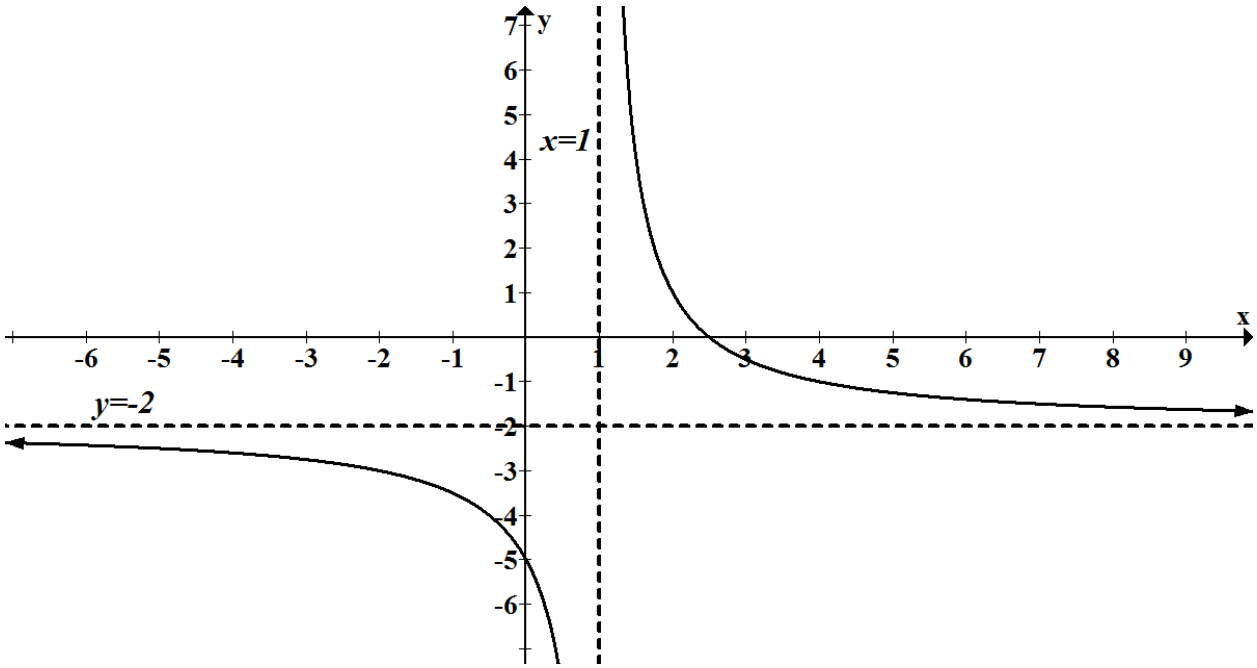


Remember, the vertical asymptote is the opposite of "p", so in this case, our vertical asymptote is x = 1 instead of "-1". The horizontal asymptote is just the value of "q", thus our horizontal asymptote is y = 2. We can clearly see these asymptotes on the graph.

Example:

$$y = \frac{3}{x - \mathbf{1}} - \mathbf{2}$$

p = -1 **q = -2**



Remember, the vertical asymptote is the opposite of "p", so in this case, our vertical asymptote is x = 1 instead of "-1". The horizontal asymptote is just the value of "q", thus our horizontal asymptote is y = -2. We can clearly see these asymptotes on the graph.

Question

Determine the asymptotes of each of the following functions.

a) $y = \frac{4}{x+2} + 5$

b) $y = \frac{4}{x-2} + 5$

c) $y = \frac{4}{x-2} - 5$

d) $y = \frac{-2}{x-0.5} + 3$

e) $y = \frac{-4}{x+\frac{1}{2}} - 20$

Solution

a) $y = \frac{4}{x+2} + 5$. The vertical asymptote is the opposite of "p", so in this case, our vertical asymptote is ***x = -2*** instead of "***2***". The horizontal asymptote is just the value of "q", thus our horizontal asymptote is ***y = 5***.

b) $y = \frac{4}{x-2} + 5$. The vertical asymptote is the opposite of "p", so in this case, our vertical asymptote is ***x = 2*** instead of "***-2***". The horizontal asymptote is just the value of "q", thus our horizontal asymptote is ***y = 5***.

c) $y = \frac{4}{x-2} - 5$. The vertical asymptote is the opposite of "p", so in this case, our vertical asymptote is ***x = 2*** instead of "***-2***". The horizontal asymptote is just the value of "q", thus our horizontal asymptote is ***y = -5***.

d) $y = \frac{-2}{x-0.5} + 3$. The vertical asymptote is the opposite of "p", so in this case, our vertical asymptote is ***x = 0.5*** instead of "***-0.5***". The horizontal asymptote is just the value of "q", thus our horizontal asymptote is ***y = 3***.

e) $y = \frac{-4}{x+\frac{1}{2}} - 20$. The vertical asymptote is the opposite of "p", so in this case, our vertical asymptote is ***x =*** $-\frac{1}{2}$ instead of "$\frac{1}{2}$". The horizontal asymptote is just the value of "q", thus our horizontal asymptote is ***y = -20***.

Question

Draw a rough sketch of each of the following functions. Clearly show the asymptotes.

a) $y = \frac{2}{x+1} + 3$

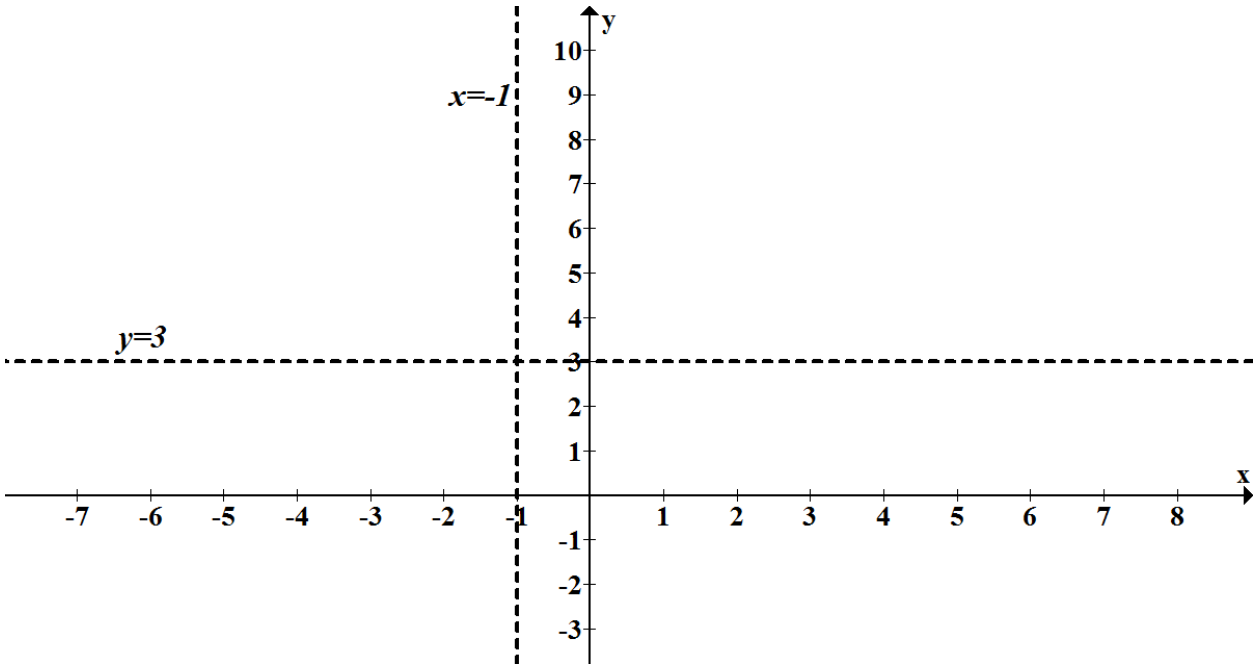
b) $y = \frac{-2}{x+1} + 3$

c) $y = \frac{2}{x-1} + 3$

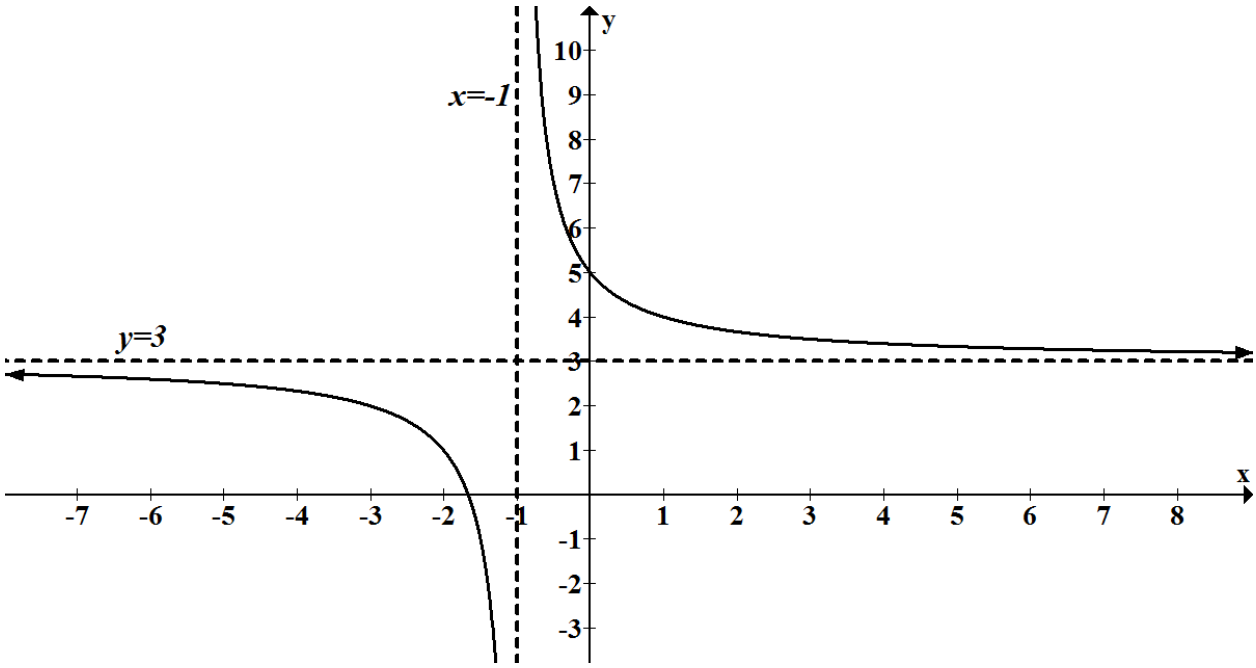
d) $y = \frac{-2}{x-1} + 3$

Solution

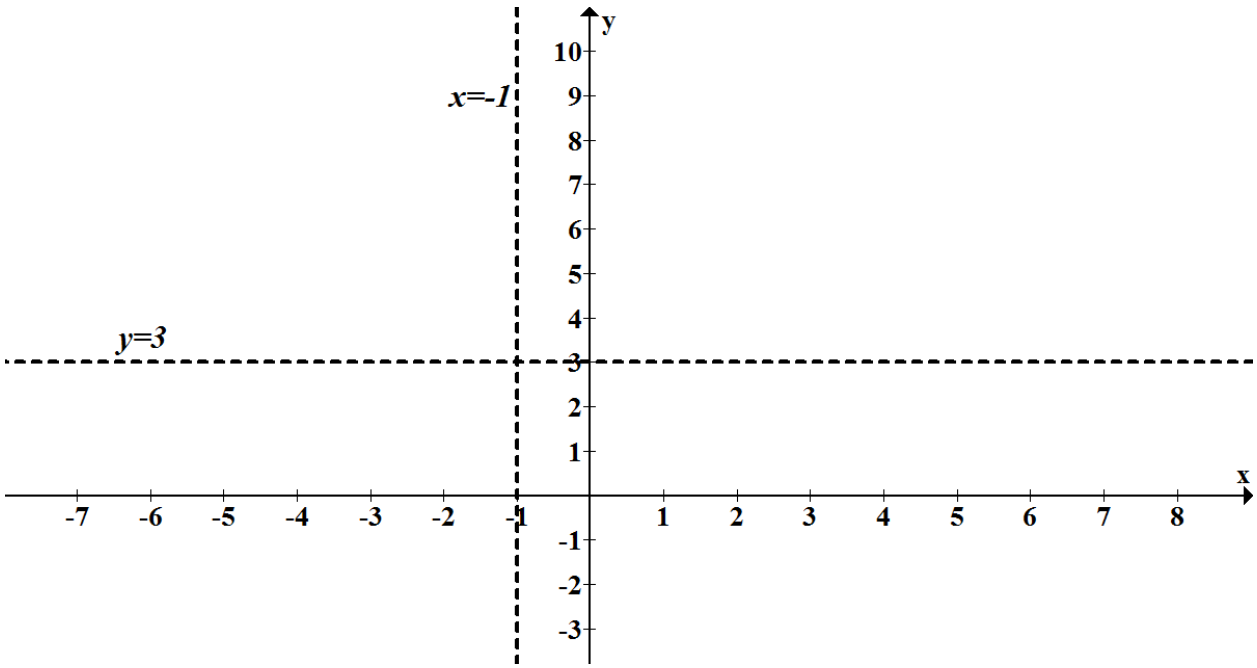
a) Let's start by writing the asymptotes from $y = \frac{2}{x+1} + 3$. The asymptotes are x = -1 and y = 3. Start by sketching the asymptotes.



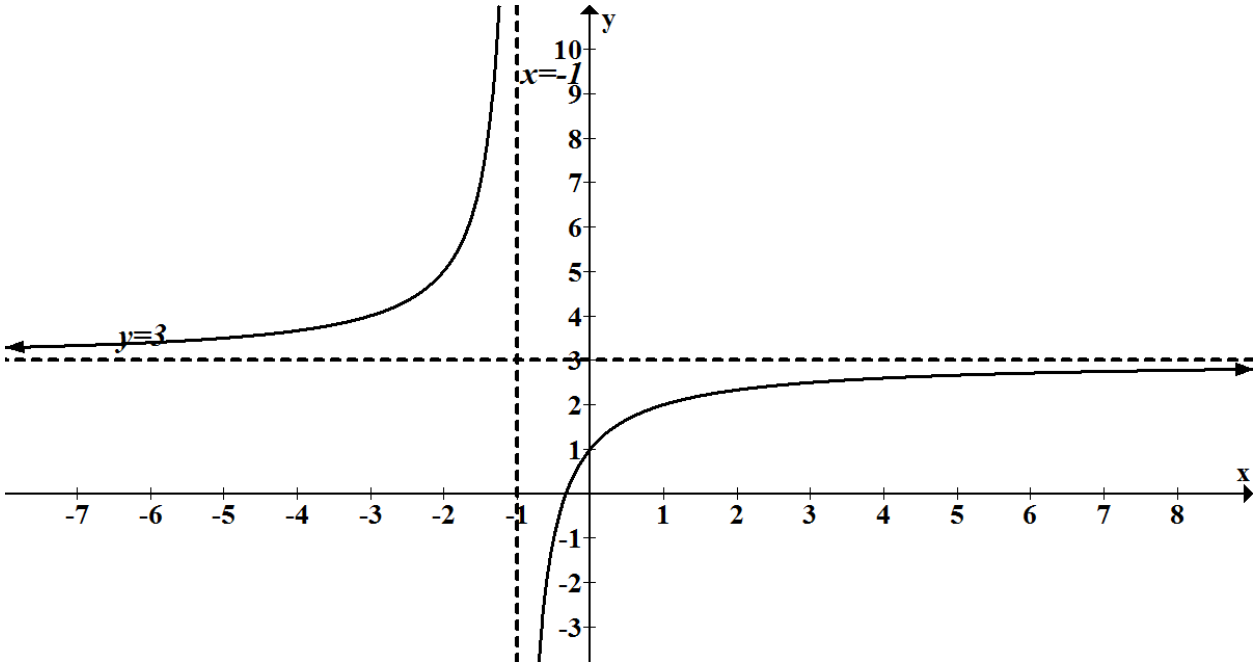
Then decide on where to draw the two parts of the graph. Since the "2" in $y = \frac{2}{x+1} + 3$ is positive, the top part will almost look like it is in the 1st quadrant, while the other part will almost be like it is in the 3rd quadrant.



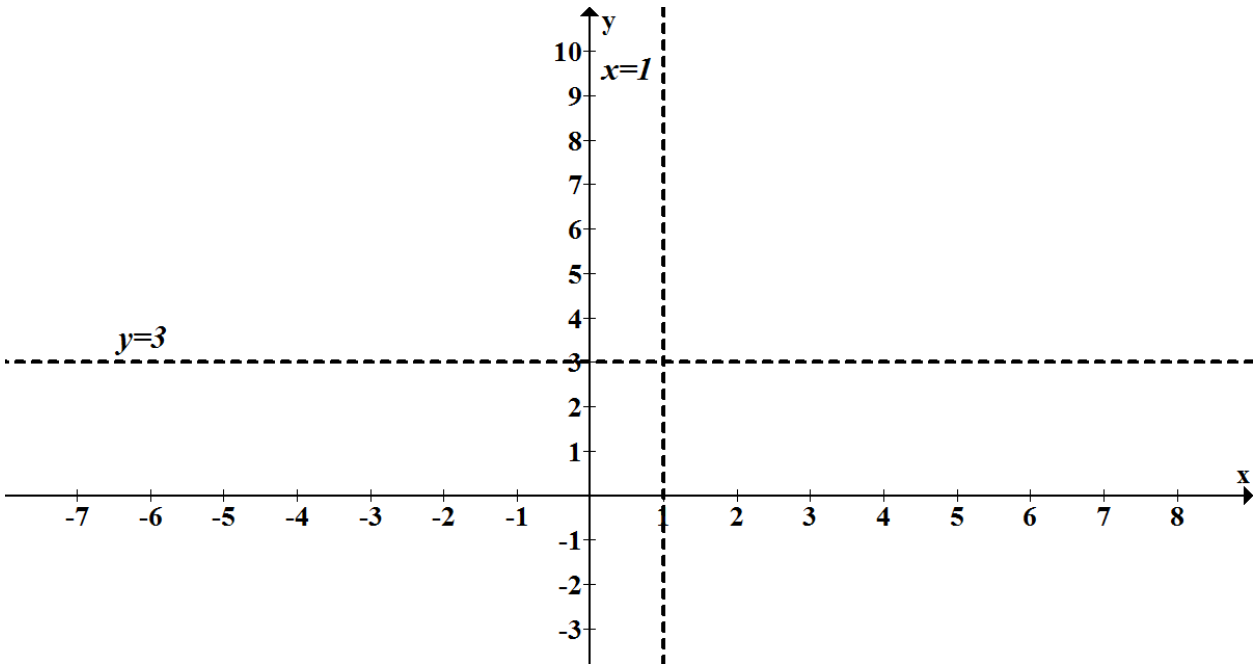
b) Let's start by writing the asymptotes from $y = \frac{-2}{x+1} + 3$. The asymptotes are x = -1 and y = 3. Start by sketching the asymptotes.



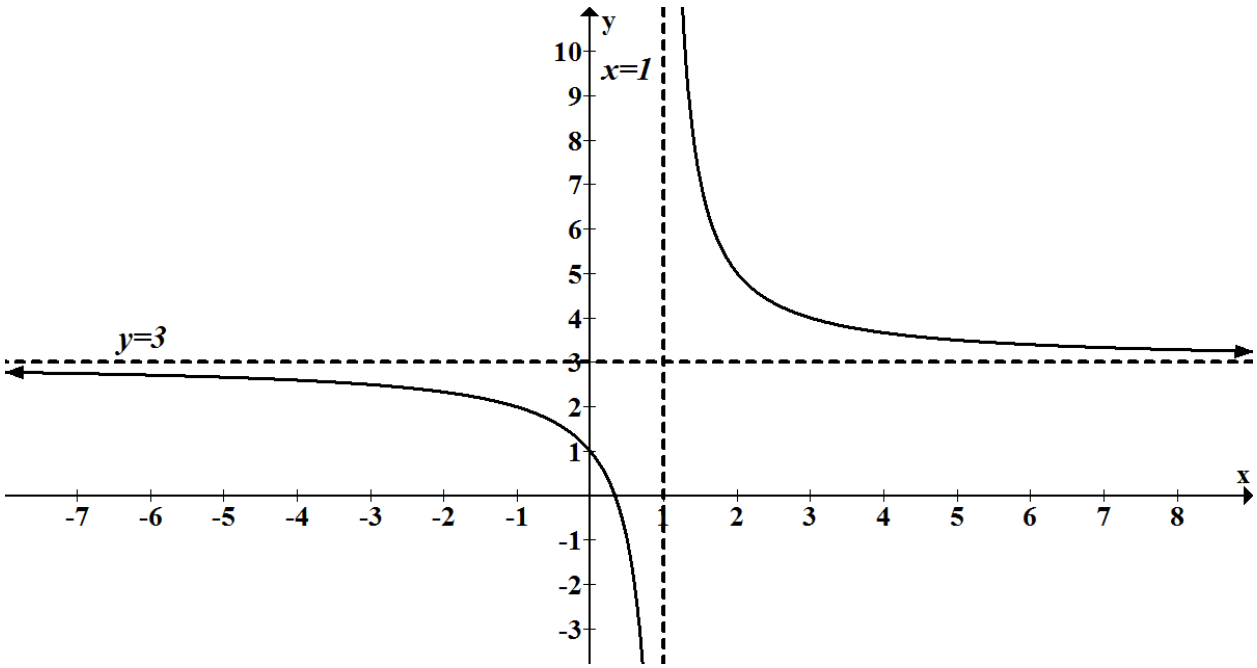
Then decide on where to draw the two parts of the graph. Since the "2" in $y = \frac{-2}{x+1} + 3$ is negative, the top part will almost look like it is in the 2nd quadrant, while the other part will almost be like it is in the 4th quadrant.



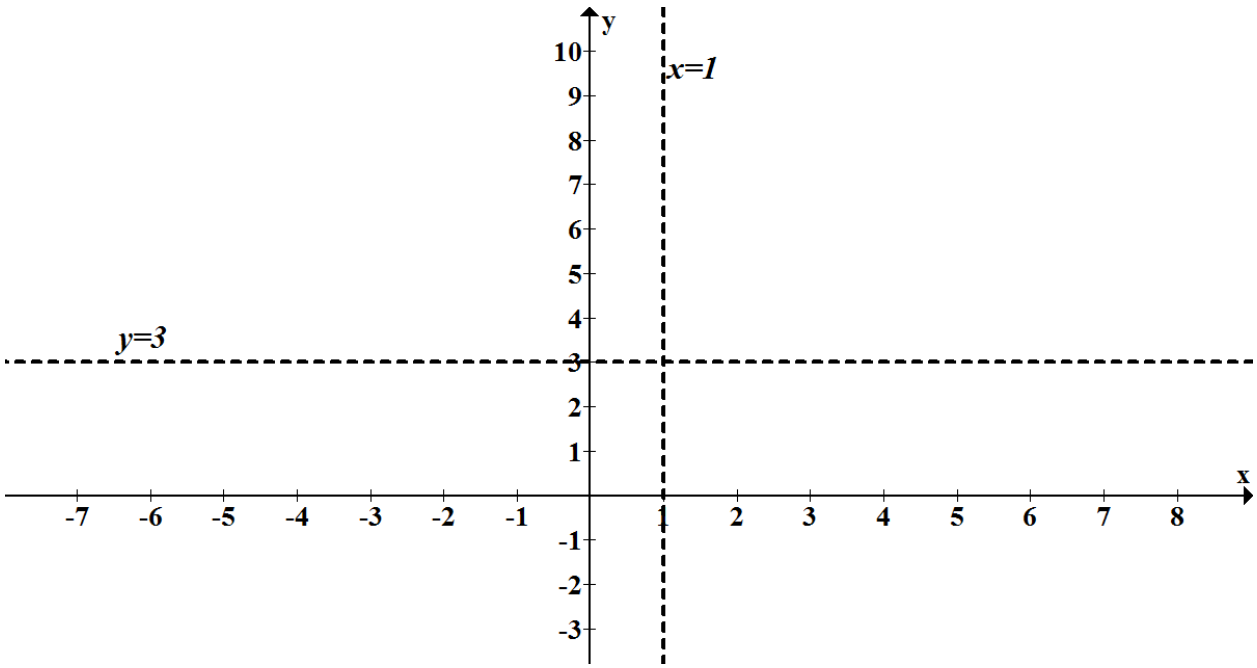
c) Let's start by writing the asymptotes from $y = \frac{2}{x-1} + 3$. The asymptotes are x = 1 and y = 3. Start by sketching the asymptotes.



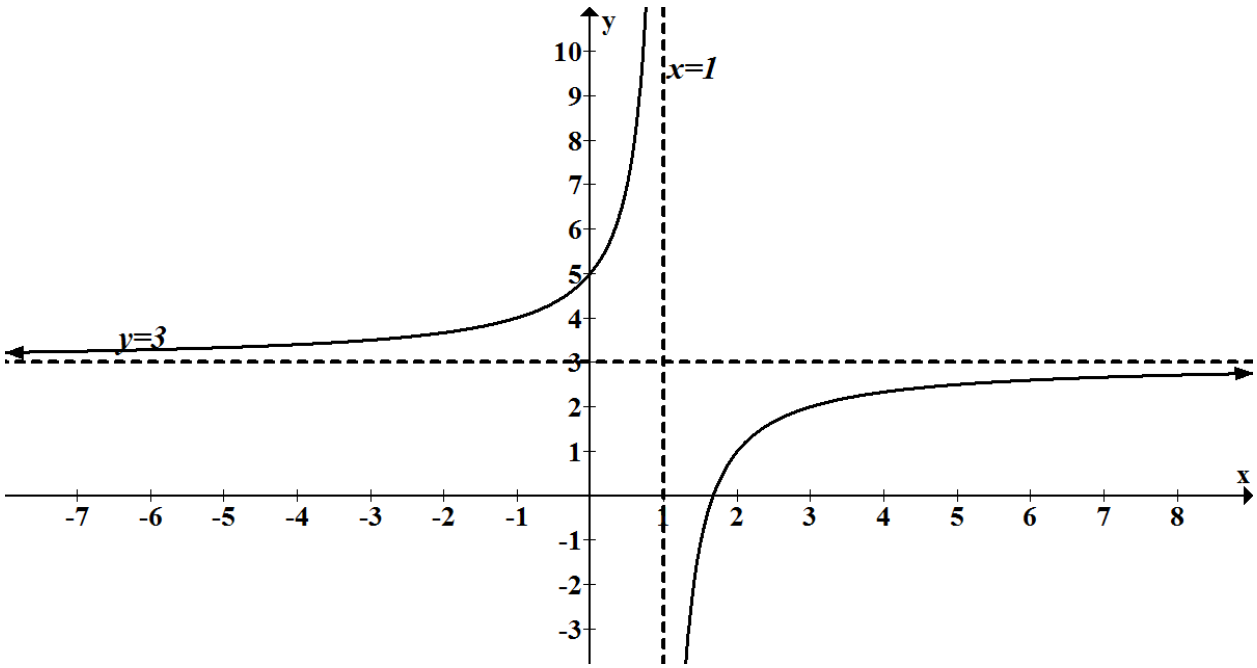
Then decide on where to draw the two parts of the graph. Since the "2" in $y = \frac{2}{x-1} + 3$ is positive, the top part will almost look like it is in the 1st quadrant, while the other part will almost be like it is in the 3rd quadrant.



d) Let's start by writing the asymptotes from $y = \frac{-2}{x-1} + 3$. The asymptotes are x = 1 and y = 3. Start by sketching the asymptotes.



Then decide on where to draw the two parts of the graph. Since the "2" in $y = \frac{-2}{x-1} - 3$ is negative, the top part will almost look like it is in the 2nd quadrant, while the other part will almost be like it is in the 4th quadrant.



Exponential Function

Consider the function $y = 2^x$

Example: Fill in the table of values and draw the graph.

x	-4	-3	-2	-1	0	1	2	3	4
y									

Solution

Our function here is $y = 2^x$. We start by substituting -4.

$y = 2^{-4}$. Punch this into your calculator to get $y = \frac{1}{16} = 0.0625$

We repeat the same other the other x values.

$y = 2^{-3} = 0.125$

$y = 2^{-2} = 0.25$

$y = 2^{-1} = 0.5$

$y = 2^0 = 1$

$y = 2^1 = 2$

$y = 2^2 = 4$

$y = 2^3 = 8$

$y = 2^4 = 16$

Fill the table and draw the graph.

x	-4	-3	-2	-1	0	1	2	3	4
y	0.0625	0.125	0.25	0.5	1	2	4	8	16

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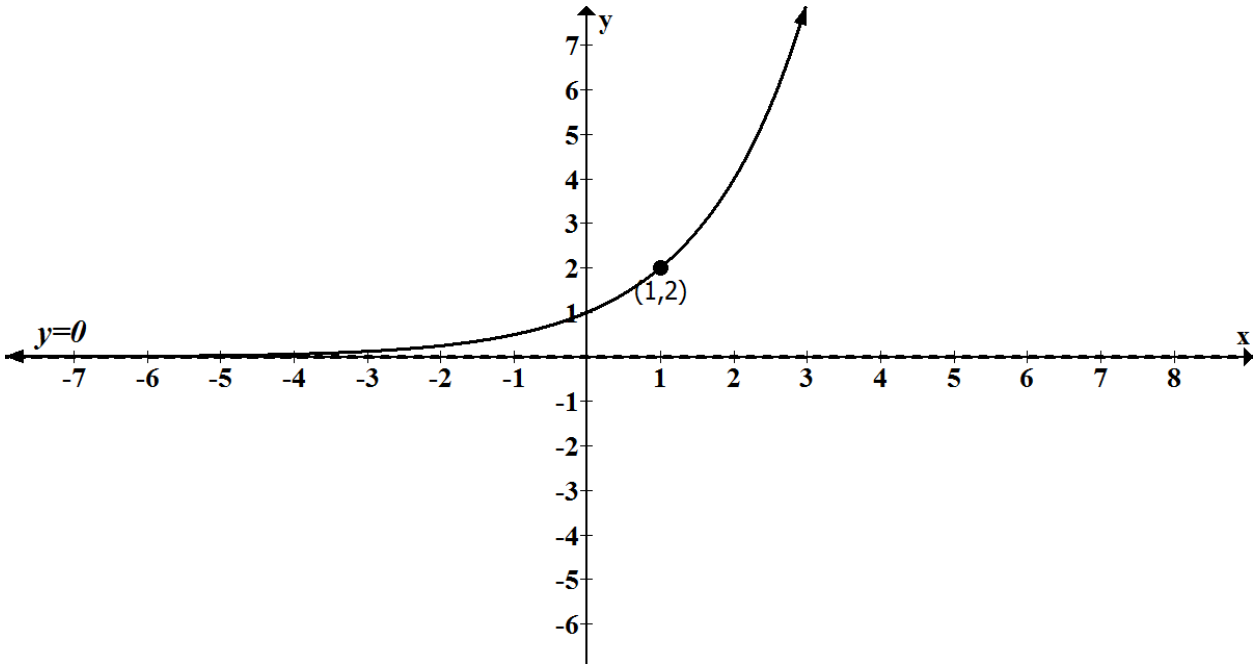


Make a quick analysis on the graph. There are four important points to make.

- The graph crosses the y-axis at (0 ; 1)
- The x-axis is the asymptote of the graph. That's because the x-axis is the line to which the graph gets close but never crosses.
- The graph is upward-slopping.
- At x = 1, y = 2, meaning that one of the coordinates is (1 ; 2)

Question

Consider the function $y = -2^x$. Fill in the table of values and draw the graph.

x	-4	-3	-2	-1	0	1	2	3	4
y									

Solution

Our function here is $y = 2^x$. We start by substituting -4.

$y = -2^{-4}$. Punch this into your calculator to get $y = -\frac{1}{16} = -0.0625$

We repeat the same other the other x values.

$y = -2^{-3} = -0.125$

$y = -2^{-2} = -0.25$

$y = -2^{-1} = -0.5$

$y = -2^{0} = -1$

$y = -2^{1} = -2$

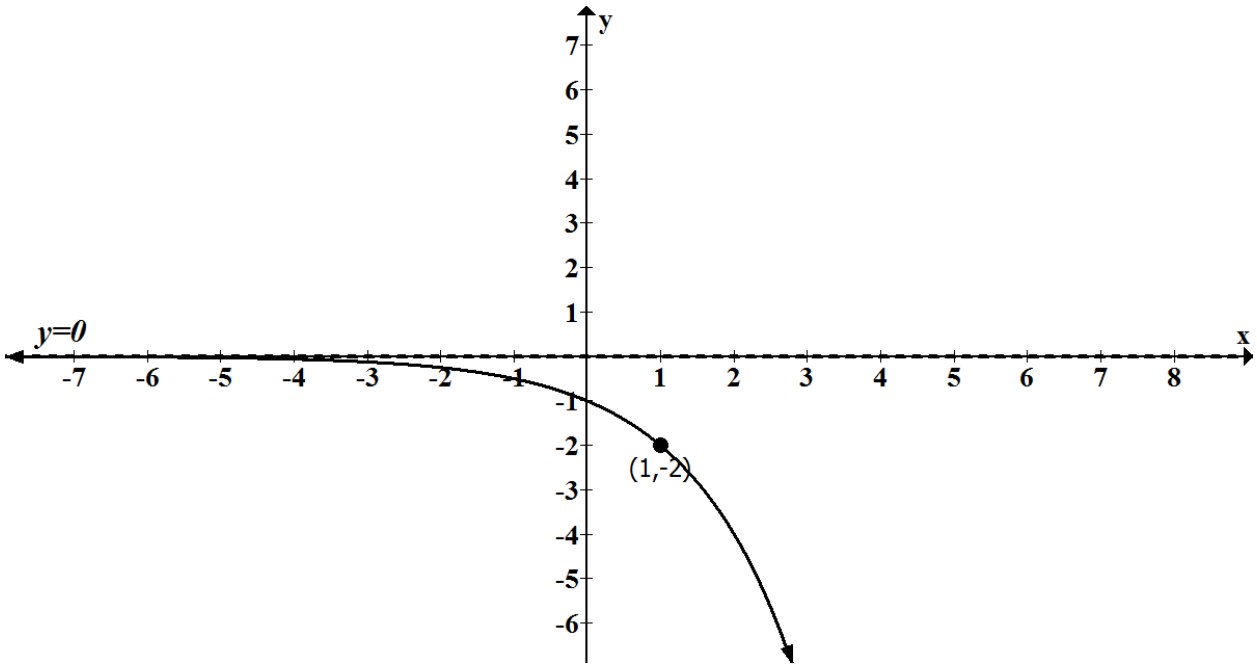
$y = -2^{2} = -4$

$y = -2^{3} = -8$

$y = -2^{4} = -16$

Fill the table and draw the graph.

x	-4	-3	-2	-1	0	1	2	3	4
y	-0.0625	-0.125	-0.25	-0.5	-1	-2	-4	-8	-16

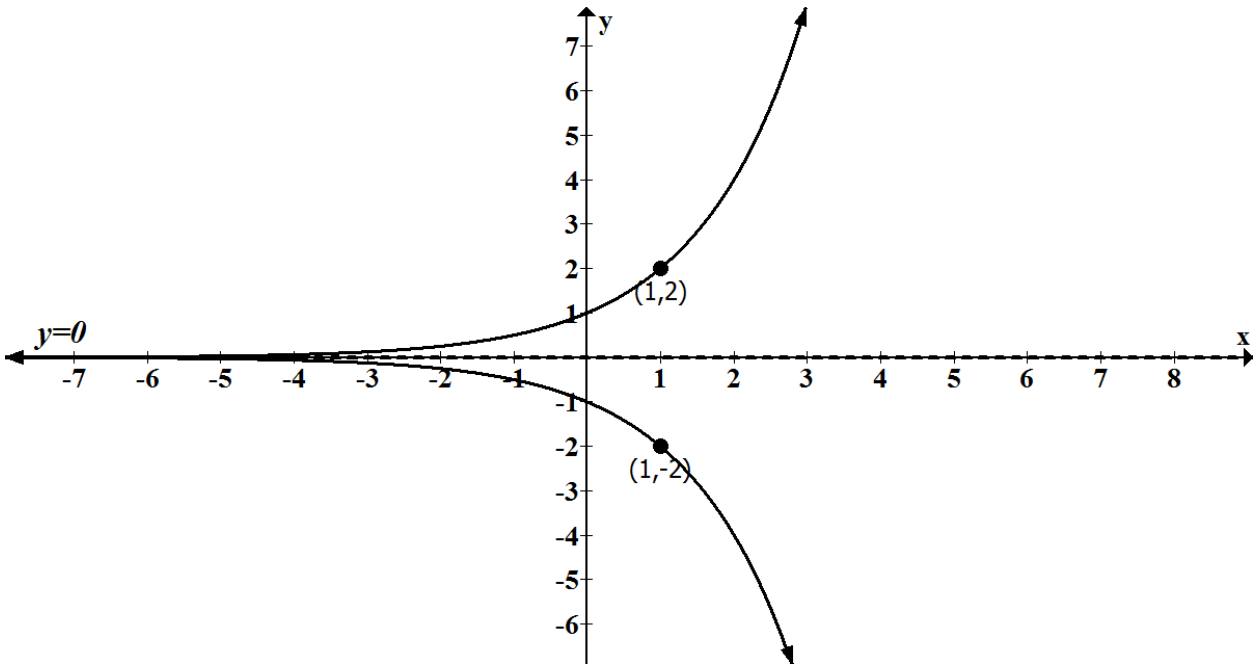


Make a quick analysis on the graph. There are four important points to make.

- The graph crosses the y-axis at (0 ; -1)

- The x-axis is the asymptote of the graph. That's because the x-axis is the line to which the graph gets close but never crosses.
- The graph is downward-slopping.
- At x = 1, y = -2, meaning that one of the coordinates is (1 ; -2)

Do notice the difference between the two graphs? Check out the graphs below.



As you can see, the function with a **positive** 2 lies above the asymptote and is upward-sloping. The function with a **negative** 2 lies below the asymptote and is downward-sloping. That's the effect of a negative base number, it reflects the original function through the asymptote.

Question

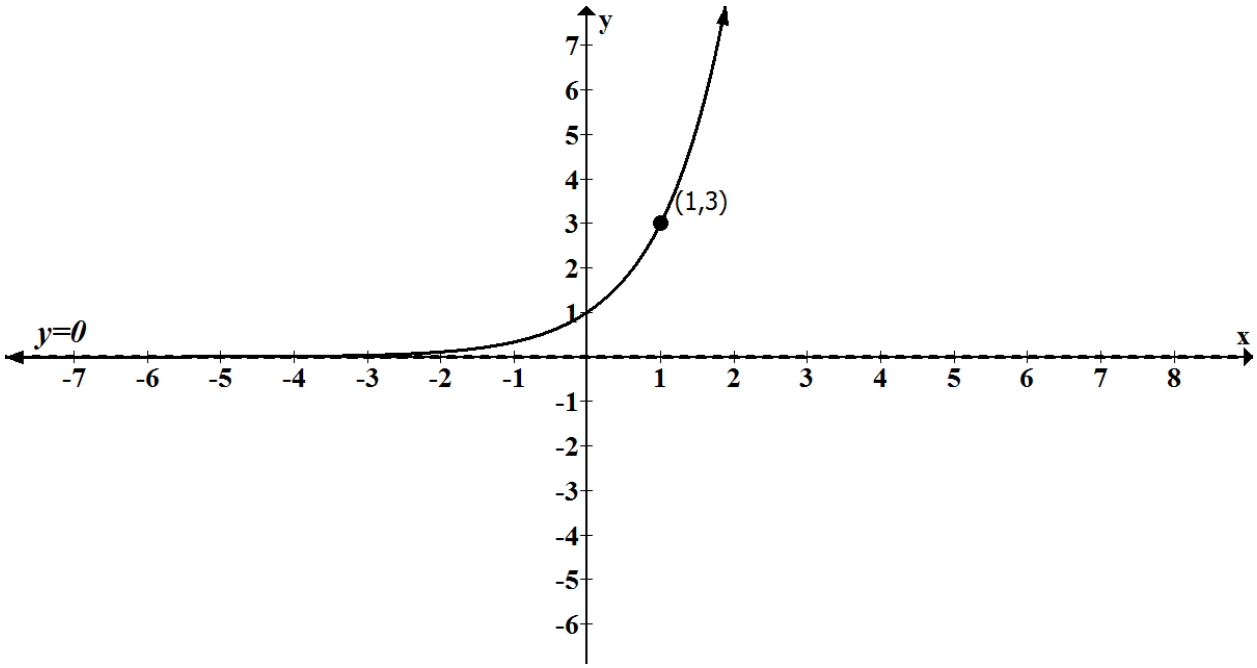
a) Draw a rough sketch of the function $y = 3^x$. You should also show the coordinates at x = 1

b) Draw a rough sketch of the function $y = -3^x$. You should also show the coordinates at x = 1

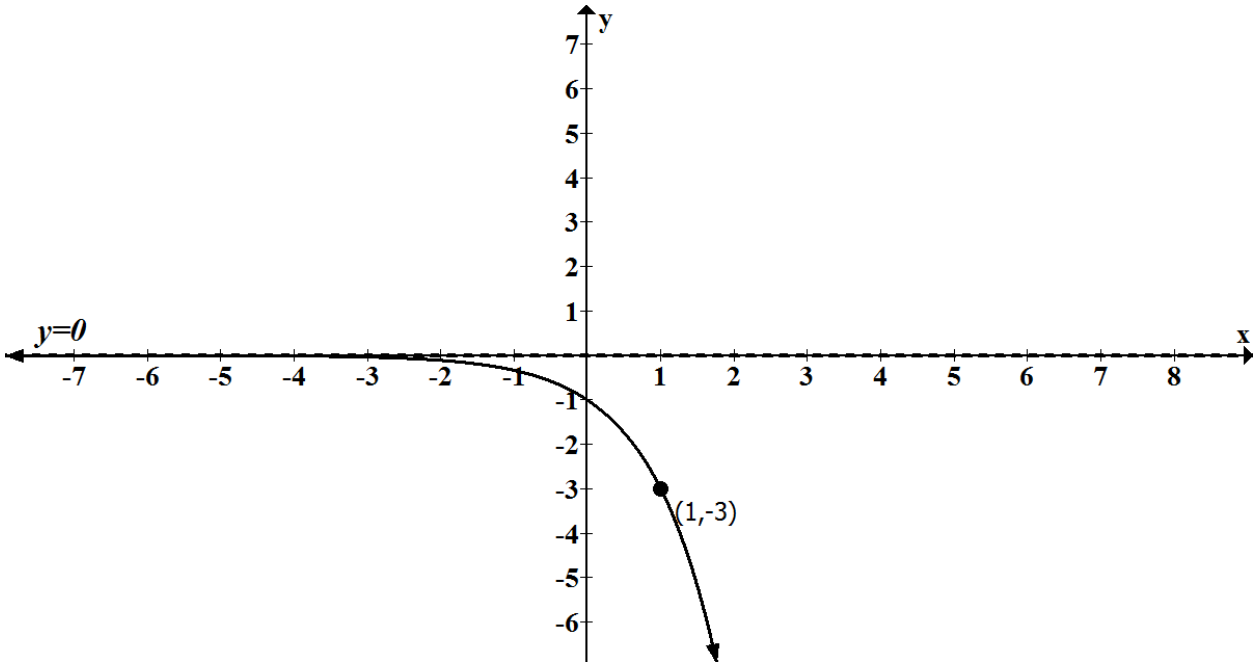
c) Sketch the two graphs on the same set of axis.

Solution

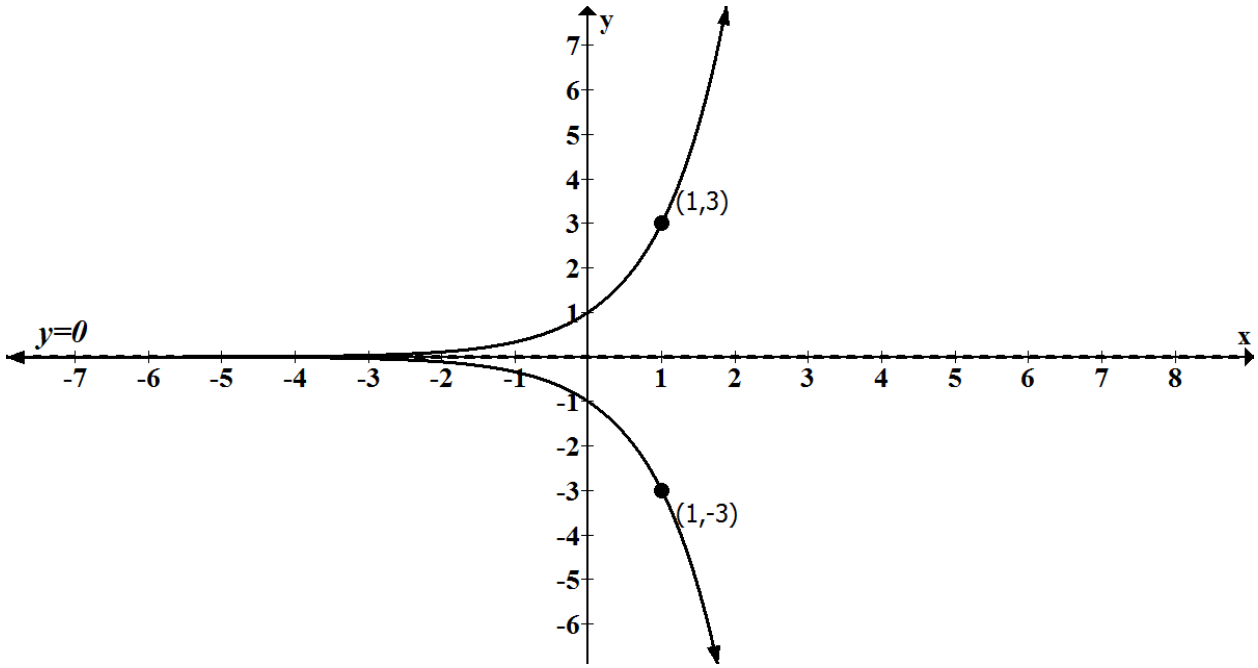
a)



b)



c)



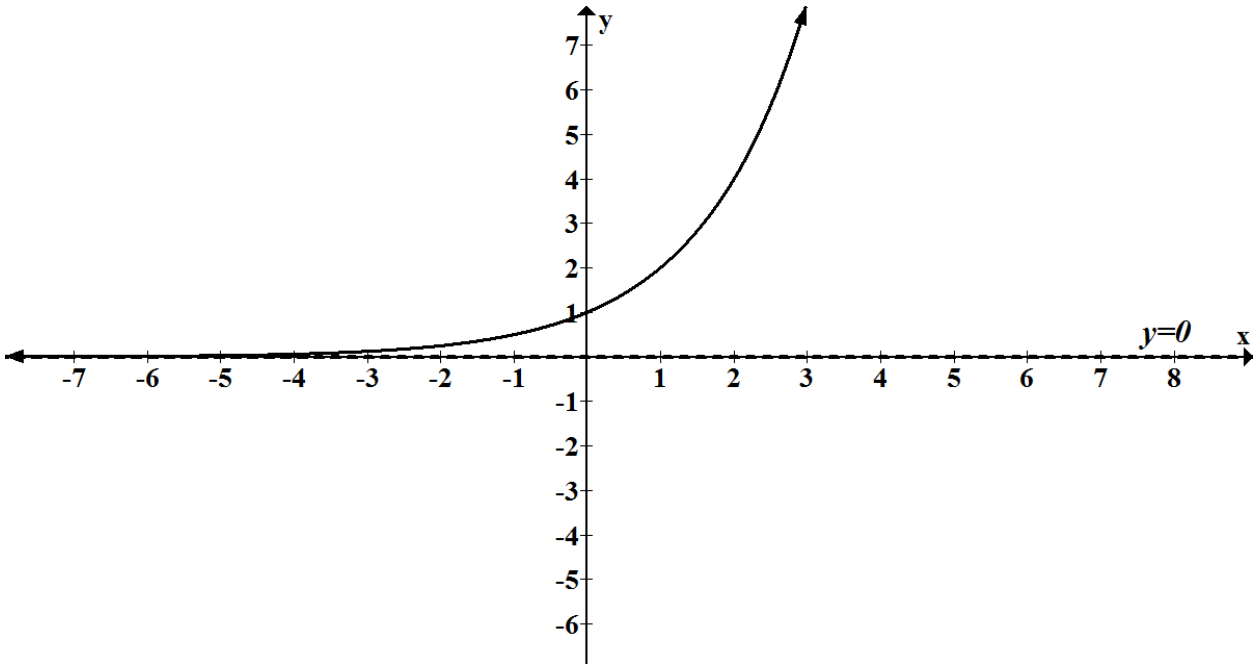
In order to fully understand the exponential function, we will relook at the function $y = 2^x$ and play around with it.

$y = 2^x + 0$, this the same thing.

Below we compare $y = 2^x + 0$ with $y_1 = 2^x + 1$. Check that out below.

$$y = 2^x + 0$$

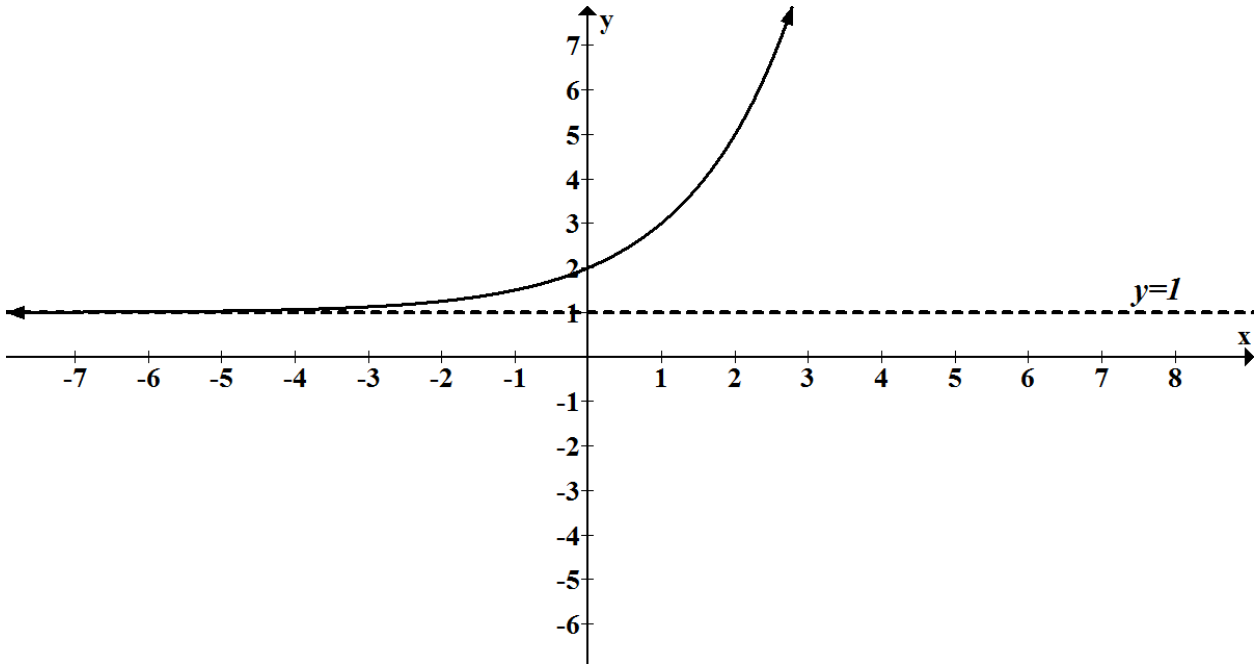
x	-4	-3	-2	-1	0	1	2	3	4
y	0.0625	0.125	0.25	0.5	1	2	4	8	16



Versus

$$y_1 = 2^x + 1$$

x	-4	-3	-2	-1	0	1	2	3	4
y	1.0625	1.125	1.25	1.5	2	3	5	9	17



Do you see the difference between the two graphs? The shapes are identical but have different asymptotes (the dotted lines in pink). For the graph of $y = 2^x + \mathbf{0}$, the asymptote is the x-axis, also called ***y* = 0**. For the graph of $y_1 = 2^x + \mathbf{1}$, the asymptote is ***y*=1**.

Therefore, we can confidently conclude that the number at the end represents the horizontal asymptote of an exponential function. Note that an exponential function doesn't have a vertical asymptote.

In general, the exponential function is in the form:

$$y = ab^{x+p} + q$$

, where "q" is the horizontal asymptote of the function.

Check out the example below.

$y = 4^x$

$y = 1.2^x + 3$, remember, 2 = 1.2 = 1×2 (The dot is a "times")

$y = a.b^x + q$

So *a = 1, b = 2, c = q, p = 0*

Check out more examples of an exponential function and their asymptotes.

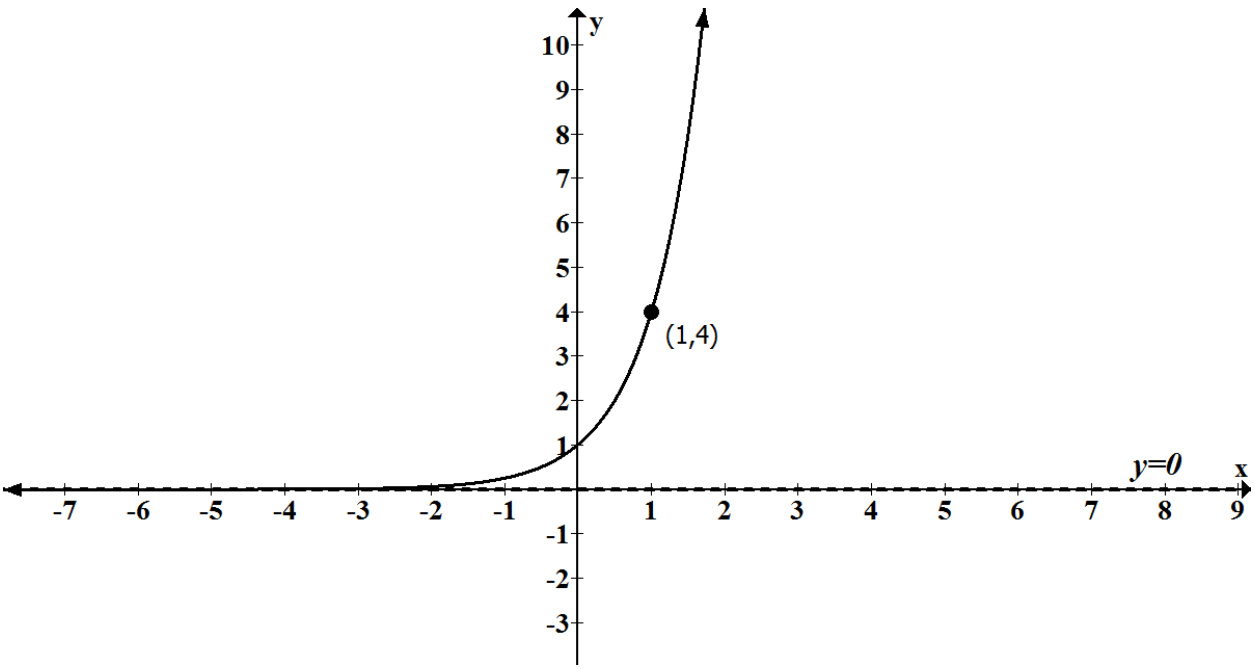
Example 1

$y = 4^x$

$y = 1.4^x + 0$

$y = a.b^x + q$

So *a = 1, b = 4, c = 0, q = 0*



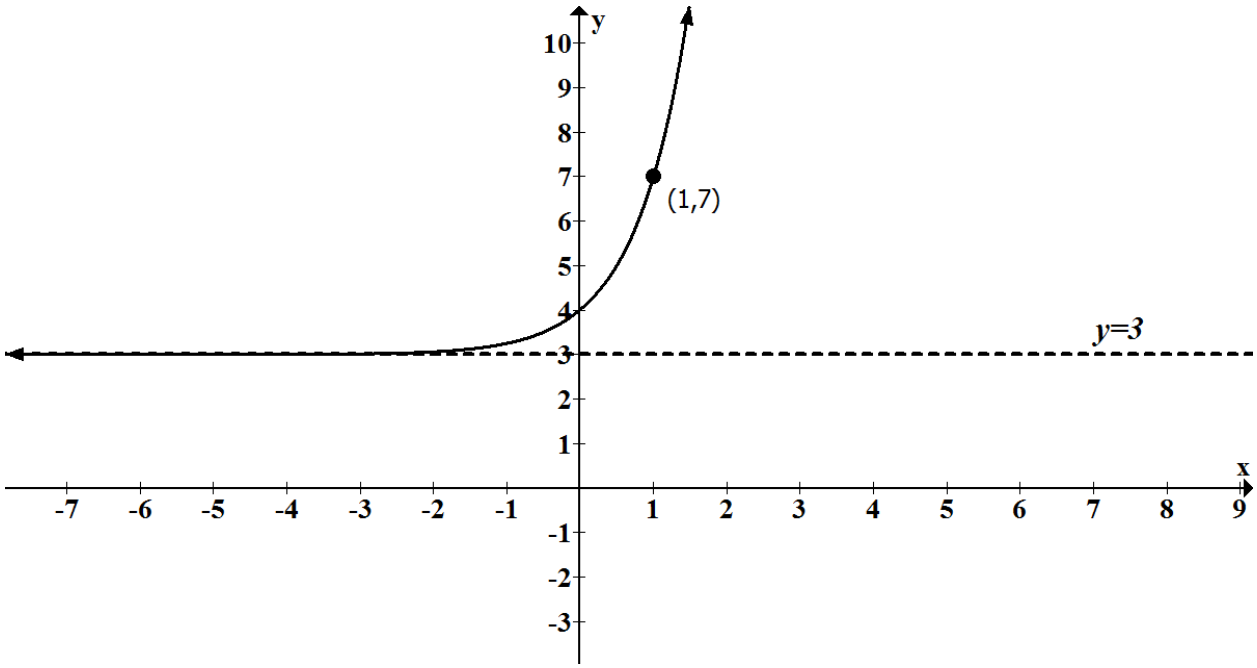
Example 2

$y = 4^x + 3$

$y = 1.4^x + 3$

$y = a.b^x + q$

So a = 1, b = 4, q = 3, p = 0



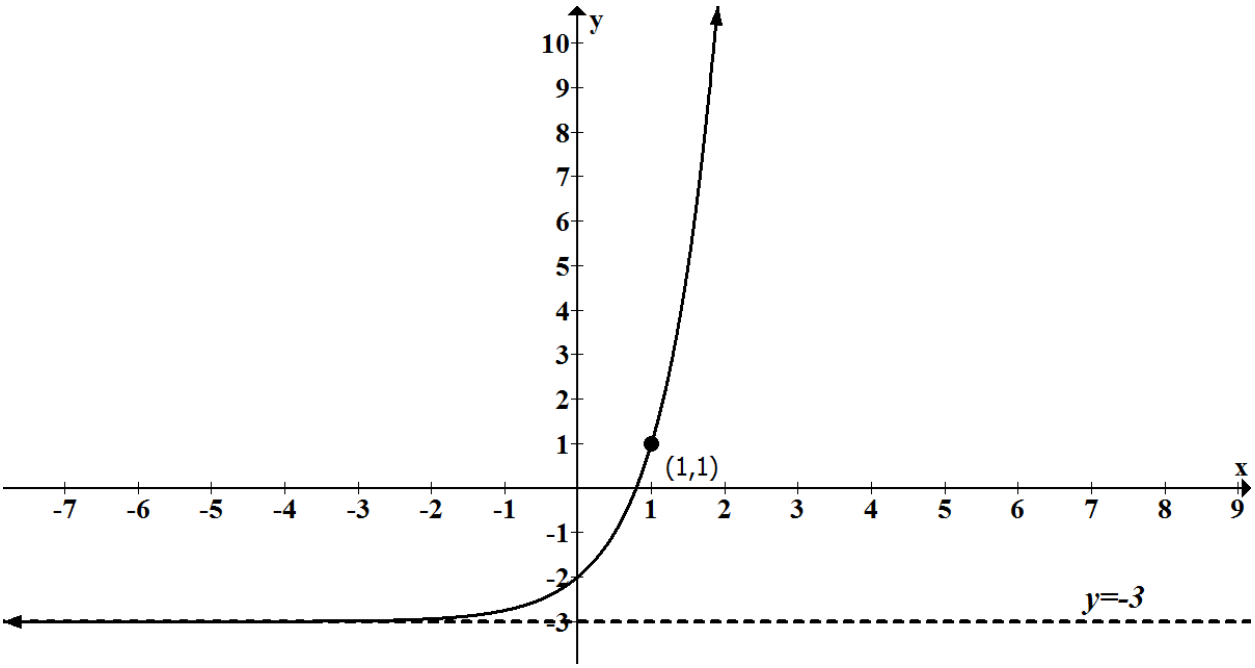
Example 3

$y = 4^x - 3$

$y = 1.4^x - 3$

$y = a.b^x + q$

So a = 1, b = 4, q = -3, p = 0



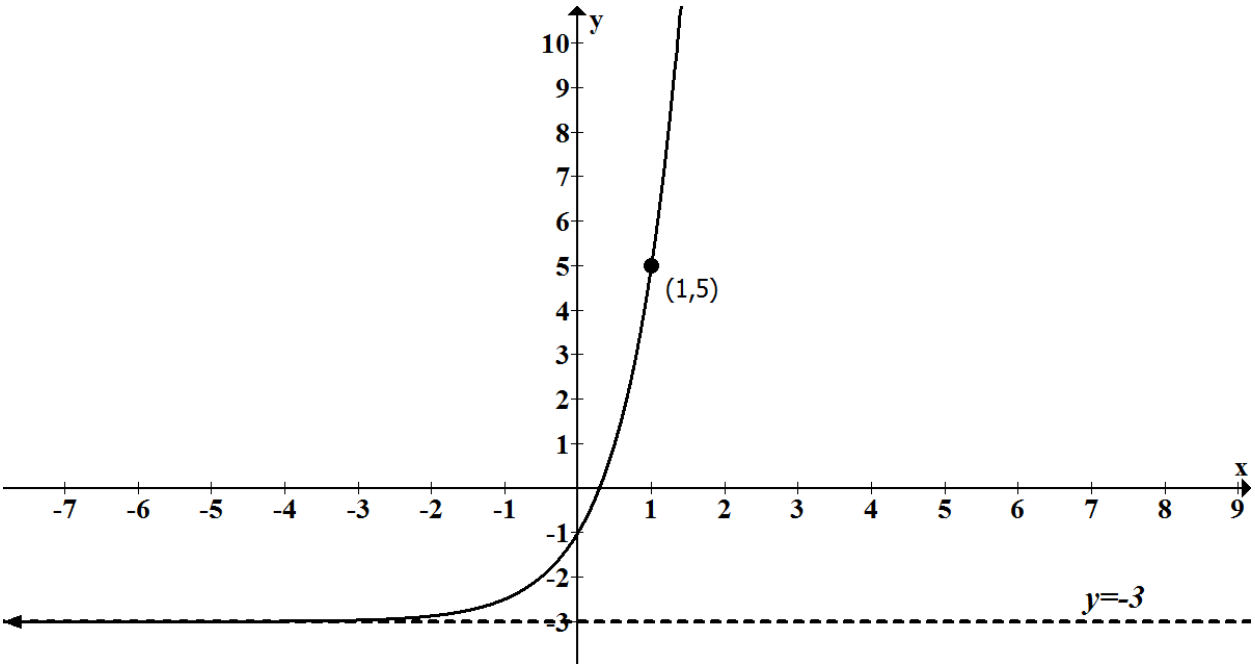
Example 4

$y = 2.4^x - 3$

$y = 2.4^x - 3$

$y = a.b^x + q$

So *a = 2, b = 4, q = -3, p = 0*



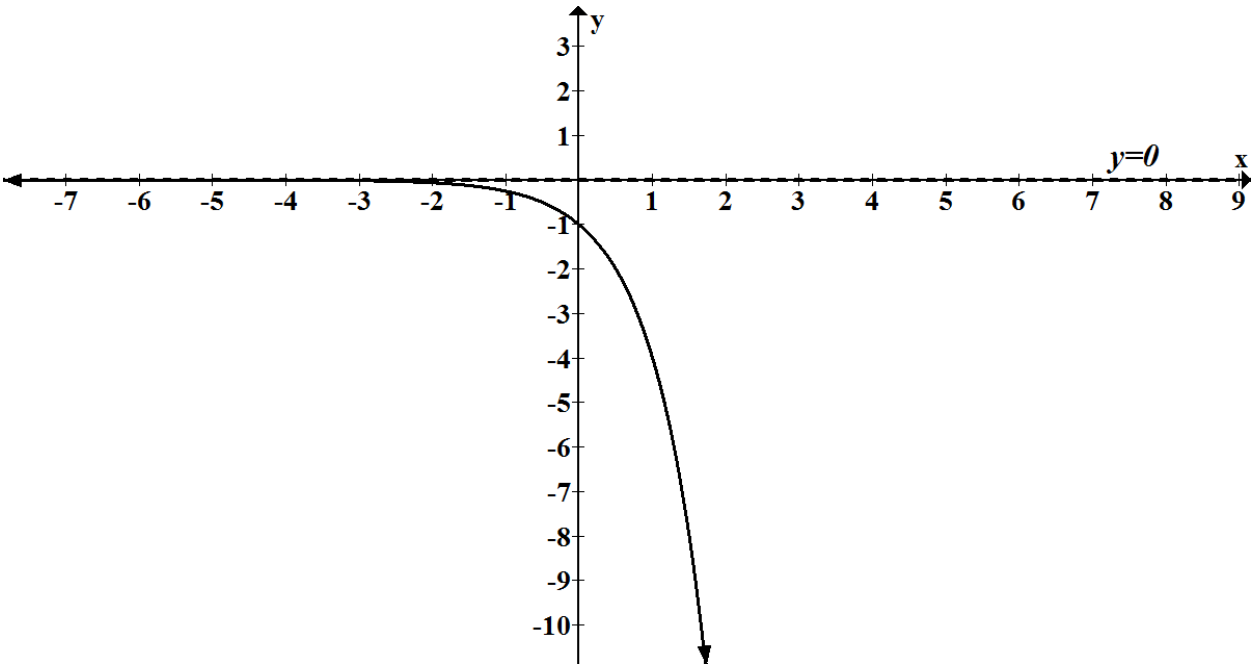
Example 5

$y = -4^x$

$y = 1.-4^x + 0$

$y = a.b^x + q$

So a = 1, b = -4, q = 0, p = 0



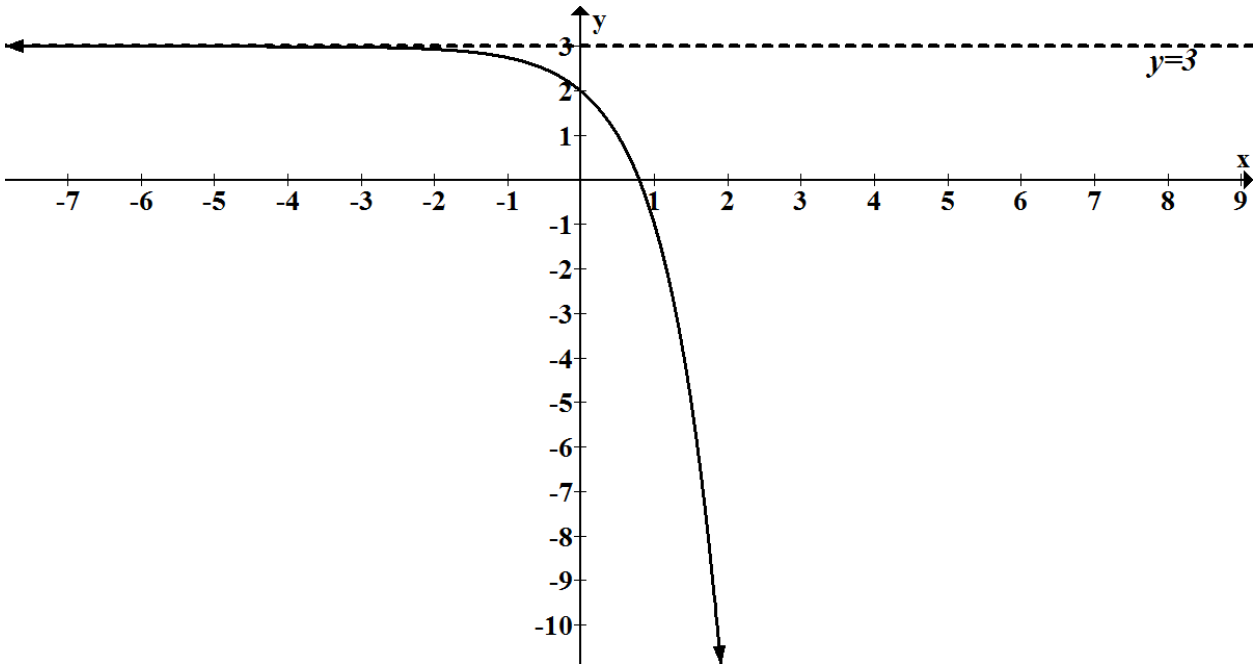
Example 6

$y = 1.-4^x + 3$

$y = 1.-4^x + 3$

$y = a.b^x + q$

So a = 1, b = -4, q = 3, p = 0



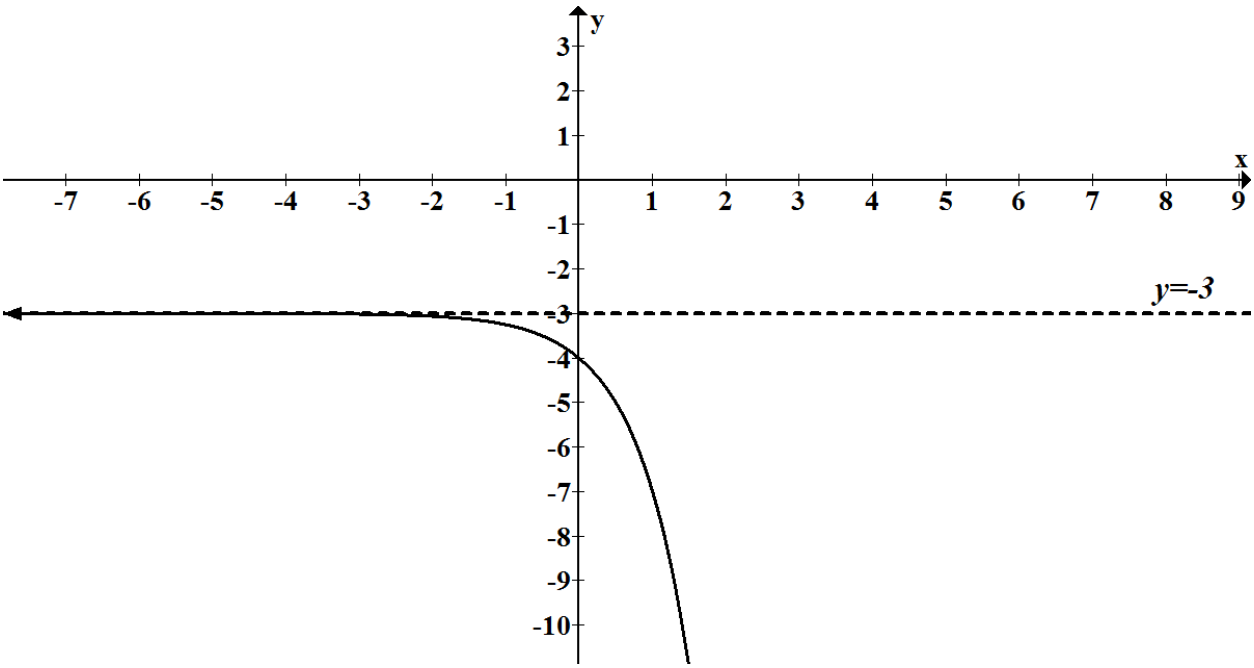
Example 7

$y = -4^x - 3$

$y = 1.-4^x - 3$

$y = a.b^x + q$

So a = 1, b = -4, q = -3, p = 0



Attempt the practice questions.

Question

Determine the asymptote of each of the following functions.

a) $y = 5^x$ **b)** $y = 5^x + 1$ **c)** $y = -5^x$ **d)** $y = -5^x + 1$

e) $y = e^x - 7$ **f)** $y = 10^x - 0.2$

Solution

a) $y = 0$ **b)** $y = 1$ **c)** $y = 0$ **d)** $y = 1$ **e)** $y = -7$ **f)** $y = -0.2$

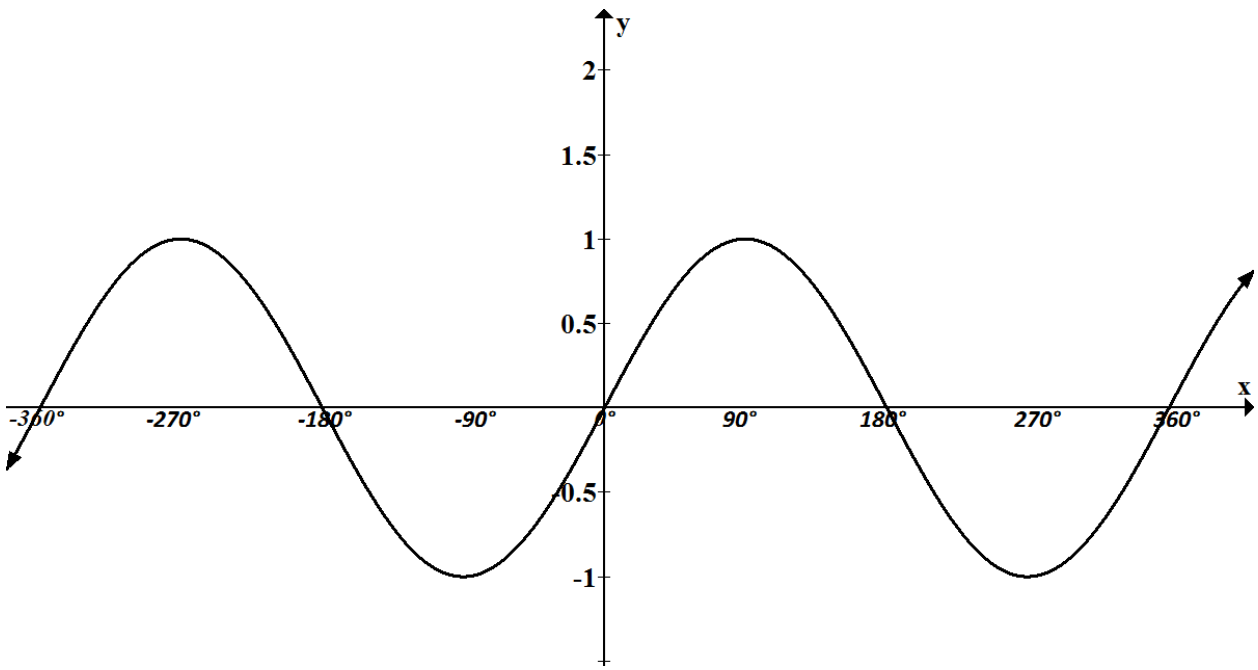
Trigonometric functions

In this section, we will be focusing on three basic trigonometric functions; the sine function, cosine function and tangent function.

Sine Function

Consider the function $y = sin\, x$, where x is an angle in degrees. Below is the table of values and graph of the function.

x	*-360*	*-315*	*-270*	*-225*	*-180*	*-135*	*-90*	*-45*	*0*	*45*	*90*	*135*	*180*	*225*	*270*	*315*	*360*
y	*0*	*0.7*	*1*	*0.7*	*0*	*-0.7*	*-1*	*-0.7*	*0*	*0.7*	*1*	*0.7*	*0*	*-0.7*	*-1*	*-0.7*	*0*



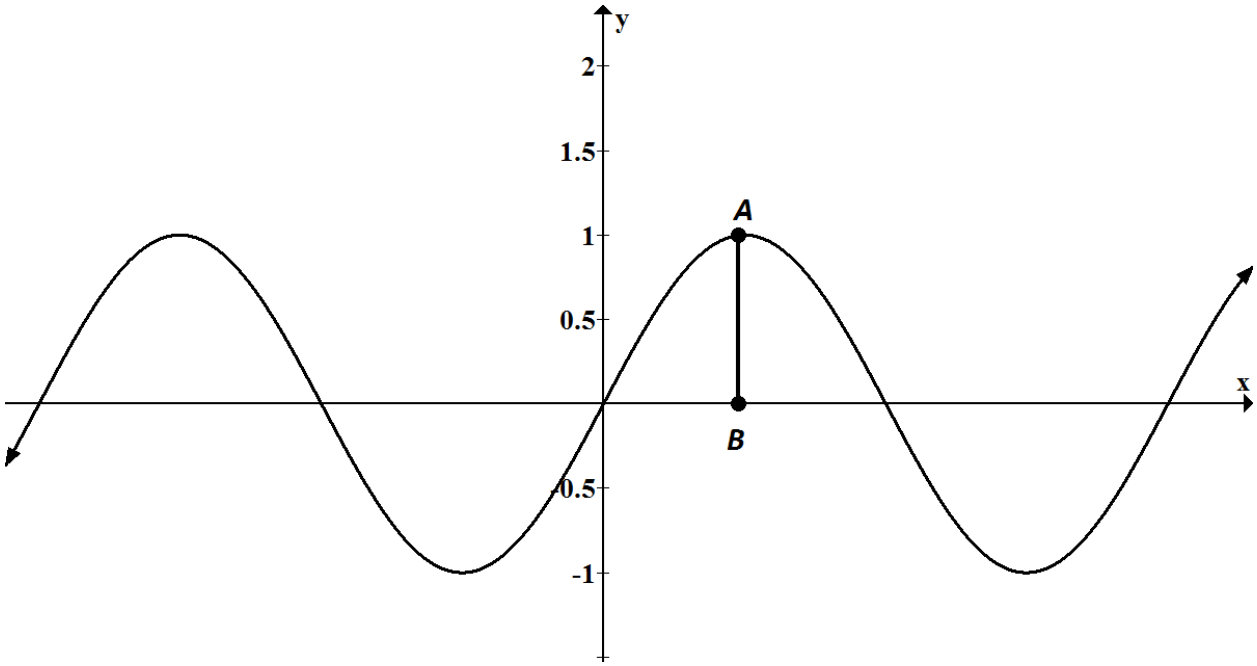
Let's get to know the sign graph in detail. There are three important properties we need to understand about the sine graph.

- Wave-like shape
- Amplitude
- Period

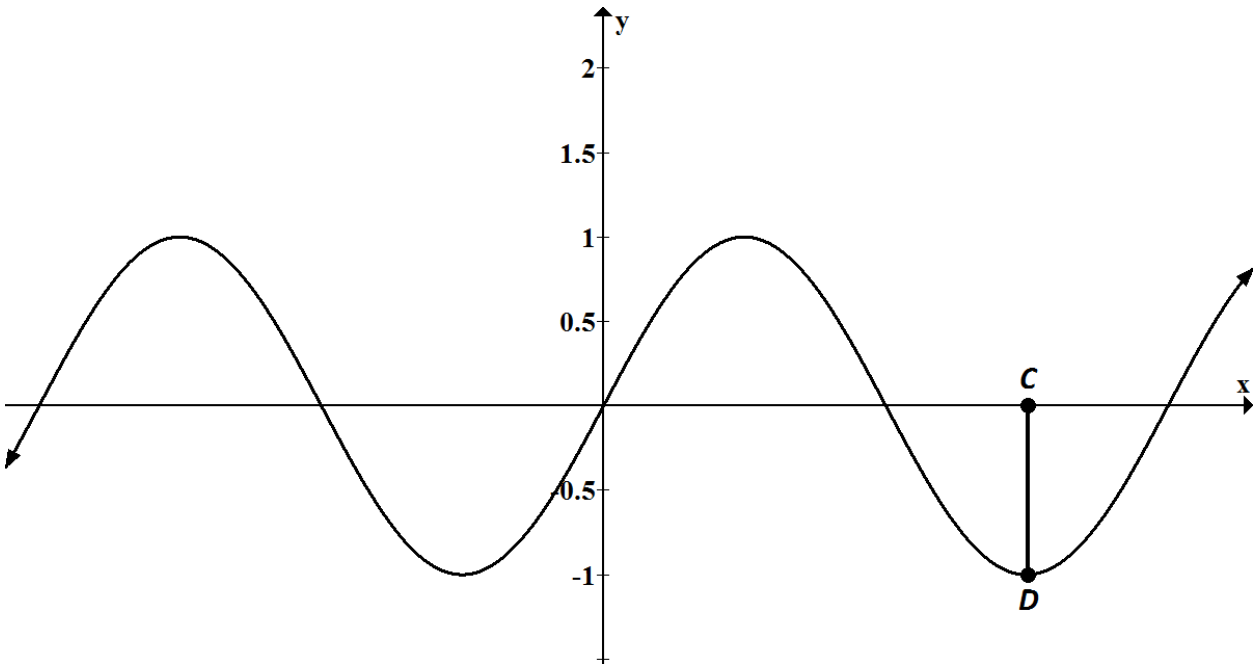
Wave-like Shape

We can clearly see that the graph looks like some sort of wave. This shape goes on and on.

Amplitude



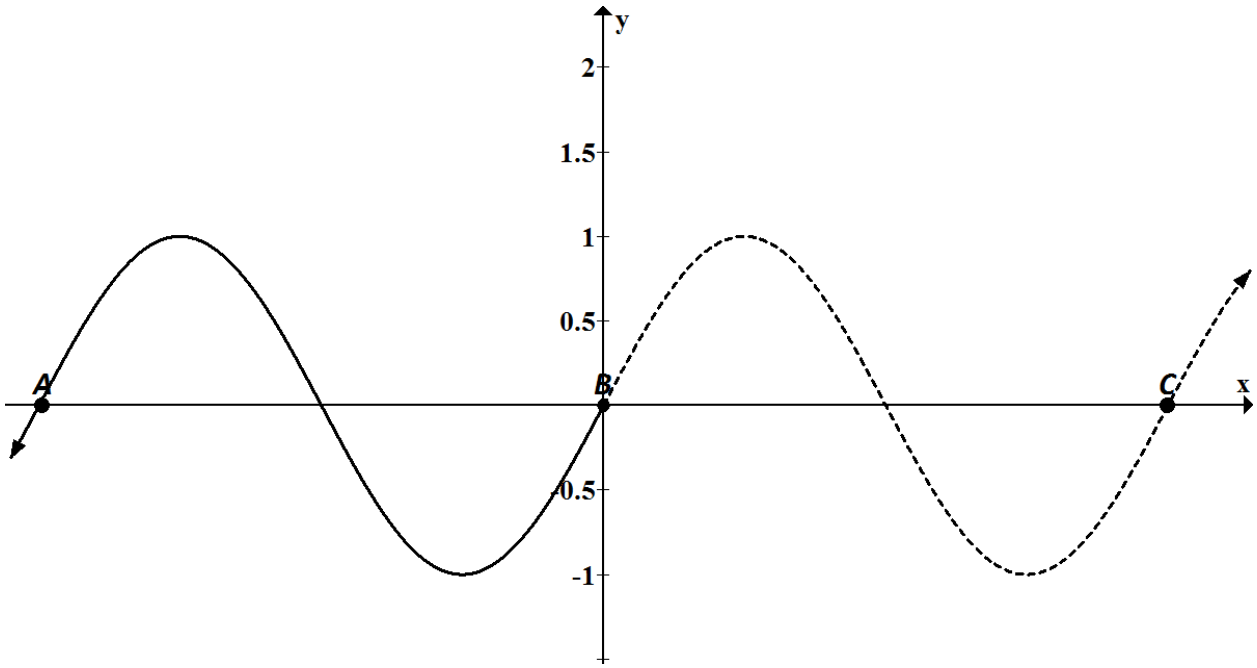
The amplitude of the graph is the length of the vertical line AB. As you can see from our graph, our amplitude is of size “1”. There’s another way of finding amplitude.



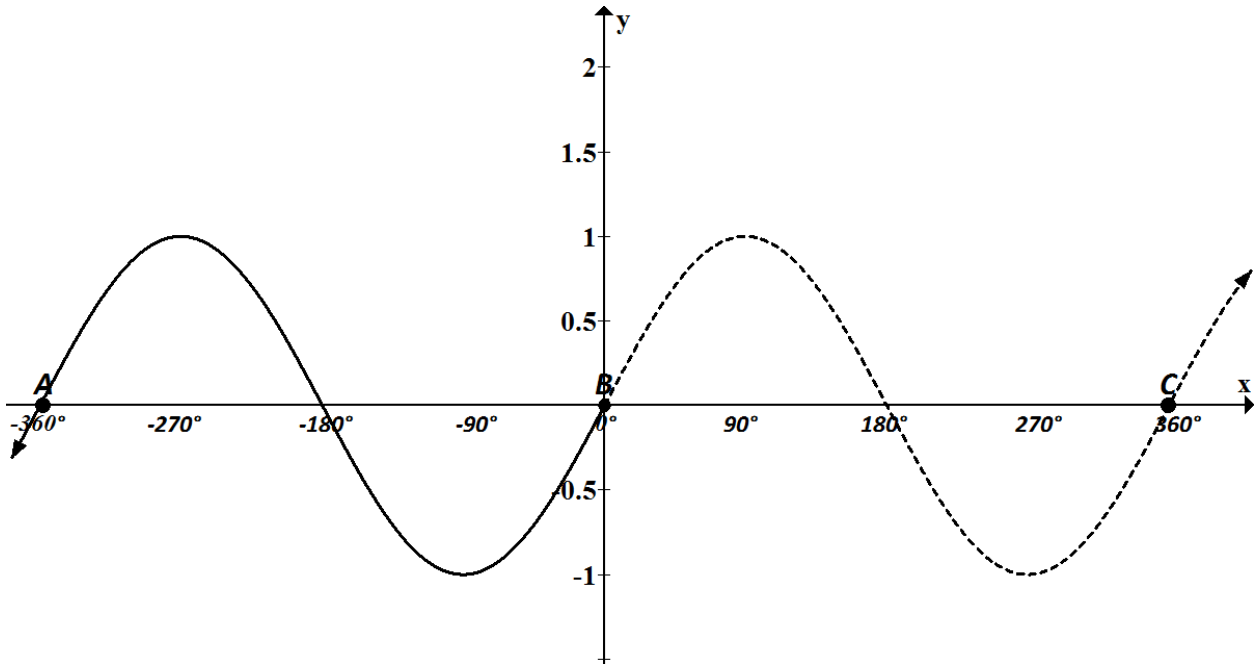
The length of CD is also the amplitude of the graph and is of size “1”. That’s just another way of finding the amplitude. So the **Amplitude** is the vertical distance from the top-most part of the graph to the “middle-part” of the graph. You can also say: The **Amplitude** is the vertical distance from the bottom-most part of the graph to the “middle-part” of the graph.

Period

Look at the graph carefully and compare the highlighted parts of the graph, the yellow part and orange part.



Consider the shape of the graph from point A to point B. Notice that the shape repeats itself from point B to C. The period of this graph is the horizontal length from point A to B, the point at which the shape will start repeating itself. As you can see, the shape starts at A and ends at B, where it will start repeating itself. Now let’s redraw the graph, label it and determine the period.



Point A is at -360°, while point B is at 0°. The horizontal length of AB is obviously 360°, therefore, the period of the graph of $y = sin\, x$ is 360°.

There's another way of finding the period. Since the part from A to B is identical to the part from B to C, you can find the period by getting the length from point B to point C. Point B is at 0°, while point C is at 360°. The horizontal length of BC is obviously 360°, therefore, the period of the graph of $y = sin\, x$ is 360°.

You will learn more about the "period" later, when we explore the other trigonometric functions, but for now, you have a decent understanding. Those are the three-most important properties of the sine function.

To derive the general form of the sine function, let's play around with the function $y = sin\, x$. We know that $y = sin\, x$ is the same as $= sin\, x + 0$. We also know that there's an invisible "1" in front of *sin x*: $y = 1.\, sin\, x + 0$

Again, we know that there's an invisible "1" in front of "x": $y = 1.\, sin 1\, x + 0$

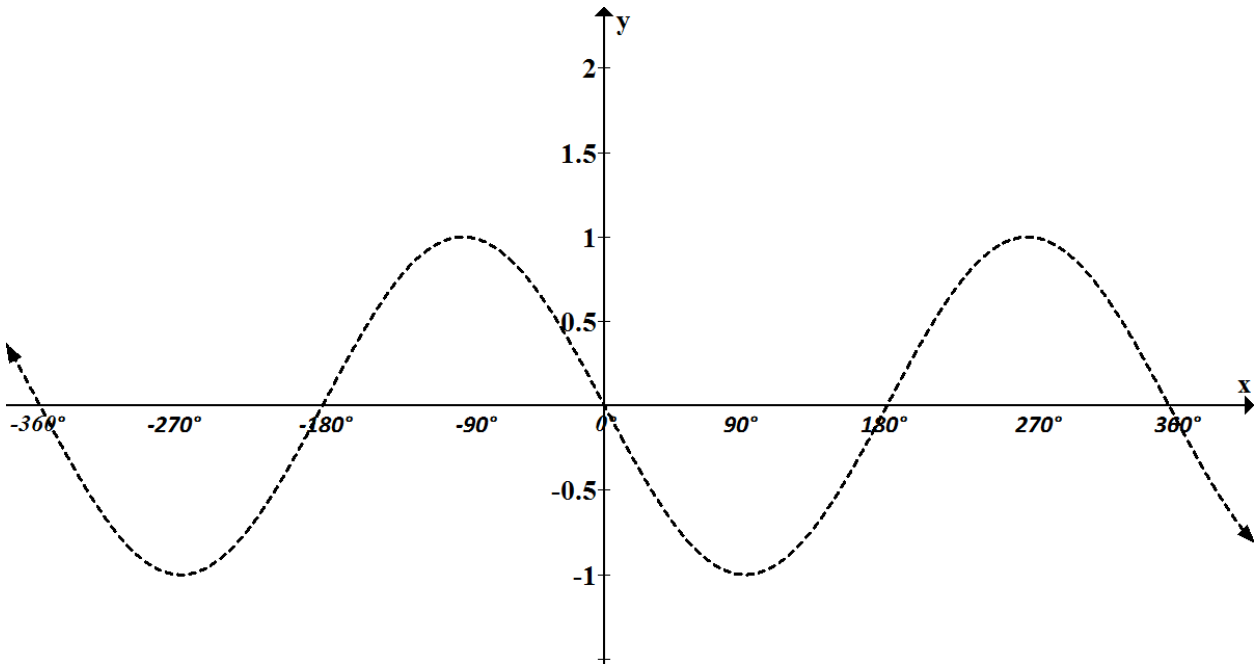
$y = 1.\, sin 1\, x + 0$

$y = a sin\, bx + c$. This is the general form of the sine function.

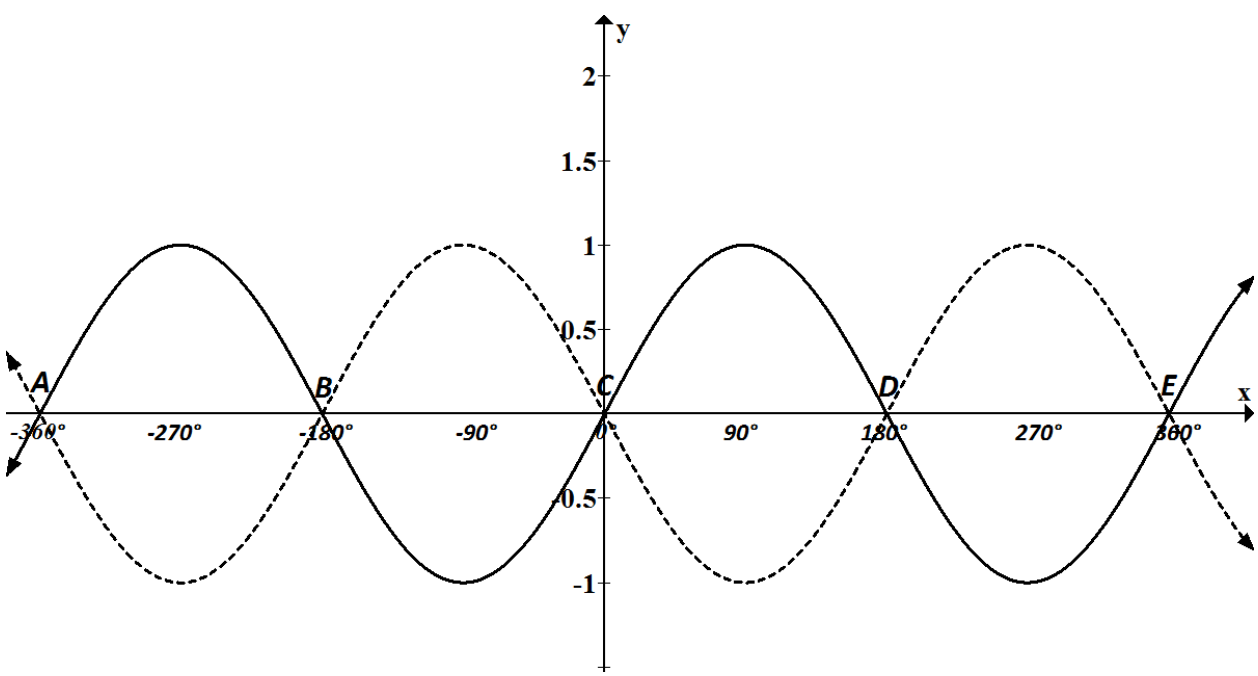
For the function $y = 1.\, sin 1\, x + 0$, a = 1, b = 1 and c = 0. We will get into more detail about a, b and c in the next grade.

Consider the function $y_1 = -sin\,x$, where x is an angle in degrees. Below is the table of values and graph of the function.

x	360°	315°	270°	225°	180°	135°	90°	45°	0°	45°	90°	135°	180°	225°	270°	315°	360°
y	-0	-0.7	-1	-0.7	0	0.7	1	0.7	0	-0.7	-1	-0.7	0	0.7	1	0.7	0



It is time for analysis. We compare the graph of $y_1 = -sin\,x$ with that of $y = sin\,x$. We have the graphs of each of the functions below. $y = sin\,x$, and $y_1 = -sin\,x$ in a dotted line.



We begin our focus on the solid graph, the function $y = sin\, x$. Look at the part from point A to B. Notice that the dotted part **below** is a mirror reflection through the ***"x-axis".***

Again, look at the solid part from point B to C. Notice that the dotted part **above** is a mirror reflection through the ***"x-axis".***

Look at the solid part from point C to D. Notice that the dotted part **below** is a mirror reflection a mirror reflection through the ***"x-axis".***

Once more, look at the solid part from point D to E. Notice that the dotted part **above** is a mirror reflection through the *"x-axis".*

Therefore, we can safely conclude that the function $y_1 = -sin\, x$ is a mirror reflection of $y = sin\, x$ through the *"x-axis".*

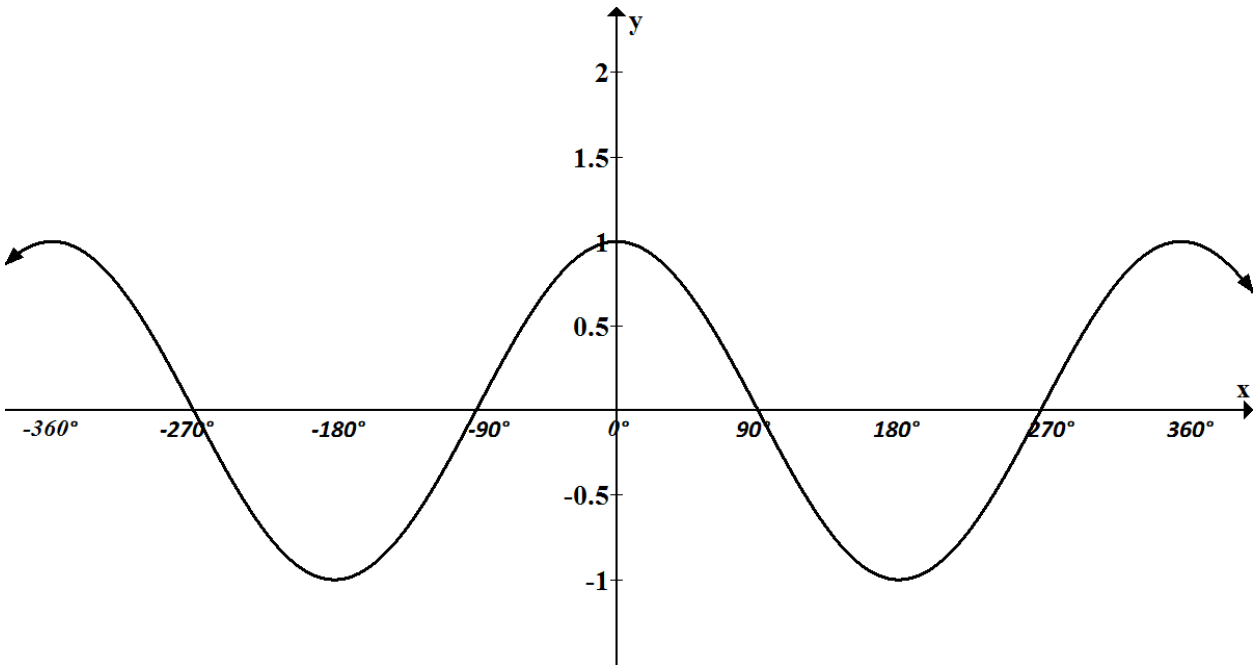
That is the sine function, we will go into more detail in the next grade.

Cosine Function

Consider the function $y = cos\, x$

, where x is an angle in degrees. Below is the table of values and graph of the function.

x	*360°*	*315°*	*270°*	*225°*	*180°*	*135°*	*90°*	*45°*	*0°*	*45°*	*90°*	*135°*	*180°*	*225°*	*270°*	*315°*	*360°*
y	*1*	*0.7*	*0*	*-0.7*	*-1*	*-0.7*	*0*	*0.7*	*1*	*0.7*	*0*	*-0.7*	*-1*	*-0.7*	*0*	*0.7*	*1*



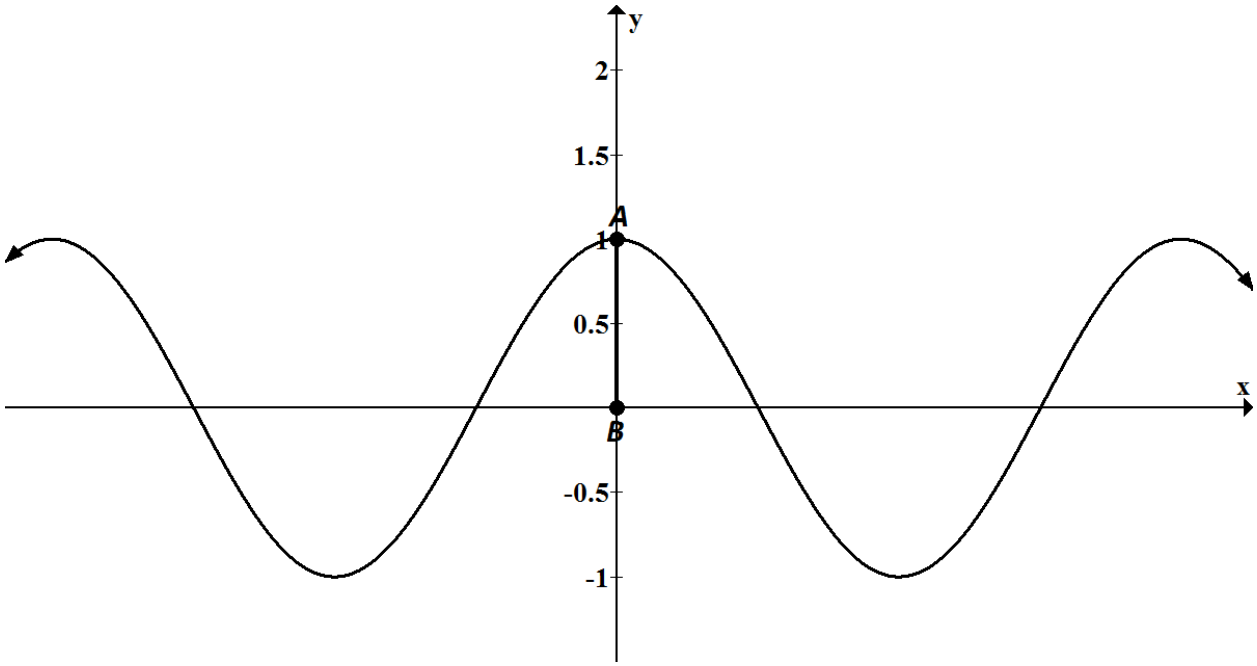
Let's get to know the sign graph in detail. There are three important properties we need to understand about the sine graph.

- Wave-like shape
- Amplitude
- Period

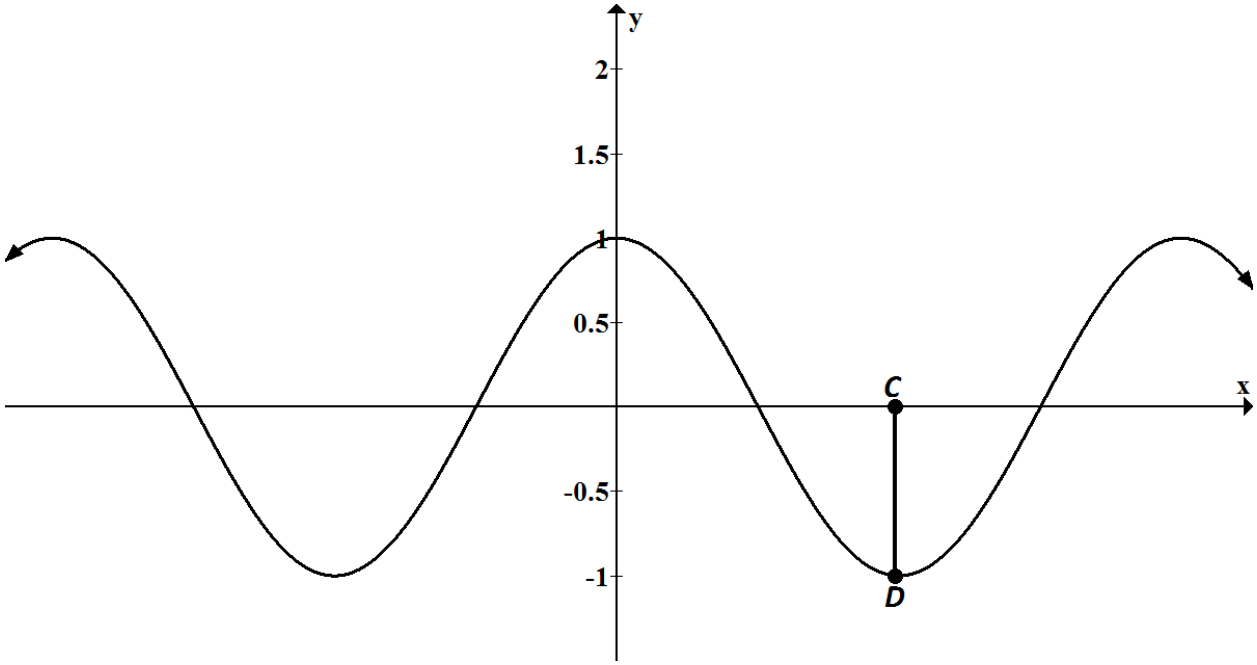
Wave-like Shape

We can clearly see that the graph looks like some sort of wave. This shape goes on and on.

Amplitude



The amplitude of the graph is the length of the vertical line AB. As you can see from our graph, our amplitude is of size “1”. There’s another way of finding amplitude.

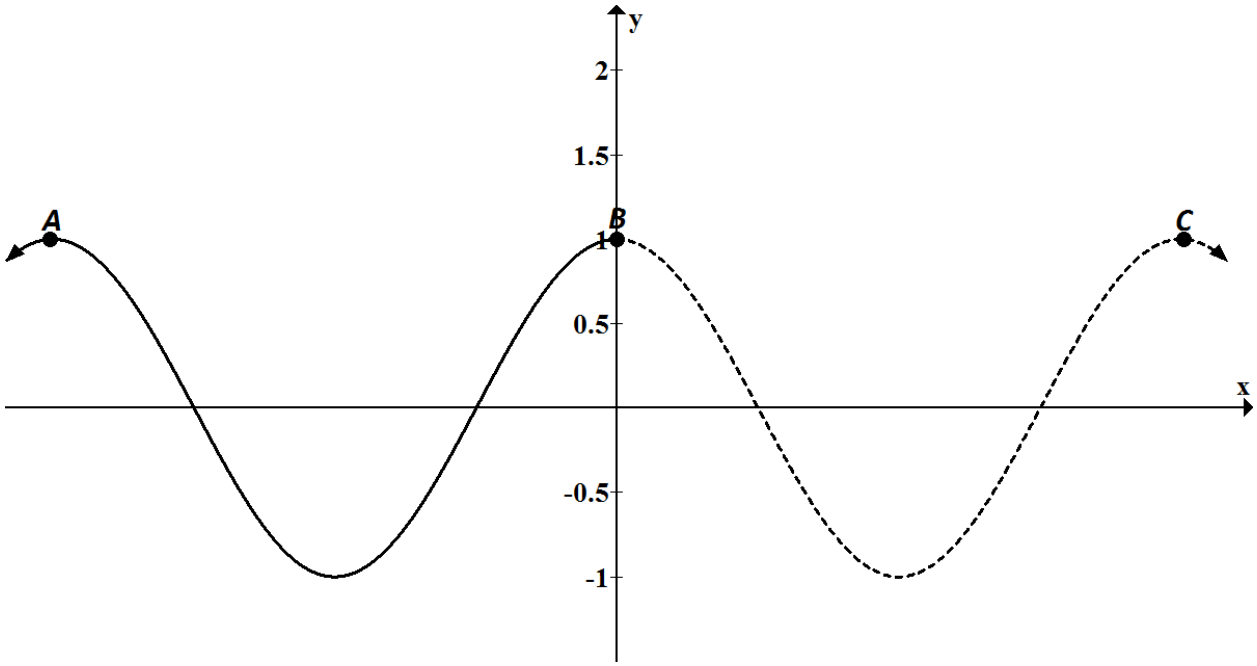


The length of CD is also the amplitude of the graph and is of size “1”. That’s just another way of finding the amplitude. So the **Amplitude** is the vertical distance from the top-most part of the

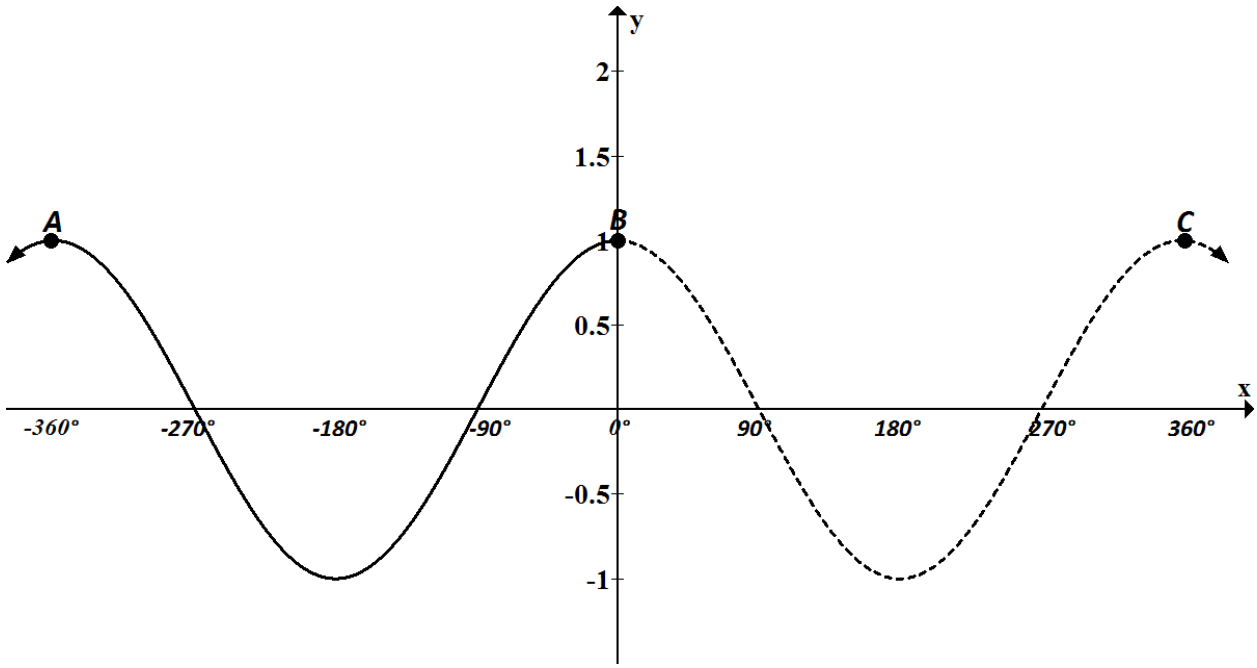
graph to the “middle-part” of the graph. You can also say: The **Amplitude** is the vertical distance from the bottom-most part of the graph to the “middle-part” of the graph.

Period

Look at the graph carefully and compare the solid part and dotted part.



Consider the shape of the graph from point A to point B. Notice that the shape repeats itself from point B to C. The period of this graph is the horizontal length from point A to B, the point at which the shape will start repeating itself. The shape starts at A and ends at B, where it will start repeating itself.



Point A is at -360°, while point B is at 0°. The horizontal length of AB is obviously 360°, therefore, the period of the graph of $y = cos\, x$ is 360°.

There's another way of finding the period. Since the dotted part is identical to the solid part, you can find the period by getting the length from point B to point C. Point B is at 0°, while point C is at 360°. The horizontal length of BC is obviously 360°, therefore, the period of the graph of $y = cos\, x$ is 360°.

You will learn more about the "period" later, when we explore the other trigonometric functions, but for now, you have a decent understanding. Those are the three-most important properties of the sine function.

To derive the general form of the cosine function, let's play around with the function $y = cos\, x$. We know that $y = cos\, x$ is the same as $= cos\, x + 0$. We also know that there's an invisible "1" in front of *cos x*: $y = 1.\, cos\, x + 0$

Again, we know that there's an invisible "1" in front of "x": $y = 1.\, cos1\, x + 0$

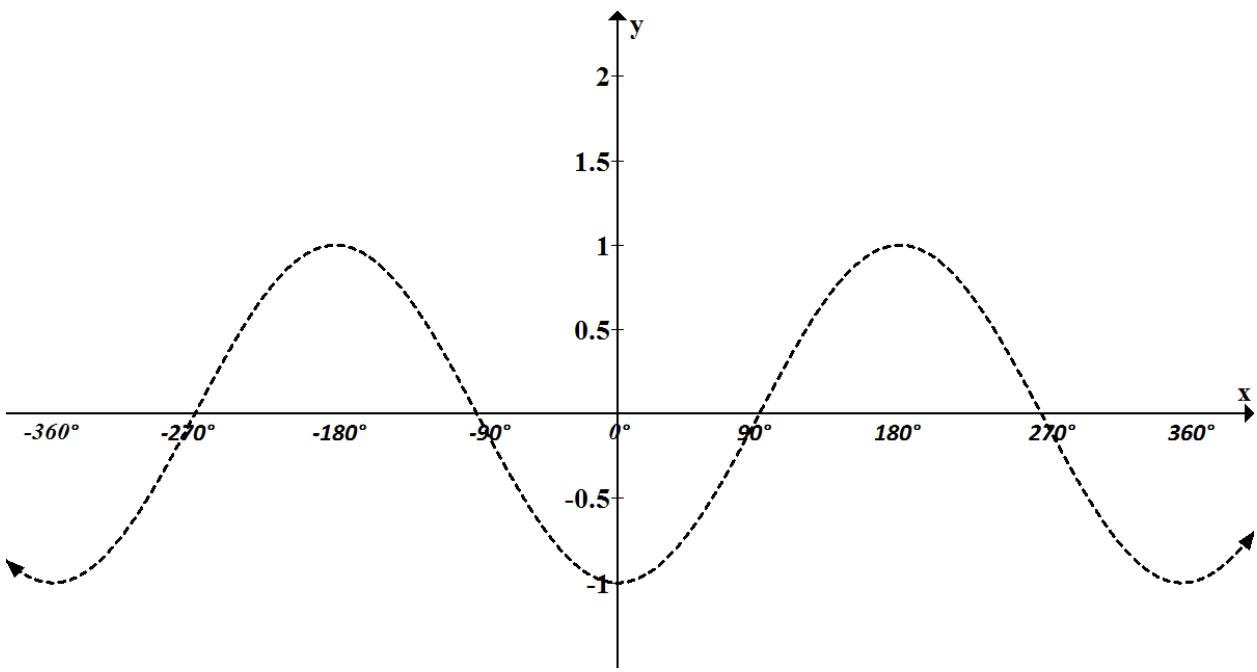
$y = 1.\, cos1\, x + 0$

$y = acos\, bx + c$. This is the general form of the cosine function.

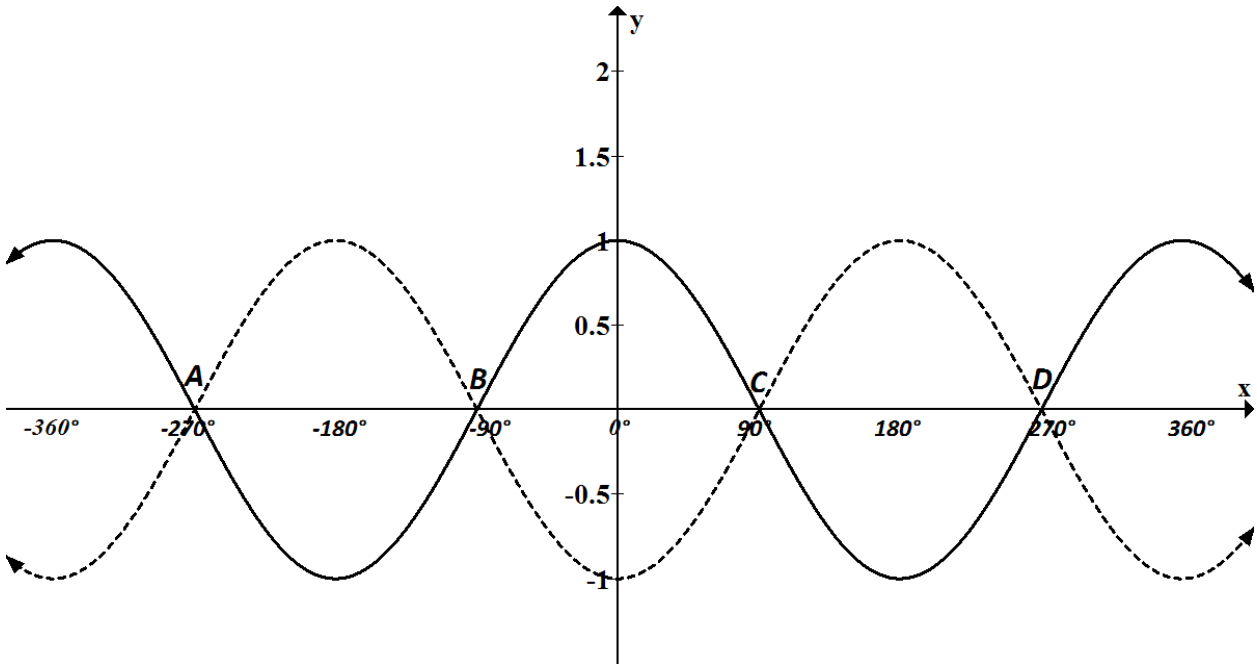
For the function $y = 1.\, cos1\, x + 0$, a = 1, b = 1 and c = 0. We will get into more detail about a, b and c in the next grade.

Consider the function $y_1 = -cos\, x$, where x is an angle in degrees. Below is the table of values and graph of the function.

x	*-360*	*-315*	*-270*	*-225*	*-180*	*-135*	*-90*	*-45*	*0*	*45*	*90*	*135*	*180*	*225*	*270*	*315*	*360*
y	*-1*	*-0.7*	*0*	*0.7*	*1*	*0.7*	*0*	*-0.7*	*-1*	*-0.7*	*0*	*0.7*	*1*	*0.7*	*0*	*-0.7*	*-1*



It is time for analysis. We compare the graph of $y_1 = -cos\, x$ with that of $y = cos\, x$. We have the graphs of each of the functions below. $y = cos\, x$ is solid, while $y_1 = -cos\, x$ is dotted.



We begin our focus on the yellow graph, the function $y = cos\, x$. Look at the solid part from point A to B. Notice that the dotted part **below** is a mirror reflection through the ***"x-axis".***

Again, look at the solid part from point B to C. Notice that the dotted part **above** is a mirror reflection through the ***"x-axis".***

Once more, look at the solid part from point C to D. Notice that the orange part **below** is a mirror reflection a mirror reflection through the ***"x-axis".***

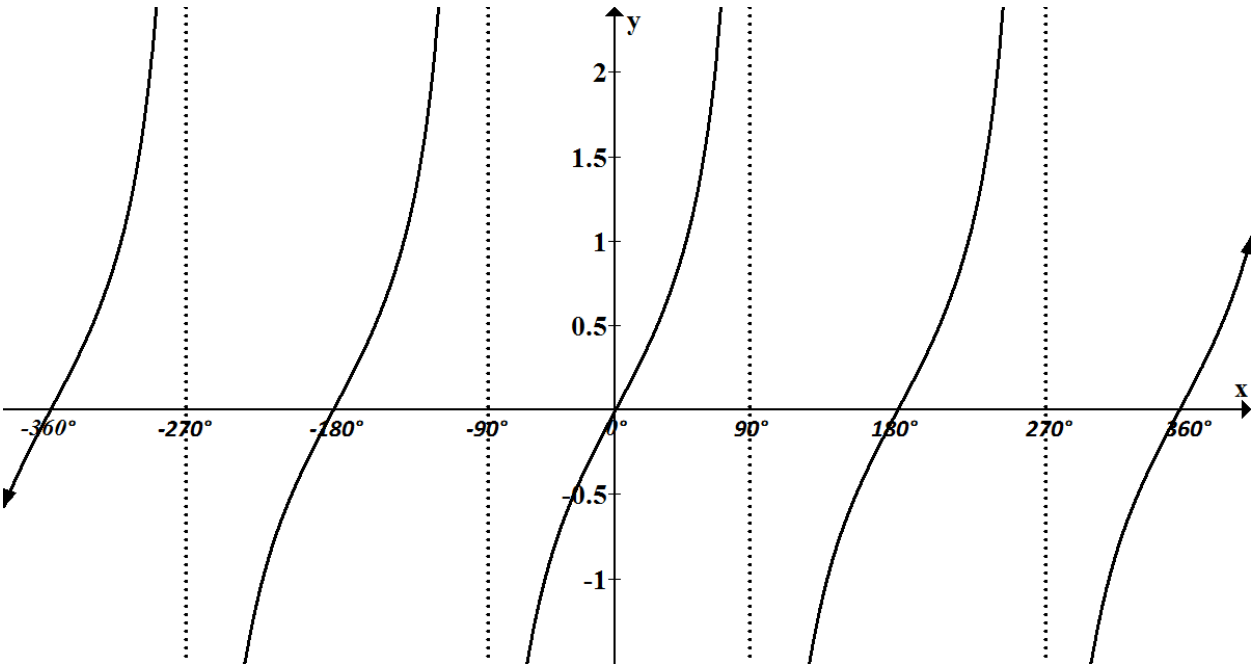
Therefore, we can safely conclude that the function

$y_1 = -cos\, x$ is a mirror reflection of $y = cos\, x$ through the ***"x-axis".***

That is the cosine function, we will go into more detail in the next grade.

Tan function

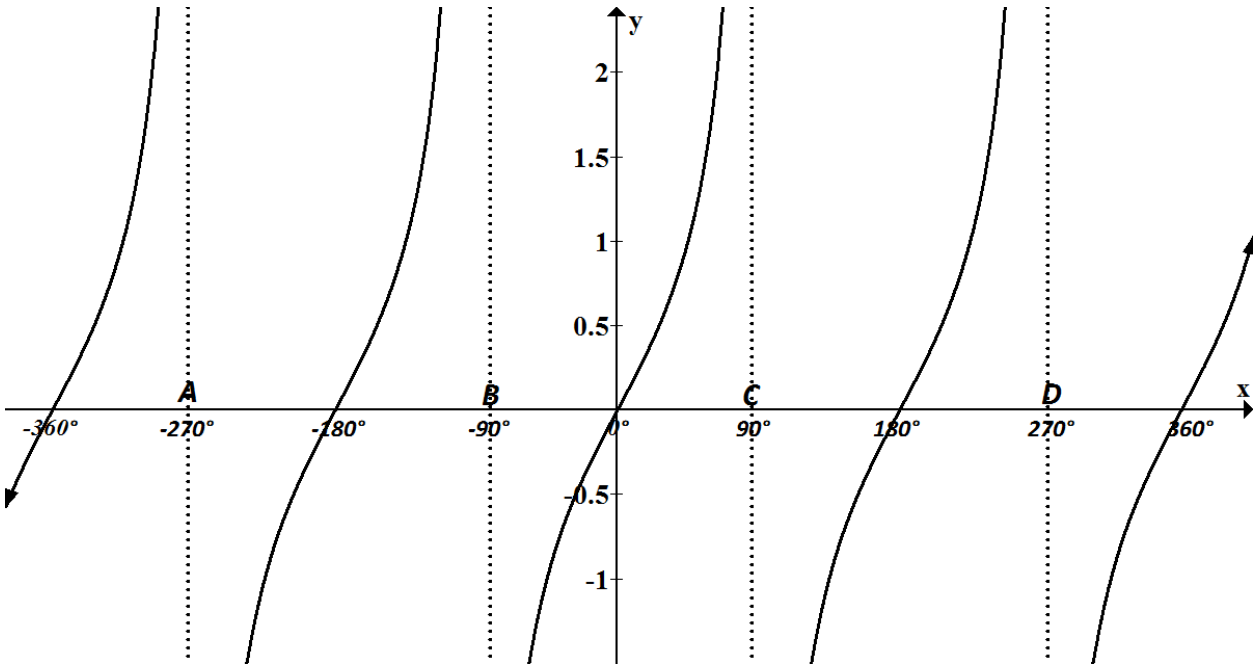
Consider the function $y = tan\, x$, where x is an angle in degrees. Below is the graph of the function.



As you can see, the graph of $tan\, x$ gets very close to the vertical dotted lines but never crosses them. These lines are the asymptotes of $tan\, x$. In the graph above, the dotted lines are $x = -270°,\ x = -90°,\ x = 90°$ and $x = 270°$

Period

Look carefully at the graph of $tan\ x$ below.



The distance between any two asymptotes is the period of the graph. For example, the length of AB is the period of this graph. Ab is $180°$ in length, so the period of the graph is $180°$.

How did we count $180°$? A is $x = -270°$ and B is at $x = -90°$, so we go:

$Period = x.value\ of\ the\ right\ point - x.value\ of\ the\ left\ point$

$Period = -90° - (-270°)$

$Period = 180°$

If you like, you may physically count and will still get the answer.

We can also use any other consecutive asymptotes to find the period. We may still use the asymptote at B and that at C.

$Period = x.value\ of\ the\ right\ point - x.value\ of\ the\ left\ point$

$Period = 90° - (-90°)$

$Period = 180°$, same answer.

So you may use any two consecutive asymptotes to find the period of the $tan\ x$ function.

To derive the general form of the "***tan***" function, let's play around with the function $y = tan\,x$. We know that $y = tan\,x$ is the same as $= tan\,x + 0$. We also know that there's an invisible "1" in front of "***tan x***" : $y = 1.tan\,x + 0$

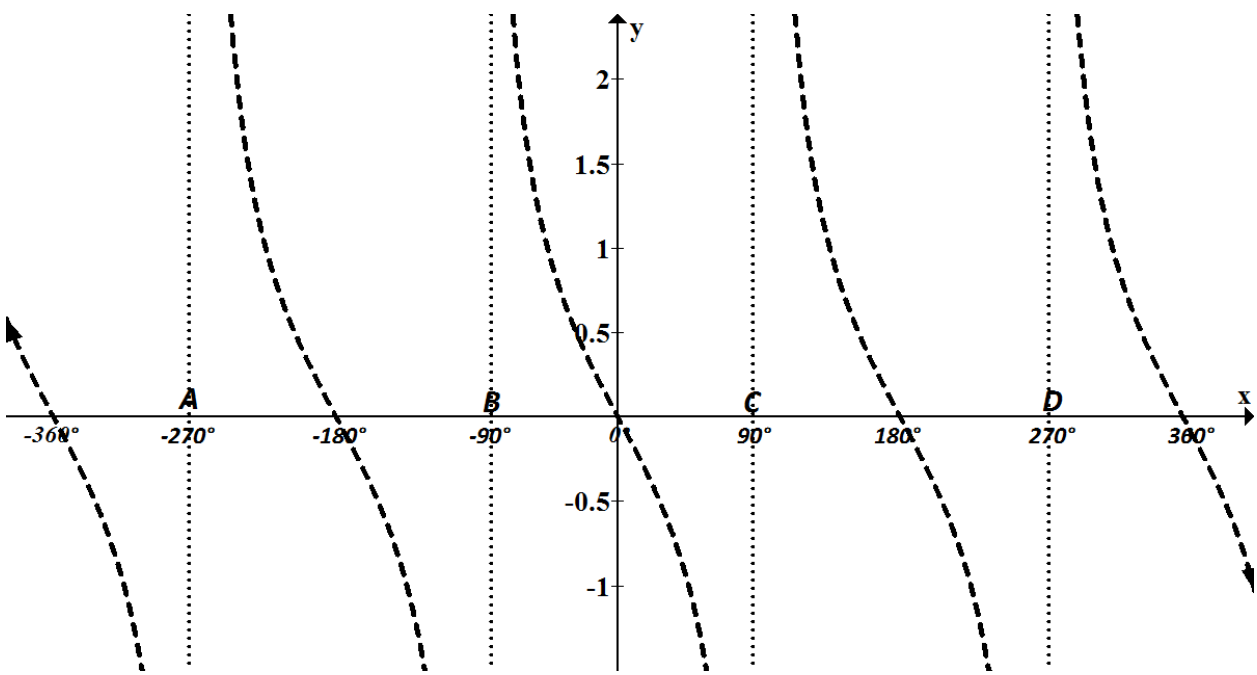
Again, we know that there's an invisible "1" in front of "x": $y = 1.tan1\,x + 0$

$y = 1.tan1\,x + 0$

$y = a\,tan\,bx + c$. This is the general form of the *"tan"* function.

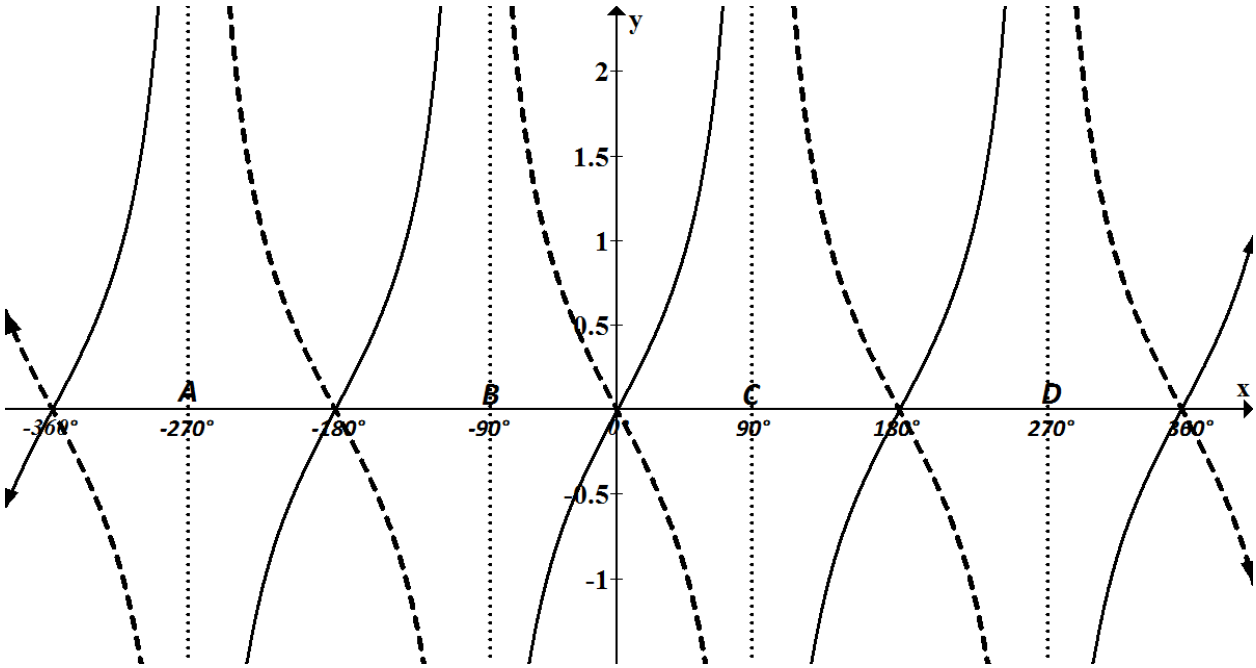
For the function $y = 1.tan1\,x + 0$, a = 1, b = 1 and c = 0. We will get into more detail about a, b and c in the next grade.

Consider the function $y_1 = -tan\,x$, where x is an angle in degrees. Below is the graph of the function.



Note that the asymptotes of $-tan\,x$ above are the same as those of $tan\,x$.

It is time for analysis. We compare the graph of $y_1 = -tan\,x$ with that of $y = tan\,x$. We have the graphs of each of the functions below. $y = tan\,x$ is solid, while $y_1 = -tan\,x$ is dotted.



We begin our focus on the **red** graph, the function $y = tan\, x$. Look at the solid part from point A to B. Notice that the dotted part **below** is a mirror reflection through the ***"x-axis".***

Again, look at the solid part from point B to C. Notice that the dotted part **above** is a mirror reflection through the ***"x-axis".***

Once more, look at the solid part from point C to D. Notice that the dotted part **below** is a mirror reflection a mirror reflection through the ***"x-axis".***

Therefore, we can safely conclude that the function.

$y_1 = -tan\, x$ is a mirror reflection of $y = tan\, x$ through the ***"x-axis".***

That is the ***"tan"*** function, we will go into more detail in the next grade.

Trigonometry

Reduction formulae

In this section, we will learn the so called ***reduction formulas***. These formulas help us in simplifying complicated trigonometric terms. The standard angles we will be working on are:

$90° - \theta$ and $90° + \theta$

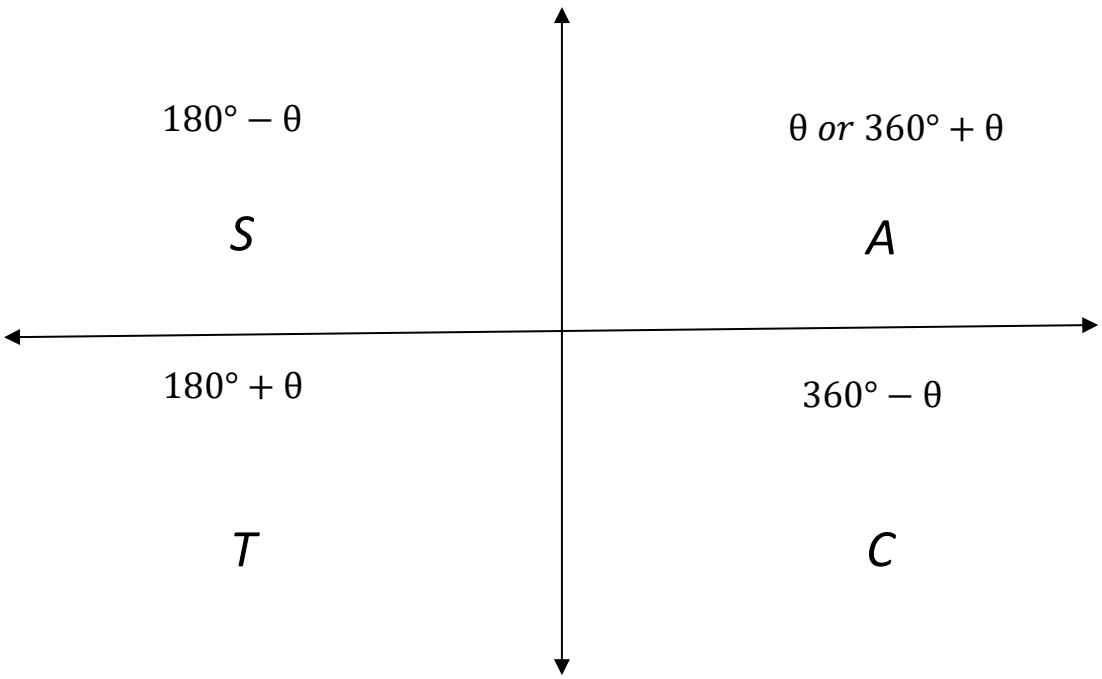
$180° - \theta$ and $180° + \theta$

$270° - \theta$ and $270° + \theta$

$360° - \theta$ and $360° + \theta$

So there is a total of 8, however, I will make it very easy for you to understand.

We will start with the most basic ones, the ones on the "CAST" diagram.



Before we get into the topic, let's first interpret each of the four quadrants above.

<u>1st Quadrant</u>

It is written "A" and $\theta \; or \; 360° + \theta$ on the *CAST* diagram.

The "A" means that **All** the three trig ratios are positive in the first quadrant, meaning $sin\,\theta$, $cos\,\theta$, $tan\,\theta$ all give positive answers, as long as the θ is a positive angle *between* $0°$ *and* $90°$.

For example, $sin\,30° = \frac{1}{2}$, $cos\,30° = \frac{\sqrt{3}}{2}$, $tan\,30° = \frac{\sqrt{3}}{3}$. Put any such acute angle, the answer will always be positive. That's what the "A" means, all the trig ratios are positive in the first quadrant.

<u>2nd Quadrant</u>

It's written "S"and $180° - \theta$ on the *CAST* diagram. The "S" means only *"sin"* is positive in the 2nd quadrant.

$180° - \theta$ means that the angles in the 2nd quadrant are greater than $90°$ but less than $180°$.

The *"sin"* of an angle between $90°$ and $180°$ always gives you a positive answer. While the *"cos" or "tan"* of an angle between $90°$ and $180°$ always gives you a negative answer.

For example;

$sin\,100° = 0.9895,$ which is positive.

$cos\,100° = -0.1736,$ which is negative.

$tan\,100° = -5.6713,$ which is negative.

Mathematicians claim that;

$sin(180° - \theta\,) = \ sin\,\theta$

$cos(180° - \theta\,) = \ -cos\,\theta$

$tan(180° - \theta\,) = \ -tan\,\theta$

Is this true? Let's test each of them below.

<u>*Test:* $sin(180° - \theta\,) = \ sin\,\theta$</u>

Punch $sin(180° - 30\,)$ into your calculator and press the equal sign.

$$sin(180° - 30°\,) = \frac{1}{2}$$

Now punch $sin\,30°$ into your calculator and press the equal sign.

$sin\,30° = \frac{1}{2}$. It gives the same answer as $sin(180° - 30°\,)$, thus $sin(180° - 30°\,) = sin\,30$

This was proven for many other values of θ. Therefore, in the 2nd quadrant, $sin(180° - \theta\,) = \ sin\,\theta$.

Let's play around a bit with $sin(180° - 30°)$

We know that $180° - 30° = 150°$

So that means $sin(180° - 30°)$ is also equal to $sin(150°)$. Simplify $sin(150°)$ as far as possible? You would first have to locate the quadrant in which the angle exists. $150°$ is located in the 2nd quadrant , this is because angles in the 2nd quadrant are between $90°$ and $180°$. The "CAST" diagram also shows that the angles in the 2nd quadrant are in the form $180° - \theta$

Now turn the $150°$ into the form $180° - \theta$

$150° = 180° - 30°$

You might be wondering how we got the $30°$ so quickly. Well, you just go $180° - 150°$ on your calculator to get $30°$. Now back to the question.

$sin(150°)$, from the above we have:

$sin(150°) = sin\ (180° - 30°)$

Remember, we also know that $sin(180° - \theta) = sin\theta$;

So $sin(180° - 30°)$

$= sin(30°)$

$= \frac{1}{2}$

Try this question: *Simplify* $sin(120°)$.

We first locate the quadrant in which $120°$ exists.

$120°$ is between $90°$ and $180°$, so it's in the 2nd quadrant. The "CAST" diagram shows that the angles in the 2nd quadrant are in the form $180° - \theta$. Let's now put the $120°$ in the form $180° - \theta$.

$120° = 180° - 60°$

You might be wondering how we got the $60°$ so quickly. Well, you just go, $180° - 120°$ on your calculator to get $60°$. Now back to the question.

$sin120°$, from the above;

$sin120° = sin\ (180° - 60°)$

Remember we also know that $(sin180° - \theta) = sin\,\theta$

So $sin(180° - 60°) = sin\ (60°)$

$$= \frac{\sqrt{3}}{2}$$

One more example: Simplify $sin\ (135°)$ without using a calculator.

$sin(135°)$

$135°$ lies in the 2nd quadrant.

$= sin(180° - 45°\ \ )$

$= sin(45°)$

$= \frac{\sqrt{2}}{2}$

Now let's test for cos.

<u>*Test:* $cos(180° - \theta\ ) = \ -cos\ \theta$</u>

Punch $cos(180° - 30°)$ into your calculator and press the equal sign.

$cos(180° - 30°) = \frac{-\sqrt{3}}{2}$

Now punch $-\cos 30°$ into your calculator and press the equal sign.

$-\cos 30° = \frac{-\sqrt{3}}{2}$. It gives the same answer as $cos(180° - 30°\ )$, thus
$cos(180° -\ 30°) = -cos\ 30°$

This was proven for many other values of θ. Therefore, in the 2nd quadrant;

$$cos(180° - \theta\ ) = \ -cos\ \theta.$$

Question

Simplify $cos(150°)$ as far as possible, without using a calculator.

Solution

$cos(150°)$, $150°$ lies in the 2nd quadrant, therefore it is between $90°$ and $180°$. So

$= cos(180° -\ 30°)$,because $180° - 30° = 150°$

$= -cos(30°)$, since $cos(180° - \theta\ ) = \ -cos\ \theta$

$= \frac{-\sqrt{3}}{2}$

Question

Simplify the following without using a calculator.

a) $\cos 120°$

b) $\cos 135°$

Solution

a) $\cos(120°)$, $120°$ lies in the 2nd quadrant, therefore it is between $90°$ and $180°$. So

$= cos(180° - 60°)$, because $180° - 60° = 120°$

$= -cos(60°)$, since $cos(180° - \theta) = -cos\,\theta$

$= -\frac{1}{2}$

b) $\cos(135°)$, $135°$ lies in the 2nd quadrant, therefore it is between $90°$ and $180°$. So

$= \cos(180° - 45°)$, because $180° - 45° = 135°$

$= -\cos(45°)$, since $\cos(180° - \theta) = -\cos\theta$

$= \frac{-\sqrt{2}}{2}$

Let's now test for "*tan*".

<u>*Test:* $tan(180° - \theta) = -tan\,\theta$</u>

Punch $tan(180° - 30)$ into your calculator and press the equal sign.

$tan(180° - 30°) = \frac{-\sqrt{3}}{3}$

Now punch $-tan\,30°$ into your calculator and press the equal sign.

$-tan\,30° = \frac{-\sqrt{3}}{3}$. It gives the same answer as $tan(180° - 30°)$, thus
$tan(180° - 30°) = -tan\,30°$

This was proven for many other values of θ. Therefore, in the 2nd quadrant;

$$tan(180° - \theta\ ) = \ -tan\,\theta$$

<u>3rd Quadrant</u>

It's written *"T"* and $180° + \theta$ on the *CAST* diagram. The *"T"* means only *"tan"* is positive in the 3rd quadrant.

$180° + \theta$ means that the angles in the 3rd quadrant are greater than $180°$ but less than $270°$.

The *"tan"* of an angle between $180°$ and $270°$ always gives you a positive answer. While the *"sin" or "cos"* of an angle between $180°$ and $270°$ always gives you a negative answer.

For example;

$sin\,190° = -0.1736,$ which is negative.

$cos\,190° = -0.9848,$ which is negative.

$tan\,190° = 0.1763,$ which is positive.

Mathematicians claim that;

$sin(180° + \theta\ ) = \ -sin\,\theta$

$cos(180° + \theta\ ) = \ -cos\,\theta$

$tan(180° + \theta\ ) = \ tan\,\theta$

Is this true? Let's test each of them below.

<u>*Test:* $sin(180° + \theta\ ) = \ -sin\,\theta$</u>

Punch $sin(180° + 30°)$ into your calculator and press the equal sign.

$$sin(180° + 30°) = -\frac{1}{2}$$

Now punch $-sin\,30°$ into your calculator and press the equal sign.

$-sin\,30° = -\frac{1}{2}$. It gives the same answer as $sin(180° + 30°)$, thus
$sin(180° +\ 30°) = -sin\,30°$

This was proven for many other values of θ. Therefore, in the 3rd quadrant;

$$sin(180° + \theta\ ) = \ -sin\,\theta.$$

Test: $cos(180° + \theta\ ) = \ -cos\,\theta$

Punch $\cos(180° + 30°)$ into your calculator and press the equal sign.

$$cos(180° + 30°) = \frac{-\sqrt{3}}{2}$$

Now punch $-\cos 30°$ into your calculator and press the equal sign.

$-\cos 30° = \frac{-\sqrt{3}}{2}$. It gives the same answer as $cos(180° + 30°)$, thus
$cos(180° + \ 30°) = -cos\,30°$

This was proven for many other values of θ. Therefore, in the 3rd quadrant;

$$\cos(180° + \theta\ ) = \ -\cos\theta.$$

Test: $tan(180° + \theta\ ) = \ tan\,\theta$

Punch $tan(180° + 30\ )$ into your calculator and press the equal sign.

$$\tan(180° + 30°) = \frac{\sqrt{3}}{3}$$

Now punch $\tan 30°$ into your calculator and press the equal sign.

$\tan 30° = \frac{\sqrt{3}}{3}$. It gives the same answer as $tan(180° + 30°\ )$, thus $tan(180° + \ 30°) = tan\,30°$

This was proven for many other values of θ. Therefore, in the 3rd quadrant;

$$tan(180° + \theta) = \ tan\,\theta.$$

4th Quadrant

It's written *"C"*and $360° - \theta$ on the *CAST* diagram. The *"C"* means only *"cos"* is positive in the 4th quadrant.

$360° - \theta$ means that the angles in the 4th quadrant are greater than $270°$ but less than $360°$.

The *"cos"* of an angle between $270°$ and $360°$ always gives you a positive answer. While the *"sin" or "tan"* of an angle between $270°$ and $360°$ always gives you a negative answer.

For example;

$sin\,280° = -0.9848$, which is negative.

$cos\,280° = 0.1736$, which is positive.

$tan\,280° = -5.6713$, which is negative.

Mathematicians claim that;

$sin(360° - \theta) = -sin\,\theta$

$cos(360° - \theta) = cos\,\theta$

$tan(360° - \theta) = -tan\,\theta$

Is this true? Let's test each of them below.

<u>*Test:* $sin(360° - \theta) = -sin\,\theta$</u>

Punch $sin(360° - 30°)$ into your calculator and press the equal sign.

$$sin(360° - 30°) = -\frac{1}{2}$$

Now punch $-sin\,30°$ into your calculator and press the equal sign.

$-sin\,30° = -\frac{1}{2}$. It gives the same answer as $sin(360° - 30°)$, thus
$sin(360° - 30°) = -sin\,30°$

This was proven for many other values of θ. Therefore, in the 4th quadrant;

$$sin(360° - \theta) = -sin\,\theta.$$

<u>*Test:* $cos(360° - \theta) = cos\,\theta$</u>

Punch $\cos(360° - 30°)$ into your calculator and press the equal sign.

$$cos(360° - 30°) = \frac{\sqrt{3}}{2}$$

Now punch $\cos 30°$ into your calculator and press the equal sign.

$\cos 30° = \frac{\sqrt{3}}{2}$. It gives the same answer as $cos(360° - 30°)$, thus
$cos(360° - 30°) = cos\,30°$

This was proven for many other values of θ. Therefore, in the 4th quadrant;

$$cos(360° - \theta) = cos\,\theta$$

<u>*Test:* $tan(360° - \theta) = -tan\,\theta$</u>

Punch $tan(360° - 30°)$ into your calculator and press the equal sign.

$$\tan(360° - 30°) = -\frac{\sqrt{3}}{3}$$

Now punch $tan\,30°$ into your calculator and press the equal sign.

$-tan\,30° = -\frac{\sqrt{3}}{3}$. It gives the same answer as $tan(360° - 30°)$, thus
$tan(360° - 30°) = -tan\,30°$

This was proven for many other values of . Therefore, in the 3rd quadrant;

$$tan(360° - \theta) = -tan\,\theta.$$

We have derived ***four of the eight*** standard angles so far. We have derived;

<u>θ or $360° + \theta$</u> **[1st quadrant]**

$sin(360° + \theta) = sin\,\theta$

$cos(360° + \theta) = cos\,\theta$ **All positive**

$tan(360° + \theta) = tan\,\theta$

<u>$180° - \theta$</u> **[2nd quadrant]**

$sin(180° - \theta) = sin\,\theta$

$cos(180° - \theta) = -cos\,\theta$ **Only sin positive**

$tan(180° - \theta) = -tan\,\theta$

<u>$180° + \theta$</u> **[3rd quadrant]**

$sin(180° + \theta) = -sin\,\theta$

$cos(180° + \theta) = -cos\,\theta$ **Only tan positive**

$tan(180° + \theta) = tan\,\theta$

$\underline{360° - \theta}$ **[4th quadrant]**

$sin(360° - \theta) = -sin\,\theta$

$cos(360° - \theta) = cos\,\theta$

$tan(360° - \theta) = -tan\,\theta$

Only cos positive

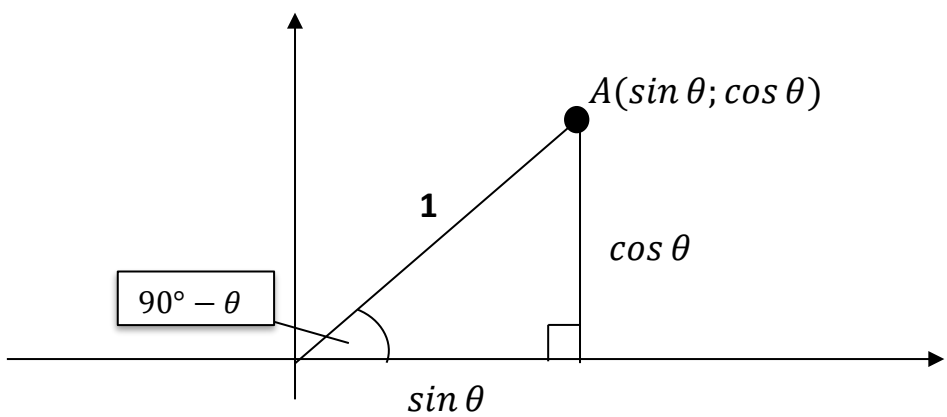
You don't have to remember this, use the CAST diagram to derive each of them when dealing with questions.

We now derive on the other four angles:

$(90° - \theta)$, $(90° + \theta)$, $(270° - \theta)$ and $(270° + \theta)$

$\underline{(90° - \theta)}$

We use the following diagram below in our derivation.



First familiarize yourself with the diagram. The point A has coordinates *($sin\,\theta$; $cos\,\theta$)* and the hypotenuse is of length 1.

A($sin\,\theta$; $cos\,\theta$)

So the x-coordinate of point A is $sin\,\theta$ and the y-coordinate is $cos\,\theta$. On the right-angled triangle, our adjacent is $sin\,\theta$ and our opposite is $cos\,\theta$. As you can see, our angle is a $(90° - \theta)$. Now let's compute the trig ratios.

$\underline{(90° - \theta)}$

For sine;

$$sin(90° - \theta) = \frac{o}{h}$$

Substitute

$sin(90° - \theta) = \frac{cos\theta}{1}$, since our opposite is $cos\theta$ and hypotenuse $= 1$

$\therefore sin(90° - \theta) = cos\theta$

For cosine;

$$cos(90° - \theta) = \frac{a}{h}$$

Substitute

$cos(90° - \theta) = \frac{sin\theta}{1}$, since our opposite is $sin\theta$ and hypotenuse $= 1$

$\therefore cos(90° - \theta) = sin\theta$

For *tan;*

$$tan(90° - \theta) = \frac{o}{a}$$

Substitute

$tan(90° - \theta) = \frac{cos\theta}{sin\theta}$, since our opposite $= cos\theta$ *and adjacent* = $sin\theta$

$$\therefore tan(90° - \theta) = \frac{cos\theta}{sin\theta}$$

Therefore, for $(90° - \theta)$

$$sin(90° - \theta) = cos\theta$$

$$cos(90° - \theta) = sin\theta$$

$$tan(90° - \theta) = \frac{cos\theta}{sin\theta}$$

<u>$(90° + \theta)$</u>

For $sin\ (90° + \theta)$

$90° + \theta$ to is obviously greater than $90°$, but less than $180°$, so it's in the 2nd quadrant. Since $90° + \theta$ is in the 2nd quadrant, we may put it in the form $180° - \theta$.

$$90° + \theta = 180° - 90° + \theta$$

Because $180° - 90° = 90°$

$90° + \theta = 180° - 90° + \theta$

$= 180° - (90° - \theta\ )$

Now put it back into sin.

$sin\ (90° + \theta) = sin\ (180° - 90° + \theta)$

$= sin\ [180° - (90° - \theta)]$

Remember then $sin(180° - \theta) = sin\theta$ in the 2nd quadrant

So $sin[180° - (90° - \theta)] = sin\ (90° - \theta)$

$sin(90° - \theta) = cos\,\theta$, *as proved in the derivations of* $90° - \theta$.

Therefore, $sin(90° + \theta) = cos\,\theta$

For $\cos\ (90° + \theta)$

$(90° + \theta)$ is in the 2nd quadrant, let's put it in the form $180° - \theta$.

$(90° + \theta) = sin\ (180° - 90° + \theta)$

$= 180° - (90° - \theta)$

Now put it back into cos.

$cos(90° + \theta) = cos\ [180 - (90° - \theta)]$

Remember that $cos(180 - \theta) = -\,cos\,\theta$ in the 2nd quadrant, so
$cos[180° - (90° - \theta)] = -\,cos(90° - \theta)$

$= -\cos(90° - \theta) = -\sin\theta$, *as proved in the derivations of* $90° - \theta$

Therefore, $\cos(90° + \theta) = -\sin\theta$

For $\tan(90° + \theta)$

From identity 1, we know that $\tan\theta = \frac{\sin\theta}{\cos\theta}$

So $\tan(90° + \theta) = \frac{\sin(90°+\theta)}{\cos(90°+\theta)}$

, and from the derivations of $90° - \theta$, we know that $\sin(90° + \theta) = \cos\theta$ and $\cos(90° + \theta) = -\sin\theta$, so we substitute and have;

$\tan(90° + \theta) = \frac{\cos\theta}{-\sin\theta}$

Therefore, $\tan(90° + \theta) = -\frac{\cos\theta}{\sin\theta}$

In summary, for $(90° + \theta)$

$$\sin(90° + \theta) = \cos\theta$$

$$\cos(90° + \theta) = -\sin\theta$$

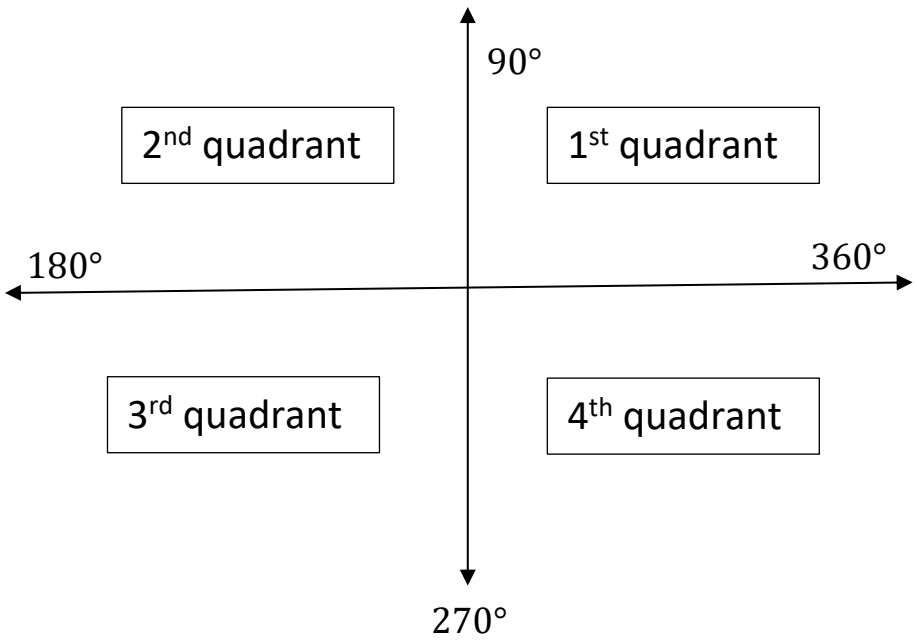
$$\tan(90° + \theta) = -\frac{\cos\theta}{\sin\theta}$$

<u>$(270° - \theta)$</u>

We are going to use some of the identities we derived for both $(90° - \theta)$ and $(90° + \theta)$ for the following derivations

For $\sin(270° - \theta)$;

The angle $(270° - \theta)$ lies in the 3rd quadrant because $(270° - \theta)$ is clearly less than $270°$ but greater than $180°$. See the diagram below.



Since $(270° - \theta)$ in the 3rd quadrant, we may put it in the form $(180° + \theta)$

$$(270° - \theta) = \underbrace{180° + 90°}_{\text{Because } 180° + 90° \text{ is } 270°} - \theta$$

So $sin(270° - \theta) = sin(180° + 90° - \theta\ )$

$= sin(180° + (90° - \theta\ ))$

Remember that $sin(180° + \theta\ ) = -sin\theta$ in the 3rd quadrant, so
$sin(180° + (90° - \theta\ )) = -sin(90° - \theta\ )$

$$\underbrace{-sin(90° - \theta\ ) = -\cos\theta}_{\text{Since } sin(90° - \theta\ ) = \cos\theta \text{ from the derivation of } 90° - \theta.}$$

Therefore, $sin(270° - \theta) = -\cos\theta$

For $cos(270° - \theta)$;

So $cos(270° - \theta) = cos(180° + 90° - \theta\ )$

$= cos(180° + (90° - \theta\ ))$

Remember that $cos(180° + \theta\ ) = -cos\theta$ in the 3rd quadrant, so
$cos(180° + (90° - \theta\ )) = -cos(90° - \theta\ )$

$$-cos(90° - \theta\ ) = -\,sin\,\theta$$

Since $cos(90° - \theta\ ) = sin\,\theta$ from the derivation of $90° - \theta$.

Therefore, $cos(270° - \theta) = -\,sin\,\theta$

For $tan(270° - \theta)$;

$$tan(270° - \theta) = \frac{sin(270° - \theta)}{cos(270° - \theta)}$$

$$= \frac{-cos\,\theta}{-sin\,\theta}$$

$$= \frac{cos\,\theta}{sin\,\theta}$$

Therefore, $tan(270° - \theta) = \frac{cos\,\theta}{sin\,\theta}$

In summary, for $(270° - \theta)$

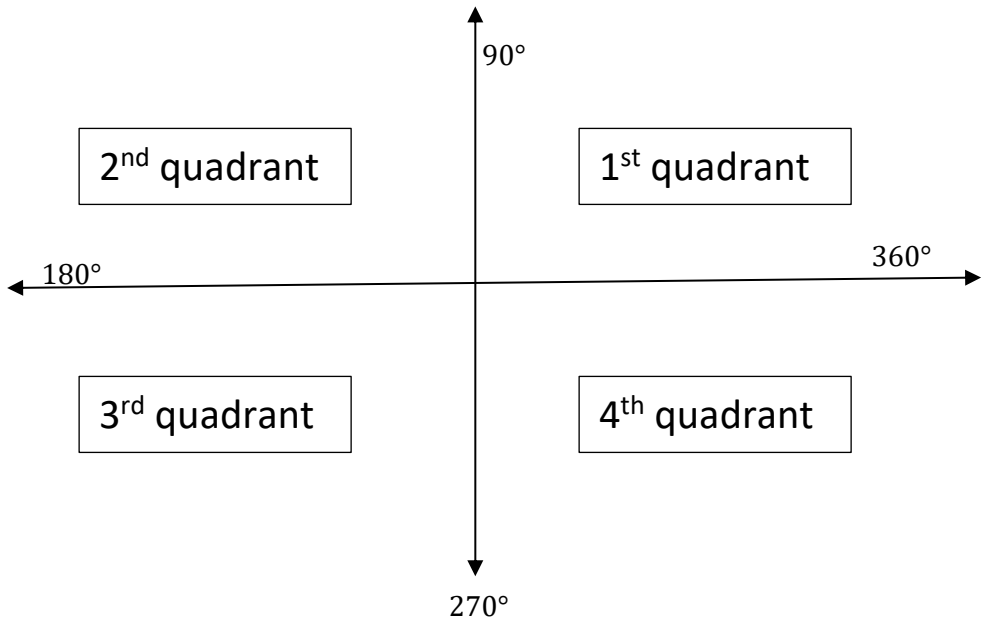
$$sin(270° - \theta) = -\,cos\,\theta$$

$$cos(270° - \theta) = -sin\,\theta$$

$$tan(270° - \theta) = \frac{cos\,\theta}{sin\,\theta}$$

$\underline{(270° + \theta)}$

The angle $(270° + \theta)$ lies in the 4th quadrant because $(270° + \theta)$ is greater than $270°$ but less than $360°$. See the diagram below.



For $sin(270° + \theta)$;

$(270° + \theta)$ is in the 4th quadrant, so we may put it in the form $(360° - \theta)$.

$(270° + \theta) = (360° - 90° + \theta)$

So $sin(270° + \theta) = sin(360° - 90° + \theta)$

$sin(270° + \theta) = sin(360° - (90° - \theta))$

Remember that $sin(360° - \theta) = -sin\,\theta$ in the 4th quadrant. We use that to simplify the above.

$sin(360° - (90° - \theta)) = -sin(90° - \theta)$ ← The identity $sin(90° - \theta) = cos\,\theta$

$= -\,cos\,\theta$

Therefore, $sin(270° + \theta) = -\,cos\,\theta$

For $cos(270° + \theta)$;

$(270° + \theta)$ is in the 4th quadrant so we may put it in the form $(360° - \theta)$.

$(270° + \theta) = (360° - 90° + \theta)$

So $cos(270° + \theta) = cos(360° - 90° + \theta)$

$cos(270° + \theta) = cos(360° - (90° - \theta))$

Remember that $cos(360° - \theta) = cos\,\theta$ in the 4th quadrant. We use that to simplify the above.

$cos(360° - (90° - \theta)) = cos(90° - \theta)$ ← The identity $cos(90° - \theta) = sin\,\theta$

$= sin\,\theta$

Therefore, $cos(270° + \theta) = sin\,\theta$

For $tan(270° + \theta)$;

$$tan(270° + \theta) = \frac{sin(270° + \theta)}{cos(270° + \theta)}$$

, these are the derivations we just got; $sin(270° + \theta) = -cos\,\theta$ and $cos(270° + \theta) = sin\,\theta$. Substitute them into the equation above.

$$tan(270° + \theta) = \frac{-cos\,\theta}{sin\,\theta}$$

$$tan(270° + \theta) = -\frac{cos\,\theta}{sin\,\theta}$$

In summary, for $(270° + \theta)$

$$sin(270° + \theta) = -cos\,\theta$$

$$cos(270° + \theta) = sin\,\theta$$

$$tan(270° + \theta) = -\frac{cos\,\theta}{sin\,\theta}$$

One more thing on $(-\theta)$

$sin(-\theta) = -\,sin\,\theta$

$cos(-\theta) = cos\,\theta$

$tan(-\theta) = tan\,\theta$

Now let's put everything together.

For $(90° - \theta)$

$$sin(90° - \theta) = cos\,\theta$$

$$cos(90° - \theta) = sin\,\theta$$

$$tan(90° - \theta) = \frac{cos\,\theta}{sin\,\theta}$$

For $(90° + \theta)$

$$sin(90° + \theta) = cos\,\theta$$

$$cos(90° + \theta) = -sin\,\theta$$

$$tan(90° + \theta) = -\frac{cos\,\theta}{sin\,\theta}$$

For $(270° - \theta)$

$$sin(270° - \theta) = -\,cos\,\theta$$

$$cos(270° - \theta) = -sin\,\theta$$

$$tan(270° - \theta) = \frac{cos\,\theta}{sin\,\theta}$$

For $(270° + \theta)$

$$sin(270° + \theta) = -\,cos\,\theta$$

$$cos(270° + \theta) = sin\,\theta$$

$$tan(270° + \theta) = -\frac{cos\,\theta}{sin\,\theta}$$

$$sin(-\theta) = -\,sin\,\theta$$

$$cos(-\theta) = cos\,\theta$$

$$tan(-\theta) = tan\,\theta$$

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Examples on $(90° - \theta)$

Example

Simplify each of the following without the use of a calculator.

a) $\frac{sin(90°-\theta)}{cos\,\theta}$ b) $\frac{cos(90°-\theta)}{sin\,\theta}$ c) $\frac{1}{tan(90°-\theta)}$

Solution

a) $\frac{sin(90°-\theta)}{cos\,\theta}$

$$= \frac{cos\,\theta}{cos\,\theta}$$

$$= \frac{\cancel{cos\,\theta}}{\cancel{cos\,\theta}}$$

$$= 1$$

b) $\frac{cos(90°-\theta)}{sin\,\theta}$

$$= \frac{sin\,\theta}{sin\,\theta}$$

$$= \frac{\cancel{sin\,\theta}}{\cancel{sin\,\theta}}$$

$$= 1$$

c) $\frac{1}{tan(90°-\theta)}$

$$= \frac{1}{\left(\frac{cos\,\theta}{sin\,\theta}\right)}$$

$$= 1 \div \frac{cos\,\theta}{sin\,\theta}$$

$$= 1 \times \frac{sin\,\theta}{cos\,\theta}$$

$$= \frac{sin\,\theta}{cos\,\theta}$$

$$= tan\,\theta$$

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Examples on $(90° + \theta)$

Example

Simplify each of the following without the use of a calculator.

a) $\frac{sin(90°+\theta)}{cos\,\theta}$ b) $\frac{cos(90°+\theta)}{sin\,\theta}$ c) $\frac{1}{tan(90°+\theta)}$

Solution

a) $\frac{sin(90°+\theta)}{cos\,\theta}$

$$= \frac{cos\,\theta}{cos\,\theta}$$

$$= \frac{\cancel{cos\,\theta}}{\cancel{cos\,\theta}}$$

$$= 1$$

b) $\frac{cos(90°+\theta)}{sin\,\theta}$

$$= \frac{-sin\,\theta}{sin\,\theta}$$

$$= \frac{-\cancel{sin\,\theta}}{\cancel{sin\,\theta}}$$

$$= -1$$

c) $\frac{1}{tan(90°+\theta)}$

$$= \frac{1}{\left(-\frac{cos\,\theta}{sin\,\theta}\right)}$$

$$= 1 \div \frac{-\,cos\,\theta}{sin\,\theta}$$

$$= 1 \times \frac{sin\,\theta}{-cos\,\theta}$$

$$= \frac{sin\,\theta}{-\,cos\,\theta}$$

$= -tan\,\theta$

Examples on $(270° - \theta)$

Example

Simplify each of the following without the use of a calculator.

a) $\frac{sin(270°-\theta)}{cos\,\theta}$ b) $\frac{cos(270°-\theta)}{sin\,\theta}$ c) $\frac{1}{tan(270°-\theta)}$

Solution

a) $\frac{sin(270°-\theta)}{cos\,\theta}$

$$= \frac{-\cos\theta}{\cos\theta}$$

$$= \frac{-\cancel{\cos\theta}}{\cancel{\cos\theta}}$$

$$= -1$$

b) $\frac{cos(270°-\theta)}{sin\,\theta}$

$$= \frac{-sin\,\theta}{sin\,\theta}$$

$$= \frac{-\cancel{sin\,\theta}}{\cancel{sin\,\theta}}$$

$$= -1$$

c) $\frac{1}{tan(270°-\theta)}$

$$= \frac{1}{\left(\frac{cos\,\theta}{sin\,\theta}\right)}$$

$$= 1 \div \frac{cos\,\theta}{sin\,\theta}$$

$$= 1 \times \frac{sin\,\theta}{cos\,\theta}$$

$$= \frac{\sin\theta}{\cos\theta}$$

$$= \tan\theta$$

Examples on $(270° + \theta)$

Example

Simplify each of the following without the use of a calculator.

a) $\frac{\sin(270°+\theta)}{\cos\theta}$ b) $\frac{\cos(270°+\theta)}{\sin\theta}$ c) $\frac{1}{\tan(270°+\theta)}$

Solution

a) $\frac{\sin(270°+\theta)}{\cos\theta}$

$$= \frac{-\cos\theta}{\cos\theta}$$

$$= \frac{-\cancel{\cos\theta}}{\cancel{\cos\theta}}$$

$$= -1$$

b) $\frac{\cos(270°+\theta)}{\sin\theta}$

$$= \frac{\sin\theta}{\sin\theta}$$

$$= \frac{\cancel{\sin\theta}}{\cancel{\sin\theta}}$$

$$= 1$$

c) $\frac{1}{\tan(270°-\theta)}$

$$= \frac{1}{\left(\frac{-\cos\theta}{\sin\theta}\right)}$$

$= 1 \div \frac{-\cos\theta}{\sin\theta}$

$= 1 \times \frac{\sin\theta}{-\cos\theta}$

$= -\frac{\sin\theta}{\cos\theta}$

$= -\tan\theta$

Examples on $(180° - \theta)$; 2nd Quadrant

Example

Simplify each of the following without the use of a calculator.

a) $\frac{\sin(180°-\theta)}{\cos\theta}$ b) $\frac{\cos(180°-\theta)}{\sin\theta}$ c) $\frac{1}{\tan(180°-\theta)}$

Solution

a) $\frac{\sin(180°-\theta)}{\cos\theta}$

$= \frac{\sin\theta}{\cos\theta}$

$= \tan\theta$

b) $\frac{\cos(180°-\theta)}{\sin\theta}$

$= \frac{-\cos\theta}{\sin\theta}$

$= -\frac{\cos\theta}{\sin\theta}$

c) $\frac{1}{\tan(180°-\theta)}$

$\frac{1}{-\tan\theta}$

$= \frac{1}{\left(\frac{-\sin\theta}{\cos\theta}\right)}$

$$= 1 \div -\frac{\sin\theta}{\cos\theta}$$

$$= 1 \times -\frac{\cos\theta}{\sin\theta}$$

$$= -\frac{\cos\theta}{\sin\theta}$$

<u>Examples on $(180° + \theta)$; 3rd Quadrant</u>

Example

Simplify each of the following without the use of a calculator.

a) $\frac{\sin(180°+\theta)}{\cos\theta}$ b) $\frac{\cos(180°+\theta)}{\sin\theta}$ c) $\frac{1}{\tan(180°+\theta)}$

Solution

a) $\frac{\sin(180°+\theta)}{\cos\theta}$

$$= \frac{-\sin\theta}{\cos\theta}$$

$$= -\tan\theta$$

b) $\frac{\cos(180°+\theta)}{\sin\theta}$

$$= \frac{-\cos\theta}{\sin\theta}$$

$$= -\frac{\cos\theta}{\sin\theta}$$

c) $\frac{1}{\tan(180°+\theta)}$

$$\frac{1}{\tan\theta}$$

$$= \frac{1}{\left(\frac{\sin\theta}{\cos\theta}\right)}$$

$$= 1 \div \frac{\sin\theta}{\cos\theta}$$

$$= 1 \times \frac{\cos\theta}{\sin\theta}$$

$$= \frac{\cos\theta}{\sin\theta}$$

Examples on $(360° - \theta)$; 4th Quadrant

Example

Simplify each of the following without the use of a calculator.

a) $\frac{\sin(360°-\theta)}{\cos\theta}$ b) $\frac{\cos(360°-\theta)}{\sin\theta}$ c) $\frac{1}{\tan(360°-\theta)}$

Solution

a) $\frac{\sin(360°-\theta)}{\cos\theta}$

$$= \frac{-\sin\theta}{\cos\theta}$$

$$= -\tan\theta$$

b) $\frac{\cos(360°-\theta)}{\sin\theta}$

$$= \frac{\cos\theta}{\sin\theta}$$

c) $\frac{1}{\tan(360°-\theta)}$

$$= \frac{1}{-\tan\theta}$$

$$= \frac{1}{\left(-\frac{\sin\theta}{\cos\theta}\right)}$$

$$= 1 \div -\frac{\sin\theta}{\cos\theta}$$

$$= 1 \times -\frac{\cos\theta}{\sin\theta}$$

$$= -\frac{cos\,\theta}{sin\,\theta}$$

Question

Simplify the following as far as possible.

$$\frac{sin(180°-30°).cos(180°-30°)}{sin(360°-30°)}$$

Solution

$$\frac{sin(180° - 30°).\,cos(180° - 30°)}{sin(360° - 30°)}$$

$$= \frac{sin(180° - 30°).\,cos(180° - 30°)}{sin(360° - 30°)}$$

- In the 2nd quadrant, $sin(180° - \theta) = sin\,\theta$, so $sin(180° - 30°) = sin\,30°$
- In the 2nd quadrant, $cos(180° - \theta) = -\,cos\,\theta$, so $cos(180° - 30°) = -cos\,30°$
- In the 4th quadrant, $sin(360° - \theta) = -\,sin\,\theta$, so $sin(360° - 30°) = -\,sin\,30°$

$$= \frac{sin\,30°.-cos\,30°}{-sin\,30°}$$

$$= \frac{\cancel{sin\,30°}.-cos\,30°}{\cancel{-sin\,30°}}$$

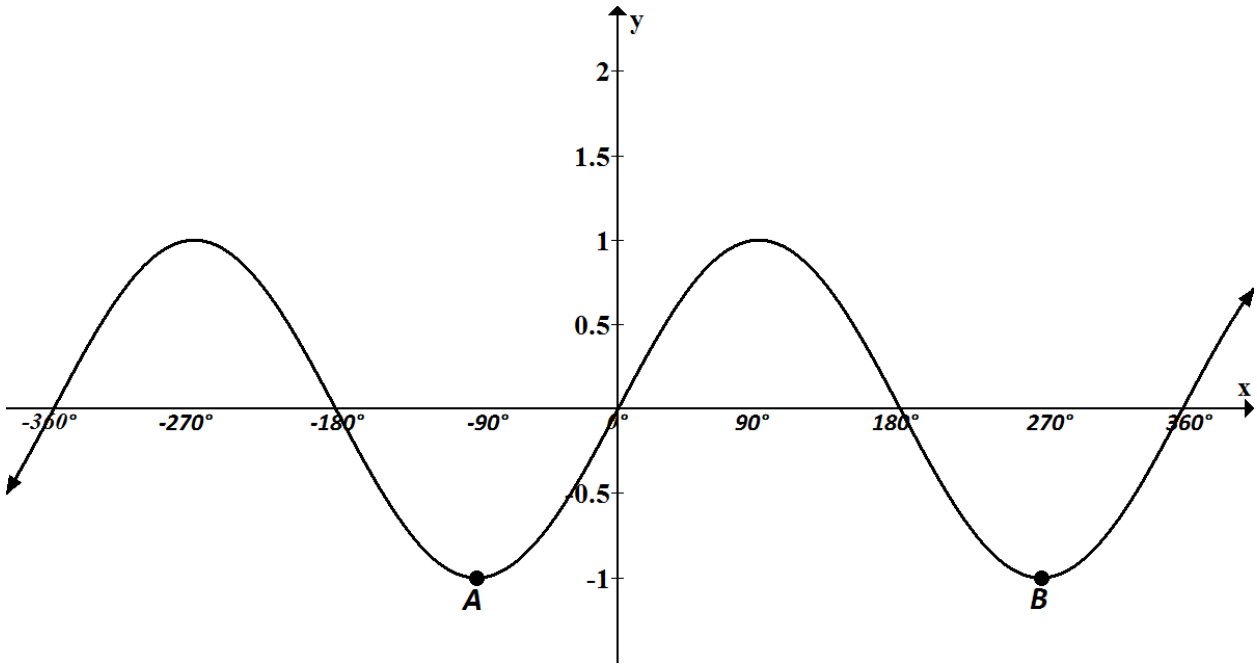
$$= \frac{-cos\,30°}{-1}$$

$$= cos\,30°$$

$$= \frac{\sqrt{3}}{2}$$

General Solutions

Consider a rough sketch graph of $f(x) = sin\, x$.



What are the coordinates of A, B? From the graph, the coordinates of the points are: $A(-90°; -1)$ and $B(270°; -1)$.

Notice that the two points have the same "***y-value***". They all have a *"y-value"* of -1. You can even test the values using your calculator.

For $A(-90°; -1)$, punch $sin(-90°)$, it gives -1, point A is true.

For $B(270°; -1)$, punch $sin(270°)$, it gives -1, point B is true.

So all of these points have a y-value of -1. Attempt the following example.

Example

Solve for x in the equation $sin\, x = -1$

Solution

Remember the aim is to remain with x "alone" on one side. In this case, we take the inverse of "sin".

$x = sin^{-1}(-1)$

$x = -90°$

Do you see the problem? The calculator is only giving $-90°$, but what about the $-450°$ from point A, what about the $270°$ from point C?

In this chapter, you are going to learn how to find such missing values. There are two formulas we should use to find all of these missing values for "sin".

$$x = angle + 360° \times k \quad \textit{and}$$

$$x = (180° - angle) + 360° \times k$$

, the "angle" is the angle you get from the calculator, and *k* is an integer (the integers we generally use are (*-1, 0, 1*). Don't panic, we will redo this to prove to you that this is super easy. Check out the steps below.

Solve for x in the equation $sin\, x = -1$

Step 1: Take the inverse.

$sin\, x = -1$

$x = sin^{-1}(-1)$

$x = -90°$

$angle = -90°$

Step 2: Apply the first formula and substitute the integers {-1, 0, 1}.

$x = angle + 360° \times k$, remember our "angle" is $-90°$, so we substitute it into the formula.

$x = -90° + 360° \times k$

Substitute each of the integers {-1, 0, 1}.

Substitute -1

$x = -90° + 360° \times -1 = -450°$

$x = -450°$

Substitute 0

$x = -90° + 360° \times 0 = -90°$

$x = -90°$

Substitute 1

$x = -90° + 360° \times 1 = 270°$

$x = 270°$

So our values are $x = -90°$, $x = 270°$ and $x = -450°$

Step 3: Apply the second formula and substitute integers {-1, 0, 1}.

$x = (180° - angle) + 360° \times k$, remember our "angle" is $-90°$, so we substitute it into the formula.

$x = (180° - (-90°)) + 360° \times k$

$x = (180° + 90°) + 360° \times k$

$x = 270° + 360° \times k$

Substitute each of the integers {-1, 0, 1}.

Substitute -1

$x = 270° + 360° \times -1 = -90°$

$x = -90°$

Substitute 0

$x = 270° + 360° \times 0 = 270°$

$x = 270°$

So our values are $x = -90°$ and $x = 270°$

Putting everything together, our values are;

$x = -450°$, $x = -90°$, $x = 270°$

That's how we get all the values. These are the $x - values$ of points A and B on the graph. If we were to continue drawing the graph further left, we would also see that coordinates $(-450°; -1)$

Example

Solve for x in the equation $sin\, x = 1$

Solution

Step 1: Take the inverse.

$sin\, x = 1$

$x = sin^{-1}(1)$

$x = 90°$

$angle = 90°$

Step 2: Apply the first formula and substitute the integers {-1, 0, 1}.

$x = angle + 360° \times k$, remember our "angle" is $90°$, so we substitute it into the formula.

$x = 90° + 360° \times k$

Substitute each of the integers {-1, 0, 1}.

Substitute -1

$x = 90° + 360° \times -1 = -270°$

$x = -270°$

Substitute 0

$x = 90° + 360° \times 0 = 90°$

$x = 90°$

Substitute 1

$x = 90° + 360° \times 1 = 450°$

$x = 450°$

So our values are $x = -270°$, $x = 90°$ and $x = 450°$

Step 3: Apply the second formula and substitute integers {-1, 0, 1}.

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$x = (180° - angle) + 360° \times k$, remember our "angle" is 90°, so we substitute it into the formula.

$x = (180° - 90°) + 360° \times k$

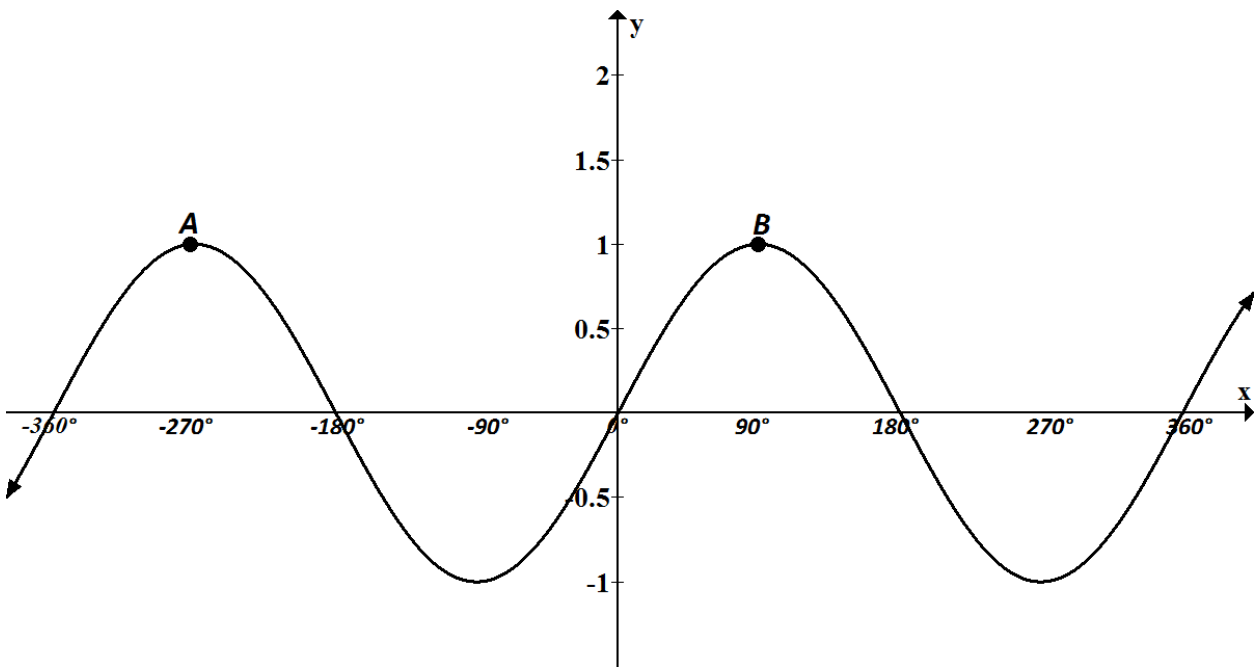
$x = 90° + 360° \times k$

This is a similar general solution to the one from the first formula, so the answers will be same.

Putting everything together, our values are;

$x = -270°, x = 90°, x = 450°$

Check if the answers are correct from the graph of $sin\, x$ below.



Focus on point $A(-270°; 1)$ and point $B(90°; 1)$

The x-coordinates of these points are the $-270°$ and $90°$ which are the answers we got in the example above. Remember that the question asked you to solve for x in $sin\, x = 1$, which means you should find all the coordinates of the points with a *y-value* of 1 on the graph of $sin\, x$. We see that the points with a y-value of 1 are A and B. These points have the x-values of the $-270°$ and $90°$, as mentioned before. So our answers are correct.

Example

Solve for x in the equation $sin\, x = \frac{1}{2}$

Solution

Step 1: Take the inverse.

$sin\, x = \frac{1}{2}$

$x = sin^{-1}\left(\frac{1}{2}\right)$

$x = 30°$

$angle = 30°$

Step 2: Apply the first formula and substitute the integers {-1, 0, 1}.

$x = angle + 360° \times k$, remember our "angle" is $30°$, so we substitute it into the formula.

$x = 30° + 360° \times k$

Substitute each of the integers {-1, 0, 1}.

Substitute -1

$x = 30° + 360° \times -1 = -330°$

$x = -330°$

Substitute 0

$x = 30° + 360° \times 0 = 30°$

$x = 30°$

Substitute 1

$x = 30° + 360° \times 1 = 390°$

$x = 390°$

So our values are $x = -330°$, $x = 30°$ and $x = 390°$

Step 3: Apply the second formula and substitute integers {-1, 0, 1}.

$x = (180° - angle) + 360° \times k$, remember our "angle" is $30°$, so we substitute it into the formula.

$x = (180° - 30°) + 360° \times k$

$x = 150 + 360° \times k$

Substitute each of the integers {-1, 0, 1}.

Substitute -1

$x = 150° + 360° \times -1 = -210°$

$x = -210°$

Substitute 0

$x = 150° + 360° \times 0 = 150°$

$x = 150°$

Substitute 1

$x = 150° + 360° \times 1 = 510°$

$x = 510°$

So our values are $x = -210°$, $x = 150°$ and $x = 510°$

Putting everything together, our values are; $x = -330°$, $x = -210°$, $x = 30°$, $x = 150°$, $x = 390°$, $x = 510°$

Another way of checking if your answers are correct is to punch your answers into your calculator, each of them should give you the answer $\frac{1}{2}$.

$$sin\, x = \frac{1}{2}$$

Punch $sin(-330°)$, it gives $\frac{1}{2}$, it is correct.

Punch $sin(-210°)$, it gives $\frac{1}{2}$, it is correct.

Punch $sin(30°)$, it gives $\frac{1}{2}$, it is correct.

Punch $sin(150°)$, it gives $\frac{1}{2}$, it is correct.

Punch $sin(390°)$, it gives $\frac{1}{2}$, it is correct.

Punch $sin(510°)$, it gives $\frac{1}{2}$, it is correct.

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Therefore, all of our answers are correct.

Sometimes a question may ask you to leave out answers that are bigger than $360°$. If that is the case, you wouldn't have to include the $390°$ and $510°$ on your answers.

Question

Solve for x in $sin\, x = \frac{\sqrt{3}}{2}$, where $-360° \leq x \leq 360°$

Solution

Note that $-360° \leq x \leq 360°$ means we only include the x-values between $-360°$ and $360°$, including the $-360°$ and $-360°$.

Step 1: Take the inverse.

$$sin\, x = \frac{\sqrt{3}}{2}$$

$$x = sin^{-1}\left(\frac{\sqrt{3}}{2}\right)$$

$$x = 60°$$

$$angle = 60°$$

Step 2: Apply the first formula and substitute the integers {-1, 0, 1}.

$x = angle + 360° \times k$, remember our "angle" is $60°$, so we substitute it into the formula.

$$x = 60° + 360° \times k$$

Substitute each of the integers {-1, 0, 1}.

Substitute -1

$$x = 60° + 360° \times -1 = -300°$$

$$x = -300°$$

Substitute 0

$$x = 60° + 360° \times 0 = 60°$$

$$x = 60°$$

Substitute 1

$x = 60° + 360° \times 1 = 420°$

$x = 420°$ (We don't include this one in our answer because it is "NOT" between $-360°$ and $360°$).

So our values are $x = -300°$, $x = 60°$

Step 3: Apply the second formula and substitute integers {-1, 0, 1}.

$x = (180° - angle) + 360° \times k$, remember our "angle" is $60°$, so we substitute it into the formula.

$x = (180° - 60°) + 360° \times k$

$x = 120 + 360° \times k$

Substitute each of the integers {-1, 0, 1}.

Substitute -1

$x = 120° + 360° \times -1 = -240°$

$x = -240°$

Substitute 0

$x = 120° + 360° \times 0 = 120°$

$x = 120°$

Substitute 1

$x = 120° + 360° \times 1 = 480°$

$x = 480°$ (We don't include this one in our answer because it is "NOT" between $-360°$ and $360°$).

So our values are $x = -240°$, $x = 120°$

Putting everything together, our values are; $x = -300°$, $x = 60°$, $x = -240°$, $x = 120°$

Formula for missing values of Cosine

$$x = angle + 360° \times k \quad \textit{and}$$

$$x = -angle + 360° \times k$$

Let's do some examples to make it clear.

Example

Solve for x in $cos\, x = -1$,

where $-360° \leq x \leq 360°$

Solution

Note that $-360° \leq x \leq 360°$ means we only include the x-values between $-360°$ and $360°$, including the $-360°$ and $-360°$.

Step 1: Take the inverse.

$cos\, x = -1$

$x = cos^{-1}(-1)$

$x = 180°$

$angle = 180°$

Step 2: Apply the first formula and substitute the integers {-1, 0, 1}.

$x = angle + 360° \times k$, remember our "angle" is $180°$, so we substitute it into the formula.

$x = 180° + 360° \times k$

Substitute each of the integers {-1, 0, 1}.

Substitute -1

$x = 180° + 360° \times -1 = -180°$

$x = -180°$

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Substitute 0

$x = 180° + 360° \times 0 = 180°$

$x = 180°$

Substitute 1

$x = 180° + 360° \times 1 = 540°$

$x = 540°$ (We don't include this one in our answer because it is "NOT" between $-360°$ and $360°$).

So our values are $x = -180°$, $x = 180°$

Step 3: Apply the second formula and substitute integers {-1, 0, 1}.

$x = -angle + 360° \times k$, remember our "angle" is $180°$, so we substitute it into the formula.

$x = -180° + 360° \times k$

Substitute each of the integers {-1, 0, 1}.

Substitute -1

$x = -180° + 360° \times -1 = -540°$

$x = -540°$ (We don't include this one in our answer because it is "NOT" between $-360°$ and $360°$).

Substitute 0

$x = -180° + 360° \times 0 = -180°$

$x = -180°$

Substitute 1

$x = -180° + 360° \times 1 = 180°$

$x = 180°$

So our values are $x = -180°$, $x = 180°$

Putting everything together, our values are; $x = -180°$, $x = 180°$

Question

Solve for x in $cos\, x = \frac{\sqrt{3}}{2}$, where $-360° \leq x \leq 360°$

Solution

Note that $-360° \leq x \leq 360°$ means we only include the x-values between $-360°$ and $360°$, including the $-360°$ and $-360°$.

Step 1: Take the inverse.

$$cos\, x = \frac{\sqrt{3}}{2}$$

$$x = \cos^{-1}\left(\frac{\sqrt{3}}{2}\right)$$

$$x = 30°$$

$$angle = 30°$$

Step 2: Apply the first formula and substitute the integers {-1, 0, 1}.

$x = angle + 360° \times k$, remember our "angle" is $30°$, so we substitute it into the formula.

$$x = 30° + 360° \times k$$

Substitute each of the integers {-1, 0, 1}.

Substitute -1

$$x = 30° + 360° \times -1 = -330°$$

$$x = -330°$$

Substitute 0

$$x = 30° + 360° \times 0 = 30°$$

$$x = 30°$$

Substitute 1

$$x = 30° + 360° \times 1 = 390°$$

$x = 390°$ (We don't include this one in our answer because it is "NOT" between $-360°$ and $360°$).

So our values are $x = -330°$, $x = 30°$

Step 3: Apply the second formula and substitute integers {-1, 0, 1}.

$x = -angle + 360° \times k$, remember our "angle" is $30°$, so we substitute it into the formula.

$x = -30° + 360° \times k$

Substitute each of the integers {-1, 0, 1}.

Substitute -1

$x = -30° + 360° \times -1 = -390°$

$x = -390°$ (We don't include this one in our answer because it is "NOT" between $-360°$ and $360°$).

Substitute 0

$x = -30° + 360° \times 0 = -30°$

$x = -30°$

Substitute 1

$x = -30° + 360° \times 1 = 330°$

$x = 330°$

So our values are $x = -330°$, $x = -30°$, $x = 30°$, $x = 330°$

Putting everything together, our values are; $x = -300°$, $x = 60°$, $x = -240°$, $x = 120°$

Formula for missing values of tan

$$\boxed{x = angle + 180° \times k}$$

The calculation of values with "**tan**" takes only 2 steps since there is only one formula to work with.

Question

Solve for x in $tan\, x = 1$, where $-360° \leq x \leq 360°$

Solution

Note that $-360° \leq x \leq 360°$ means we only include the x-values between $-360°$ and $360°$, including the $-360°$ and $-360°$.

Step 1: Take the inverse.

$tan\, x = 1$

$x = tan^{-1}(1)$

$x = 45°$

$angle = 45°$

Step 2: Apply the first formula and substitute the integers {-1, 0, 1}.

$x = angle + 180° \times k$, remember our "angle" is $45°$, so we substitute it into the formula.

$x = 45° + 180° \times k$

Substitute each of the integers {-1, 0, 1}.

Substitute -1

$x = 45° + 180° \times -1 = -135°$

$x = -135°$

Substitute 0

$x = 45° + 180° \times 0 = 45°$

$x = 45°$

Substitute 1

$x = 45° + 180° \times 1 = 225°$

$x = 225°$

So our values are $x = -135°$, $x = 45°$, $x = 225°$,

Before we move any further, let's quickly revise step 1 of the solution above.

We have $x = 45° + 180° \times k$

This is called the general solution of the equation
$tan\, x = 1$ because we use it to get all the required $x - values$.

For **cosine**, we had something like;

$x = 180° + 360° \times k$ and $= -180° + 360° \times k$, these are the two general solutions of $cos\, x = -1$ because we used them to find all the required values of x. Attempt the questions below to check if you fully understand what a general solution is.

Question

Find the general solution(s) of each of the following questions.

a) $sin\, x = -1$

b) $cos\, x = -1$

c) $tan\, x = 2$

d) $sin\, x = 1$

Solution

a) $sin\, x = -1$

$x = sin^{-1}(-1)$

$x = -90°$

$angle = -90°$

We now make use of the two formulas of the missing values of ***sin***.

$x = angle + 360° \times k$ and $x = (180° - angle) + 360° \times k$

We then substitute the angle into both formulas.

$x = -90° + 360° \times k$ and $x = (180° - (-90°)) + 360° \times k$

$x = -90° + 360° \times k$ and $x = 270° + 360° \times k$

, where k is an integer

The two are the general solutions for $sin\, x = -1$

b) $cos\, x = -1$

$x = cos^{-1}(-1)$

$x = 180°$

$angle = 180°$

We now make use of the two formulas of the missing values of ***cos***.

$x = angle + 360° \times k$ and $x = -angle + 360° \times k$

We then substitute the angle into both formulas.

$x = 180° + 360° \times k$ and $x = -180° + 360° \times k$

, where k is an integer

The two are the general solutions for $cos\, x = -1$

c) $tan\, x = 2$

$x = tan^{-1}(2)$

$x = 63.4°$

$angle = 63.43°$

The "**tan**" has only one formula for missing values.

$x = angle + 360° \times k$

We then substitute the angle into both formulas.

$x = 63.43° + 360° \times k$, where k is an integer

This is the general solution for $tan\, x = 2$

d) $sin\, x = 1$

$x = sin^{-1}(1)$

$x = 90°$

$angle = 90°$

We now make use of the two formulas of the missing values of ***sin***.

$x = angle + 360° \times k$ and $x = (180° - angle) + 360° \times k$

We then substitute the angle into both formulas.

$x = 90° + 360° \times k$ and $x = (180° - 90°) + 360° \times k$

$x = 90° + 360° \times k$ and $x = 90° + 360° \times k$

, where k is an integer

Since the two general solutions are similar, we just take one of them.

$x = 90° + 360° \times k$, where k is an integer

That's the general solution for $sin\, x = 1$

Question

Find the general solution(s) of each of the following and solve for x over the interval $-360° \leq x \leq 360°$

a) $cos(x) + 1 = \frac{1}{2}$

b) $cos(2x) = -1$

c) $cos\left(\frac{x}{2}\right) = -1$

d) $2sin(x) = 1$

e) $2sin(2x) = 1$

f) $3sin(x) + 1 = 3$

g) $2tan\left(\frac{x}{3}\right) = 5$

Solution

a) $cos(x) + 1 = \frac{1}{2}$

$$cos(x) = \frac{1}{2} - 1$$

$$cos(x) = -\frac{1}{2}$$

$$x = cos^{-1}\left(-\frac{1}{2}\right)$$

$$x = 120°$$

$$angle = 120°$$

We now make use of the two formulas.

$x = angle + 360° \times k \;\; and \; x = -angle + 360° \times k$

We then substitute the angle into both formulas.

$\boldsymbol{x = 120° + 360° \times k}$ and $\boldsymbol{x = -120° + 360° \times k}$

, where k is an integer

Those are the two general solutions.

Then the x-values are;

First Formula: $x = 120° + 360° \times k$

For $k = -1$, $x = 120° + 360° \times -1 = -240°$

For $k = 0$, $x = 120° + 360° \times 0 = 120°$

For $k = 1$, $x = 120° + 360° \times 1 = 480°$ (We don't include this one in our answer because it is "NOT" between $-360°$ and $360°$).

Second Formula: $x = -120° + 360° \times k$

For $k = -1$, $x = -120° + 360° \times -1 = -480°$ (We don't include this one in our answer because it is "NOT" between $-360°$ and $360°$).

For $k = 0$, $x = -120° + 360° \times 0 = -120°$

For $k = 1$, $x = -120° + 360° \times 1 = 240°$

$x = -240°, x = -120°, x = 120°, x = 240°,$

b) $cos(2x) = -1$

$2x = cos^{-1}(-1)$

$2x = 180°$

$x = \frac{180°}{2}$

$angle = 90°$

We now make use of the two formulas.

$x = angle + 360° \times k \;\; and \; x = -angle + 360° \times k$

We then substitute the angle into both formulas.

$x = 90° + 360° \times k$ and $x = -90° + 360° \times k$

, where k is an integer

These are the two general solutions.

Then the x-values are;

First Formula: $x = 90° + 360° \times k$

For $k = -1$, $x = 90° + 360° \times -1 = -270°$

For $k = 0$, $x = 90° + 360° \times 0 = 90°$

For $k = 1$, $x = 90° + 360° \times 1 = 450°$ (We don't include this one in our answer because it is "NOT" between $-360°$ and $360°$).

Second Formula: $x = -90° + 360° \times k$

For $k = -1$, $x = -90° + 360° \times -1 = -450°$ (We don't include this one in our answer because it is "NOT" between $-360°$ and $360°$).

For $k = 0$, $x = -90° + 360° \times 0 = -90°$

For $k = 1$, $x = -90° + 360° \times 1 = 270°$

$x = -270°$, $x = -90°$, $x = 90°$, $x = 270°$,

c) $cos\left(\frac{x}{2}\right) = -1$

$\frac{x}{2} = cos^{-1}(-1)$

$\frac{x}{2} = 180°$

$x \div 2 = 180°$

$x = 180° \times 2$

$angle = 360°$

We now make use of the two formulas.

$x = angle + 360° \times k \;\; and \; x = -angle + 360° \times k$

We then substitute the angle into both formulas.

$x = 360° + 360° \times k$ and $x = -360° + 360° \times k$

, where k is an integer

These are the two general solutions.

Then the x-values are;

First Formula: $x = 360° + 360° \times k$

For $k = -1$, $x = 360° + 360° \times -1 = 0°$

For $k = 0$, $x = 360° + 360° \times 0 = 360°$

For $k = 1$, $x = 360° + 360° \times 1 = 720°$ *(Excluded)*

Second Formula: $x = -360° + 360° \times k$

For $k = -1$, $x = -360° + 360° \times -1 = -720°$*(Excluded)*

For $k = 0$, $x = -360° + 360° \times 0 = -360°$

For $k = 1$, $x = -360° + 360° \times 1 = 0°$

$x = -360°$, $x = 0°$, $x = 360°$

d) $2sin(x) = 1$

$sin(x) = \frac{1}{2}$

$x = sin^{-1}\left(\frac{1}{2}\right)$

$x = 30°$

$angle = 30°$

We now make use of the two formulas.

$x = angle + 360° \times k$ and $x = (180° - angle) + 360° \times k$

We then substitute the angle into both formulas.

$\mathrm{x} = 30° + 360° \times \mathrm{k}$ and $\mathrm{x} = (180° - 30°) + 360° \times \mathrm{k}$

, where k is an integer

$x = 30° + 360° \times k$ and $x = 150° + 360° \times k$

, where k is an integer

These are the two general solutions.

Then the x-values are;

First Formula: $x = 30° + 360° \times k$

For $k = -1$, $x = 30° + 360° \times -1 = -330°$

For $k = 0$, $x = 30° + 360° \times 0 = 30°$

For $k = 1$, $x = 30° + 360° \times 1 = 390°$ *(Excluded)*

Second Formula: $x = 150° + 360° \times k$

For $k = -1$, $x = 150° + 360° \times -1 = -210°$

For $k = 0$, $x = 150° + 360° \times 0 = 150°$

For $k = 1$, $x = 150° + 360° \times 1 = 510°$ *(Excluded)*

$x = -330°$, $x = -210°$, $x = 30°$, $x = 150°$

e) $2sin(2x) = 1$

$sin(2x) = \frac{1}{2}$

$2x = sin^{-1}\left(\frac{1}{2}\right)$

$2x = 30°$

$x = \frac{30°}{2}$

$x = 15°$

We now make use of the two formulas.

$x = angle + 360° \times k$ and $x = (180° - angle) + 360° \times k$

We then substitute the angle into both formulas.

$x = 15° + 360° \times k$ and $x = 165° + 360° \times k$

, where k is an integer

These are the two general solutions.

Then the x-values are;

First Formula: $x = 15° + 360° \times k$

For $k = -1$, $x = 15° + 360° \times -1 = -345°$

For $k = 0$, $x = 15° + 360° \times 0 = 15°$

For $k = 1$, $x = 15° + 360° \times 1 = 375°$ *(Excluded)*

Second Formula: $x = 165° + 360° \times k$

For $k = -1$, $x = 165° + 360° \times -1 = -195°$

For $k = 0$, $x = 165° + 360° \times 0 = 165°$

For $k = 1$, $x = 165° + 360° \times 1 = 525°$ *(Excluded)*

$x = -345°$, $x = -195°$, $x = 15°$, $x = 165°$

e) $3sin(x) + 1 = 3$

$3sin(x) = 3 - 1$

$3sin(x) = 2$

$sin(x) = \frac{2}{3}$

$x = sin^{-1}\left(\frac{2}{3}\right)$

$x = 41.81°$

$angle = 41.81°$

We now make use of the two formulas.

$x = angle + 360° \times k$ and $x = (180° - angle) + 360° \times k$

We then substitute the angle into both formulas.

$x = 41.81° + 360° \times k$ and $x = (180° - 41.81°) + 360° \times k$

$x = 41.81° + 360° \times k$ and $x = 138.19° + 360° \times k$

, where k is an integer

These are the two general solutions.

Then the x-values are;

First Formula: $x = 41.81° + 360° \times k$

For $k = -1$, $x = 41.81° + 360° \times -1 = -318.19°$

For $k = 0$, $x = 41.81° + 360° \times 0 = 41.81°$

For $k = 1$, $x = 41.81° + 360° \times 1 = 401.81°$ *(Excluded)*

Second Formula: $x = 138.19° + 360° \times k$

For $k = -1$, $x = 138.18° + 360° \times -1 = -221.81°$

For $k = 0$, $x = 138.19° + 360° \times 0 = 138.19°$

For $k = 1$, $x = 138.19° + 360° \times 1 = 498.19°$ *(Excluded)*

$x = -318.19°$, $x = -221.81°$, $x = 41.81°$, $x = 138.19°$

f) $2tan\left(\frac{x}{3}\right) = 5$

$$tan\left(\frac{x}{3}\right) = \frac{5}{2}$$

$$\frac{x}{3} = tan^{-1}\left(\frac{5}{2}\right)$$

$$\frac{x}{3} = 68.2°$$

$x = 68.2° \times 3$

$x = 204.6°$

$angle = 204.6°$

$x = angle + 180° \times k$

We then substitute the angle into the formula.

$\boldsymbol{x = 204.6° + 180° \times k}$, where k is an integer

That's the general solution. Remember "***tan***" has got one solution.

$x = 204.6° + 180° \times k$

For $k = -1$, $x = 204.6° + 180° \times -1 = 24.6°$

For $k = 0$, $x = 204.6° + 180° \times 0 = 204.6°$

For $k = 1$, $x = 204.6° + 180° \times 1 = 384.6°$ *(Excluded)*

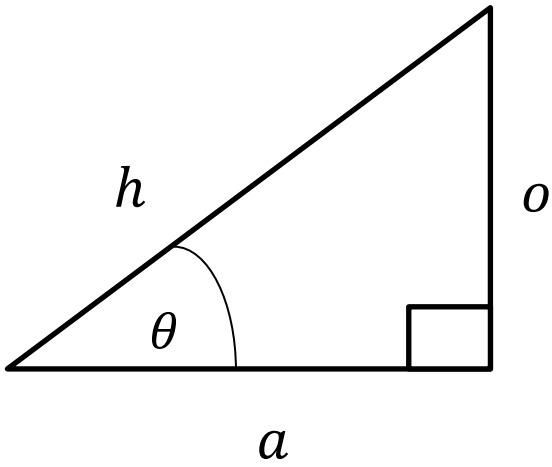
$x = 24.6°$, $x = 204.6°$

Proving Trig Identities, Sine, Cosine and Area Rules

In this section, we derive the identity:

$$tan\,\theta = \frac{sin\,\theta}{cos\,\theta}$$

We are going to use the right-angled triangle below.



$sin\,\theta = \frac{o}{h}$ $cos\,\theta = \frac{a}{h}$ $tan\,\theta = \frac{o}{a}$

We are going to divide $sin\,\theta$ by $cos\,\theta$ to see if it gives us $\frac{o}{a}$, similar to $tan\,\theta$ above.

$$\frac{sin\,\theta}{cos\,\theta} = \frac{\left(\frac{o}{h}\right)}{\left(\frac{a}{h}\right)}$$

Remember that this "over" sign means divide.

$$\frac{sin\,\theta}{cos\,\theta} = \left(\frac{o}{h}\right) \div \left(\frac{a}{h}\right)$$

$$\frac{sin\,\theta}{cos\,\theta} = \left(\frac{o}{h}\right) \times \left(\frac{h}{a}\right)$$

$$\frac{sin\,\theta}{cos\,\theta} = \left(\frac{o}{\cancel{h}}\right) \times \left(\frac{\cancel{h}}{a}\right)$$

$$\frac{sin\,\theta}{cos\,\theta} = \frac{o}{a}$$

This is similar to $tan\,\theta = \frac{o}{a}$, therefore $tan\,\theta = \frac{sin\,\theta}{cos\,\theta}$

Please know this identity by heart, it will help you solve bigger problems. Let's name it identity 1.

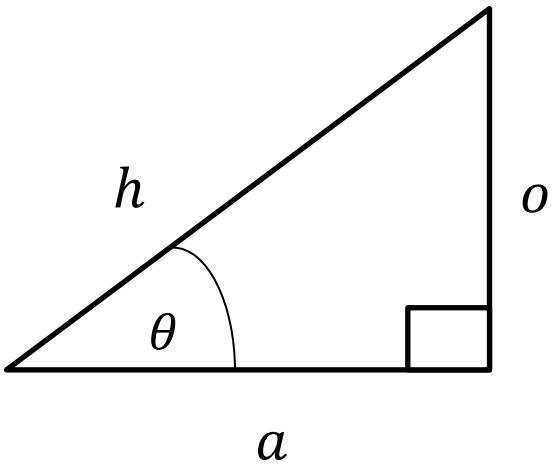
<u>Identity 1</u>

$$\boxed{tan\,\theta = \frac{sin\,\theta}{cos\,\theta}}$$

<u>Identity 2</u>

We now derive identity 2:

Again, we use the right-angled triangle.



$sin\,\theta = \frac{o}{h}$, square both sides.

$$(sin\,\theta)^2 = \left(\frac{o}{h}\right)^2$$

$$sin^2\,\theta = \frac{o^2}{h^2}$$

We do the same with *"cos".*

$cos\,\theta = \frac{a}{h}$, square both sides.

$(cos\,\theta)^2 = \left(\frac{a}{h}\right)^2$

$cos^2\,\theta = \frac{a^2}{h^2}$

So we have $sin^2\,\theta = \frac{o^2}{h^2}$ and $cos^2\,\theta = \frac{a^2}{h^2}$

Add them to together and see what happens.

$= sin^2\,\theta + cos^2\,\theta$

$$= \frac{o^2}{h^2} + \frac{a^2}{h^2}$$

$= \frac{o^2+a^2}{h^2}$

We know that $o^2 + a^2$ equals h^2 by the theorem of Pythagoras.

$$= \frac{h^2}{h^2} = 1$$

Therefore, $sin^2\,\theta + cos^2\,\theta = 1$. Please know this identity by heart, it will help you solve bigger problems.

Identity 2

$$sin^2\,\theta + cos^2\,\theta = 1$$

We have learnt two identities so far.

Identity 1

$$tan\,\theta = \frac{sin\,\theta}{cos\,\theta}$$

Identity 2

$$sin^2\,\theta + cos^2\,\theta = 1$$

Below are questions based on identity 1.

Question

Simplify the following as far as possible.

a) $tan\,\theta \times \frac{1}{sin\,\theta}$

b) $tan\,\theta \times cos\,\theta$

Solution

a) $tan\,\theta \times \frac{1}{sin\,\theta}$, use identity 1 on $tan\,\theta$.

$$= \frac{sin\,\theta}{cos\,\theta} \times \frac{1}{sin\,\theta}$$

$$= \frac{sin\,\theta}{cos\,\theta \times sin\,\theta}$$

$$= \frac{\cancel{sin\,\theta}}{cos\,\theta \times \cancel{sin\,\theta}}$$

$$= \frac{1}{cos\,\theta}$$

b) $tan\,\theta \times cos\,\theta$

Use identity 1 on $tan\,\theta$.

$$= \frac{sin\,\theta}{cos\,\theta} \times cos\,\theta$$

$$= \frac{sin\,\theta}{\cancel{cos\,\theta}} \times \cancel{cos\,\theta}$$

$$= sin\,\theta$$

Such questions may be asked in various other ways.

Question

Prove the following equations.

a) $tan\,\theta \times \frac{1}{sin\,\theta} = \frac{1}{cos\,\theta}$

b) $tan\,\theta \times cos\,\theta = sin\,\theta$

Solution

In questions such as proving equations, you basically choose one of the sides and simplify it until it gives what's on the other side. Moreover, we don't just choose any side, we choose the more complicated one.

a) $tan\,\theta \times \frac{1}{sin\,\theta} = \frac{1}{cos\,\theta}$

Choose the Left-Hand Side (LHS) and simplify it until it gives us what's on the Right-Hand Side (RHS).

We choose it because it likes the more complicated side.

<u>Choose LHS</u>

$tan\,\theta \times \frac{1}{sin\,\theta}$, use identity 1 on $tan\,\theta$

$$= \frac{sin\,\theta}{cos\,\theta} \times \frac{1}{sin\,\theta}$$

$$= \frac{sin\,\theta}{cos\,\theta \times sin\,\theta}$$

$$= \frac{\cancel{sin\,\theta}}{cos\,\theta \times \cancel{sin\,\theta}}$$

$$= \frac{1}{cos\,\theta}$$

, which gives the Right-Hand Side (RHS). So we have proved that $tan\,\theta \times \frac{1}{sin\,\theta} = \frac{1}{cos\,\theta}$

b) $tan\,\theta \times cos\,\theta = sin\,\theta$

Choose the Left-Hand Side (LHS) and simplify it until it gives us what's on the Right-Hand Side (RHS).

We choose it because it likes the more complicated side.

<u>Choose LHS</u>

$= tan\,\theta \times cos\,\theta$, use identity 1 on $tan\,\theta$.

$$= \frac{sin\,\theta}{cos\,\theta} \times cos\,\theta$$

$$= \frac{sin\,\theta}{\cancel{cos\,\theta}} \times \cancel{cos\,\theta}$$

$= sin\,\theta$

, which gives the Right-Hand Side (RHS). So we have proved that $tan\,\theta \times cos\,\theta = sin\,\theta$

Practice more of these questions, this time we include identity 2. Please attempt the questions before you check the answers.

Question

Prove the following equations.

a) $\frac{sin\,\theta(\,sin^2\,\theta + cos^2\,\theta)}{cos\,\theta} = tan\,\theta$

b) $cos\,\theta = cos\,\theta.\ sin^2\,\theta + cos\,\theta\,.\ cos^2\,\theta$

Solution

a) $\frac{sin\,\theta(\,sin^2\,\theta + cos^2\,\theta)}{cos\,\theta} = tan\,\theta$

Choose the Left-Hand Side (LHS), it's more complicated.

$\frac{sin\,\theta(\,sin^2\,\theta + cos^2\,\theta)}{cos\,\theta}$, use identity 2 on $sin^2\,\theta + cos^2\,\theta$

$$= \frac{sin\,\theta(1)}{cos\,\theta}$$

$$= \frac{sin\,\theta}{cos\,\theta}$$

, we then use identity 1 on $\frac{sin\,\theta}{cos\,\theta} = tan\,\theta$, which equals the Right-Hand Side (RHS), we have proved that $\frac{sin\,\theta(\,sin^2\,\theta + cos^2\,\theta)}{cos\,\theta} = tan\,\theta$

b) $cos\,\theta = cos\,\theta.\ sin^2\,\theta + cos\,\theta\,.\ cos^2\,\theta$

Choose the Right-Hand Side (LHS), it's more complicated.

$= cos\,\theta.\ sin^2\,\theta + cos\,\theta\,.\ cos^2\,\theta$

Notice that the RHS is in the form:

$$a\ b + a\ c$$

$$cos\,\theta.\ \overbrace{sin^2\,\theta} + cos\,\theta\,.\ \overbrace{cos^2\,\theta}$$

Factorizing $a\ b + a\ c = a(b + c)$, so we can do the same with the RHS, factor out $cos\,\theta$.

$= cos\,\theta(sin^2\,\theta + cos\,\theta\,.\ cos^2\,\theta)$

$= cos\,\theta(\ sin^2\,\theta +\ cos^2\,\theta)$, use identity 2 on $sin^2\,\theta +\ cos^2\,\theta$

$= cos\,\theta(1)$

$= cos\,\theta$, which equals the Left-Hand Side (LHS), we have proved that $cos\,\theta = cos\,\theta.\ sin^2\,\theta + cos\,\theta\,.\ cos^2\,\theta$

Factorization is a very important skill in solving such equations, you should always be ready to use it.

Question

Prove the following equation.

$cos\,\theta.\ sin^2\,\theta +\ cos^3\,\theta = cos\,\theta$

Solution

$cos\,\theta.\ sin^2\,\theta +\ cos^3\,\theta = cos\,\theta$

Choose the LHS, it's more complicated.

$cos\,\theta.\ sin^2\,\theta +\ cos^3\,\theta$

Notice that $cos^3\,\theta$ is basically $(cos\,\theta)^3$

$(cos\,\theta)^3 = cos\,\theta\,.cos\,\theta\,.cos\,\theta$

Substitute this into the "LHS".

$= cos\,\theta.\ sin^2\,\theta + cos\,\theta\,.cos\,\theta\,.cos\,\theta$

Now factor out $cos\,\theta$.

$= cos\,\theta.\ sin^2\,\theta + cos\,\theta\,.cos\,\theta\,.cos\,\theta$

$= cos\,\theta(\,sin^2\,\theta + cos\,\theta\,.cos\,\theta)$

We know that $cos\,\theta\,.cos\,\theta = (cos\,\theta)^2 = \ cos^2\,\theta$

$= cos\,\theta(\,sin^2\,\theta + \ cos^2\,\theta)$

$= cos\,\theta(1)$

$= cos\,\theta$, which equals the RHS. Proved!

Notice that this question is similar to the previous one, the only difference is that it was presented differently.

$cos\,\theta.\ sin^2\,\theta + \ cos^3\,\theta \qquad vs \qquad cos\,\theta.\ sin^2\,\theta + cos\,\theta\,.\ cos^2\,\theta$

Be on the alert when it comes to terms like $cos^3\,\theta,\ sin^3\,\theta,\ sin^4\,\theta,\ cos^4\,\theta$ and so forth, all you have to do is separate.

For example;

$sin^3\,\theta = sin\,\theta.\ sin^2\,\theta \qquad cos^3\,\theta = cos\,\theta.\ cos^2\,\theta$

$sin^4\,\theta = sin^2\,\theta.\ sin^2\,\theta \qquad cos^4\,\theta = cos^2\,\theta.\ cos^2\,\theta$

Since you will be using the above a lot in many proofs, let's name the set of equations *Tool 1*.

Tool 1

$sin^3\,\theta = sin\,\theta.\ sin^2\,\theta \qquad cos^3\,\theta = cos\,\theta.\ cos^2\,\theta$

$sin^4\,\theta = sin^2\,\theta.\ sin^2\,\theta \qquad cos^4\,\theta = cos^2\,\theta.\ cos^2\,\theta$

Question

$sin\,\theta = \ sin^3\,\theta + sin\,\theta.\ cos^2\,\theta$

Solution

Choose LHS

$= \underbrace{sin^3\,\theta} + sin\,\theta.\ cos^2\,\theta$

Use tool 1 here.

$= \boldsymbol{sin\,\theta\,.sin^2\,\theta} + sin\,\theta.\ cos^2\,\theta$

Notice that this is in the form:

$$a\ b + a\ c$$

$$sin\,\theta.\ \overbrace{sin^2\,\theta} + sin\,\theta\,.\ \overbrace{cos^2\,\theta}$$

Factorizing $a\ b + a\ c = a(b + c)$, so we can do the same with the RHS, factor out $sin\,\theta$.

$= sin\,\theta(sin^2\,\theta + \ cos^2\,\theta)$

$= sin\,\theta(sin^2\,\theta + \ cos^2\,\theta)$, use identity 2 on $sin^2\,\theta + \ cos^2\,\theta$

$= sin\,\theta(1)$

$= sin\,\theta,$ which equals the Left-Hand Side (LHS), we have proved that $sin\,\theta =$
$sin\,\theta.\ sin^2\,\theta + sin\,\theta\,.\ cos^2\,\theta$

Do you see how important tool 1 is? Factorization is also important. Let's create a second tool based on factorization.

Tool 2

$$ab + ac = a(b + c)$$

$$ab + ac + ad = a(b + c + d)$$

Question

Prove

$$\frac{cos^2\,\theta\,.\,sin\,\theta + \ sin^3\,\theta}{sin\,\theta} = 1$$

Solution

LHS

$$= \frac{cos^2\,\theta.sin\,\theta + \underbrace{sin^3\,\theta}_{\text{Use Tool 1}}}{sin\,\theta}$$

$$= \frac{cos^2\,\theta\,.\,sin\,\theta + sin\,\theta\,.\,sin^2\,\theta}{sin\,\theta}$$

$$= \frac{sin\,\theta\,(\,\boldsymbol{cos^2\,\theta} + \ \boldsymbol{sin^2\,\theta})}{sin\,\theta}$$

$$= \frac{sin\,\theta\,(1)}{sin\,\theta}$$

$$= \frac{sin\,\theta}{sin\,\theta} = 1 = RHS$$

Sometimes you will be required to prove equations involving terms like $(sin\,\theta + cos\,\theta)^2$ and $(sin\,\theta - cos\,\theta)^2$. You will be required to expand them. For convenience, we will expand them now and learn their expansions once and for all.

Note that $(sin\,\theta + cos\,\theta)^2$ is in the form $(x + y)^2$. We now simplify it similar to $(x + y)^2$.

$$(x + y)^2$$

$$(x + y)(x + y)$$

$$x^2 + xy + xy + \ y^2$$

$$x^2 + 2xy + \ y^2$$

$$(sin\,\theta + cos\,\theta)^2$$

$$(sin\,\theta + cos\,\theta)(sin\,\theta + cos\,\theta)$$

$$sin^2\,\theta + sin\,\theta\,.\,cos\,\theta + sin\,\theta\,.\,cos\,\theta + \ cos^2\,\theta$$

$$\boldsymbol{sin^2\,\theta} + 2\,sin\,\theta\,.\,cos\,\theta + \ \boldsymbol{cos^2\,\theta}$$

$$\boldsymbol{sin^2\,\theta} + \ \boldsymbol{cos^2\,\theta} + 2\,sin\,\theta\,.\,cos\,\theta$$

$$1 + 2\,sin\,\theta\,.\,cos\,\theta$$

ducatio

Thus we have;

$$(sin\,\theta + cos\,\theta)^2 = 1 + 2\,sin\,\theta\,.\,cos\,\theta$$

We do the same for $(sin\,\theta - cos\,\theta)^2$.

$$(x - y)^2$$

$$(x - y)(x - y)$$

$$x^2 - xy - xy + \ y^2$$

$$x^2 - 2xy + \ y^2$$

$$(sin\,\theta - cos\,\theta)^2$$

$$(sin\,\theta - cos\,\theta)(sin\,\theta - cos\,\theta)$$

$$sin^2\,\theta - sin\,\theta\,.\,cos\,\theta - sin\,\theta\,.\,cos\,\theta + \ cos^2\,\theta$$

$$\boldsymbol{sin^2\,\theta} - 2\,sin\,\theta\,.\,cos\,\theta + \ \boldsymbol{cos^2\,\theta}$$

$$\boldsymbol{sin^2\,\theta} + \ \boldsymbol{cos^2\,\theta} - 2\,sin\,\theta\,.\,cos\,\theta$$

$$1 - 2\,sin\,\theta\,.\,cos\,\theta$$

We put that together and call it tool 3.

Tool 3

$$(sin\,\theta + cos\,\theta)^2 = 1 + 2sin\,\theta.\,cos\,\theta$$

$$(sin\,\theta - cos\,\theta)^2 = 1 - 2sin\,\theta.\,cos\,\theta$$

Question

Prove

$$\frac{(sin\,\theta + cos\,\theta)^2 - 1}{2cos^2\,\theta} = tan\,\theta$$

Solution

LHS

$$= \frac{(sin\,\theta + cos\,\theta)^2 - 1}{2cos^2\,\theta}$$

Expand using tool 3.

$$= \frac{1 + 2\,sin\,\theta\,.\,cos\,\theta - 1}{2cos^2\,\theta}$$

$$= \frac{2\,sin\,\theta\,.\,cos\,\theta}{2cos^2\,\theta}$$

$$= \frac{2\,sin\,\theta\,.\,cos\,\theta}{2cos\,\theta\,.\,cos\,\theta}$$

$$= \frac{\cancel{2}\,sin\,\theta\,.\,\cancel{cos\,\theta}}{\cancel{2}cos\,\theta\,.\,\cancel{cos\,\theta}}$$

$$= \frac{sin\,\theta}{cos\,\theta}$$ The use the identity 1

$= tan\,\theta$, which equals the RHS.

Question

Prove:

$\frac{1}{tan\,\theta} = \frac{cos\,\theta}{sin\,\theta}$

Solution

$$\frac{1}{tan\,\theta} = \frac{cos\,\theta}{sin\,\theta}$$

LHS

$$= \frac{1}{tan\,\theta}$$

$$= \frac{1}{\left(\frac{sin\,\theta}{cos\,\theta}\right)}$$

$$= 1 \div \left(\frac{sin\,\theta}{cos\,\theta}\right)$$

$= 1 \div \frac{sin\,\theta}{cos\,\theta}$

$= 1 \times \frac{cos\,\theta}{sin\,\theta}$

$= \frac{cos\,\theta}{sin\,\theta} = RHS$, thus we have proved that $\frac{1}{tan\,\theta} = \frac{cos\,\theta}{sin\,\theta}$

This relationship appears a lot in such problems, so it is important to know it by heart.

We will name it Identity 3

Identity 3

$$\frac{1}{tan\,\theta} = \frac{cos\,\theta}{sin\,\theta}$$

To make things easier, let's make a summary of Identities and Tools so far.

Identities

Identity 1

$$tan\,\theta = \frac{sin\,\theta}{cos\,\theta}$$

Identity 2

$$sin^2\,\theta + cos^2\,\theta = 1$$

Identity 3

$$\frac{1}{tan\,\theta} = \frac{cos\,\theta}{sin\,\theta}$$

Tools

Tool 1

$$sin^3\,\theta = sin\,\theta.\ sin^2\,\theta \qquad cos^3\,\theta = cos\,\theta.\ cos^2\,\theta$$

$$sin^4\,\theta = sin^2\,\theta.\ sin^2\,\theta \qquad cos^4\,\theta = cos^2\,\theta.\ cos^2\,\theta$$

Tool 2

$$ab + ac = a(b + c)$$

$$ab + ac + ad = a(b + c + d)$$

Tool 3

$$(sin\,\theta + cos\,\theta)^2 = 1 + 2sin\,\theta.cos\,\theta$$

$$(sin\,\theta - cos\,\theta)^2 = 1 - 2sin\,\theta.cos\,\theta$$

Example

Prove:

$\frac{cos^2\,\theta}{1-sin\,\theta} = 1 + sin\,\theta$

LHS

$$\frac{cos^2\,\theta}{1 - sin\,\theta}$$

This one is a unique situation because of the $1 - sin\,\theta$ as a **denominator**.

Here you just have to multiply both the numerator and denominator by $\mathbf{1 + \boldsymbol{sin\,\theta}}$, and then simplify.

$$= \frac{cos^2\,\theta}{1 - sin\,\theta} \times \frac{\mathbf{1 + sin\,\theta}}{\mathbf{1 + sin\,\theta}}$$

Notice that multiplying by $\frac{\mathbf{1+sin\,\theta}}{\mathbf{1+sin\,\theta}}$ is the same a multiplying by $\frac{1}{1} = 1$, because $\frac{\mathbf{1+sin\,\theta}}{\mathbf{1+sin\,\theta}} = 1$, so it **doesn't** change anything.

$$= \frac{cos^2\,\theta\,(\mathbf{1 + sin\,\theta})}{(1 - sin\,\theta)(\mathbf{1 + sin\,\theta})}$$

$$= \frac{cos^2\,\theta(\mathbf{1+sin\,\theta})}{(1-sin\,\theta)(\mathbf{1+sin\,\theta})}$$

$$= \frac{cos^2\,\theta\,(\mathbf{1 + sin\,\theta})}{1 + sin\,\theta - sin\,\theta - \ sin^2\,\theta}$$

Don't rush to simplify the numerator.

$$= \frac{cos^2\,\theta\,(1 + sin\,\theta)}{1 - \ sin^2\,\theta}$$

Note that we can rearrange identity 2: $sin^2\,\theta + cos^2\,\theta = 1$ in any way we like. For this problem, we would need to rearrange identity 2 to replace our denominator. Take $sin^2\,\theta$ to the right-hand side of identity 2: $\boldsymbol{cos^2\,\theta} = 1 - \ sin^2\,\theta$. We can substitute this into our denominator above.

$$= \frac{cos^2\,\theta\,(1 + sin\,\theta)}{cos^2\,\theta}$$

$$= \frac{\cancel{cos^2\,\theta}\,(1 + sin\,\theta)}{\cancel{cos^2\,\theta}}$$

$= 1 + sin\,\theta$, which equals the RHS

The multiplying- trick we just used has proven to be quite helpful in proving "bigger" equations.

We had $1 - sin\,\theta$ on the denominator, so we multiplied everything by $\frac{1+sin\,\theta}{1+sin\,\theta}$, so we simply change the signs.

When we have $1 + sin\,\theta$ on the denominator, we multiply by $\frac{1-sin\,\theta}{1-sin\,\theta}$

For a denominator of $1 - cos\,\theta$, we multiply by $\frac{1+cos\,\theta}{1+cos\,\theta}$

For a denominator of $1 + cos\,\theta$, we multiply by $\frac{1-cos\,\theta}{1-cos\,\theta}$

Let's name this set tool 4; we will be using this a lot too.

Tool 4

$$1 - sin\,\theta \longrightarrow \frac{1+sin\,\theta}{1+sin\,\theta} \qquad 1 - cos\,\theta \longrightarrow \frac{1+cos\,\theta}{1+cos\,\theta}$$

$$1 + sin\,\theta \longrightarrow \frac{1-sin\,\theta}{1-sin\,\theta} \qquad 1 + cos\,\theta \longrightarrow \frac{1-cos\,\theta}{1-cos\,\theta}$$

You don't always have to use Tool 4, if any of the other tools or identities work out, use them instead.

Playing with identity 2 also helps. From $sin^2\,\theta + cos^2\,\theta = 1$, you can rearrange to $cos^2\,\theta = 1 - sin^2\,\theta$ or to $sin^2\,\theta = 1 - cos^2\,\theta$. **This is a very important "tool" as well.**

Example

Prove.

$$\frac{1}{1 + cos\,\theta} = \frac{1 - cos\,\theta}{sin^2\,\theta}$$

Choose any one of the sides, they are both look complicated.

$$\frac{1}{1+cos\,\theta}$$

$$= \frac{1}{1 + cos\,\theta} \times \frac{1 - cos\,\theta}{1 - cos\,\theta}$$

Since we have $1 + cos\,\theta$ on the denominator, we multiply by $\frac{1-cos\,\theta}{1-cos\,\theta}$

$$= \frac{1 - cos\,\theta}{(1 + cos\,\theta)(1 - cos\,\theta)}$$

$$= \frac{1 - cos\,\theta}{1 - cos\,\theta + cos\,\theta - cos^2\,\theta}$$

$$= \frac{1-\cos\theta}{\mathbf{1-\ \cos^2\theta}}$$

Use a rearranged identity 2;
$\sin^2\theta = \mathbf{1-\ \cos^2\theta}$

$$= \frac{1-\cos\theta}{\sin^2\theta}$$

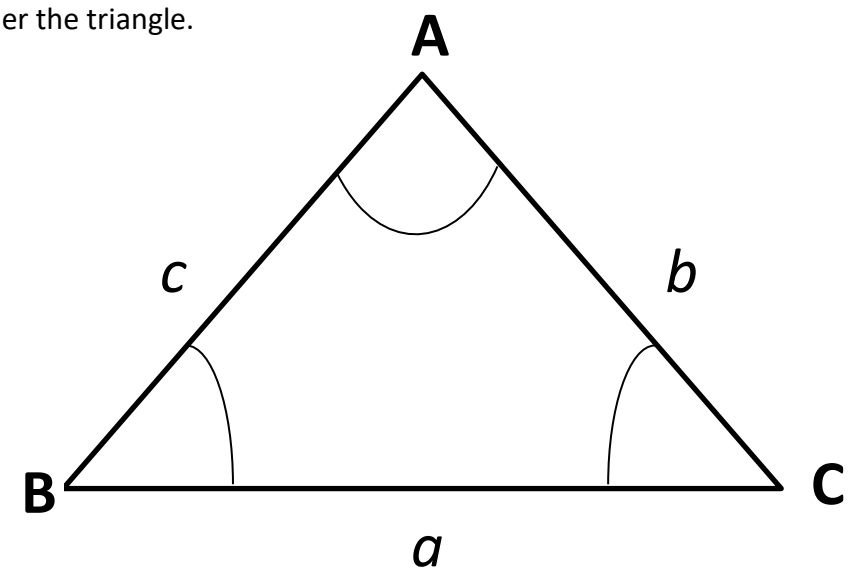
, which equals the RHS.

Sine Rule, Cosine Rule and Area Rule

In this section, we will learn three important formulae; the Sine rule, Cosine rule and Area rule.

Sine Rule

Consider the triangle.



The Sine Rule entails that;

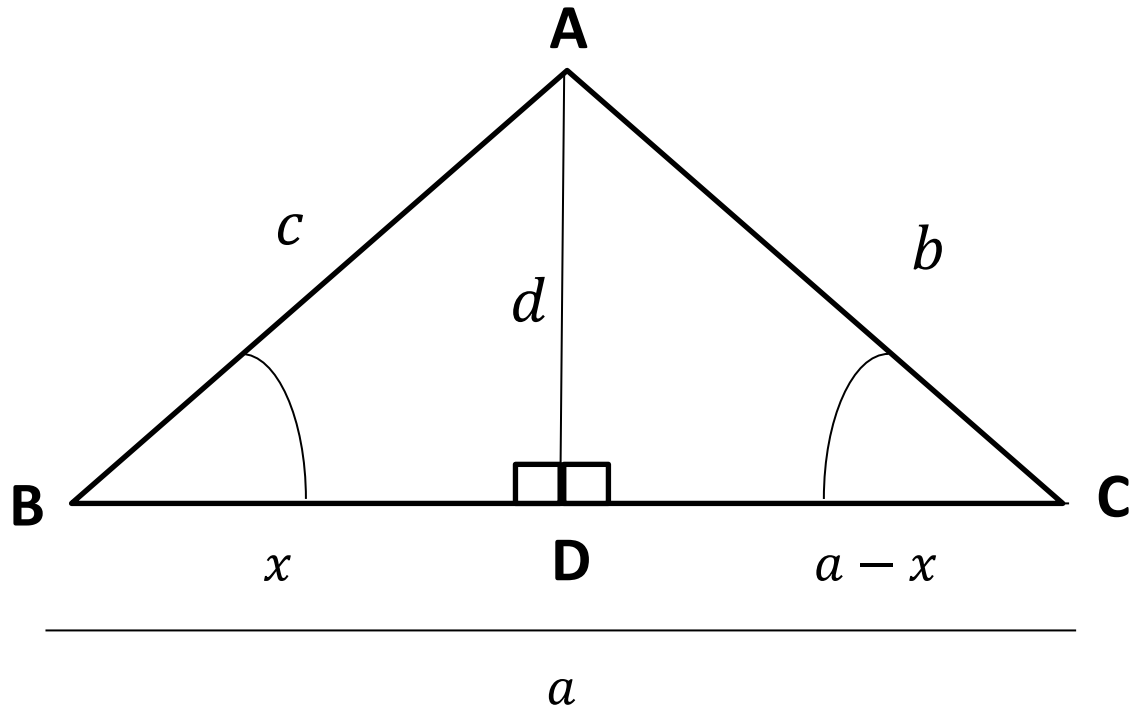
$$\frac{\sin\hat{A}}{a} = \frac{\sin\hat{B}}{b} = \frac{\sin\hat{C}}{c}$$

or

$$\frac{a}{\sin\hat{A}} = \frac{b}{\sin\hat{B}} = \frac{c}{\sin\hat{C}}$$

Note that $\hat{A}$ is the angled at *A*, $\hat{B}$ is the angle at *B* and $\hat{C}$ is the angle at *C*.

Now let's work out how mathematicians found the formulas above.



Carefully read the labels of all the sides and angles, especially side BC. BC is the length "a".

On triangle ABD:

$$\sin \hat{B} = \frac{o}{h}$$

$$\sin \hat{B} = \frac{d}{c}$$

On triangle ACD:

$$\sin \hat{C} = \frac{o}{h}$$

$$\sin \hat{C} = \frac{d}{b}$$

Let's make d be on its own on one side for both

$$c.\sin \hat{B} = d$$

$$b.\sin \hat{C} = d$$

$$d = \underbrace{c.\sin \hat{B}}$$

$$d = \underbrace{b.\sin \hat{C}}$$

$$c.\sin \hat{B} = b.\sin \hat{C}$$

Call this equation 1

Let's divide both sides by c

$$\frac{c.\sin \hat{B}}{c} = \frac{b.\sin \hat{C}}{c}$$

$$\frac{\cancel{c}.\sin \hat{B}}{\cancel{c}} = \frac{b.\sin \hat{C}}{c}$$

$$\sin \hat{B} = \frac{b.\sin \hat{C}}{c}$$

Now let's divide both sides by b;

$$\frac{\sin \hat{B}}{b} = \frac{b.\sin \hat{C}}{b.c}$$

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$$\frac{\sin\hat{B}}{b} = \frac{\cancel{b}.\sin\hat{C}}{\cancel{b}.c}$$

$$\frac{\sin\hat{B}}{b} = \frac{b.\sin\hat{C}}{b.c}$$

$$\frac{\sin\hat{B}}{b} = \frac{\sin\hat{C}}{c}$$

It can also be shown that; $\frac{\sin\hat{A}}{a} = \frac{\sin\hat{B}}{b} = \frac{\sin\hat{C}}{c}$

Again, it can also be proven that :

$$\frac{a}{\sin\hat{A}} = \frac{b}{\sin\hat{B}} = \frac{c}{\sin\hat{C}}$$

$$c.\sin\hat{B} = b.\sin\hat{C}$$

We use equation 1 again.

Let's divide both sides by $\sin\hat{C}$

$$\frac{c.\sin\hat{B}}{\sin\hat{C}} = \frac{b.\sin\hat{C}}{\sin\hat{C}}$$

$$\frac{c.\sin\hat{B}}{\sin\hat{C}} = \frac{b.\sin\hat{C}}{\sin\hat{C}}$$

$$\frac{c.\sin\hat{B}}{\sin\hat{C}} = b$$

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Now let's divide both sides by $sin\,\hat{B}$;

$$\frac{c.\,sin\,\hat{B}}{sin\,\hat{C}.\,sin\,\hat{B}} = \frac{b}{sin\,\hat{B}}$$

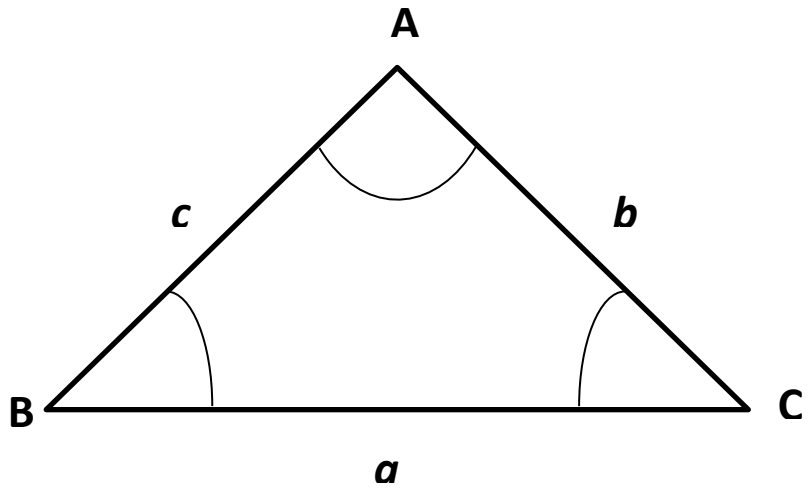
$$\frac{c.\,\cancel{sin\,\hat{B}}}{sin\,\hat{C}\,\cancel{sin\,\hat{B}}} = \frac{b}{sin\,\hat{B}}$$

$$\frac{c}{sin\,\hat{C}} = \frac{b}{sin\,\hat{B}}$$

Hence, it can also be proven that;

$$\frac{a}{sin\,\hat{A}} = \frac{c}{sin\,\hat{C}} = \frac{b}{sin\,\hat{B}}$$

That's our ***Sine Rule*** in a nutshell.



$$\frac{sin\,\hat{A}}{a} = \frac{sin\,\hat{B}}{b} = \frac{sin\,\hat{C}}{c}$$

$$\frac{a}{sin\,\hat{A}} = \frac{b}{sin\,\hat{B}} = \frac{c}{sin\,\hat{C}.}$$

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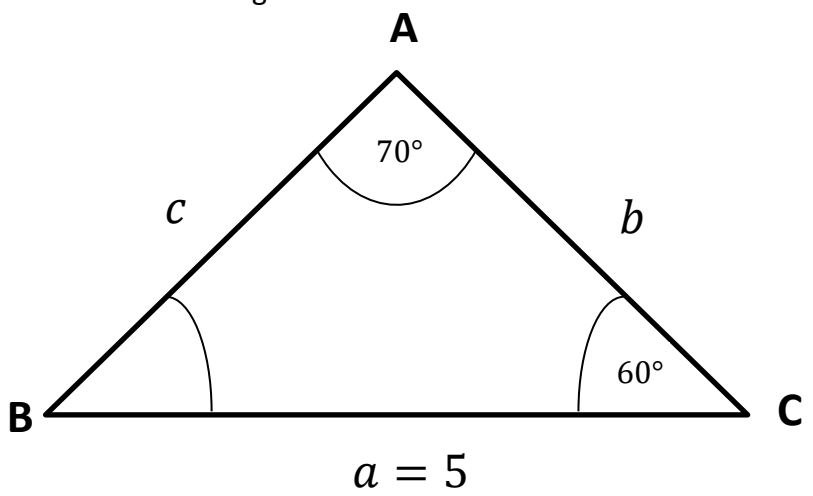
Question

Consider the triangle ABC.



Use the sine rule to find the length of side c.

Solution

It is important to know when to apply the Sine Rule. We apply the sine rule if any of the situations below is true;

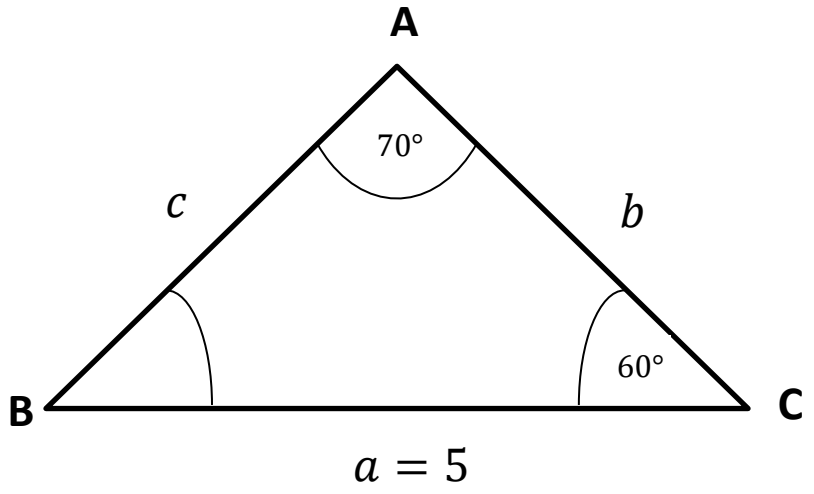
a) If you're given *2 angles* and *1 side* corresponding to one of the given angles.

Or

b) If you are given *2 sides* and *1 angle* corresponding to one of the given sides.

Please make sure you fully understand what these two conditions mean.

Now let's check if the information on the triangle above satisfies one of the conditions.



We are given *2 angles,* $(\hat{A} = 70°, \hat{C} = 60°)$ and *1 side* corresponding to one of the given angles, $a = 5$, corresponding to angle $\hat{A}$. So we can apply the sine rule because the first condition is met.

$$\frac{a}{sin\,\hat{A}} = \frac{c}{sin\,\hat{C}}.$$

Substitute

$$\frac{5}{sin\,70°} = \frac{c}{sin\,60°}$$

Cross multiply.

$$5.sin\,60° = c.sin\,70°$$

Divide both sides by $sin\,70°$ so that you remain with c on its own.

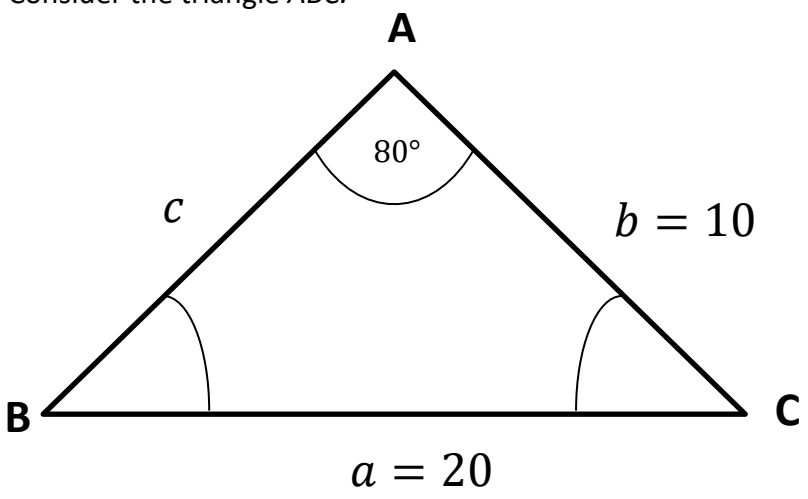
$$\frac{5\,.sin\,60°}{sin\,70°} = c$$

$$c = 4.61$$

Note that we chose the second sine rule; we do so when we calculate a side for convenience. They give the same answer.

Question

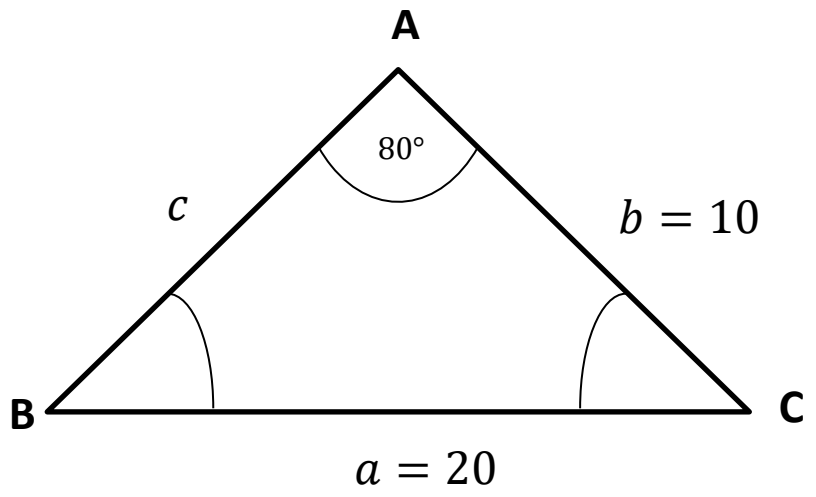
Consider the triangle *ABC.*



Use the sine rule to find the length of side c.

Solution

Now let's check if the information on the triangle above satisfies one of the conditions.



We are given *2 sides,* $(a = 20, b = 10)$ and *1 angle* corresponding to one of the given sides, $\hat{A} = 80°$, corresponding to side a. So we can apply the sine rule because the first condition is met.

$$\frac{sin\,\hat{A}}{a} = \frac{sin\,\hat{B}}{b}$$

We choose to use the first formula because it's more user friendly when we are looking for an angle.

Substitute.

$$\frac{sin\,80°}{20} = \frac{sin\,\hat{B}}{10}$$

Cross multiply.

$$10.\,sin\,80° = 20.\,sin\,\hat{B}$$

Divide both sides by 20 because we want to have $sin\,\hat{B}$ left on its own.

$$\frac{10.\,sin\,80°}{20} = sin\,\hat{B}$$

$$sin\,\hat{B} = \frac{10.\,sin\,80°}{20}$$

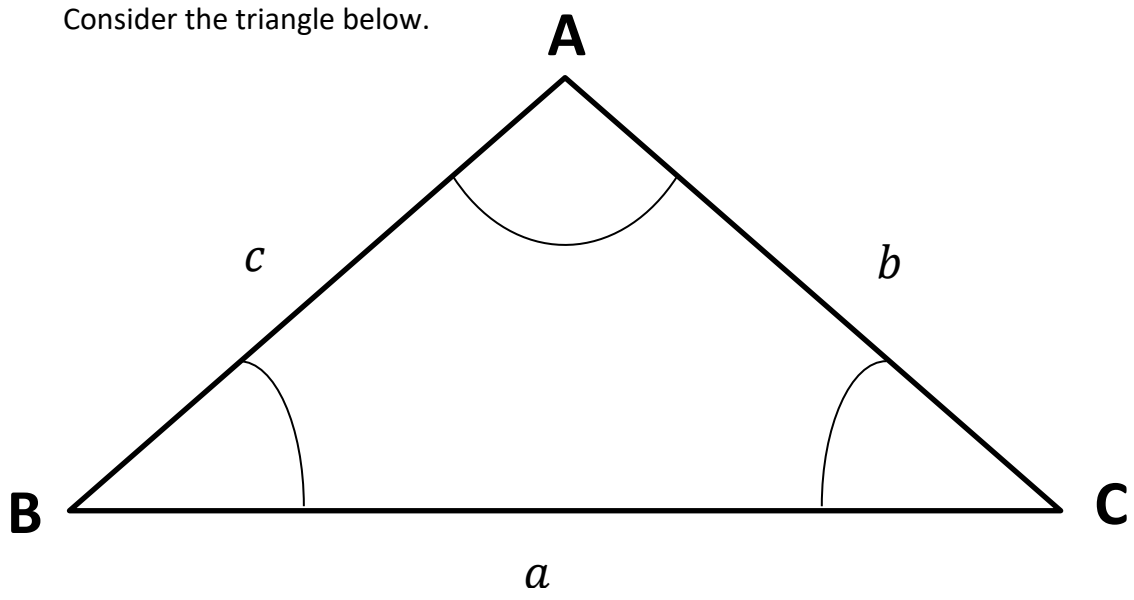
Take the inverse.

$\hat{B} = sin^{-1}\left(\frac{10.\,sin\,80°}{20}\right)$

$\hat{B} = 29.5°$

Area Rule

Consider the triangle below.



The area of triangle *ABC* can be found in following 3 ways.

$$Area\ triangle\ ABC = \frac{1}{2}ab\,sin\,\hat{C}$$

$$Area\ triangle\ ABC = \frac{1}{2}ac\,sin\,\hat{B}$$

$$Area\ triangle\ ABC = \frac{1}{2}bc\,sin\,\hat{A}$$

Study these formulas correctly to see how they relate to one another.

Now let's work out how mathematicians found these formulas.

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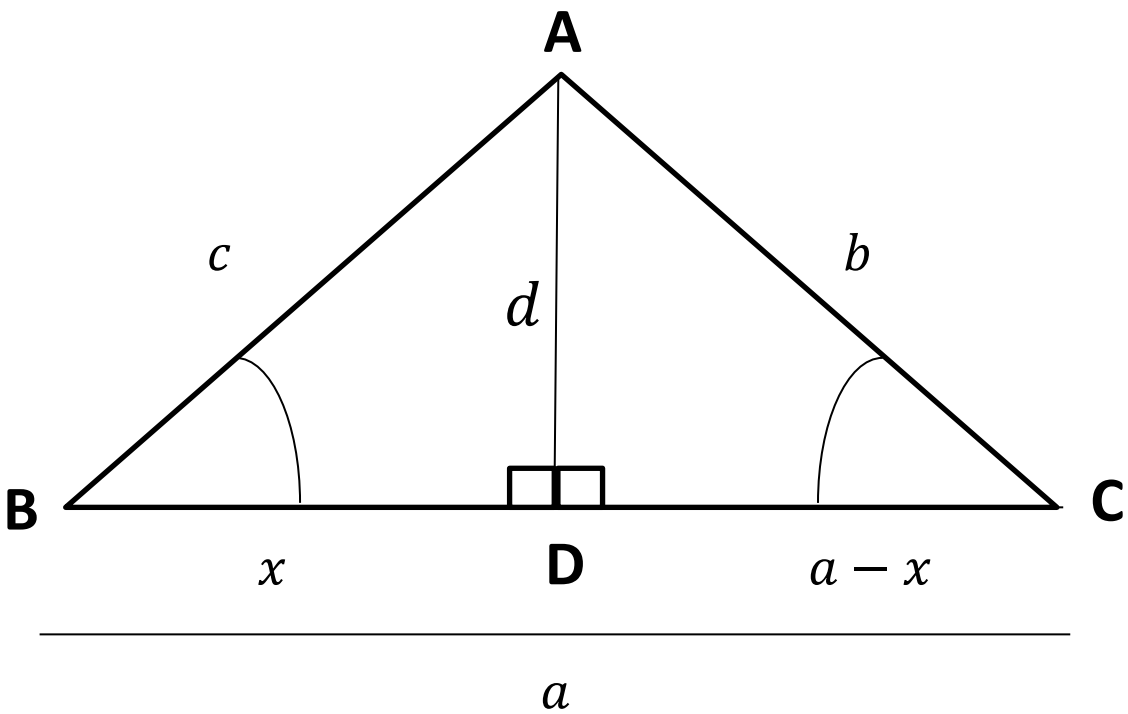
Consider the triangle below.



On triangle *ABD*;

$$Area\ of\ \ triangle\ ABD = \frac{1}{2}bh$$

$$Area\ of\ triangle\ ABD = \frac{1}{2}x.\boldsymbol{d}$$

On triangle *ACD*;

$$Area\ of\ ACD = \frac{1}{2}bh$$

$$Area\ of\ triangle\ ACD = \frac{1}{2}(a-x).\boldsymbol{d}$$

$$Area\ of\ triangle\ ACD = \frac{1}{2}(a\boldsymbol{d} - x\boldsymbol{d})$$

We are actually interested in finding the area of the whole triangle, that is, triangle *ABC*. We know that the area of triangle *ABD* plus area of triangle *ACD* gives the area of triangle *ABC*.

$$Area\ of\ triangle\ ABC = Area\ of\ triangle\ ABD + Area\ of\ triangle\ ACD$$

$$= \frac{1}{2}x.\boldsymbol{d} + \frac{1}{2}(a\boldsymbol{d} - x\boldsymbol{d})$$

Factorize $\frac{1}{2}$

$$= \frac{1}{2}(x\boldsymbol{d} + (a\boldsymbol{d} - x\boldsymbol{d}))$$

$$= \frac{1}{2}(x\boldsymbol{d} - x\boldsymbol{d} + a\boldsymbol{d}))$$

$Area\ of\ triangle\ ABC = \frac{1}{2}a\boldsymbol{d}$ ← Let's call this equation 1.

On triangle ABD,

$sin\,\hat{B} = \frac{o}{h}$

$sin\,\hat{B} = \frac{\boldsymbol{d}}{c}$

Make $\boldsymbol{d}$ be on its own by cross multiplication.

$c.sin\,\hat{B} = \boldsymbol{d}$

$\boldsymbol{d} = c.sin\,\hat{B}$

Substitute this value of $\boldsymbol{d}$ into equation 1.

$Area\ of\ triangle\ ABC = \frac{1}{2}a\boldsymbol{d}$

$Area\ of\ triangle\ ABC = \frac{1}{2}a\,c.sin\,\hat{B}$

Proved!

Let's find a second formula for area. We go back to equation 1.

$Area\ of\ triangle\ ABC = \frac{1}{2}a\boldsymbol{d}$

On triangle ACD,

$sin\,\hat{C} = \frac{o}{h}$

$sin\,\hat{C} = \frac{\boldsymbol{d}}{b}$

Make $\boldsymbol{d}$ be on its own by cross multiplication.

$b.\sin\hat{C} = \boldsymbol{d}$

$\boldsymbol{d} = b.\sin\hat{C}$

Substitute this value of $\boldsymbol{d}$ into equation 1.

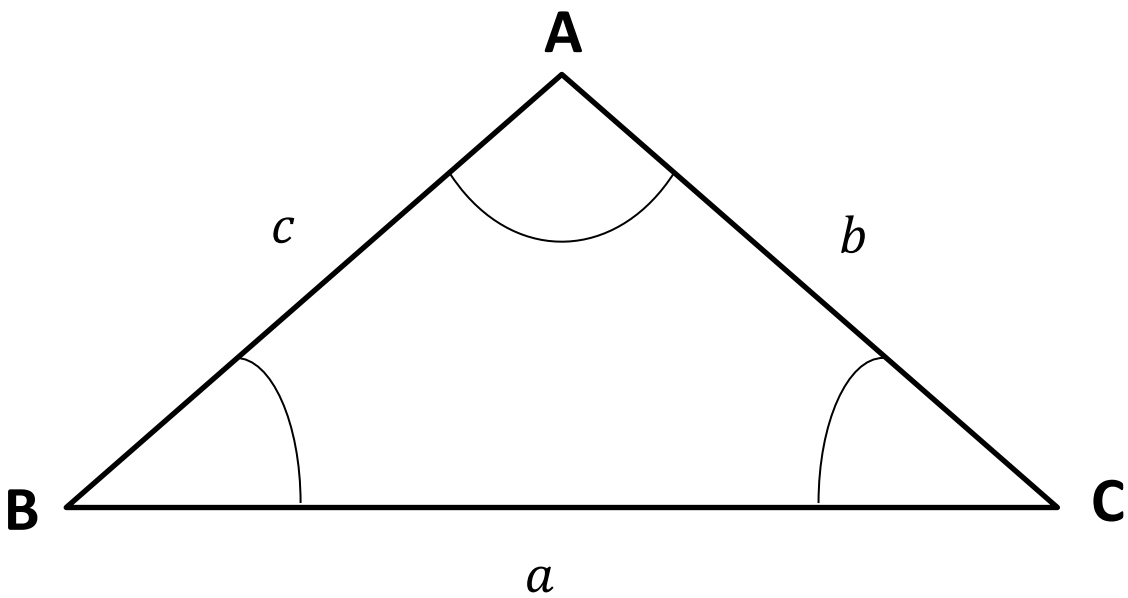
$Area\ of\ triangle\ ABC = \frac{1}{2}a\boldsymbol{d}$

$Area\ of\ triangle\ ABC = \frac{1}{2}a\,b.\sin\hat{C}$

Proved!

The third formula can be proved in a similar way. That's the *Area Rule.*

Area Rule



$$Area\ triangle\ ABC = \frac{1}{2}ab\sin\hat{C}$$

$$Area\ triangle\ ABC = \frac{1}{2}ac\sin\hat{B}$$

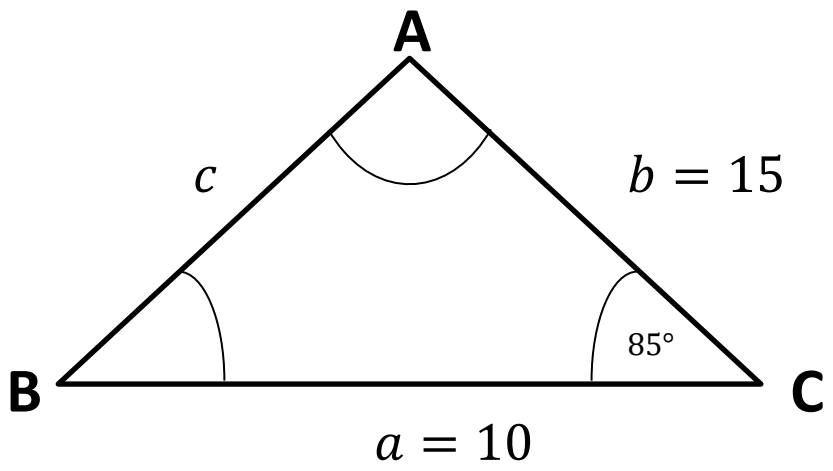
$$Area\ triangle\ ABC = \frac{1}{2}bc\sin\hat{A}$$

Condition on which the area rule is applied

If you're given *2 sides* of a triangle and *1 angle* which ***doesn't*** correspond with the given sides, then you have all the tools to find the area of a triangle using the area rule.

Question

Calculate the area of the triangle below.



Solution

We are given *2 sides*, $a = 10, b = 15$, and *1 angle* $(85°)$ which doesn't correspond to any of the 2 given sides. So we can apply the area rule based on what we are given. We apply the one involving a and b.

$$Area\ triangle\ ABC = \frac{1}{2} ab\ sin\ \hat{C}$$

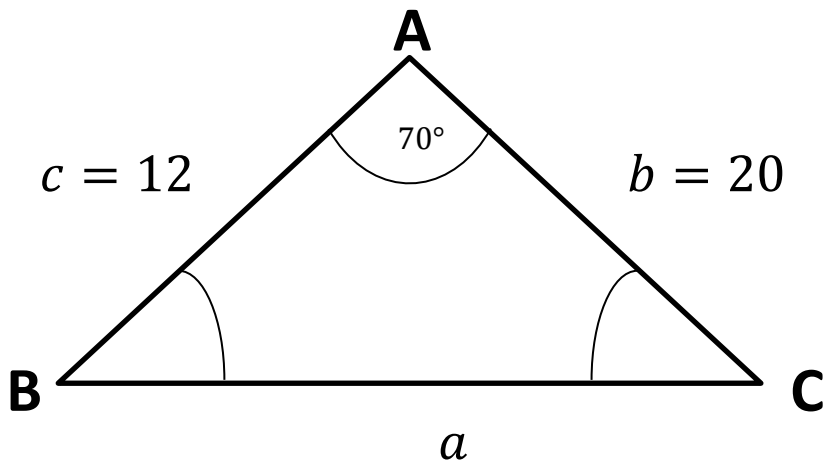
$$= \frac{1}{2} 10.15.\ sin\ 85°$$

$$= 74.71$$

Thus the area of the triangle is *74.71* square units.

Question

Calculate the area of the triangle below.



Solution

We are given *2 sides*, $b = 20, c = 12$, and *1 angle* $(70°)$ which doesn't correspond to any of the 2 given sides. So we can apply the area rule based on what we are given. We apply the one involving b and c.

$$Area\ triangle\ ABC = \frac{1}{2} bc\ sin\ \hat{A}$$

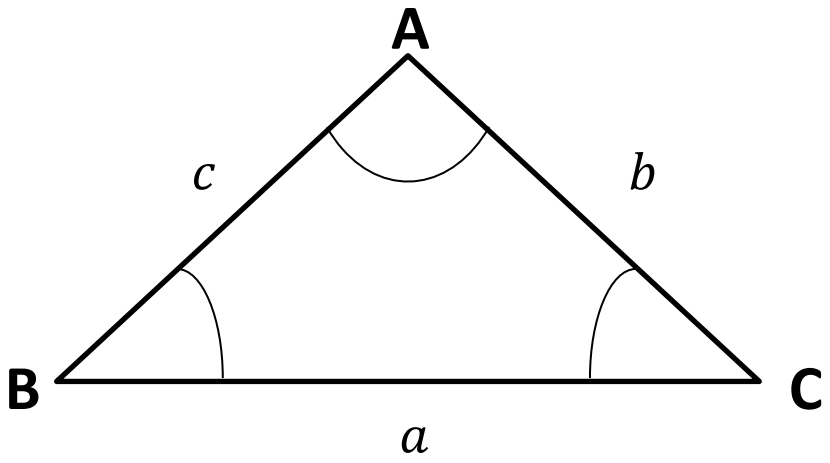
$$= \frac{1}{2} 20.12\ sin\ 70°$$

$$= 112.76$$

Thus the area of the triangle is *112.76* square units.

Cosine Rule

Consider the triangle below.



The Cosine rule states that;

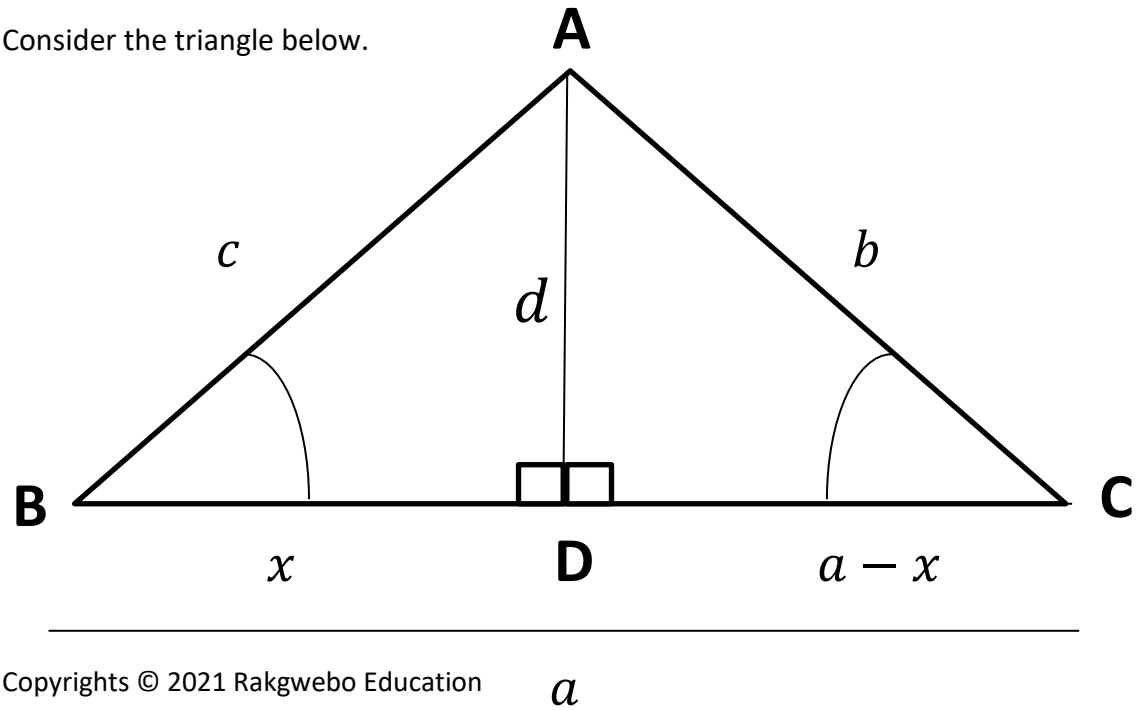
$a^2 = b^2 + c^2 - 2bc.\cos\hat{A}$

$b^2 = a^2 + c^2 - 2ac.\cos\hat{B}$

$c^2 = a^2 + b^2 - 2ab.\cos\hat{C}$

Now let's work out how mathematicians found the formulas.

Consider the triangle below.



On triangle *ABD, the Theorem of Pythagoras gives*;

$c^2 = x^2 + d^2$

c is the hypotenuse

x is the adjacent

d is the opposite

On triangle *ACD, the Theorem of Pythagoras gives*;

$b^2 = (a - x)^2 + d^2$

b is the hypotenuse

$a - x$ is the adjacent

d is the opposite

$$c^2 = x^2 + d^2$$

Move x^2 to the left-hand side.

$$c^2 - x^2 = d^2$$

$$b^2 = (a - x)^2 + d^2$$

Move $(a - x)^2$ to the left-hand side.

$$b^2 - (a - x)^2 = d^2$$

Since both equal d^2

$$c^2 - x^2 = b^2 - (a - x)^2$$

$$c^2 - x^2 = b^2 - (a - x)(a - x)$$

$$c^2 - x^2 = b^2 - (a^2 - ax - ax + x^2)$$

$$c^2 - x^2 = b^2 - a^2 + ax + ax - x^2$$

$$c^2 - \boldsymbol{x^2} = b^2 - a^2 + 2ax - \boldsymbol{x^2}$$

$$c^2 - \boldsymbol{x^2} + \boldsymbol{x^2} = b^2 - a^2 + 2ax$$

$$c^2 = b^2 - a^2 + 2ax$$

Make b^2 to be on its own.

$$c^2 + a^2 - 2ax = b^2$$

$$b^2 = a^2 + c^2 - 2ax$$

Equation 2

We now have to substitute the value of the x above. We get the value from triangle *ABD*.

$$cos\,\hat{B} = \frac{adjacent}{hypotenuse}$$

$$cos\,\hat{B} = \frac{x}{\boldsymbol{c}}$$

Cross multiply to have x on its own.

$$\boldsymbol{c}.cos\,\hat{B} = x$$

$$x = \boldsymbol{c}.cos\,\hat{B}$$

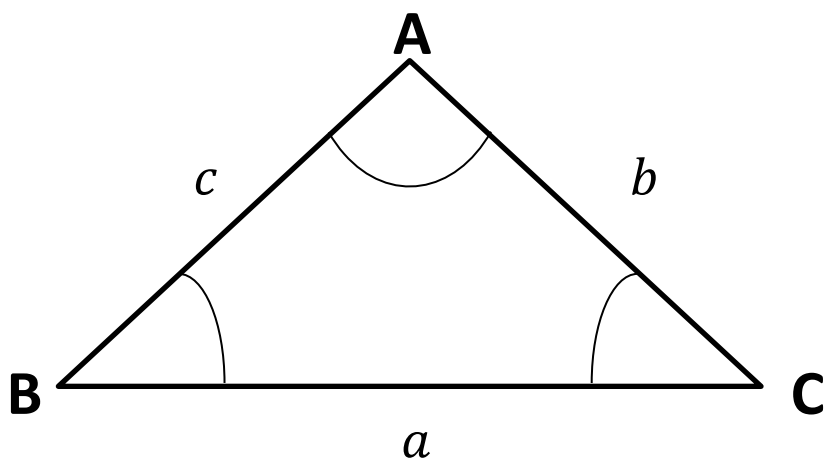
You may now substitute this into equation 2.

$$b^2 = a^2 + c^2 - 2a.\boldsymbol{c}.cos\,\hat{B}$$

$$b^2 = a^2 + c^2 - 2ac.cos\,\hat{B}$$

The other two formulas can be derived in a similar way.

Cosine Rule



The Cosine rule states that;

$$a^2 = b^2 + c^2 - 2bc.cos\,\hat{A}$$

$$b^2 = a^2 + c^2 - 2ac.cos\,\hat{B}$$

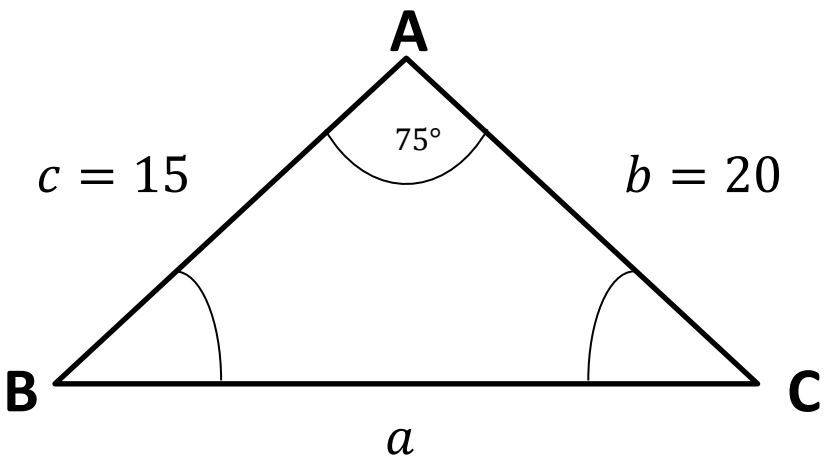
$$c^2 = a^2 + b^2 - 2ab.cos\,\hat{C}$$

Conditions on which the Cosine rule is applied

If you're given *2 sides* of a triangle and *1 angle* which ***doesn't*** correspond with the given sides, then you have all the tools to apply rule.

Question

Consider the triangle below.



What is the length of "side a"?

Solution

Apply the Cosine Rule

$a^2 = b^2 + c^2 - 2bc.cos\,\hat{A}$

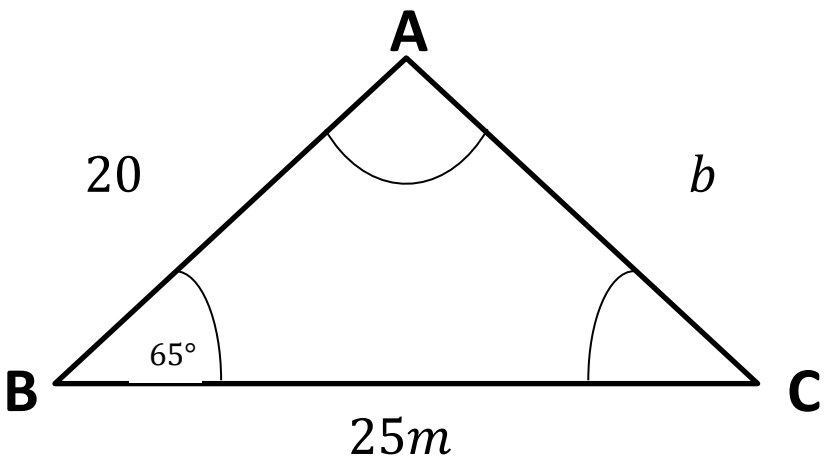
$a^2 = 20^2 + 15^2 - 2.20.15.cos\,75°$

$a = \sqrt{20^2 + 15^2 - 2.20.15.cos\,75°}$

$a = 21.67$

Question

Consider the triangle below.



What is the length of "side b"?

Solution

Apply the Cosine Rule

$b^2 = a^2 + c^2 - 2ac.cos\,\hat{B}$

$a^2 = 25^2 + 20^2 - 2.25.20.cos\,65°$

$a = \sqrt{25^2 + 20^2 - 2.25.20.cos\,65°}$

$a = 24.54m$

Analytical Geometry

Analytical geometry

Finding the equation of a straight line

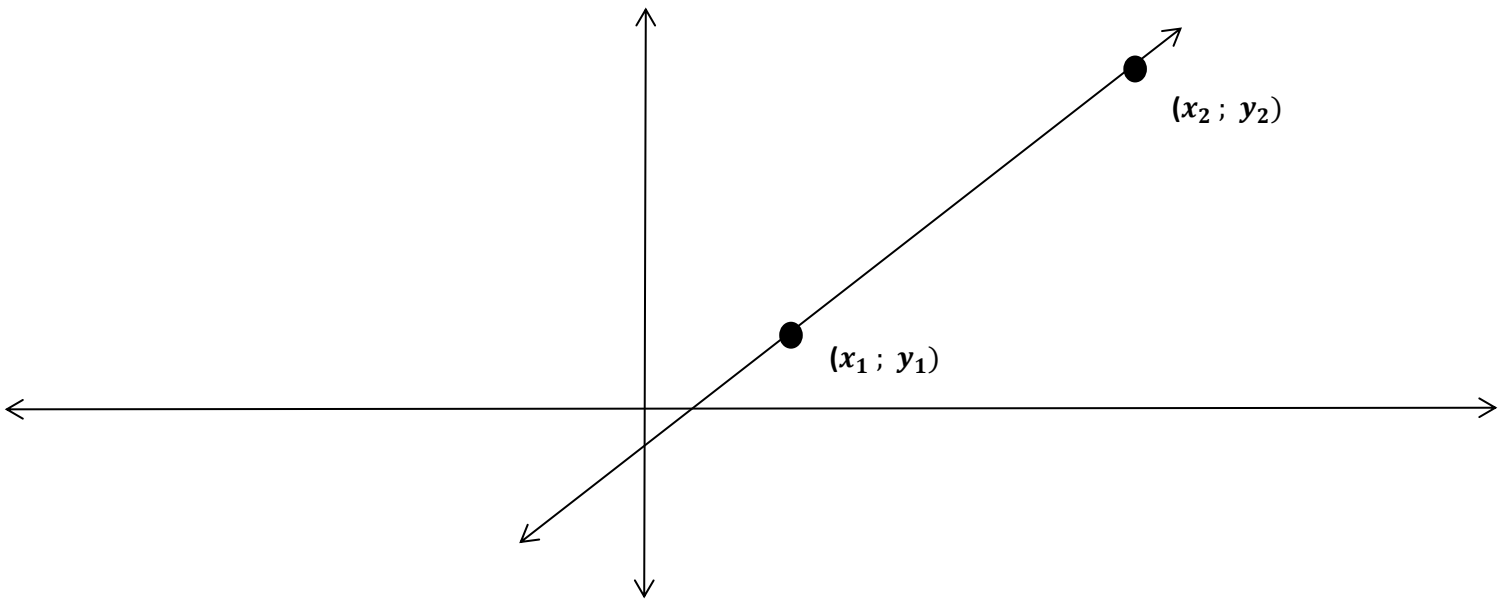
Finding the equation of a line is one of the easiest things to do in mathematics. The general form of the equation of a line is:

y = mx + c , where the ***m*** represents the gradient of the line, while ***c*** represents the y-cut. So when asked to find the equation of a line, your main aim is to calculate the values of **m** and **c**. For example, if you calculate the gradient to be ***m = 2*** and calculate the y-cut to be ***c = 1***, then you can substitute these values into the general form ***y = mx + c*** to get ***y = 2x + 1***. That's the whole aim.

If you are not given the m and c in a question, you would have to use some other methods to find the equation of a particular line. There are 3 basic methods to do so, which one to use depends on what you are given in a question.

Method 1

You are given two points that lie on the straight line.



In this instance, you first calculate the gradient using the gradient formula $\boldsymbol{m} = \dfrac{y_2 - y_1}{x_2 - x_1}$

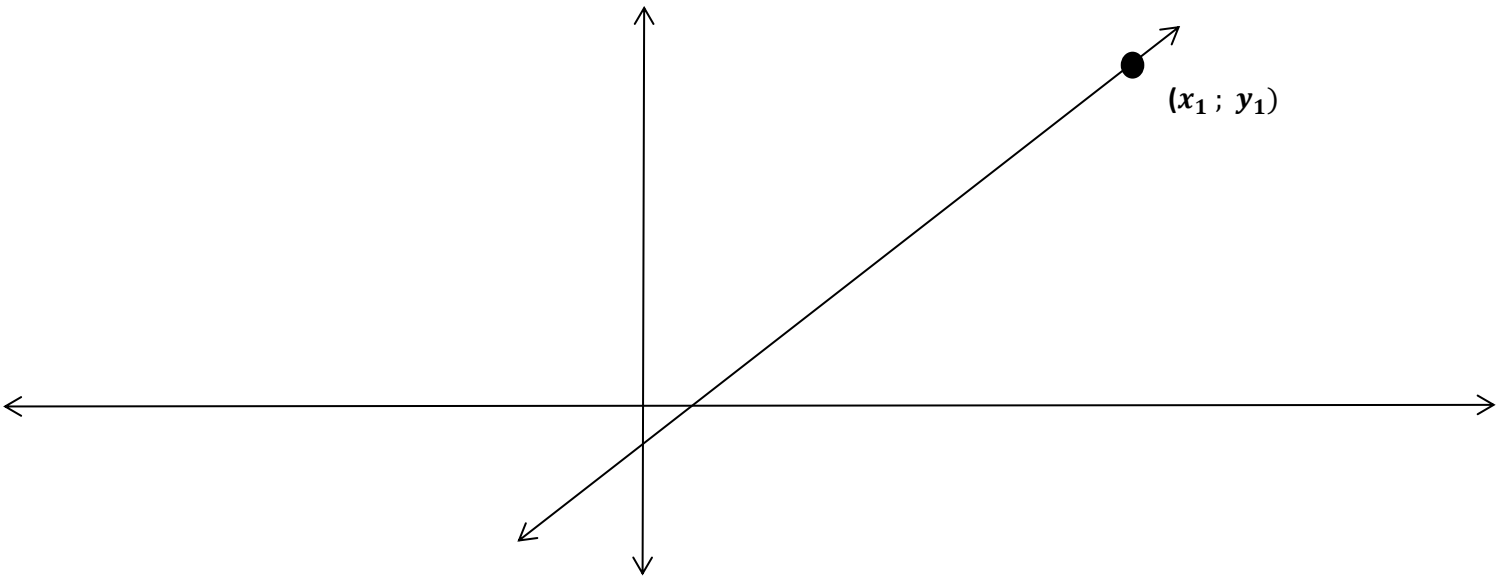
After finding the gradient, you use the value of the gradient to find **c**. You then finally substitute into the general form of a straight line:

y = mx + c

Details of how this is done will be given shortly, but first, let's highlight the other two methods.

Method 2

You are given one point on the line and the value of the value of the gradient **m**.



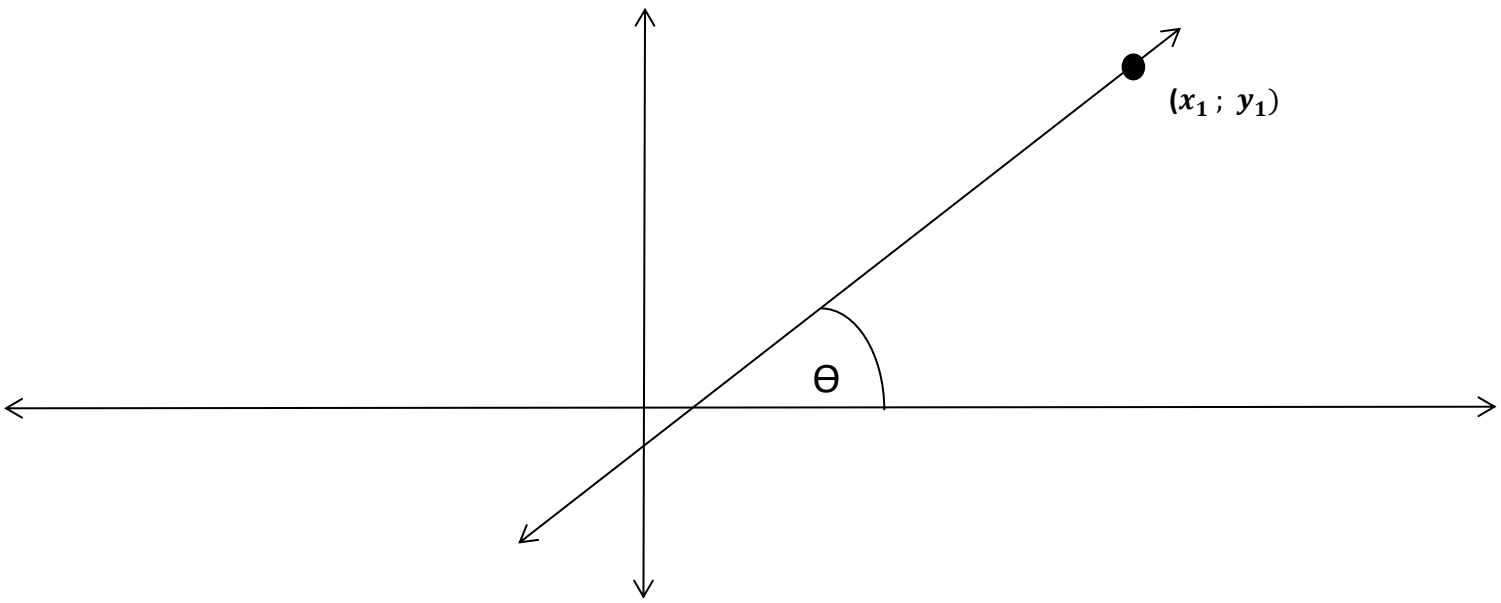
Here you use the value of the **gradient** together with the **given point** to find **c**. You then finally substitute into the general form of a straight line:

y = mx + c

Details of how this is done will be given shortly.

Method 3

You are given one point on the line and also given the angle that the line makes with the x-axis.



In this instance, you first calculate the gradient using the gradient formula $\boldsymbol{m} = tan\ \theta$

After finding the gradient, you use the value of the **gradient** together with the **given point** to find **c**. You then finally substitute into the general form of a straight line:

y = mx + c

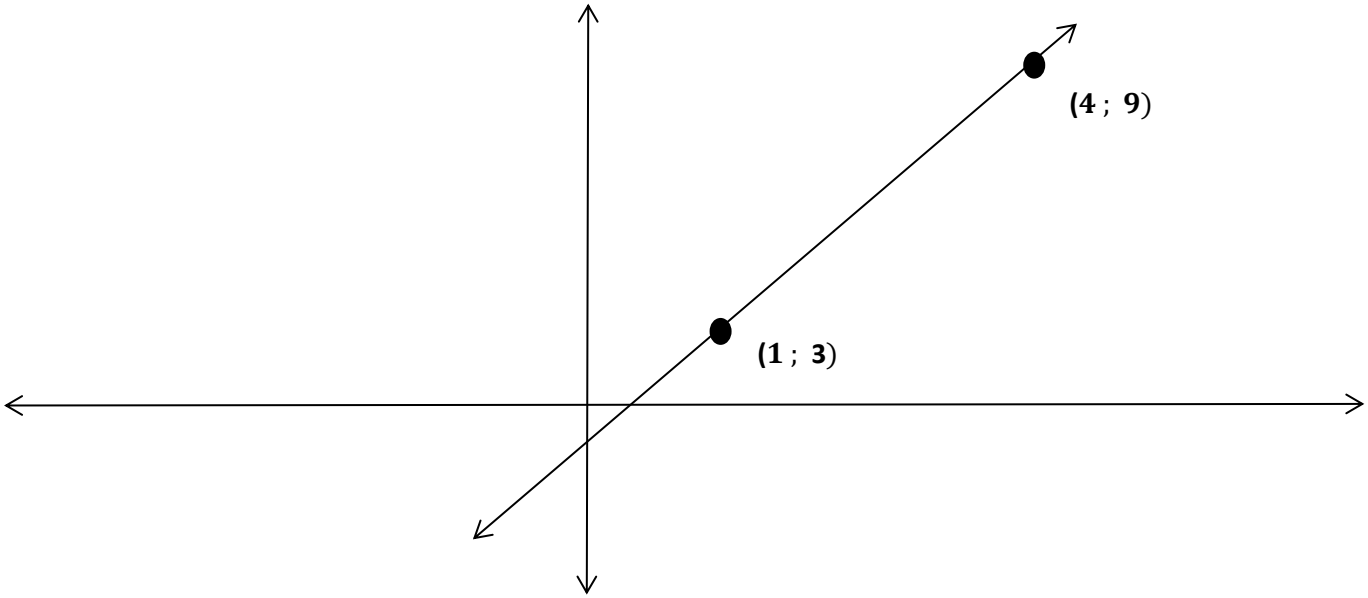
Details of how this is done will be given shortly.

As mentioned, your choice of method depends on what you are given in the question. Let's start with the explanation on Method 1. The other two will be dealt with a bit later.

Method 1

This method is about using the given two points to find the equation of a line. I'll use an example to explain this method.

Example: Find the equation of the line that passes through the points (1 ; 3) and (4 ; 9).



Solution: Remember, the general form of a straight line is ***y = mx + c***

We seek to find the value of **m** (gradient) and **c** (y-cut). We start by calculating gradient using the formula:

$$\boldsymbol{m} = \frac{y_2 - y_1}{x_2 - x_1}$$

(1 ; 3) *(4 ; 9)*

(x_1 ; y_1) *(x_2 ; y_2)*

Now substitute

$$m = \frac{9-3}{4-1}$$

$m = \frac{6}{3} = 2$. We have found **m** and are now left with finding **c**.

On the general form ***y = mx + c***, substitute m and any one of the points to calculate c.

Substitute *m = 2* point *(1 ; 3)*

y = mx + c

3 = 2 (1) + c

3 = 2 + c

3 – 2 = c

1 = c

c = 1

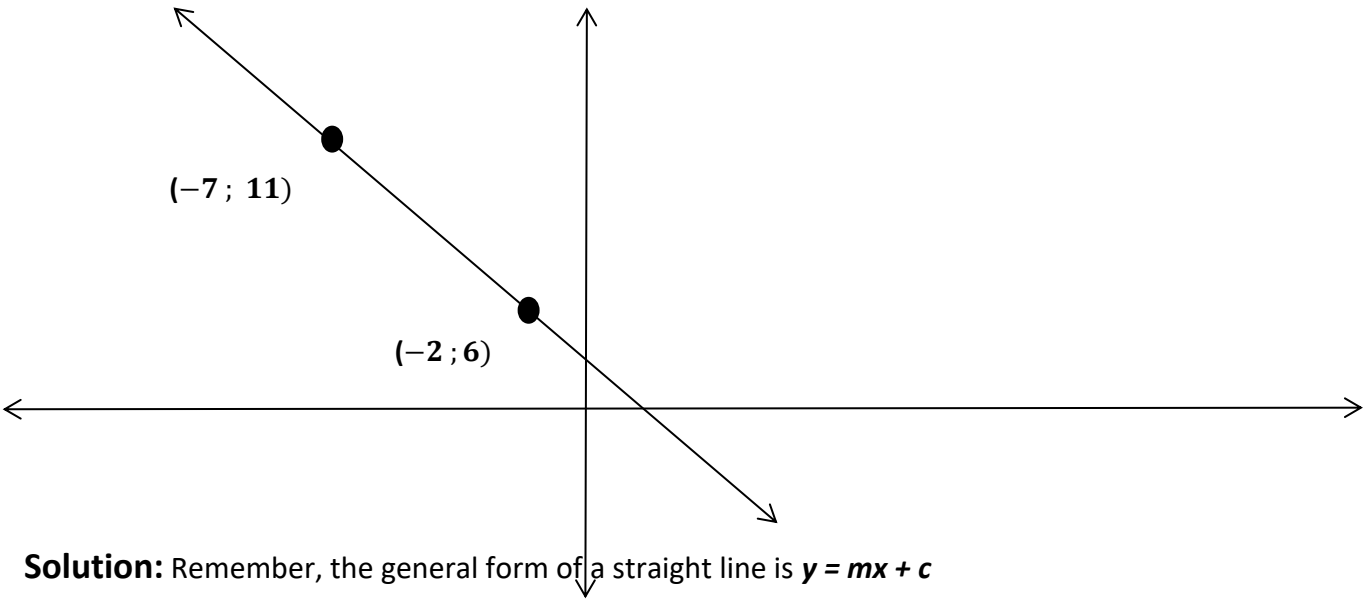
We now have the values of both m and c. Substitute them into the general form.

m = 2 c = 1

y = mx + c

y = 2x + 1. That's the equation of the line.

Example: Find the equation of the line that passes through the points (-2 ; 6) and (-7 ; 11).



Solution: Remember, the general form of a straight line is ***y = mx + c***

We seek to find the value of **m** (gradient) and **c** (y-cut). We start by calculating gradient using the formula:

$$\boldsymbol{m} = \frac{y_2 - y_1}{x_2 - x_1}$$

(-2 ; 6) *(-7 ; 11)*

(x_1 ; y_1) *(x_2 ; y_2)*

Now substitute

$$m = \frac{11 - 6}{-7 - (-2)}$$

$m = \frac{5}{-5} = -1$. We have found **m** and are now left with finding **c**.

On the general form ***y = mx + c***, substitute m and any one of the points to calculate c.

Substitute *m = -1* point *(-2 ; 6)*

y = mx + c

6 = -1 (-2) + c

6 = 2 + c

6 – 2 = c

4 = c

c = 4

We now have the values of both m and c. Substitute them into the general form.

m = -1 c = 4

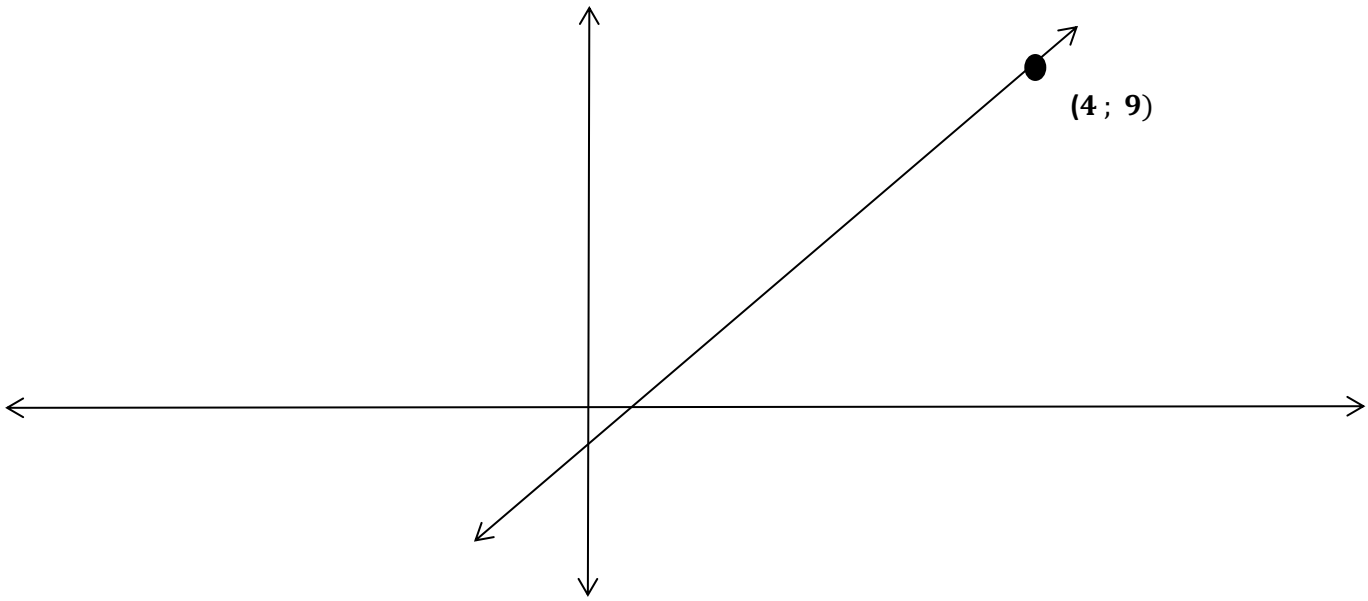
y = mx + c

y = -1x + 4 . That's the equation of the line.

Method 2

This method is about using the given **point** and given **gradient** to find the equation of a line. I'll use an example to explain this method.

Example: Find the equation of the line that passes through the point (4 ; 9) and has gradient ***m*** *= 2*



Solution: Remember, the general form of a straight line is ***y = mx + c***

We have the value of **m** already, so we only need to find the value of **c** (y-cut).

On the general form ***y = mx + c***, substitute m and point (4 ; 9) to calculate c.

Substitute *m = 2* point *(4 ; 9)*

y = mx + c

9 = 2 (4) + c

9 = 8 + c

9 – 8 = c

1 = c

c = 1

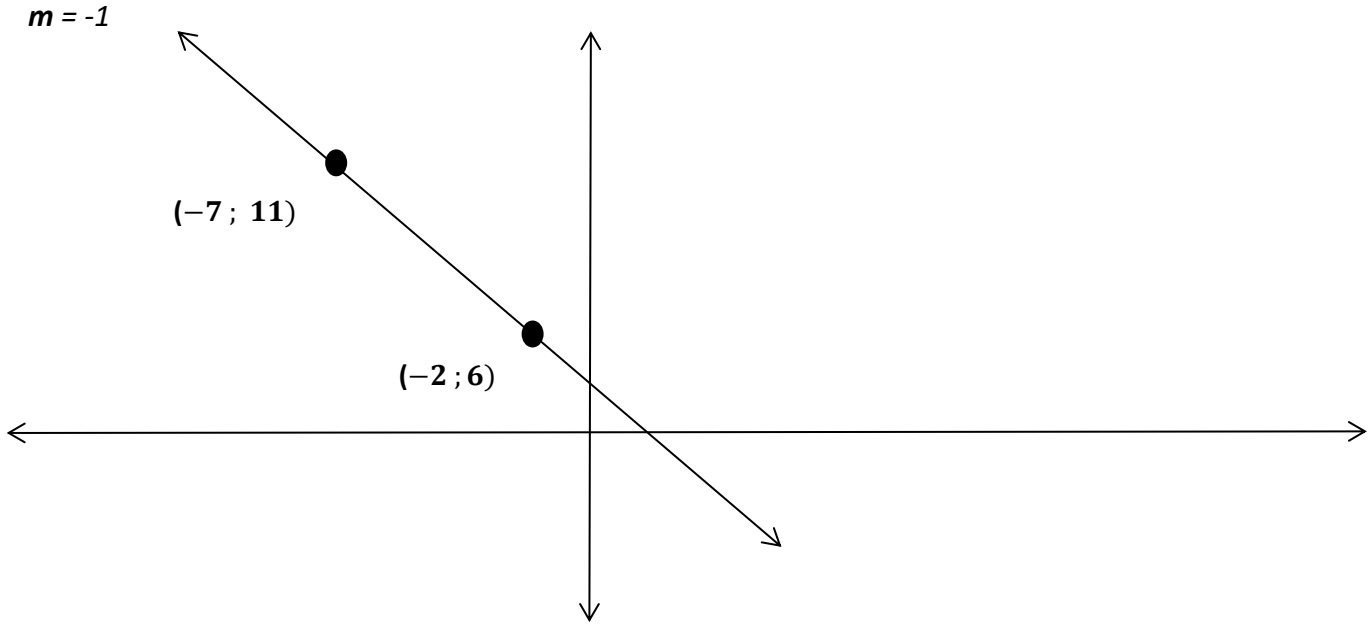
We now have the values of both m and c. Substitute them into the general form.

m = 2 c = 1

y = mx + c

y = 2x + 1. That's the equation of the line.

Example: Find the equation of the line that passes through the point (-2 ; 6) and has gradient ***m*** *= -1*



Solution: Remember, the general form of a straight line is ***y = mx + c***

We have the value of **m** already, so we only need to find the value of **c** (y-cut).

On the general form ***y = mx + c***, substitute m and point (-2 ; 6) to calculate c.

Substitute *m = -1* point *(-2 ; 6)*

y = mx + c

6 = -1 (-2) + c

6 = 2 + c

6 – 2 = c

4 = c

c = 4

We now have the values of both m and c. Substitute them into the general form.

m = -1 c = 4

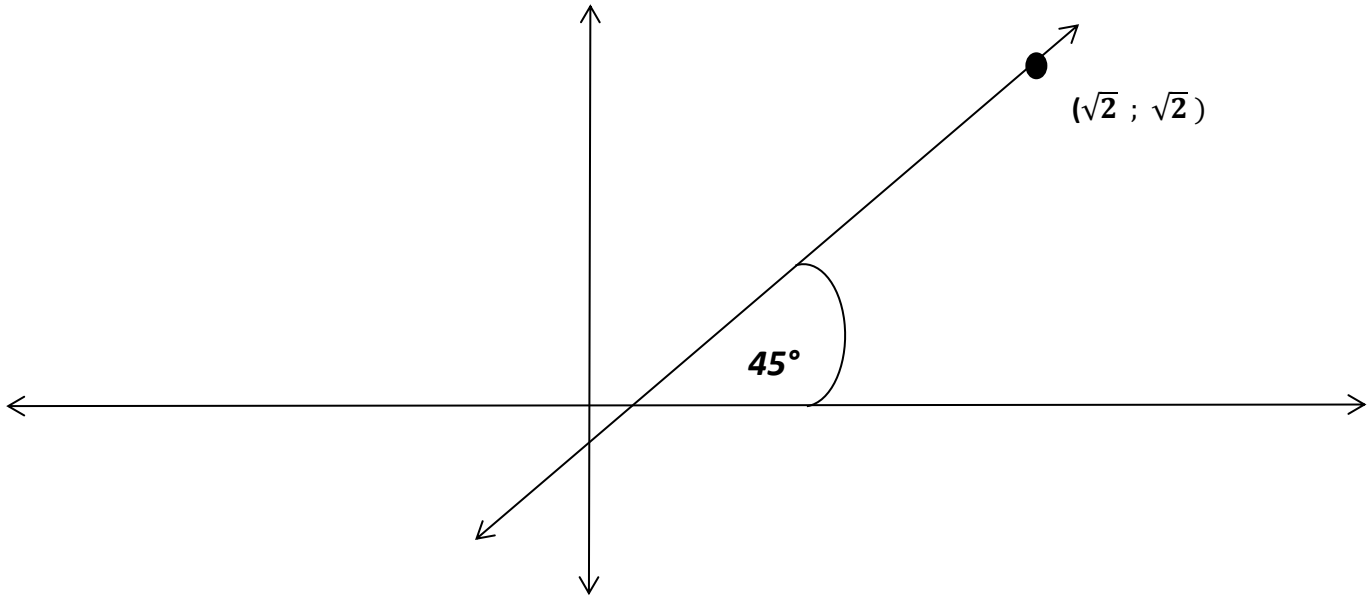
y = mx + c

y = -1x + 4 **.** That's the equation of the line.

Method 3

This method is about using the given **point** and the given **angle** to find the equation of a line. The angle we are talking about is the one the line makes with the x-axis. As always, I'll use an example to explain this method.

Example: Find the equation of the line that passes through the point $(\sqrt{2}\ ;\sqrt{2})$. The line makes an angle of $\boldsymbol{\theta} = 45°$ with the x-axis.



Solution: Remember, the general form of a straight line is ***y = mx + c***

We seek to find the value of **m** (gradient) and **c** (y-cut). We start by calculating gradient using the formula:

$\boldsymbol{m} = tan\,\theta$

Now substitute ***θ*** *= 45°*

$\boldsymbol{m} = \tan 45°$

$\boldsymbol{m} = 1$

Substitute *m = 1* point $(\sqrt{2}; \sqrt{2})$

y = mx + c

$\sqrt{2} = 1\,(\sqrt{2}) + c$

$\sqrt{2} = \sqrt{2} + c$

$\sqrt{2} - \sqrt{2} = c$

$0 = c$

$c = 0$

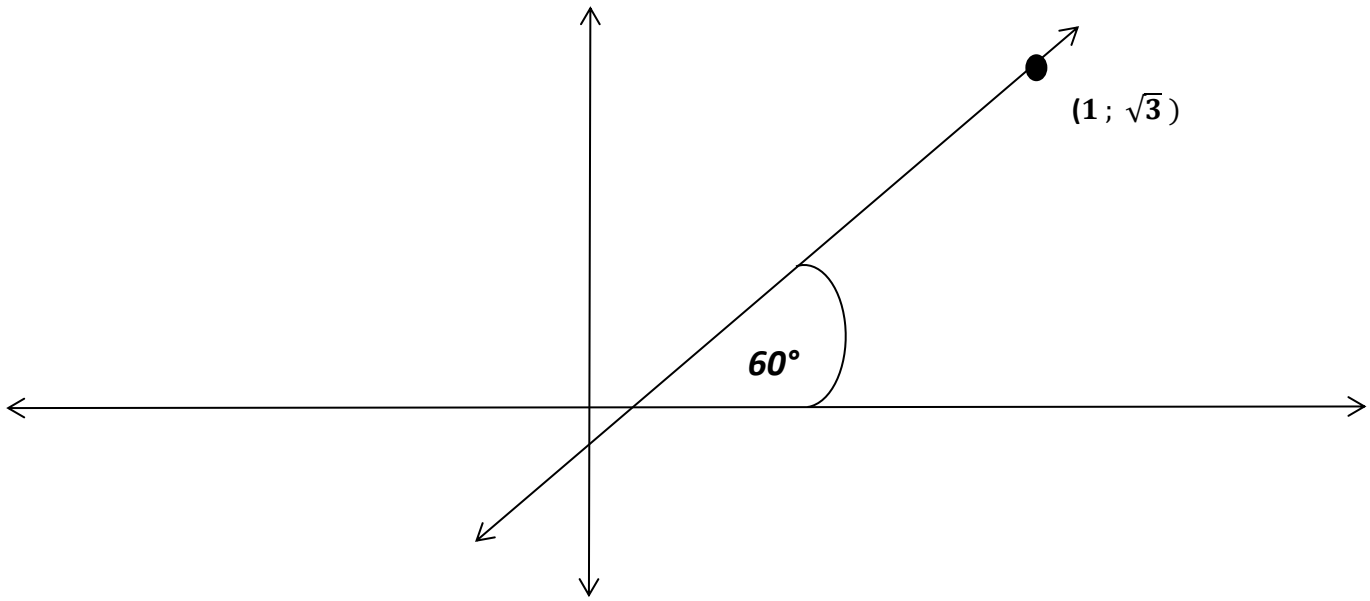
We now have the values of both m and c. Substitute them into the general form.

m = 1 c = 0

y = mx + c

y = 1x + 0, which then gives ***y = 1x***. That's the equation of the line.

Example: Find the equation of the line that passes through the point $(1; \sqrt{3})$. The line makes an angle of ***θ*** *= 60°* with the x-axis.



Solution: Remember, the general form of a straight line is ***y = mx + c***

We seek to find the value of **m** (gradient) and **c** (y-cut). We start by calculating gradient using the formula:

$\boldsymbol{m} = tan\,\theta$

Now substitute ***θ*** *= 60°*

$\boldsymbol{m} = \tan 60°$

$\boldsymbol{m} = \sqrt{3}$

Substitute $m = \sqrt{3}$ point $(1 ; \sqrt{3})$

y = mx + c

$\sqrt{3} = \sqrt{3}\,(1) + c$

$\sqrt{3} = \sqrt{3} + c$

$\sqrt{3} - \sqrt{3} = c$

$0 = c$

$c = 0$

We now have the values of both m and c. Substitute them into the general form.

m = $\sqrt{3}$ c = 0

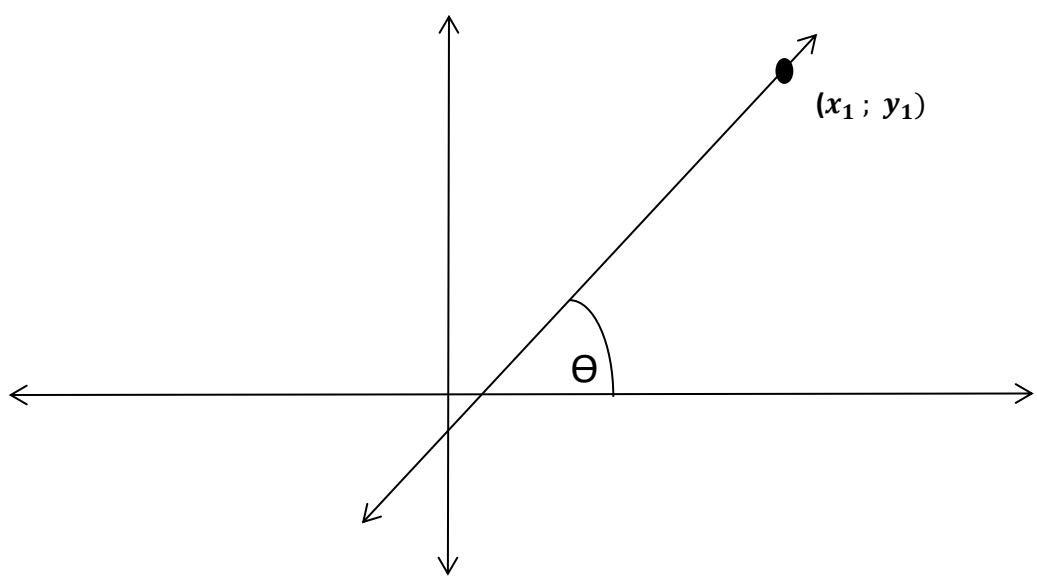
$$y = mx + c$$

$y = \sqrt{3}\,x + 0$, which then gives $y = \sqrt{3}\,x$. That's the equation of the line.

As you just saw, the examples we did on method 3 are based only on positive gradients. The formula $m = \tan\theta$ does apply on a negative gradient too, but needs to be treated with more care. I am going to exphylain the whole concept behind the formula. Mathematicians call this concept the "line of inclination".

Line of Inclination

Look at the diagram.

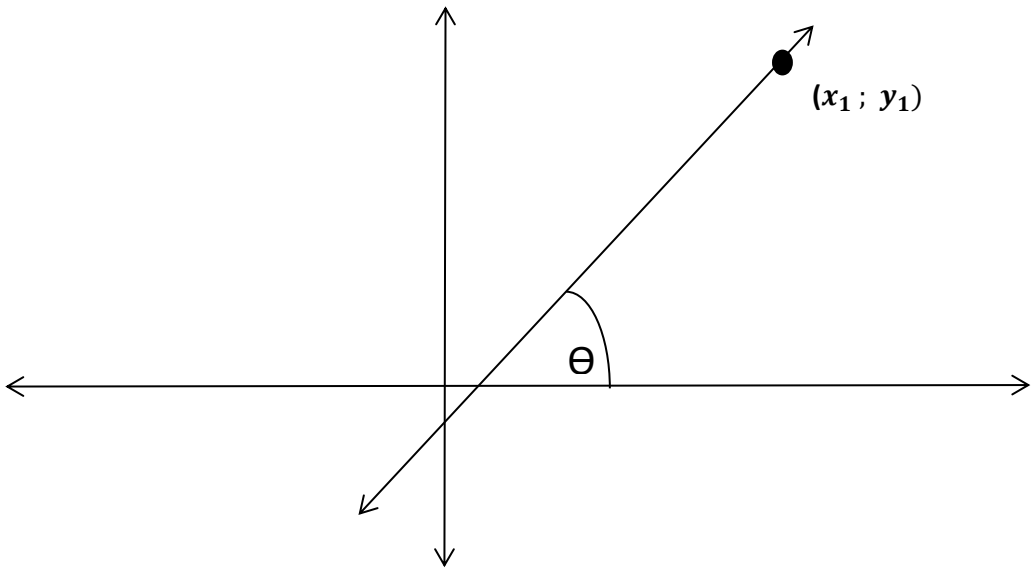


The concept of the "inclination of a line" gives us that $\boxed{\boldsymbol{m} = tan\,\theta}$

This means that the gradient of a line equals the "***tan***" of the angle the line makes with the x-axis.

In order to simplify the explanations, we can split the concept into two parts: one involving inclination for upward-sloping lines and the other for downward-sloping.

Line with positive gradient (upward-sloping line)

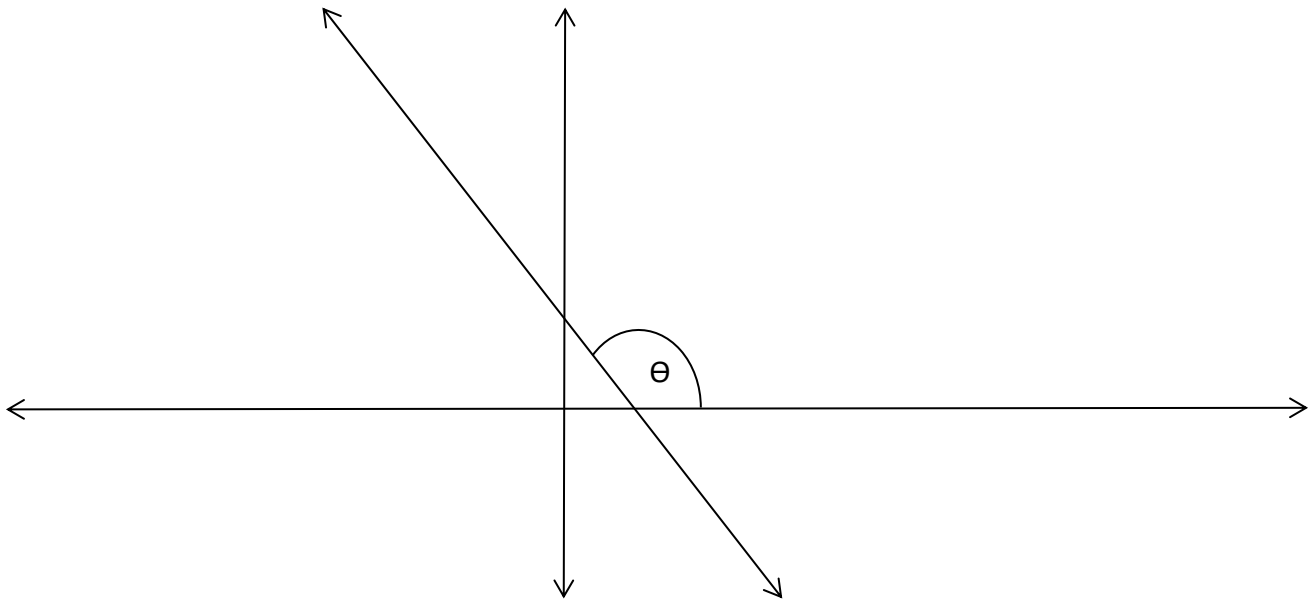


In the case of an upward-sloping line (positive ***m***), the formula is still: $\boldsymbol{m} = tan\,\theta$

, where 0° ≤ θ < 90°.

The size of the angle is very important, it has to be between 0° and 90°, but **never** equal to 90°.

Line with negative gradient (downward-sloping line)

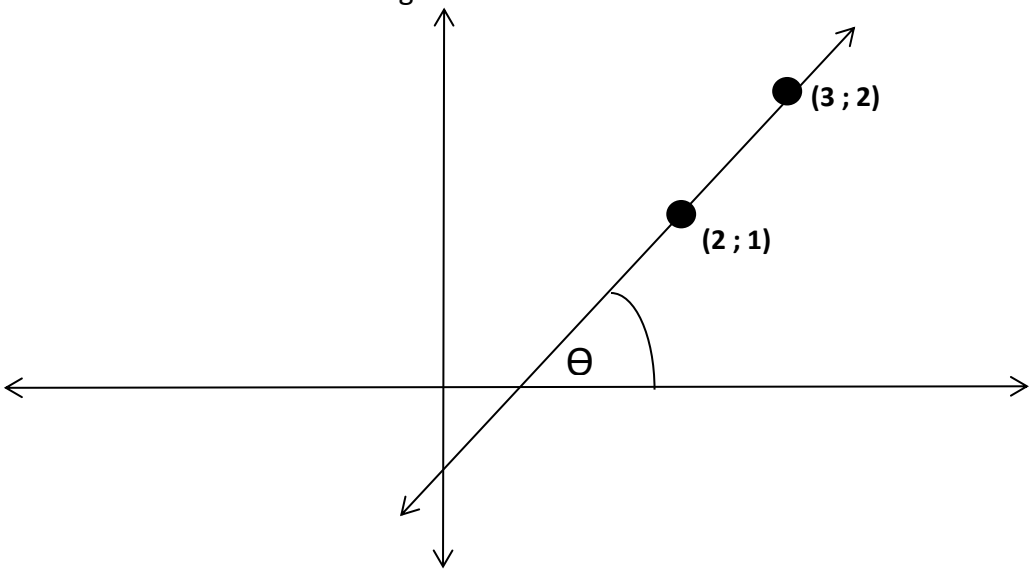


In the case of a downward-sloping line (negative ***m***), the formula is still: $\boldsymbol{m} = tan(\Theta - 180)$

, where 90° < Θ < 180°. The size of the angle is very important, it has to be between 90° and 180°, but never equal to 90° **nor** 180°.

Attempt the some examples to check your understanding. The examples below are based only on upward-sloping lines (positive gradient) for now.

Example: A straight line passes through points (2 ; 1) and (3 ; 2). The line forms the angle Θ with the x-axis. Calculate angle Θ.



Answer: Here we use the formula m = tan*Θ* to find the value of *Θ.* We do this in two easy steps.

Step 1: Find the value of gradient m using the gradient formula.

$m = \frac{y_2 - y_1}{x_2 - x_1}$, substitute (2 ; 1) and (3 ; 2)

$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{2-1}{3-2} = \frac{1}{1} = 1$, so m = **1**.

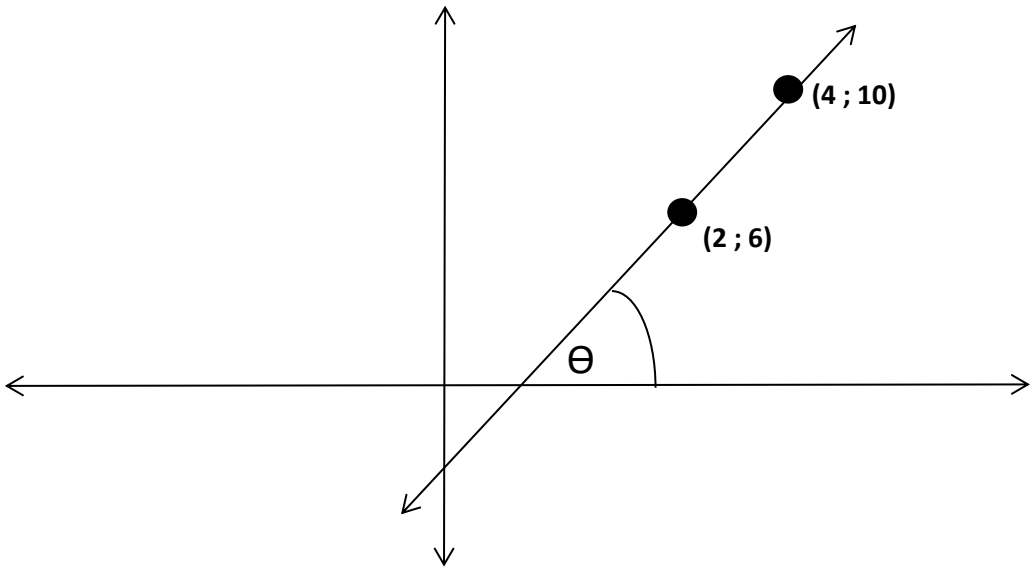
Step 2: Substitute the value of m into the formula, *m = tanΘ*

tan*Θ = m*

tan*Θ = 1*

Θ = 45°, therefore, the angle the line makes with the x-axis is 45°.

Example: A straight line passes through points (2 ; 6) and (4 ; 10). The line forms the angle Θ with the x-axis. Calculate angle Θ.



Answer: Here we use the formula m = tan*Θ* to find the value of *Θ.* We do this in two easy steps.

Step 1: Find the value of gradient m using the gradient formula.

$m = \frac{y_2 - y_1}{x_2 - x_1}$, substitute (2 ; 6) and (4 ; 10)

$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{10-6}{4-2} = \frac{4}{2} = 2$, so m = **2**.

Step 2: Substitute the value of m into the formula, m = tan*Θ*

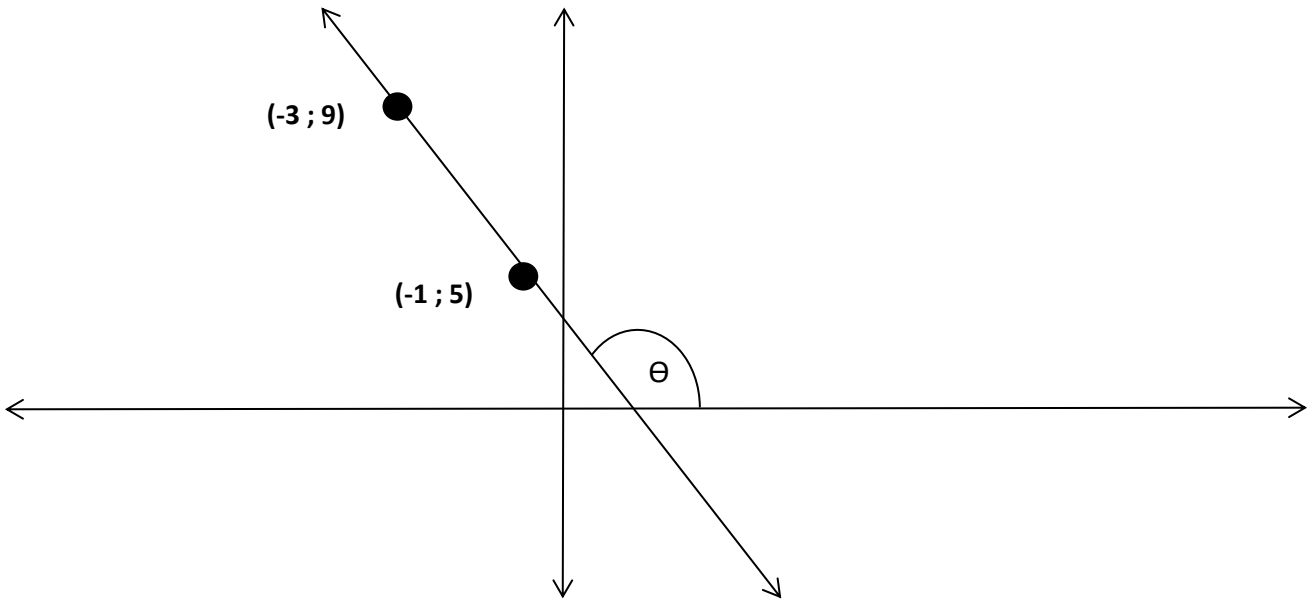
tan*Θ = m*

tan*Θ = 2*

Θ = 63.43°, therefore, the angle the line makes with the x-axis is 63.43°

So far we have done examples based on lines with a positive gradient. We know attempt examples on lines with a negative gradient (downward-sloping lines).

Example: A straight line passes through points (2 ; 1) and (3 ; 2). The line forms the angle Θ with the x-axis. Calculate angle Θ.



Answer: Here we use the formula m = tan(*Θ - 180°)* to find the value of *Θ.* We do this in two easy steps.

Step 1: Find the value of gradient m using the gradient formula.

$m = \frac{y_2 - y_1}{x_2 - x_1}$, substitute (-1 ; 5) and (-3 ; 9)

$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{9-5}{-3-(-1)} = \frac{4}{-2} = -2$, so m = **-2**

Step 2: Substitute the value of m into the formula, *m = tanΘ*

tan(*Θ - 180°) = m*

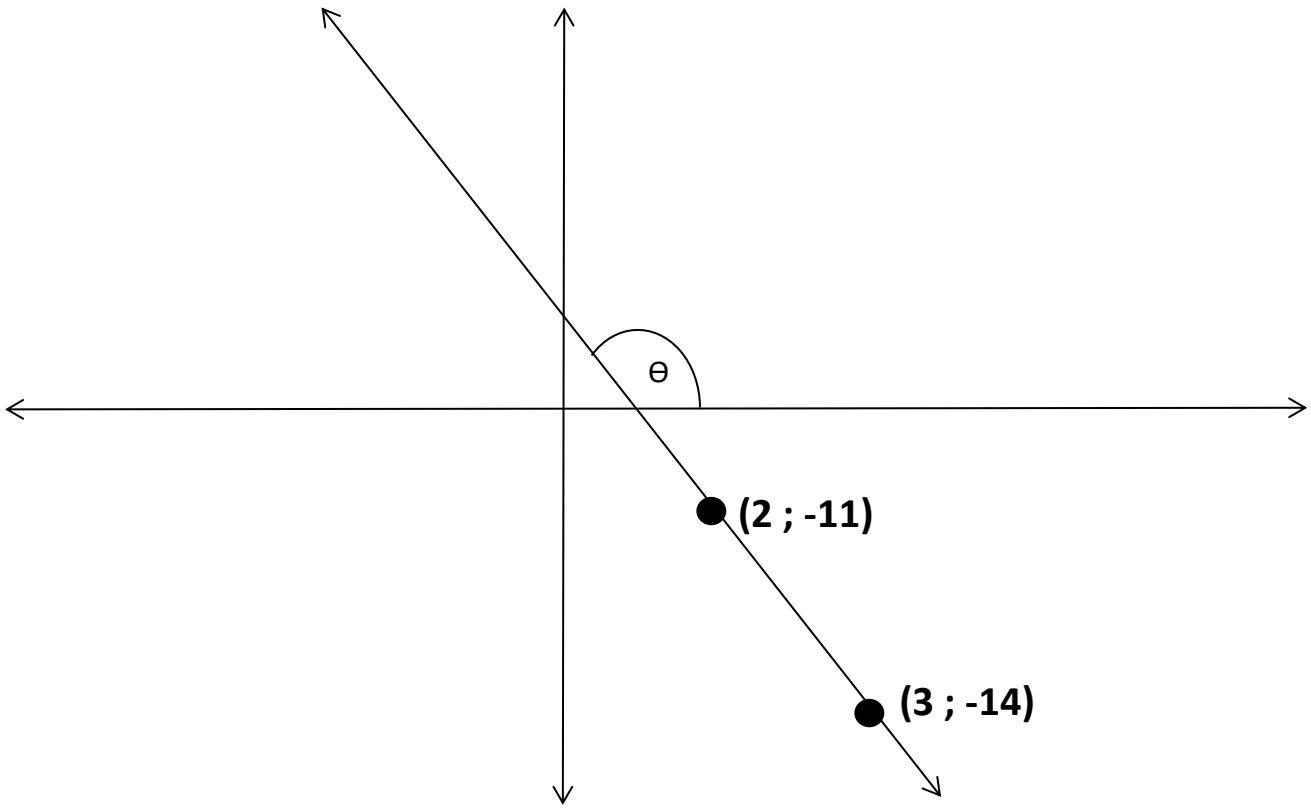
tan(*Θ -180°) = -2*

Θ -180° = -63.43°

Θ = -63.43°+180°

Θ = 116.57°, therefore, the angle the line makes with the x-axis is *116.57°*.

Example: A straight line passes through points *(2 ; -11)* and *(3 ; -14).* The line forms the angle θ with the x-axis. Calculate angle θ.



Answer: Here we use the formula m = tan(*Θ - 180°)* to find the value of *Θ.* We do this in two easy steps.

Step 1: Find the value of gradient m using the gradient formula.

$m = \frac{y_2 - y_1}{x_2 - x_1}$, substitute (2 ; -11) and (3 ; -14)

$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{-14-(-11)}{3-2} = \frac{-3}{1} = -3$, so m = **-3**

Step 2: Substitute the value of m into the formula, m = tanθ

tan(θ - *180°) = m*

tan(θ *-180°) = -3*

θ *-180° = -71.57°*

θ *= -71.57°+180°*

θ *= 108.43°*, therefore, the angle the line makes with the x-axis is *108.43°*.

You have learnt the fundamental techniques of dealing with straight lines. Every time you come across straight lines, you may use the methods you just learnt. Along with these methods, you may use the concepts of parallel and perpendicular lines.

- Remember, when two lines are parallel to each other, they have the same gradient.
- And, when two lines are perpendicular to each other, the product of their gradients equals ***-1***:

$$m_1 \times m_2 = -1$$

Gradient of first line **Gradient of second line**

Attempt the following questions using the methods you learnt, along with the concepts of parallel and perpendicular lines.

Probability

Quick Revision

Joseph throws a fair "dice".

Let's list all the possible outcomes the "die" could land on.

The "die" may land 1 up, or 2 up, or 3 up or 4 or 5 or 6. These are all the possible outcomes of throwing a "die". Mathematicians call this the sample space, and it is represented by the letter "S".

S = {1 , 2 , 3 , 4 , 5 , 6}

So these are all the six possible outcomes when Joseph throws the "die". The die can only land on one of them.

When Joseph throws the die, what is the probability of getting 4 as the outcome?

The outcome of interest here is **4**. Mathematicians call this the "event". Let's represent of event (getting a **4**) by the letter ***E***.

E *= { 4 }*

Now let's answer the question.

S *= {1 , 2 , 3 ,* ***4*** *, 5 , 6}*

As you can see, out of all the six possible outcomes, there's only one "***4***". Therefore, the probability of getting "***4***" is **one** out of **six** outcomes: $\frac{1}{6}$

P(4) = $\frac{1}{6}$

If you like, you could've also used the basic formula for probability to get the answer.

P(event) = $\frac{number\ of\ outcomes\ in\ the\ event}{number\ of\ all\ the\ possible\ outcomes}$

We know that the "number of outcomes in the event" is **one**, since "***4***" is our only event of interest. We also know that the "number of all possible outcomes" is six.

number of outcomes in the event = 1

number of all possible outcomes = 6

Then substitute.

P(4) = $\frac{1}{6}$

Do one more example to sharpen your understanding.

Example

Joseph throws a fair "die". What is the probability of getting an even number as the outcome?

Solution

The event of interest here is an "even number". First identify the even numbers from the sample space.

S *= {1 ,* ***2*** *, 3 ,* ***4*** *, 5 ,* ***6****}* . So our even numbers are 2, 4 , 6. Let's use the letter ***E*** to represent them.

E *= {* ***2*** *,* ***4*** *,* ***6****}*

So out of all the **six** possible outcomes, there are **three** even numbers. Therefore, the probability of getting an even number is **three** out of **six**: $\frac{3}{6}$

P("even number") = $\frac{3}{6}$

Again, you could've also used the basic formula for probability to get the answer.

P(event) = $\frac{number\ of\ outcomes\ in\ the\ event}{number\ of\ all\ the\ possible\ outcomes}$

We know that the "number of outcomes in the event" is **three**, since we are interested in the event ***E*** *= {* ***2*** *,* ***4*** *,* ***6****}*.

We also know that the "number of all possible outcomes" is **six**.

number of outcomes in the event = 3

number of all possible outcomes = 6

Then substitute.

P(4) = $\frac{3}{6}$

Independent Events

Scenario 1

Prince throws a fair "die". There are obviously six possible outcomes. We represent all of these outcomes with the letter ***S***.

S *= {1 , 2 , 3 , 4 , 5 , 6}*

What is the probability of getting a perfect square as an outcome?

Answer

Well, the event of interest is the "square number". Let's identify and highlight all the square numbers from the sample space S.

***S** = {**1** , 2 , 3 , **4** , 5 , 6}*

So our perfect squares are 1 and 4. We represent them with the letter E.

E = {1 , 4}

Remember that a perfect square is a number whose square root gives a whole number. Check the examples below.

$\sqrt{1} = 1$, the square root of **1** gives a whole number, so **1** is a perfect square.

$\sqrt{4} = 2$, the square root of **4** gives a whole number, so 4 is a perfect square.

$\sqrt{2} = 1.41$, the square root of **2** is "not" a whole number, so **2** is "not" a perfect square.

$\sqrt{3} = 1.73$, the square root of **3** is "not" a whole number, so **3** is "not" a perfect square.

$\sqrt{9} = 3$, the square root of **9** gives a whole number, so **9** is a perfect square.

Now let's go back to our question. *What is the probability of getting a perfect square as an outcome?*

We have already identified the perfect squares from the six outcomes, we highlighted them in green.

***S** = {**1** , 2 , 3 , **4** , 5 , 6}*

As I have mentioned before, our perfect squares are 1 and 4. We represent them with the letter ***E***.

***E** = {**1** , **4**}*

So out of the **six** possible outcomes, there are **two** "perfect squares". Therefore, the probability of getting a perfect square when Prince throws the "die" is **two** out of **six**: $\frac{2}{6}$

P("perfect square") = $\frac{2}{6}$

$= \frac{1}{3}$

Now let's go back to the beginning and play around with this scenario.

Scenario 2

Prince throws the "die". After the die lands, he hides the outcome from you and asks to guess what it is. However, he gives you a clue. He tells you that the outcome is one of the "even numbers".

What is the probability that the number is a perfect square?

Our event of interest here is a "perfect square". We represent that with the letter ***E***.

E = {1 , 4}

Since he tells you that the number he got is one of the "even numbers", it is either 2 or 4 or 6. Let's call this list event ***A***.

A = {2 , 4 , 6}

Since Prince gave you a clue that the number he got is one of the even numbers above, identify the "perfect square" numbers from these three possibilities above.

A = {2 , 4 , 6}

The ***4*** highlighted above is the only perfect square in this list. So out of the three possible outcomes, **4** is the only perfect square. Therefore, the probability of getting a perfect square is **one** out of **three**: $\frac{1}{3}$

This probability is the same as that of scenario 1. Even though Prince told you that the outcome was an even number, it didn't affect the probability of getting a perfect square, it is still $\frac{1}{3}$

That means the probability of getting a perfect square (event ***E***) is not affected by that the outcome was already an even number (event ***A***).

So we say that ***E*** is independent of event ***A***. We can also say that ***E*** and ***A*** are independent, just means the same thing.

Sometimes it may be difficult to see whether or not two events are independent. Luckily, mathematicians have found a "math-way" of checking that.

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Any two events ***A*** and ***B*** are independent if and only if ***P(A and B) = P(A) × P(B)***

Since we said that the events ***A*** and ***E*** in scenario 2 are independent, let's use the definition above to check if that is actually true.

For scenario 2, event ***A*** is ***A = {2 , 4 , 6}*** and event ***E*** is ***E = {1 , 4}***. To check if they are independent, we prove the equation ***P(A and B) = P(A) × P(B)***

Let's start with ***P(A and B)***

P(A and B) means the probability of the outcome A and E share.

A = {2 , 4 , 6} ***E = {1 , 4}***

As you can see, ***A*** and ***E*** both have a ***4***, thus the we say they share a ***4***. So we write ***P(A and B)***, we are simply asking for the probability of ***4***.

Out of all the initial six possible outcomes

{1, 2 , 3 , 4 , 5 , 6} , the is only one ***4***. Therefore, the probability of ***(A and E)*** is **one** out of **six**: $\frac{1}{6}$

P(A and B) = $\frac{1}{6}$

Now let's check if this equals ***P(A) × P(B)***

P(A) is the probability of getting an even number as the outcome. In our sample space, ***S = {1 , 2 , 3 , 4 , 5 , 6}*** , there are t**hree** even numbers, ***2*** , ***4*** and ***6***. So out of the ***six*** possible outcomes, there are ***three*** even numbers. Therefore, the probability is **three** out of **six**: $\frac{3}{6}$

P(A)= $\frac{3}{6}$

P(E) is the probability of getting a perfect square as the outcome. In our sample space,

S = {1 , 2 , 3 , 4 , 5 , 6} , there are two perfect squares, **1** and **4**. So out of the **six** possible outcomes, there are **two** even numbers. Therefore, the probability is two out of six: $\frac{2}{6}$

P(E)= $\frac{2}{6}$

Then multiply P(A) and P(E).

P(A) × P(E)

= $\frac{3}{6} \times \frac{2}{6}$

= $\frac{3 \times 2}{6 \times 6}$

= $\frac{6}{36}$

= $\frac{1}{6}$

This equals ***P(A and B) =*** $\frac{1}{6}$

These satisfy the formula: ***P(A and B) = P(A) × P(B)***

Therefore, ***A*** and ***E*** are independent.

Do more examples to boost your understanding.

Example

Prince throws a fair "die".

Let S be the sample space, S = {1 , 2 , 3 , 4 , 5 , 6}

Let A be the event that the "die" lands as an even number, A = {2 , 4 , 6}

Let be the event that the "die" lands as a 4, B = {4}

Are events A and B independent?

Solution

All we do is to check if the formula

P(A and B) = P(A) × P(B) is true?

We start with ***P(A and B)***

P(A and B) is the probability of the outcome A and B share.

A = {2 , ***4*** , 6} B = {***4***}

As you can see, both A and B have **4**, thus they share 4. So when we write ***P(A and B)***, we are asking for the probability of **4**. Out of the six possible outcomes, there's only one **4**. Therefore, the probability of ***(A and B***) is **one** out of **six**: $\frac{1}{6}$

P(A and B) = $\frac{1}{6}$

Now let's check if this equals ***P(A) × P(B)***

P(A) is the probability of getting an even number as the outcome. In our sample space, ***S = {1 , 2 , 3 , 4 , 5 , 6}*** , there are t**hree** even numbers, ***2*** , ***4*** and ***6***. So out of the ***six*** possible outcomes, there are ***three*** even numbers. Therefore, the probability is **three** out of **six**: $\frac{3}{6}$

P(A)= $\frac{3}{6}$

P(E) is the probability of getting a ***4*** as the outcome. In our sample space,

S = {1 , 2 , 3 , 4 , 5 , 6} , there is only **one** ***4*** out of the **six** possible outcomes. Therefore, the probability is **one** out of **six**: $\frac{1}{6}$

P(E)= $\frac{1}{6}$

Then multiply ***P(A)*** and ***P(B)***.

P(A) × P(B)

= $\frac{3}{6} \times \frac{1}{6}$

= $\frac{3 \times 1}{6 \times 6}$

= $\frac{3}{36}$

= $\frac{1}{12}$

This is "**Not**" equal to ***P(A and B) =*** $\frac{1}{6}$

Therefore, A and B are "**NOT**" independent.

Example

P(A) = 0.3 P(B) = 0.5 and *P(A and B) = 0.15*

Are A and B independent events?

Solution

Test if *P(A and B) = P(A) × P(B)*

P(A and B) = 0.15 as given.

Check P(A) × P(B)

P(A) × P(B)

= 0.3 *× 0.5*

= 0.15 , which equals P(A and B)

Therefore A and B are independent events

Example

P(A) = 0.3 P(B) = 0.8 and *P(A and B) = 0.15*

Are A and B independent events?

Solution

Test if *P(A and B) = P(A) × P(B)*

P(A and B) = 0.15 as given.

Check P(A) × P(B)

P(A) × P(B)

= 0.3 *× 0.8*

*= 0.24 , which is "**Not**" equal to P(A and B)*

Therefore A and B are "**Not**" independent.

Example

$P(A) = 0.3$ $P(B) = 0.8$ and $P(A \cup B) = 0.9$

Are A and B independent events?

Solution

Before we solve, remember that a shortened way of representing *P(A and B) is* $P(A \cap B)$.

Now back to the solution:

Test if $P(A \cap B) = P(A) \times P(B)$

But we have a problem, we are given the value of $P(A \cup B)$ instead of $P(A \cap B)$.

However, you will be pleased to know that we can use the value of $P(A \cup B)$ to get the value of $P(A \cap B)$ using the formula below.

$P(A \cup B) = P(A) + P(B) - P(A \cap B)$

Substitute the given values $P(A) = 0.3$, $P(B) = 0.8$ and $P(A \cup B) = 0.9$

$0.9 = 0.3 + 0.8 - P(A \cap B)$

$0.9 = 1.1 - P(A \cap B)$

Move 1.1 to the left-hand side

$0.9 - 1.1 = - P(A \cap B)$

$- 0.2 = - P(A \cap B)$

Move – 0.2 to the left-hand side, on which it will change signs to 0.2

$0 = - P(A \cap B) + 0.2$

Then move $- P(A \cap B)$ to the left-hand side, on which it will change signs to $P(A \cap B)$

$P(A \cap B) = + 0.2$

$P(A \cap B) = 0.2$

Now check P(A) × P(B)

P(A) × P(B)

= 0.3 *× 0.8*

*= 0.24 , which is "**Not**" equal to P(A ∩ B)*

Therefore A and B are "**Not**" independent.

Example

P(A) = 0.4 P(B) = 0.5 and *P(A***U***B) = 0.8*

Are A and B independent events?

Solution

Before we solve, remember that a shortened way of representing *P(A and B) is P(A ∩ B) .*

Now back to the solution:

Test if *P(A ∩ B) = P(A) × P(B)*

But we have a problem, we are given the value of *P(A***U***B*) instead of *P(A ∩ B).*

However, you will be pleased to know that we can use the value of *P(A***U***B*) to get the value of *P(A ∩ B)* using the formula below.

*P(A***U***B*) *= P(A) + P(B) – P(A ∩ B)*

Substitute the given values *P(A)* = 0.4, *P(B)* = 0.5 and *P(A***U***B*) = 0.8

0.8 = 0.4 + 0.5 – *P(A ∩ B)*

0.8 = 0.9 – *P(A ∩ B)*

Move 0.9 to the left-hand side, on which it changes signs to – 0.9

0.8 – 0.9 = – *P(A ∩ B)*

– 0.1 = – *P(A ∩ B)*

Move – 0.*1* to the left-hand side, on which it will change signs to 0.1

0 = – *P(A ∩ B)* + 0.1

Then move – *P(A ∩ B)* to the left-hand side, on which it will change signs to *P(A ∩ B)*

P(A ∩ B) = + 0.1

P(A ∩ B) = 0.1

Now check P(A) × P(B)

P(A) × P(B)

= 0.4 *× 0.5*

*= 0.2 , which is "**Not**" equal to P(A ∩ B)*

Therefore A and B are "**Not**" independent.

Important Formula
P(A∪B) = P(A) + P(B) – P(A ∩ B)

Know this formula by heart.

Tree Diagram

A sack contains 3 Red balls and 5 Blue balls. Let's assume that you can't see inside the sack.

Eric takes out one ball from the sack without looking. What is the color of the ball he took out?

Well, we can't be 100% sure of the color of the ball, but we know that there are two possibilities, possibility 1 and possibility 2.

Possibility 1

The ball he takes out may be Red. This means that there would then be **2 Red balls** and still **5 Blue balls** remaining in the sack.

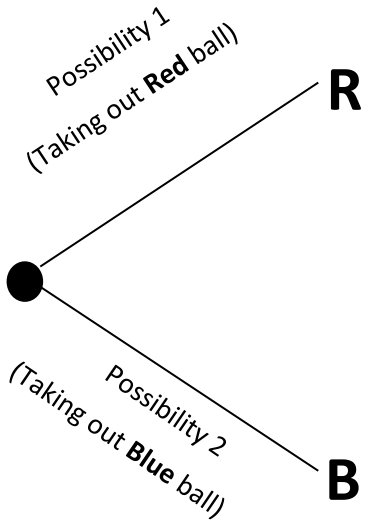
Possibility 2

Or the ball he takes out may be Blue. This means that there would then still be **3 Red balls** but **4 Blue balls** remaining in the sack.

Let's summarize the two possibilities above in a diagram.

Let R refer to taking out a Red ball.

Let B refer to taking out a Blue ball.



a) What is the probability of possibility 1, that is, what is the probability that the ball Eric takes out is **Red**?

b) What is the probability of possibility 2, that is, what is the probability that the ball Eric takes out is Blue?

Solution

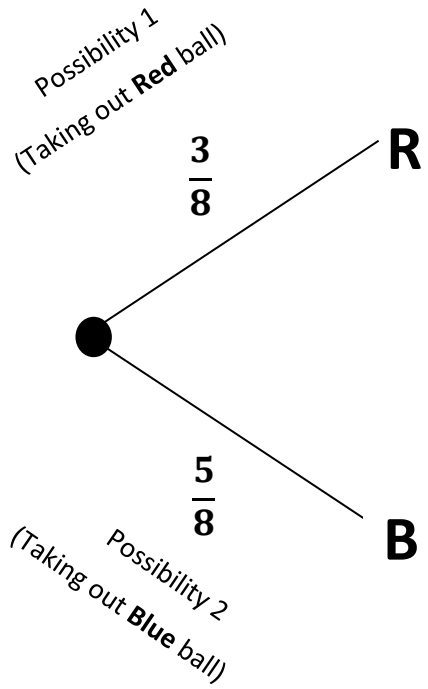
a) Since there are **3 Red** balls out of 8 balls in total, the probability that Eric takes out a Red ball is $\frac{3}{8}$

P(R)= $\frac{3}{8}$

a) Since there are **5 Blue** balls out of 8 balls in total, the probability that Eric takes out a Red ball is $\frac{5}{8}$

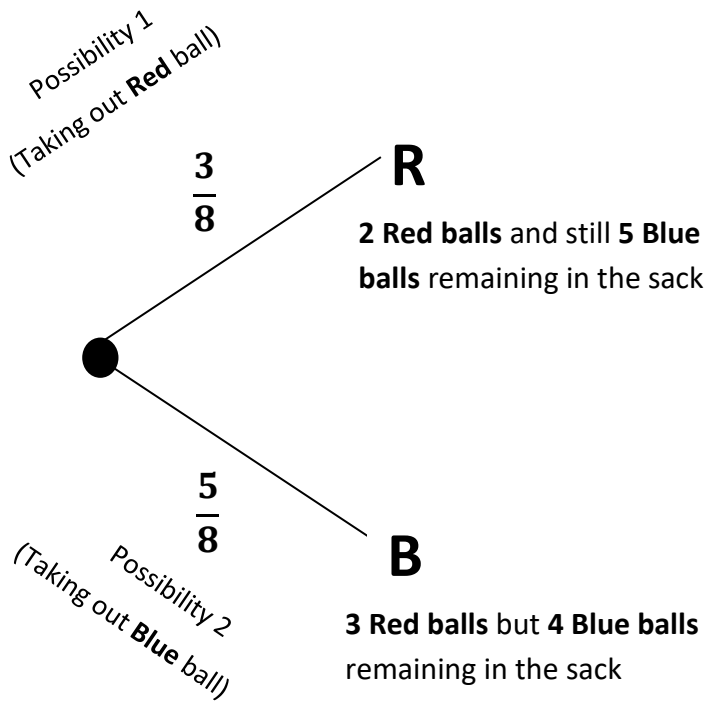
P(B)= $\frac{5}{8}$

Let's include these probabilities on our diagram.



Remember that if possibility 1 happens (the ball he takes out is Red), there would be **2 Red balls** and still **5 Blue balls** remaining in the sack.

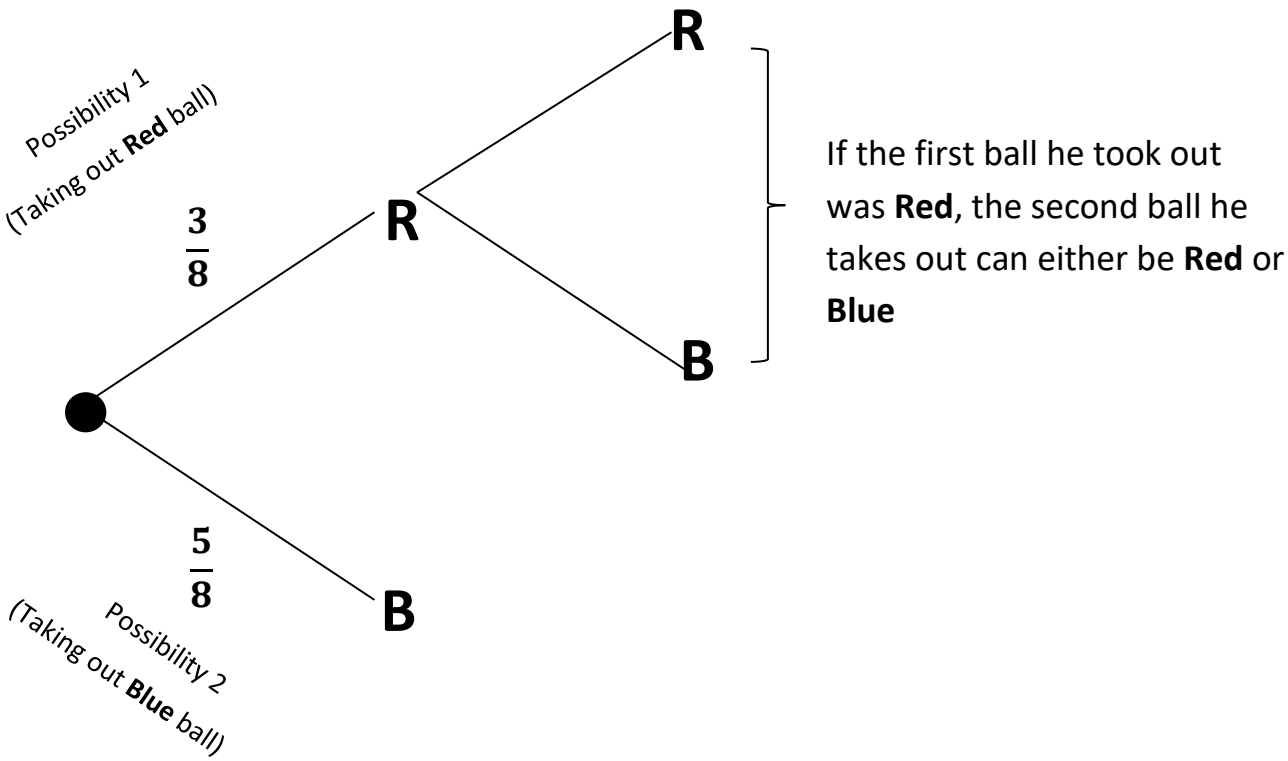
And if possibility 2 happens (the ball he takes out is Blue), there still be **3 Red balls** but **4 Blue balls** remaining in the sack. Let's add this information on our diagram.



Mathematicians call such a diagram a Tree-diagram.

After taking out the first ball, Eric decides to take out another ball.

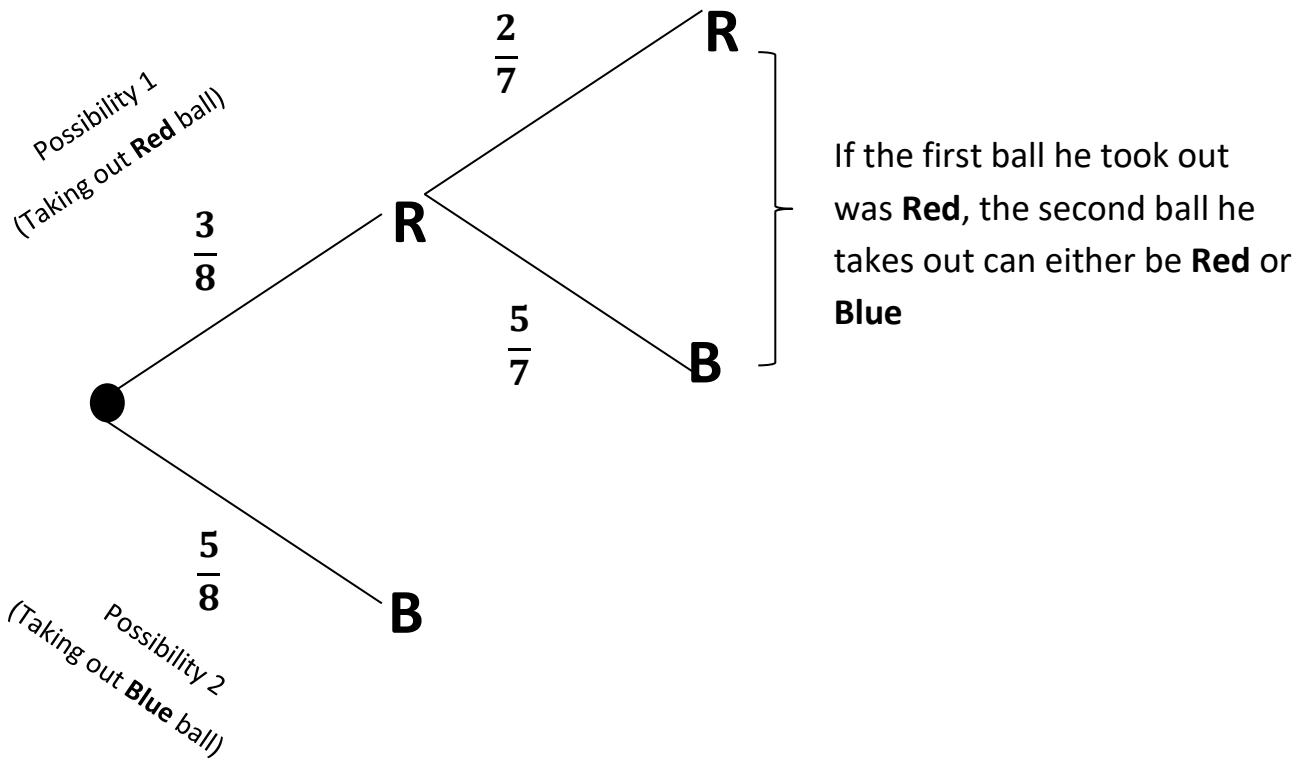
If Eric takes out a Red ball the first time of "taking out", he can then either take out a **Red** ball or **Blue** ball the second time of "taking out". We include this information on our tree-diagram below.



So if the ball he takes out the first time is **Red**, the probability that he takes out a **Red** ball the second time of taking out is $\frac{2}{7}$. This is because there were **2 Red** balls left in the sack out of a total of **7** balls.

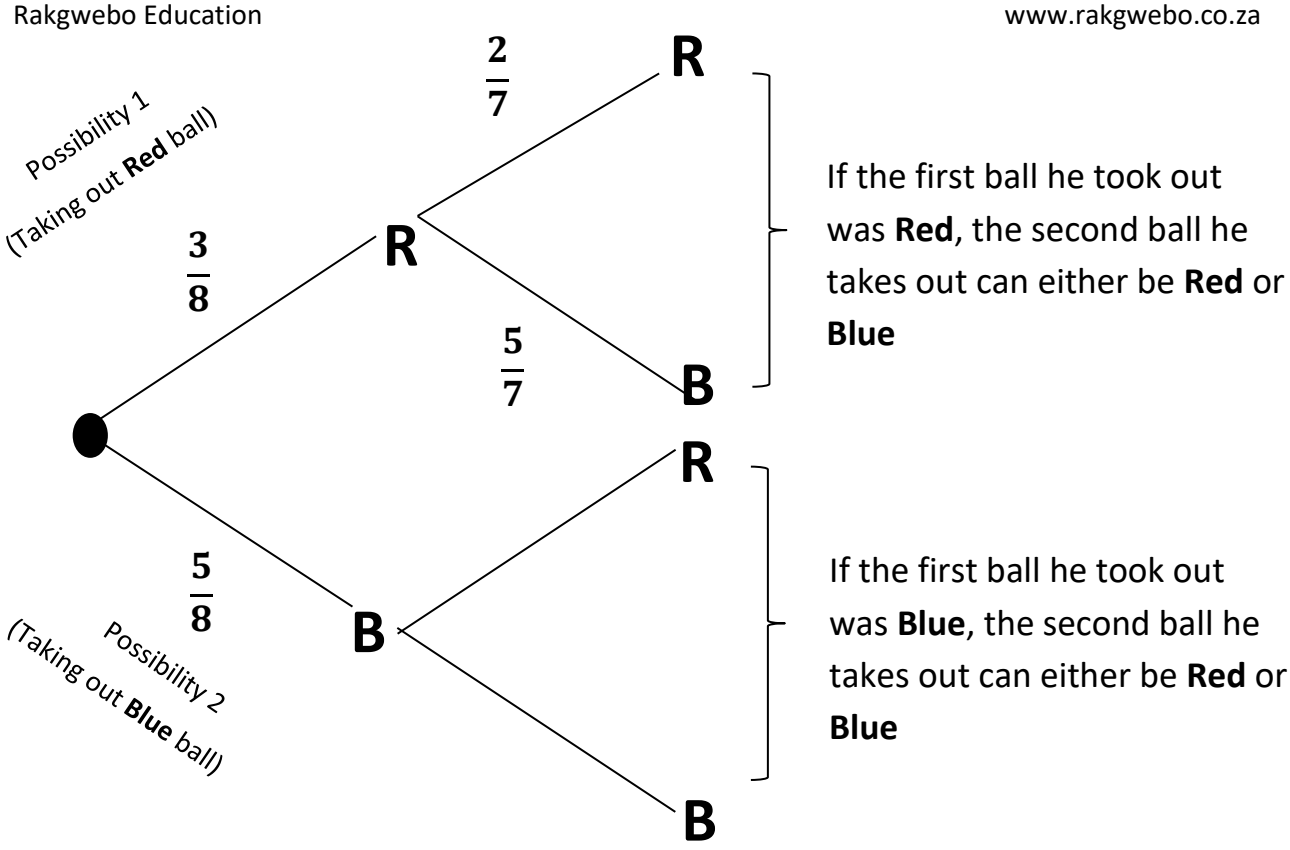
Or, if the ball he takes out the first time is **Red**, the probability that he takes out a **Blue** ball the second time of taking out is $\frac{5}{7}$. This is because there were still **5 Blue** balls left in the sack out of a total of **7** balls.

We add this information on our diagram.



But, what if Eric takes out a **Blue** ball the first time of "taking out"?

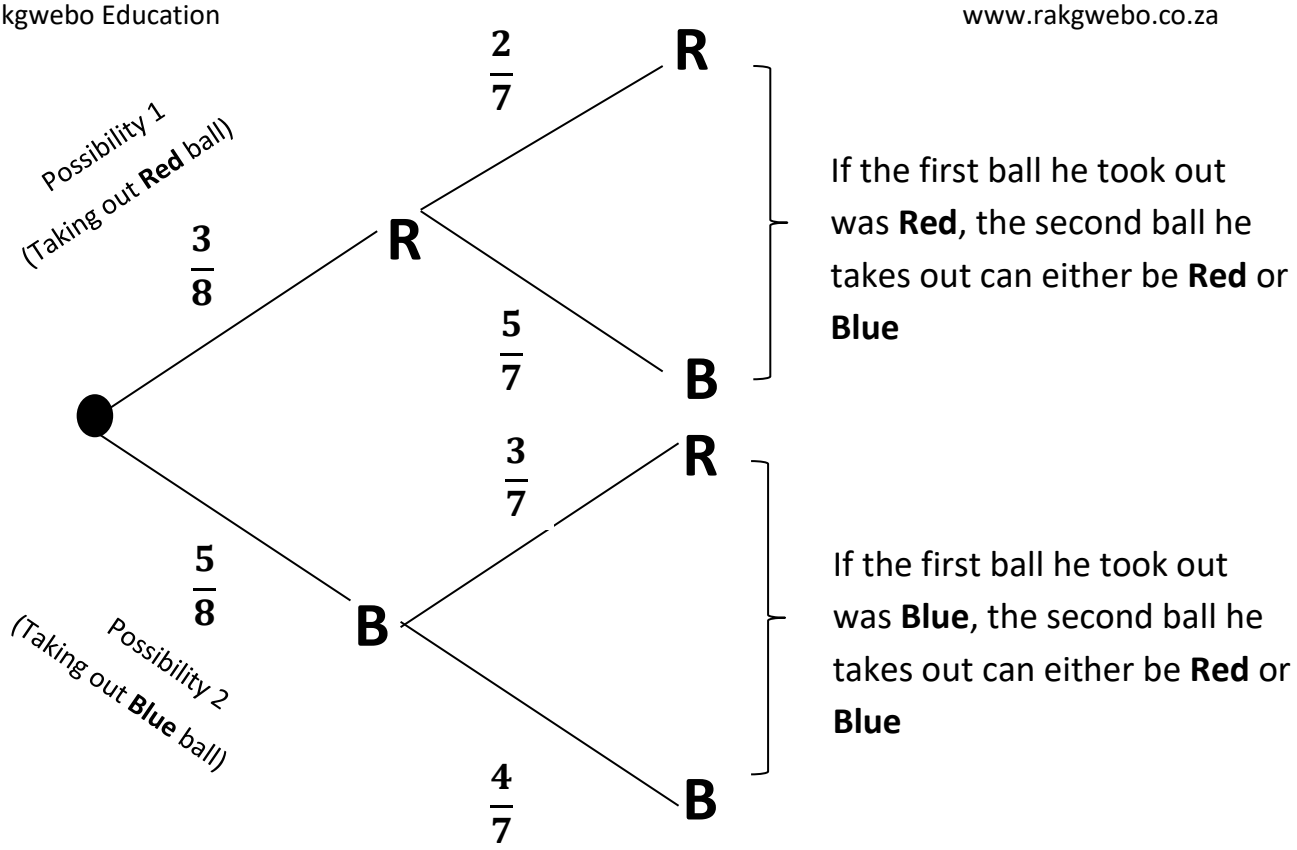
If that is the case and he decides to take out a second ball, the second ball he takes out can either be **Red** or **Blue**. We add this information on our diagram.



So if the ball he takes out the first time is **Blue**, the probability that he takes out a **Red** ball the second time of "taking out" is $\frac{3}{7}$. This is because there were still **3 Red** balls left in the sack out of a total of **7** balls.

Or, if the ball he takes out the first time is **Blue**, the probability that he takes out a **Blue** ball the second time of "taking out" is $\frac{4}{7}$. This is because there were **4 Blue** balls left in the sack out of a total of **7** balls.

We add this information on our diagram.



The Tree-diagram above shows all the possible outcomes and probabilities when Eric takes out a ball the first time and then later take out a second ball. A tree-diagram is important in such situations because it gives us a clear picture of a series of events of probability.

At the end of the first "taking out" and the second "taking out", there are four possible outcomes.

P1: **Red** ball first and **Red** ball second: **RR**

P2: **Red** ball first and **Blue** ball second: **RB**

P1: **Blue** ball first and **Red** ball second: **BR**

P1: **Blue** ball first and **Blue** ball second: **BB**

Let's include this information on our diagram.

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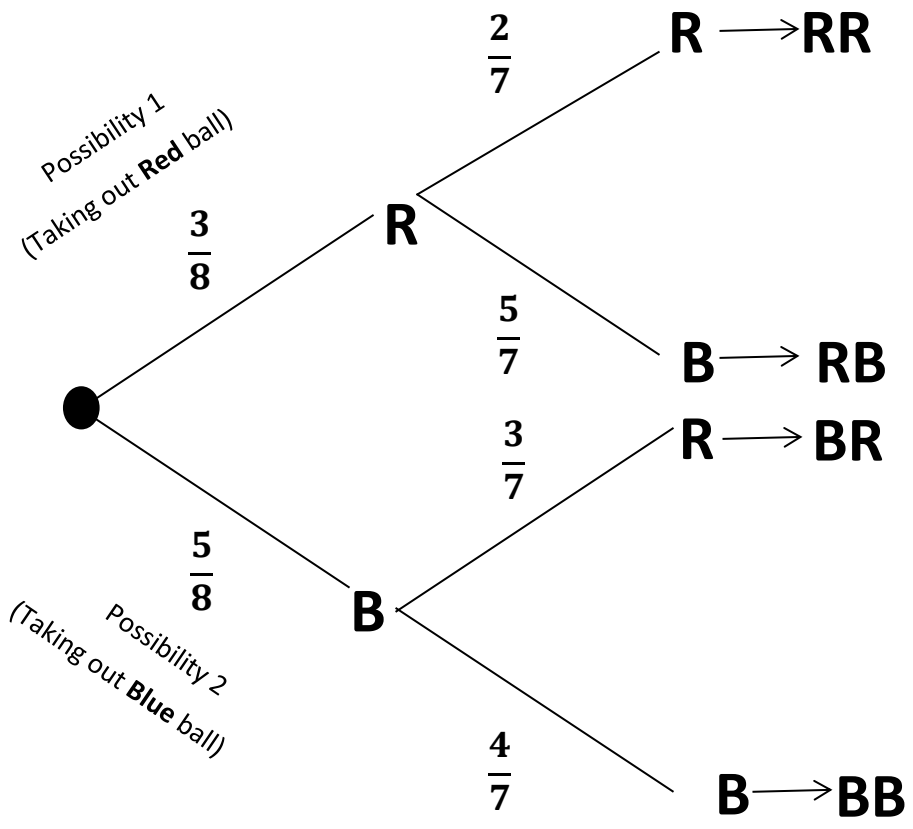
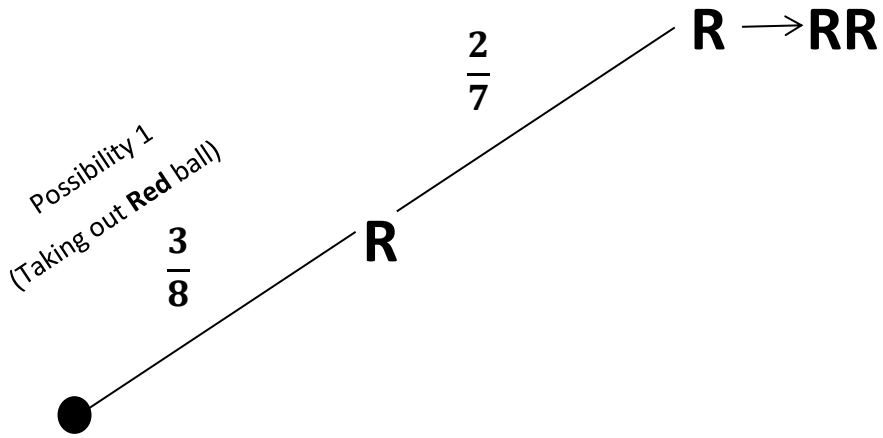


What is the probability that Eric takes out a **Red** ball the first time and another **Red** ball the second time of taking out?

Here we are looking for the probability of **RR**. Check the portion of **RR** on the tree-diagram above.



P(RR)= ? We simply multiply the probabilities above.

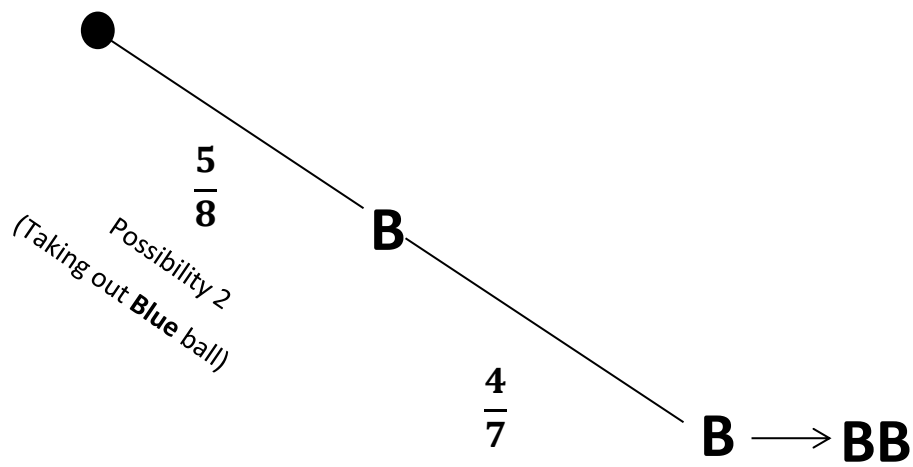
P(RR)= $\frac{3}{8} \times \frac{2}{7}$

$= \frac{6}{35}$

Therefore, the probability that Eric takes out a **Red** ball the first time and another **Red** ball the second time of taking out is $\frac{6}{35}$

What is the probability that Eric takes out a **Blue** ball the first time and another **Blue** ball the second time of taking out?

Here we are looking for the probability of **BB**. Check the portion of **BB** on the tree-diagram above.



P(BB)= ? We simply multiply the probabilities above.

P(BB)= $\frac{5}{8} \times \frac{4}{7}$

$= \frac{5}{14}$

Therefore, the probability that Eric takes out a **Blue** ball the first time and another **Blue** ball the second time of taking out is $\frac{5}{14}$

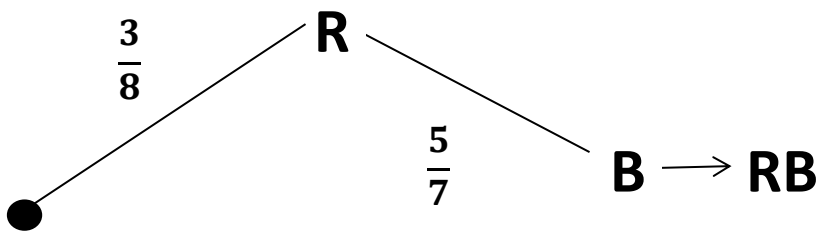
Question

a) What is the probability that Eric takes out a **Red** ball the first time and then a **Blue** ball the second time of taking out?

b) What is the probability that Eric takes out a **Blue** ball the first time and then a **Red** ball the second time of taking out?

Solution

a) Here we are looking for the probability of **RB**. Check the portion of **RB** on the tree-diagram.



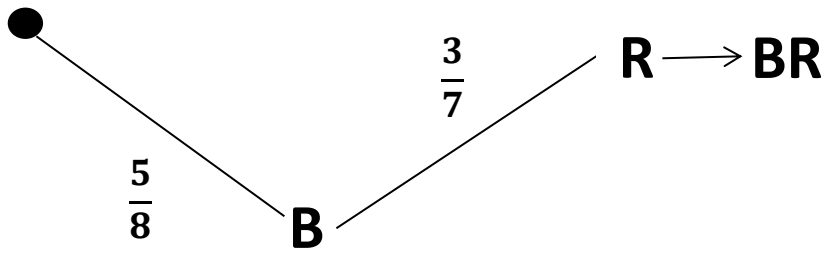
P(RB)= ? We simply multiply the probabilities above.

P(RB)= $\frac{3}{8} \times \frac{5}{7}$

$= \frac{15}{56}$

Therefore, the probability that Eric takes out a **Red** ball the first time and another **Blue** ball the second time of taking out is $\frac{15}{56}$

b) Here we are looking for the probability of **BR**. Check the portion of **BR** on the tree-diagram above.



P(BR)= ? We simply multiply the probabilities above.

P(BR)= $\frac{5}{8} \times \frac{3}{7}$

$= \frac{15}{56}$

Therefore, the probability that Eric takes out a **Blue** ball the first time and another **Red** ball the second time of taking out is $\frac{15}{56}$

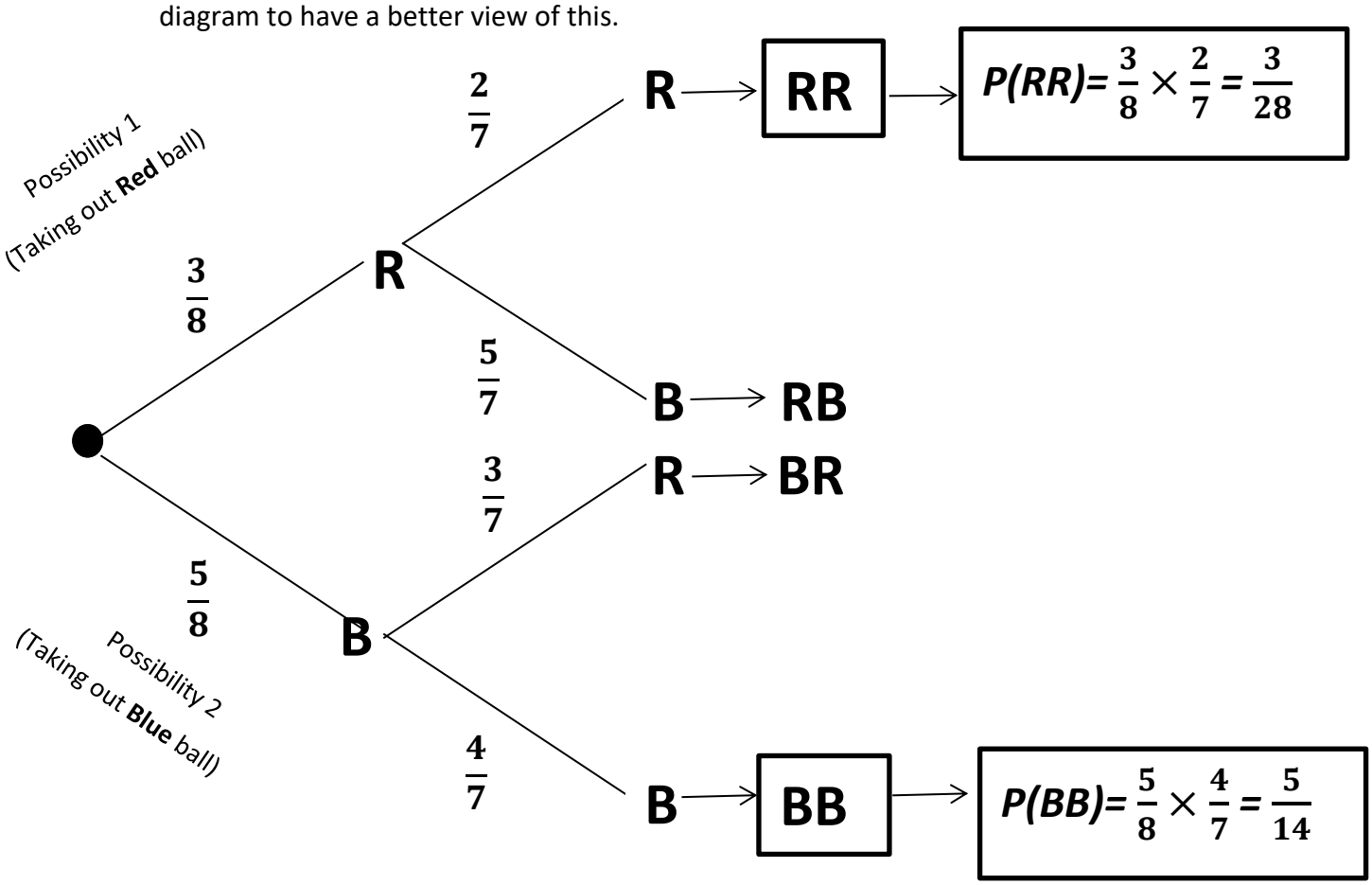
Question

a) What is the probability that Eric gets two balls of the same color at the end?

b) What is the probability that Eric gets one of each color at the end?

Solution

a) The question asking for the probability of getting either **RR** or **BB**. Let's redraw our tree-diagram to have a better view of this.



P(RR or BB)= ?

The ***P(RR)=*** $\frac{3}{28}$ and ***P(BB)=*** $\frac{5}{14}$ as you can see from the diagram above.

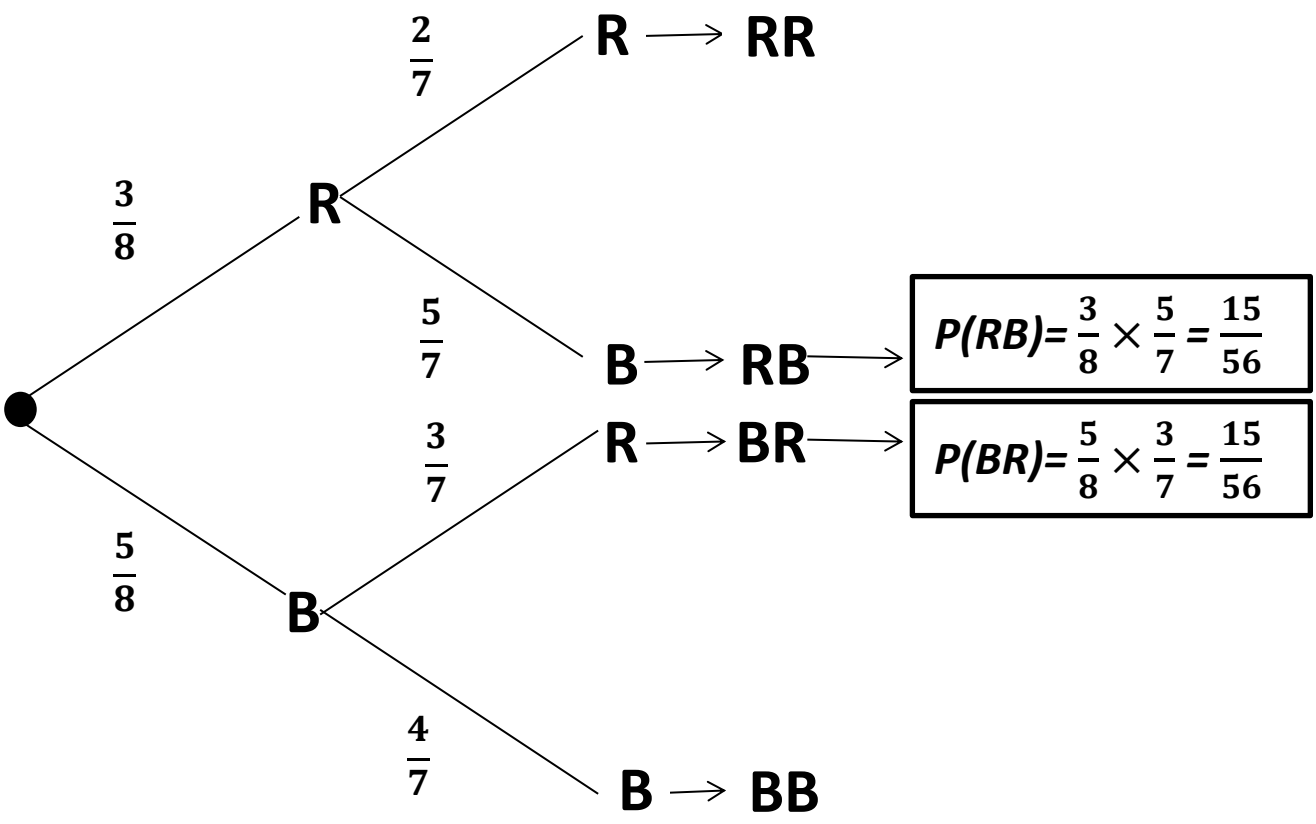
To get ***P(RR or BB),*** you simply add them.

P(RR or BB)= $\frac{3}{28} + \frac{5}{14}$

$= \frac{13}{28}$

b)

The question is asking for the probability of getting either **RB** or **BR**. Let's redraw our tree-diagram to have a better view of this.



P(RB or BR)= ?

The ***P(RB)=*** $\frac{15}{56}$ and ***P(BR)=*** $\frac{15}{56}$ as you can see from the diagram above.

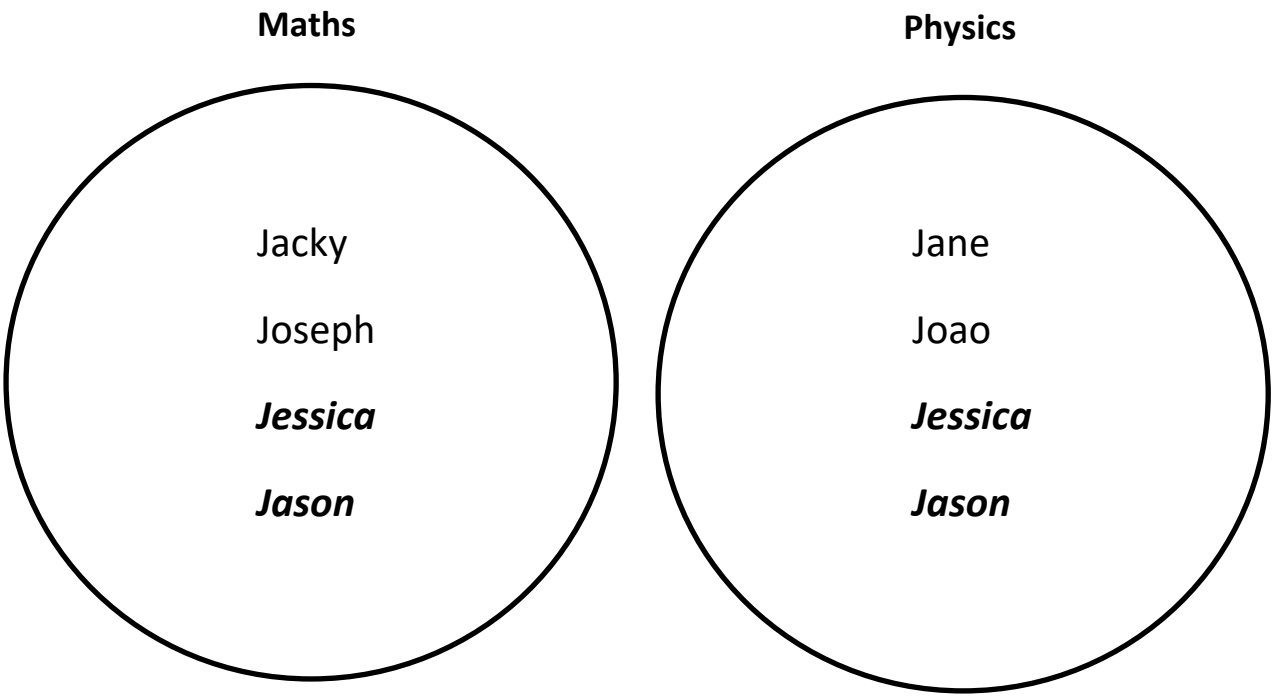
To get ***P(RB or BR),*** you simply add them.

P(RB or BR) = $\frac{15}{56} + \frac{15}{56} = \frac{15}{28}$

Venn-Diagrams

A High School has got only 6 students. Their names are Jacky, Joseph, Jessica, Jason, Jane and Joao. There are only two subjects in the school, Maths and Physics.

Some students do maths only, some do physics only, and some do both maths and physics. Below is a list of the students with the subjects they chose.



Note that Jessica and Jason do both Maths and Physics. That's why their names are listed in both classes.

Note that Jacky and Joseph are doing **maths only**.

Note that Jane and Joao are doing **physics only**.

There's a better way in which we can show that Jessica and Jason do both maths and physics. We see that below.

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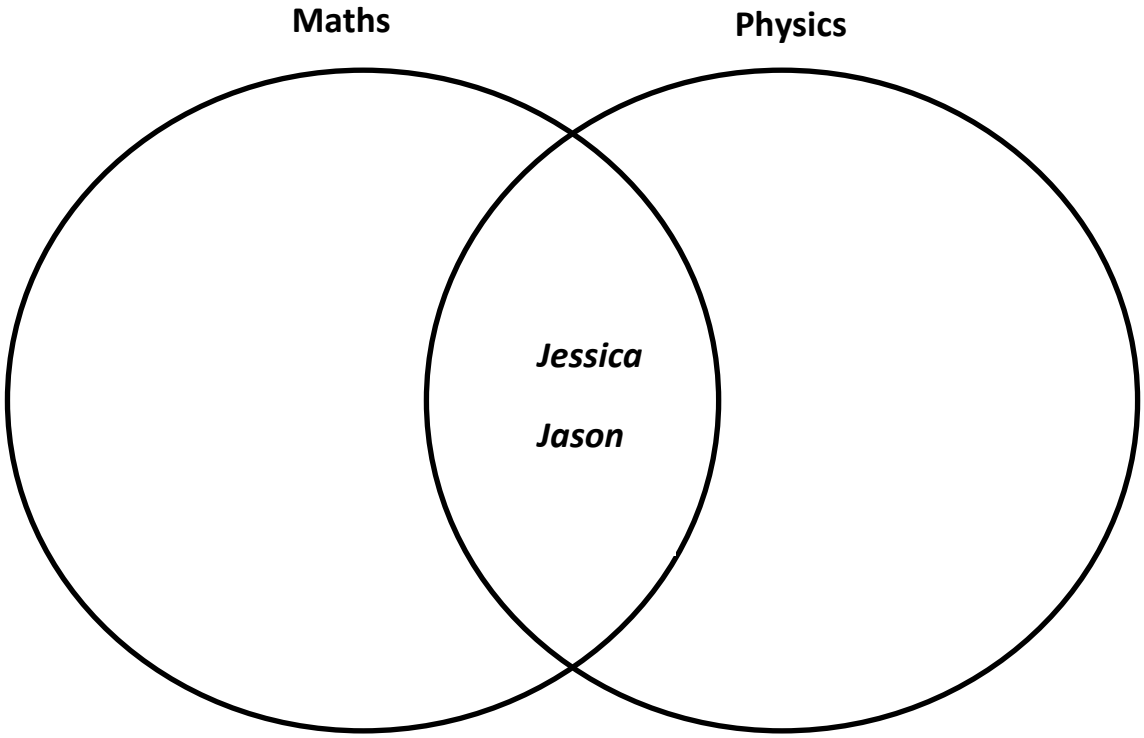
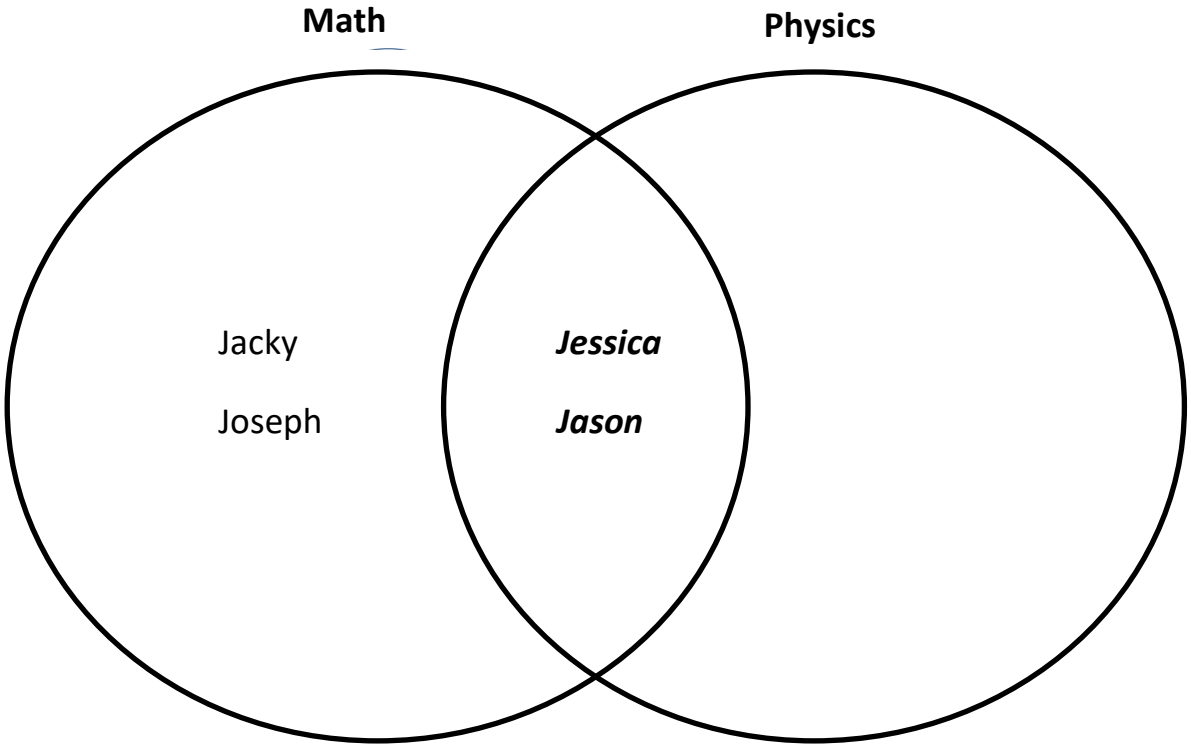
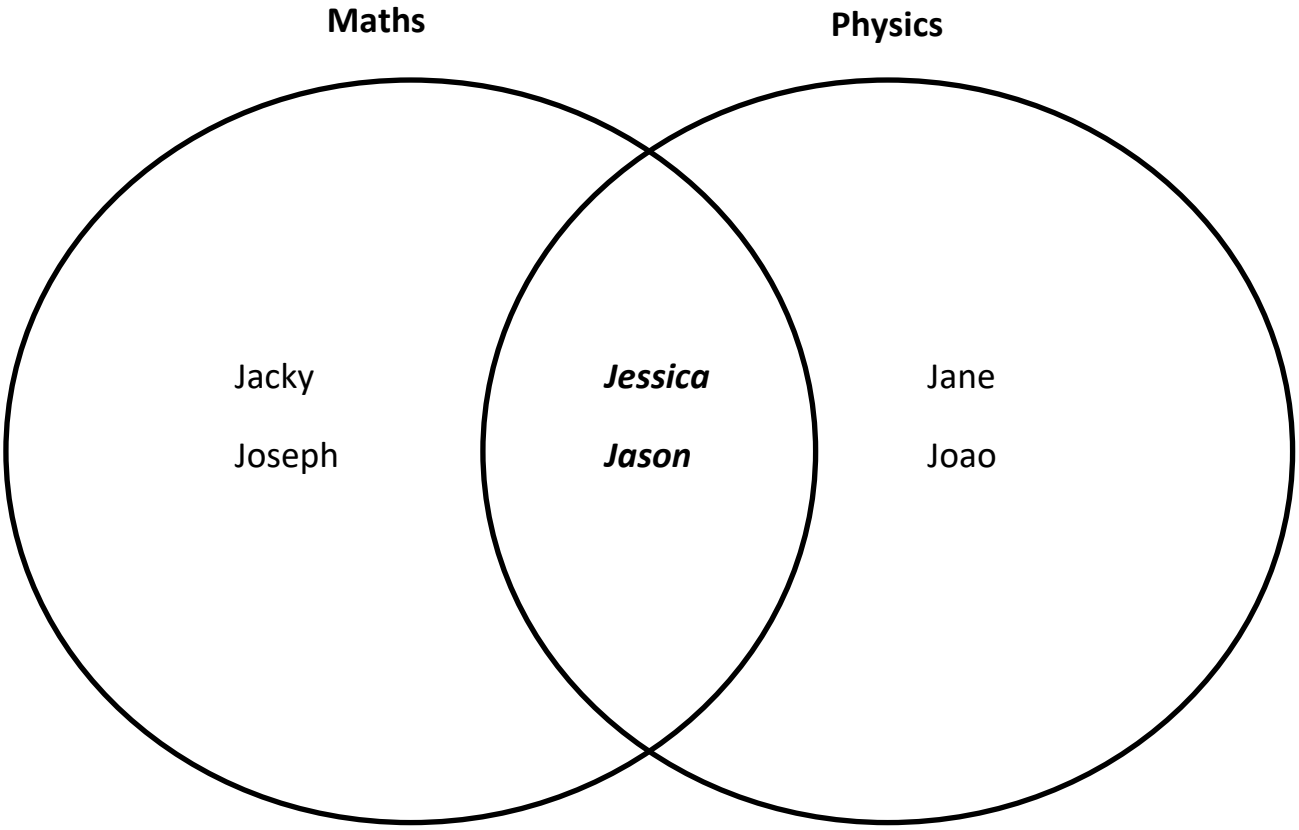


This diagram shows that Jessica and Jason belong to both the math class and the physics class.

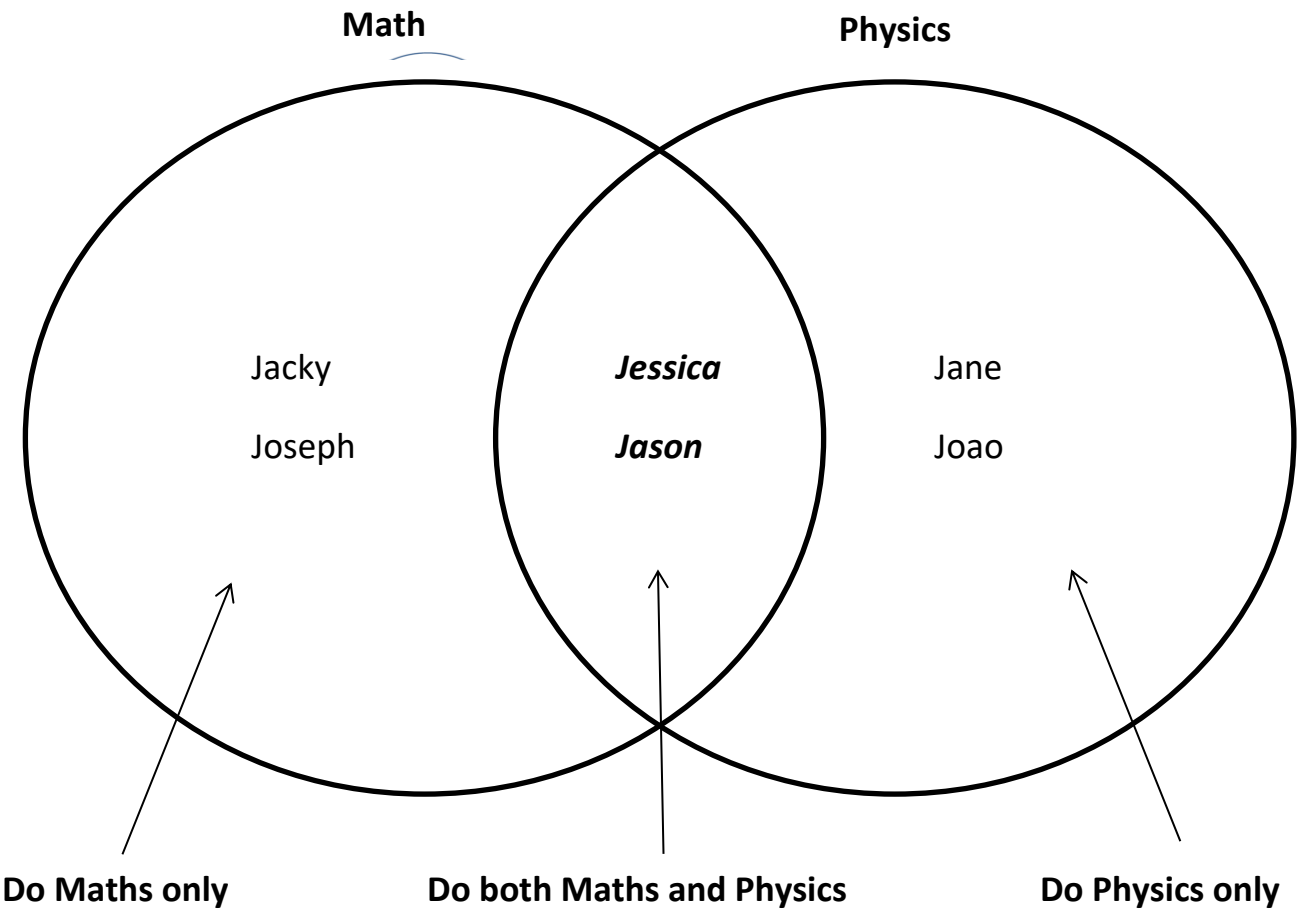
Now let's fill in the other names into this diagram. Jacky and Joseph do maths only, so list them in the maths "circle".



Now fill in for the students doing physics only. Jane and Joao do physics only.



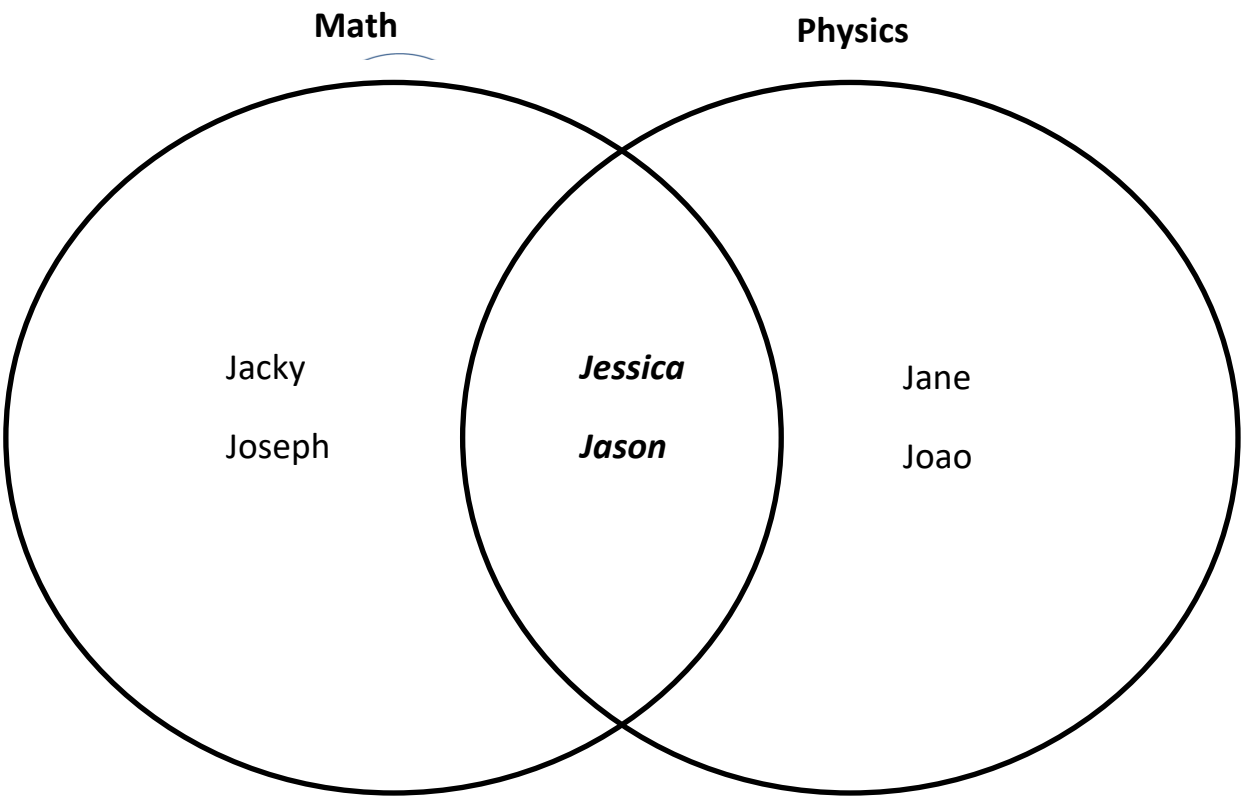
Let's further label the diagram to make it even clearer.



Mathematicians call such a diagram a Venn-diagram. Probability questions may be asked based on such diagrams. Attempt the example below.

Example

Again, consider the Venn-diagram on the school.



One student is chosen at random from the school, what is the probability that the student does **both** maths and physics?

Answer

Firstly, note that there is a total of "six" students in the school. Out of the "six" students, "two" students are doing both maths and physics (***Jessica and Jason***). Therefore, the probability that a student chosen at random does both maths and physics is "two" out of "six": $\frac{2}{6}$

Example

a) One student is chosen at random from the entire school. What is the probability that the student does maths "**only"**?

b) One student is chosen at random from the entire school. What is the probability that the student does maths?

Answer

a) Firstly, note that there is a total of "six" students in the school. Out of the "six" students, "two" students do maths "**only**" (Jacky and Joseph). We don't include Jessica and Jane because

they are doing both maths and physics, we are only interested in those doing maths **only**. Therefore, the probability that a student chosen at random does maths only is "two" out of "six": $\frac{2}{6}$

b) Note the difference between this question and question **a)** above. "a)" is interested in students doing "maths **only**", without doing physics too, while **b)** is interested in students doing maths in general, it doesn't matter whether or not they are doing physics too.

Out of a total of "six" students in the school, "four" students are doing maths (Jacky, Joseph, Jessica, Jason). Therefore, the probability that a student chosen at random is doing maths is "four" out of "six": $\frac{4}{6}$

Example

a) One student is chosen at random from the entire school. What is the probability that the student does physics "**only"**?

b) One student is chosen at random from the entire school. What is the probability that the student does physics?

Answer

a) Firstly, note that there is a total of "six" students in the school. Out of the "six" students, "two" students do physics "**only**" (Jane and Joao). We don't include Jessica and Jane because they are doing both maths and physics, we are only interested in those doing physics **only**. Therefore, the probability that a student chosen at random does physics only is "two" out of "six": $\frac{2}{6}$

b) Note the difference between this question and question **a)** above. "a)" is interested in students doing "physics **only**", without doing maths too, while **b)** is interested in students doing physics in general, it doesn't matter whether or not they are doing maths too.

Out of a total of "six" students in the school, "four" students are doing maths (Jane, Joao, Jessica, Jason). Therefore, the probability that a student chosen at random is doing physics is "four" out of "six": $\frac{4}{6}$

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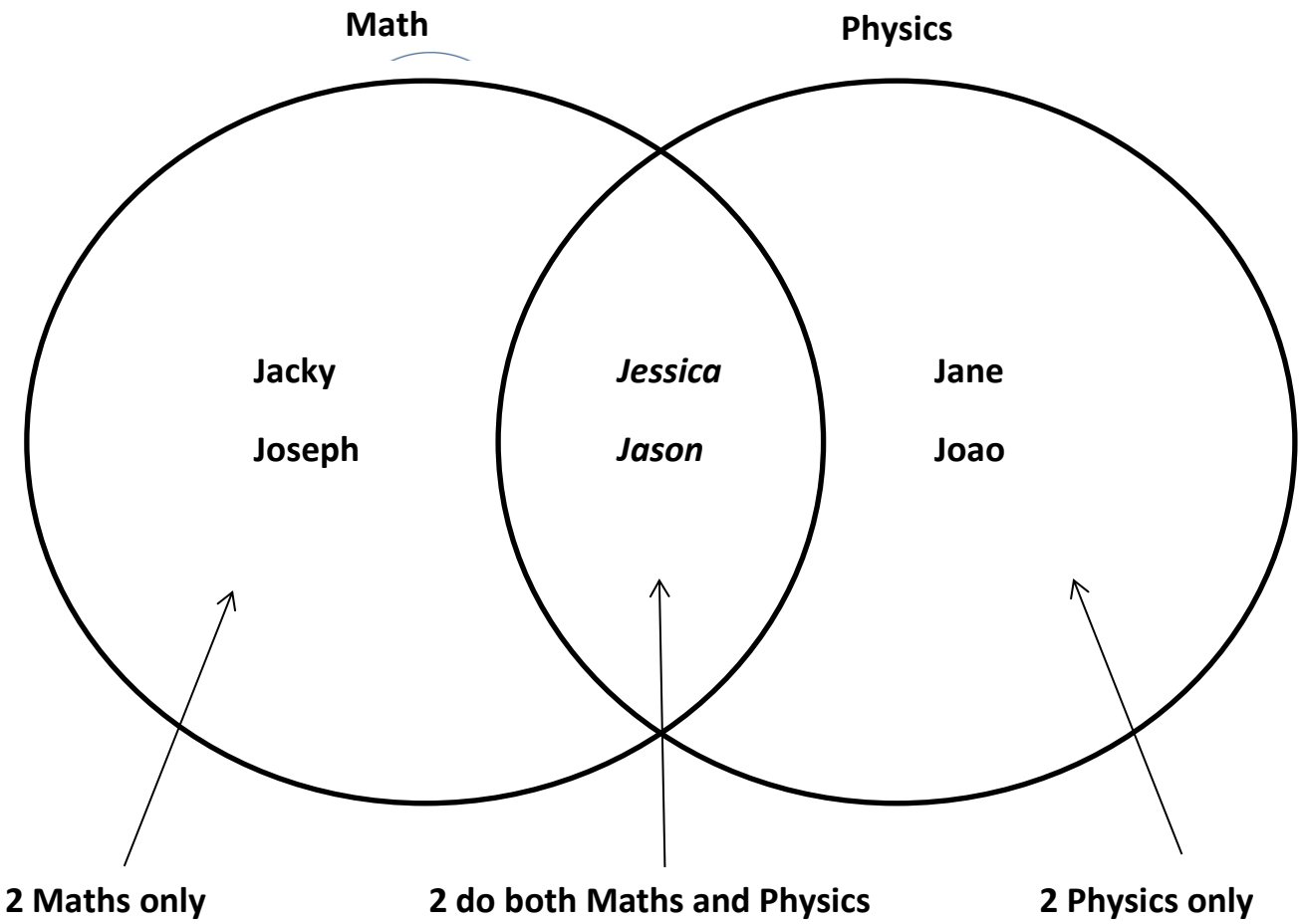
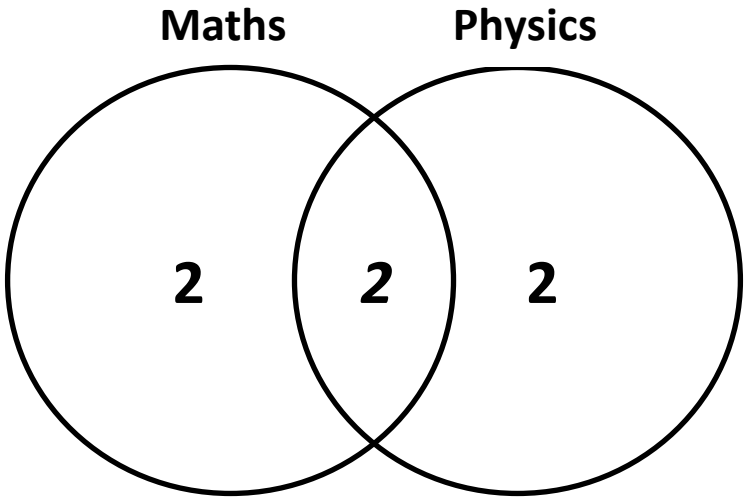
In the next section, we play around with our Venn-diagram by including numbers in the diagram instead of names.



Put numbers instead of the names on the diagram.

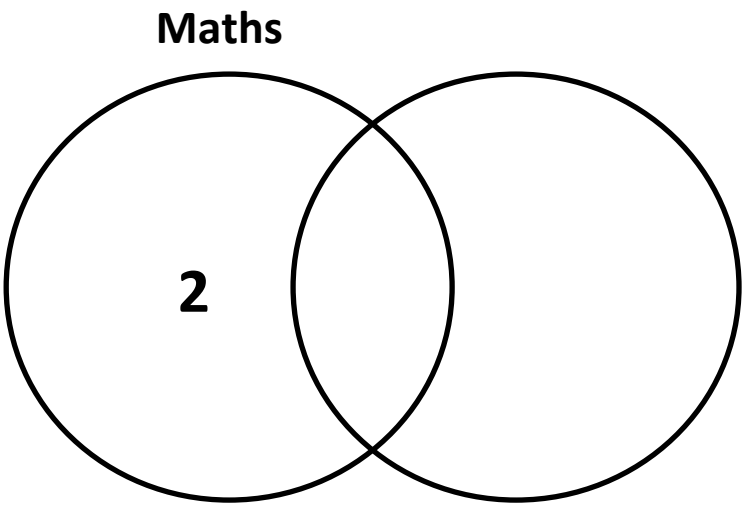


We now break this down to ensure that you fully understand it.

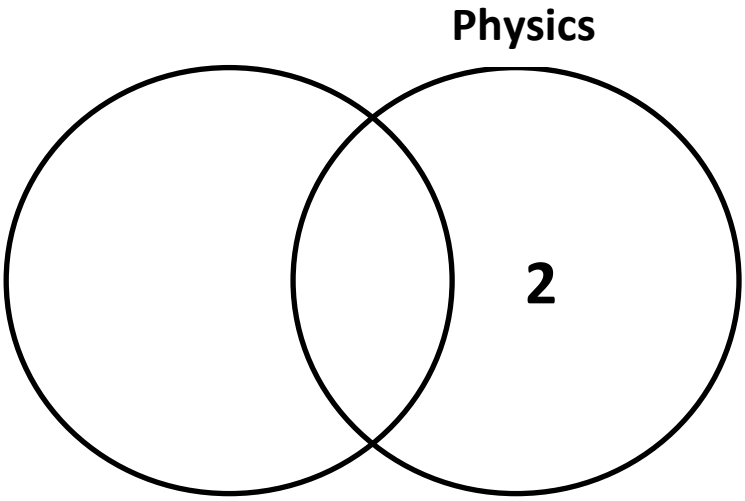
There is a total of “six” students in the school.

2 + 2 + 2 = 6

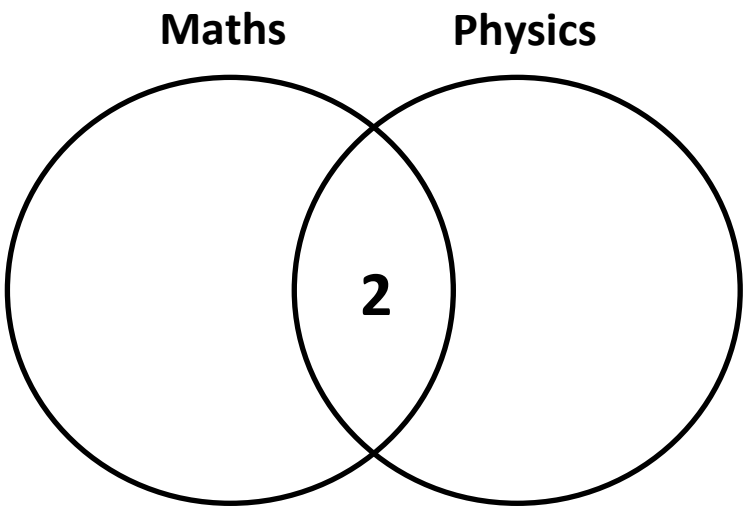
2 students are doing maths only and **no** physics, as shown below.



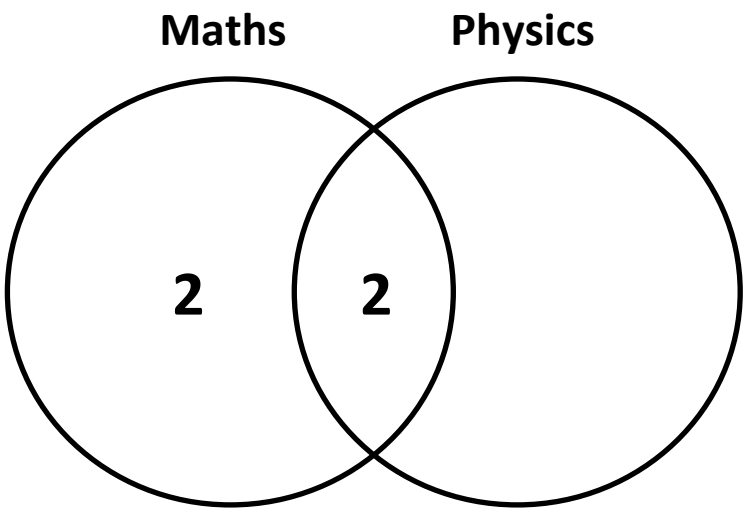
2 students are doing physics only and **no** maths, as shown below**.**



2 students are doing both maths and physics, as shown below.

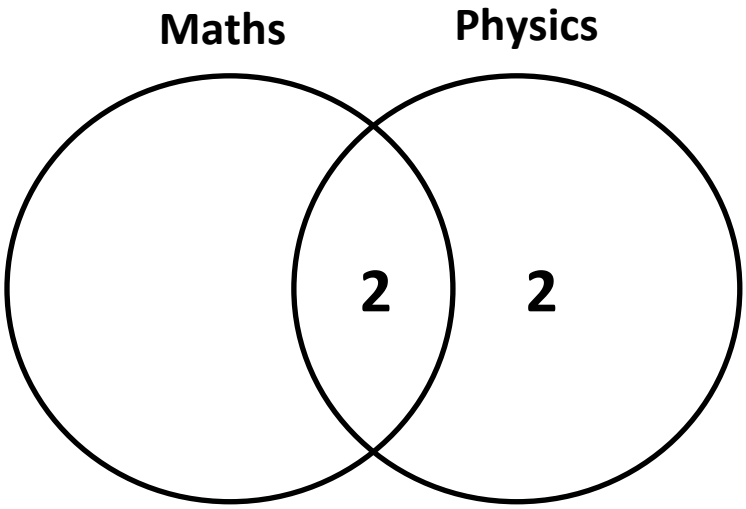


4 students are doing maths, as shown below.



2 + 2 = 4

4 students are doing physics.

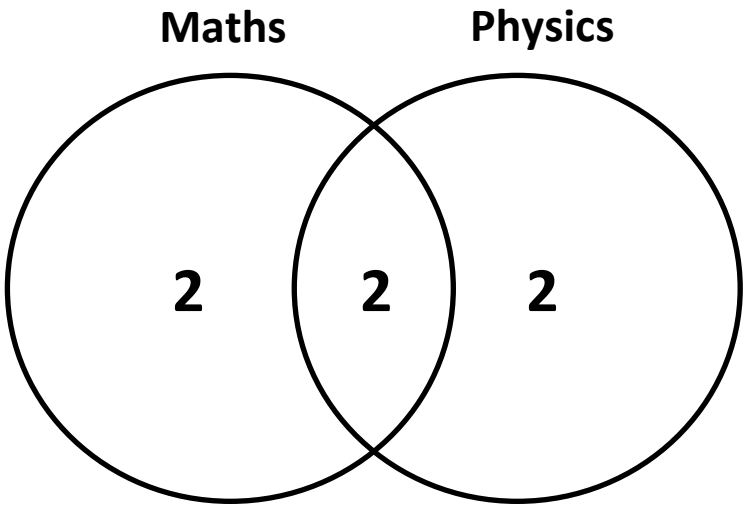


2 + 2 = 4

Attempt the following questions to check your understanding.

Question

Consider the Venn-diagram from the school.



a) A student is chosen at random. What is the probability that the student does **both** maths and physics?

b) A student is chosen at random. What is the probability that the student does **maths only**?

c) A student is chosen at random. What is the probability that the student does **physics only**?

Solution

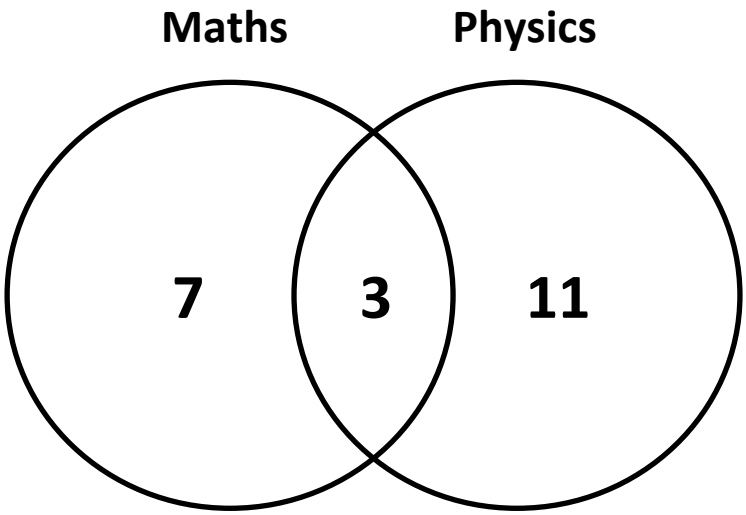
a) Firstly, note that there is a total of "six" students in the school. Out of the "six" students, "two" students are doing both maths and physics. Therefore, the probability that a student chosen at random does **both** maths and physics is "two" out of "six": $\frac{2}{6}$

b) Firstly, note that there is a total of "six" students in the school. Out of the "six" students, "two" students do maths "**only**". Remember, we are only interested in those doing maths **only**. Therefore, the probability that a student chosen at random does maths only is "two" out of "six": $\frac{2}{6}$

c) Firstly, note that there is a total of "six" students in the school. Out of the "six" students, "two" students do physics "**only**". Remember, we are only interested in those doing physics **only**. Therefore, the probability that a student chosen at random does physics only is "two" out of "six": $\frac{2}{6}$

Question

Consider the Venn-diagram below on some high school. The diagram shows the number of students doing **maths only**, those **physics only** and those doing **both** maths and physics.



a) What is the total number of students in the school?

b) What is the number of students doing **maths only**?

c) What is the number of students doing **physics only**?

d) What is the number of students doing both maths and physics?

Solution

a) **7** + **3** + **11** = **21**. There is a total of **21** students in the school.

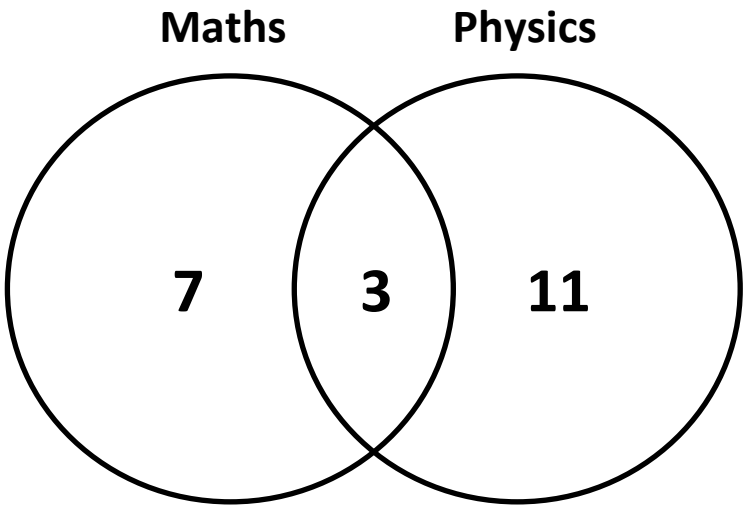
b) There are **7** students doing maths only.

c) There are **11** students doing physics only.

d) There are **3** students doing both maths and physics.

Question

Consider the same diagram.



a) A student is chosen at random, what is the probability that the student does **maths only**?

b) What is the probability that the student does **physics only**?

c) What is the probability that the student does **both** maths and physics?

Solution

a) Out of the 21 students in the school, 7 are doing maths only. Therefore, the probability that the student does maths only is 7 out of 21: $\frac{7}{21}$

b) Out of the 21 students in the school, 11 are doing physics only. Therefore, the probability that the student does maths only is 11 out of 21: $\frac{11}{21}$

c) Out of the 21 students in the school, 3 are doing **both** maths and physics. Therefore, the probability that the student does maths only is 3 out of 21: $\frac{3}{21}$

Contingency tables

A high school has got 31 students. There are only **two** subjects in the school, **Physics** and **Accounting**. A student can only do one of the subjects, you can't do both at the same time. The school has both males and females.

There are 12 males in the school. Of the 12 males, **5** do Physics and **7** do Accounting. This put this information into a table.

	Physics	Accounting
Male	5	7

5 males do Physics **7 males do Accounting**

There are 19 males in the school. Of the 12 males, **9** do Physics and **10** do Accounting. This put this information into a table.

	Physics	Accounting
Female	9	10

9 females do Physics **10 females do Accounting**

Let's combine these two tables above in to one table, without changing anything.

	Physics	Accounting
Male	5	7
Female	9	10

We now include the totals on males and females.

	Physics	Accounting	Total
Male	5	7	***12***
Female	9	10	***19***

Include the totals on each subject.

	Physics	Accounting	**Total**
Male	5	7	***12***
Female	9	10	***19***
Total	***14***	***17***	**31**

5+9=14 students do physics

7+10=17 students do accounting

12+19=31. There are 31 students in the school

Mathematicians call such a table a Contingency table. Probability questions can be asked based on such a table.

Example

Consider the table again.

	Physics	Accounting	**Total**
Male	5	7	***12***
Female	9	10	***19***
Total	***14***	***17***	**31**

a) Out of the total of 31 students in school, a student is chosen at random. What is the probability that the student is a male?

b) Out of the total of 31 students in school, a student is chosen at random. What is the probability that the student is a female?

Answer

a)

	Physics	Accounting	**Total**
Male	5	7	***12***
Female	9	10	***19***
Total	***14***	***17***	**31**

There are 31 students in the school

There is a total of 12 males

Out of the 31 students, there are 12 males. Therefore, the probability that a student chosen at random is a male is 12 out of 31: $\frac{\mathbf{12}}{\mathbf{31}}$

$$\boldsymbol{P(male) = \frac{12}{31}}$$

b)

	Physics	Accounting	**Total**
Male	5	7	***12***
Female	9	10	***19***
Total	***14***	***17***	**31**

There are 31 students in the school

There is a total of 19 females

Out of the 31 students, there are 19 females. Therefore, the probability that a student chosen at random is a female is 19 out of 31: $\frac{\mathbf{19}}{\mathbf{31}}$

$$\boldsymbol{P(female) = \frac{19}{31}}$$

Question

Consider the contingency table again.

	Physics	Accounting	**Total**
Male	5	7	***12***
Female	9	10	***19***
Total	***14***	***17***	**31**

a) A student is chosen at random. What is the probability that the student does **physics**?

b) A student is chosen at random. What is the probability that the student does **accounting**?

c) A student is chosen at random. What is the probability that the student is a **male** who does **physics**?

d) A student is chosen at random. What is the probability that the student is a **female** who does **physics**?

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e) A student is chosen at random. What is the probability that the student is a **male** who does **accounting**?

f) A student is chosen at random. What is the probability that the student is a **female** who does **accounting**?

Solution

a)

	Physics	Accounting	**Total**
Male	5	7	***12***
Female	9	10	***19***
Total	***14***	***17***	**31**

There are 14 students doing physics

Out of the 31 students, there are 14 students doing physics. Therefore, the probability that a student chosen at random does physics is 14 out of 31: $\frac{\mathbf{14}}{\mathbf{31}}$

$$\boldsymbol{P(physics) = \frac{14}{31}}$$

b)

	Physics	Accounting	**Total**
Male	5	7	***12***
Female	9	10	***19***
Total	***14***	***17***	**31**

There are 17 students doing accounting

Out of the 31 students, there are 17 students doing accounting. Therefore, the probability that a student chosen at random does accounting is 17 out of 31: $\frac{17}{31}$

$$P(accounting) = \frac{17}{31}$$

c)

	Physics	Accounting	**Total**
Male	5	7	***12***
Female	9	10	***19***
Total	***14***	***17***	**31**

There are 5 males doing physics

Out of the 31 students, there are 5 males doing physics. Therefore, the probability that a student chosen at random is a male doing physics is 5 out of 31: $\frac{5}{31}$

$$P(male\ and\ does\ physics) = \frac{5}{31}$$

d)

	Physics	Accounting	**Total**
Male	5	7	***12***
Female	9	10	***19***
Total	***14***	***17***	**31**

There are 9 females doing physics

Out of the 31 students, there are 9 females doing physics. Therefore, the probability that a student chosen at random is a female doing physics is 9 out of 31: $\frac{9}{31}$

$$P(female\ and\ does\ physics) = \frac{9}{31}$$

e)

	Physics	Accounting	**Total**
Male	5	7	***12***
Female	9	10	***19***
Total	***14***	***17***	**31**

There are 7 males doing accounting

Out of the 31 students, there are 7 males doing accounting. Therefore, the probability that a student chosen at random is a male doing accounting is 7 out of 31: $\frac{7}{31}$

$$\boldsymbol{P(male\ and\ does\ accounting) = \frac{7}{31}}$$

f)

	Physics	Accounting	**Total**
Male	5	7	***12***
Female	9	10	***19***
Total	***14***	***17***	**31**

There are 10 females doing accounting

Out of the 31 students, there are 10 females doing accounting. Therefore, the probability that a student chosen at random is a female doing accounting is 10 out of 31: $\frac{10}{31}$

$$\boldsymbol{P(female\ and\ does\ accounting) = \frac{10}{31}}$$

Statistics

Statistics

10 students write a maths test and later have their marks published as shown below.

Name	Mark
Jason	***24***
Jacky	***35***
Jadon	***53***
Eric	***52***
Jane	***57***
Mbali	***60***
Sipho	***67***
Jessica	***78***
Lebo	***79***
Rakgwebo	***95***

Question

a) How many students got a mark between 20 and 40?
b) How many students got a mark between 40 and 60?
c) How many students got a mark between 60 and 80?
d) How many students got a mark between 80 and 100?

Solution

a) There are 2 students who got a mark between 20 and 40, that's Jason and Jacky. Note that when we say a mark between 20 and 40 we exclude the 20 but include the 40. For example, if someone gets a mark of 20, they don't fall "between 20 and 40", but if someone gets a mark of 40, they do fall "between 20 and 40". A mathematical way of writing this is; ***(20 ; 40].***

b) There are 4 students who got a mark between 40 and 60, that's Jadon, Eric, Jane, Mbali. This interval is represented by ***(40 ; 60]***. Remember that we exclude 40 but include 60. For example, if someone gets a mark of 40, they don't fall "between 40 and 60", but if someone gets a mark of 60, they do fall "between 40 and 60".

c) There are 3 students who got a mark between 60 and 80, that's Sipho, Jessica, Lebo. This interval is represented by ***(60 ; 80]***.

d) There is only 1 student who got a mark between 80 and 100, that's Rakgwebo, highlighted in red on the table above. This interval is represented by ***(80 ; 100]***.

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In order to have a better view of what's going on, let's put these numbers in a table. For example, we put 2 next to "between 20 and 40" because 2 students got a mark between 20 and 40, we do the same for others.

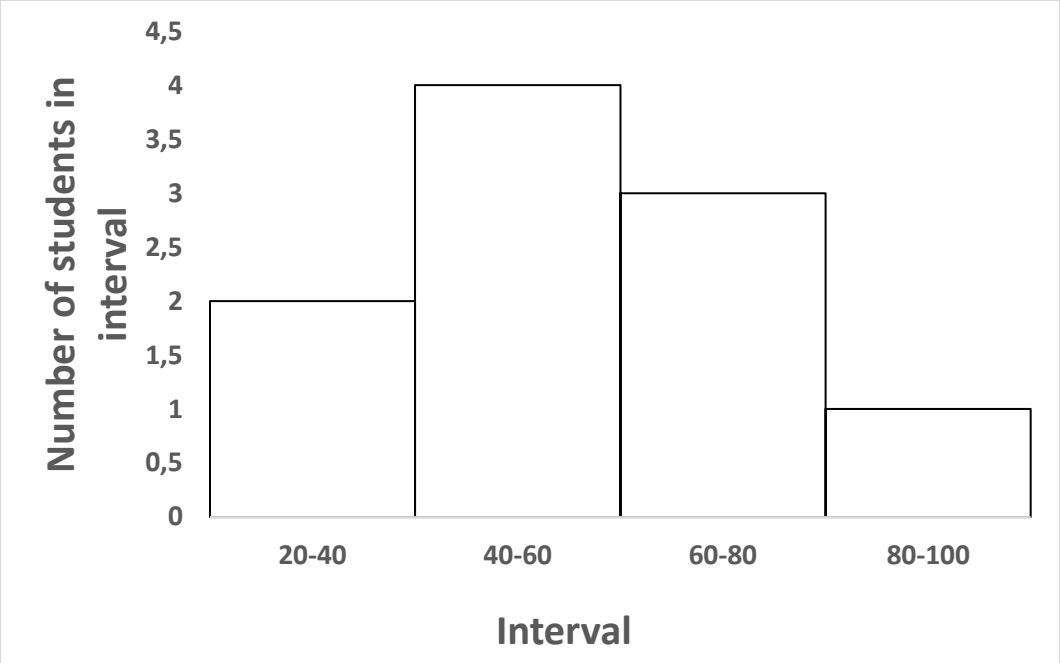
Interval	**Number of students in the interval**
Between **20** and **40**	**2**
Between **40** and **60**	**4**
Between **60** and **80**	**3**
Between **80** and **100**	**1**

I'm sure you'd agree with me that writing ***(20 ; 40]*** is better than writing "Between 20 and 40" in the table. So let's replace all of them to make the table look better.

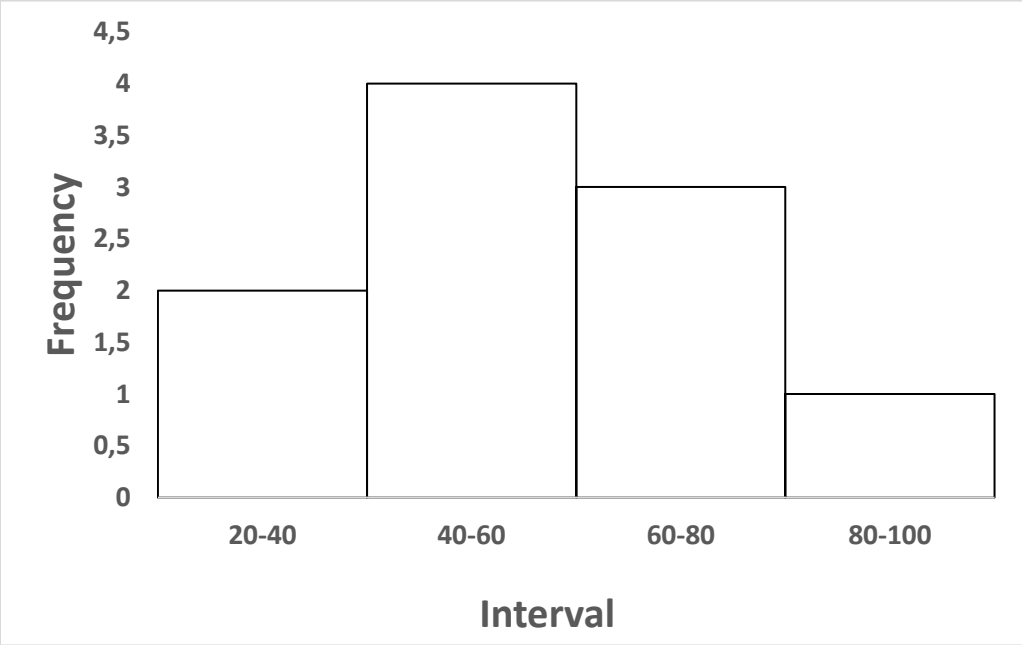
Interval	**Number of students in the interval**
(20 ; 40]	**2**
(40 ; 60]	**4**
(60 ; 80]	**3**
(80 ; 100]	**1**

As a maths student, you should always have the urge to draw a graph every time you see a table of numbers. This gives you a better picture of what's going.

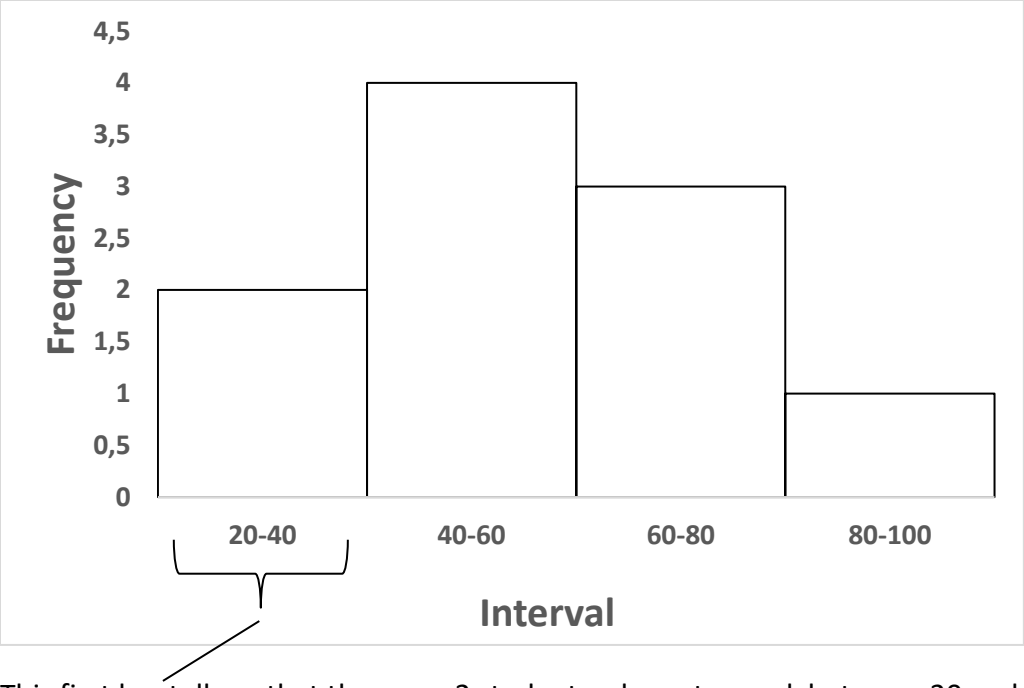
We put the "interval values" on the x-axis and the "number of students in the interval" on the y-axis. A bar graph of this information is drawn below.



The label of the y-axis is too wordy, the word frequency is used in place of "number of students in interval". Frequency generally means the number of times something happens. So we relabel the graph.



Now that you have an idea of how the graph looks like, let's do a quick breakdown of how the graph is set up.



This first bar tells us that there are 2 students who got a mark between 20 and 40.

The second bar tells us that there are 4 students who got a mark between 40 and 60.

The third bar tells us that there are 3 students who got a mark between 60 and 80.

The fourth bar tells us that there is only 1 student who got a mark between 80 and 100.

This type of bar graph is called a **histogram.** As you can see, this graph gives a clear picture of what's going on.

Again, there's another form of graph that gives a nicer picture of what's going on. We get this kind of graph by drawing dots at the top of each bar.

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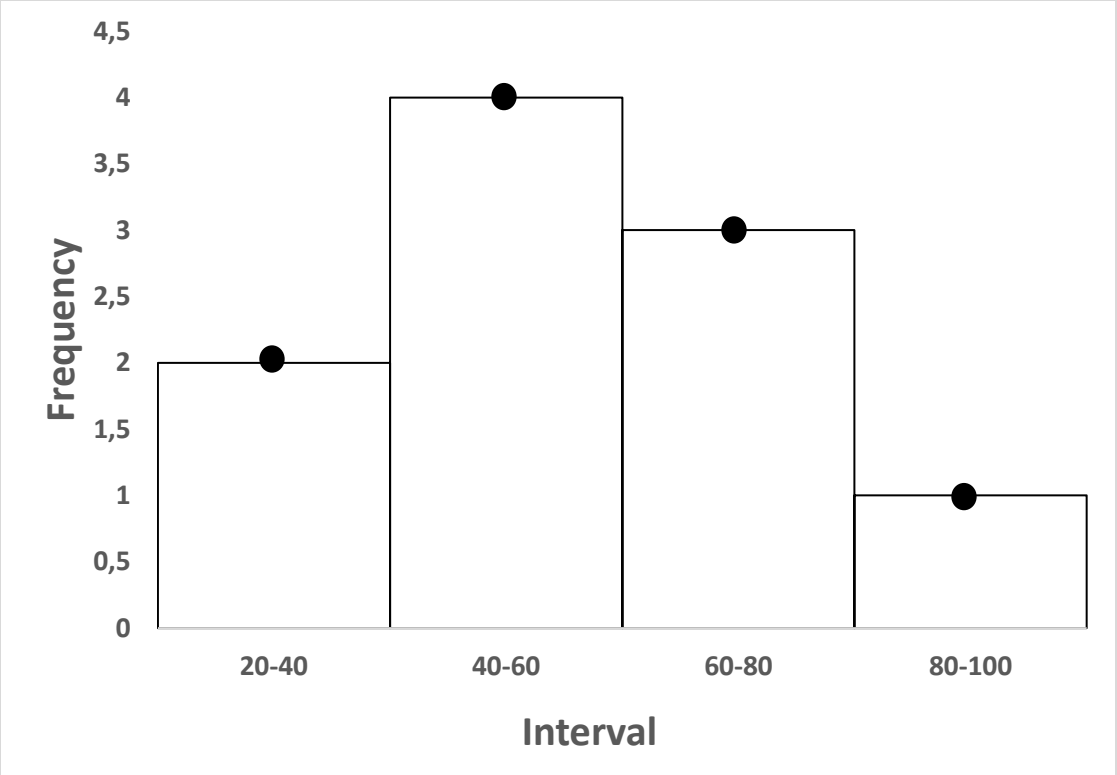
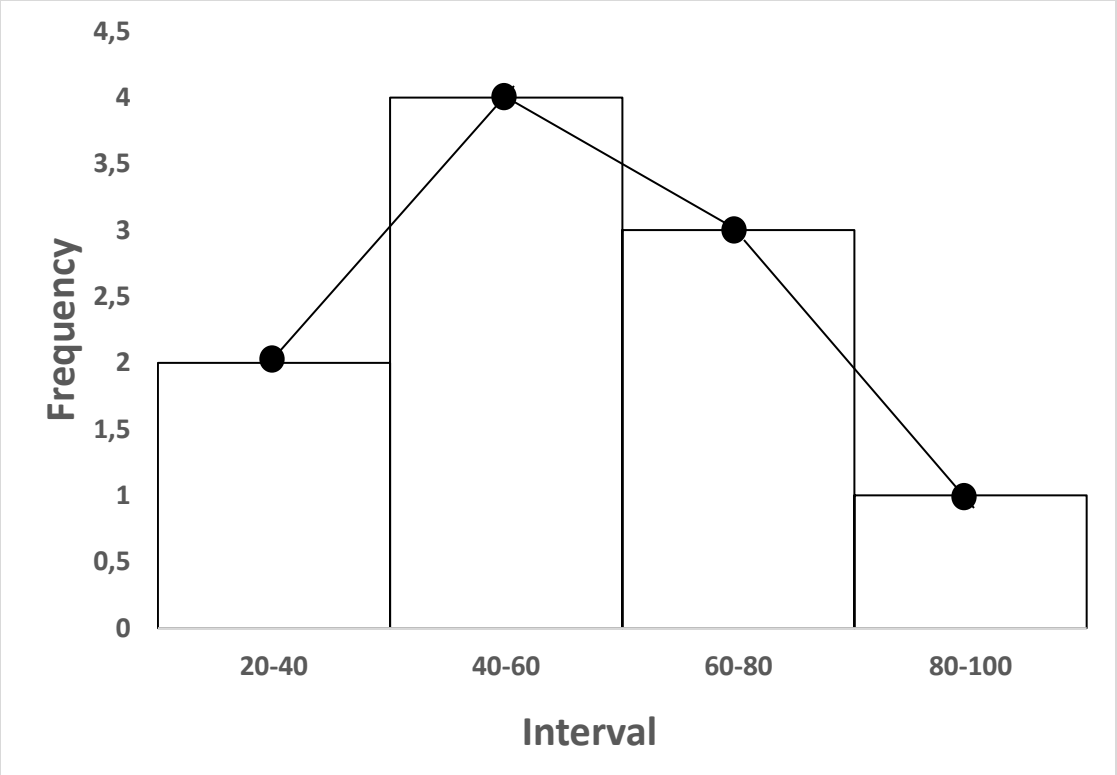
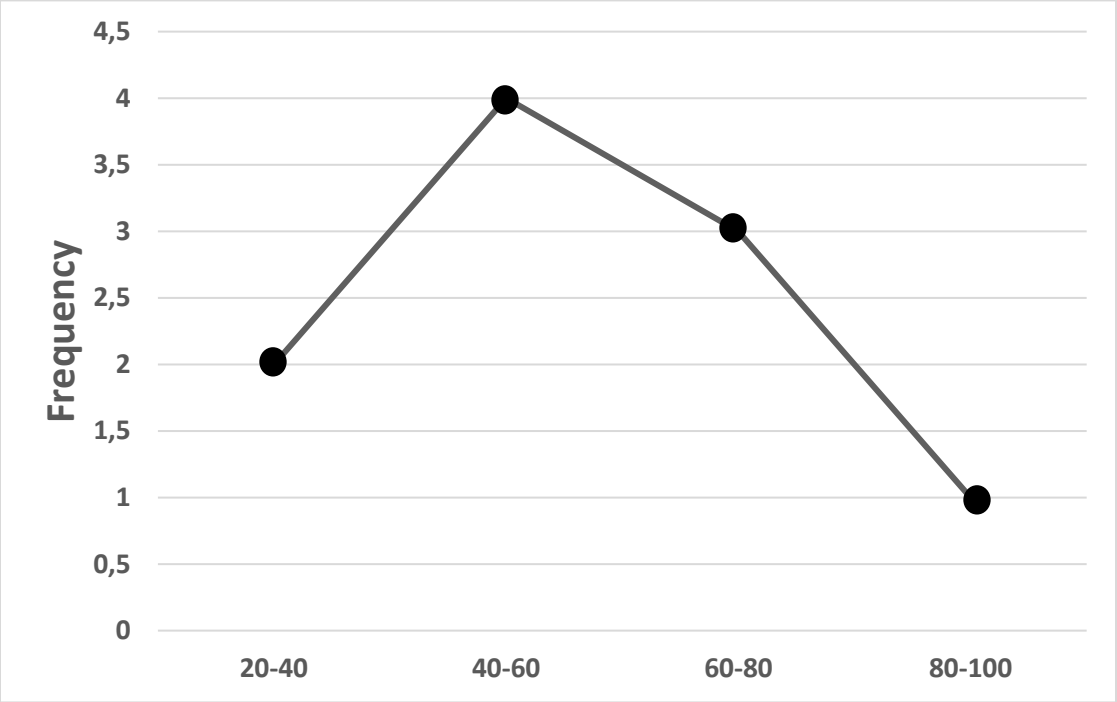


Notice that these dots are literally put in the top and middle of each bar. We then join the dots with straight lines between them.

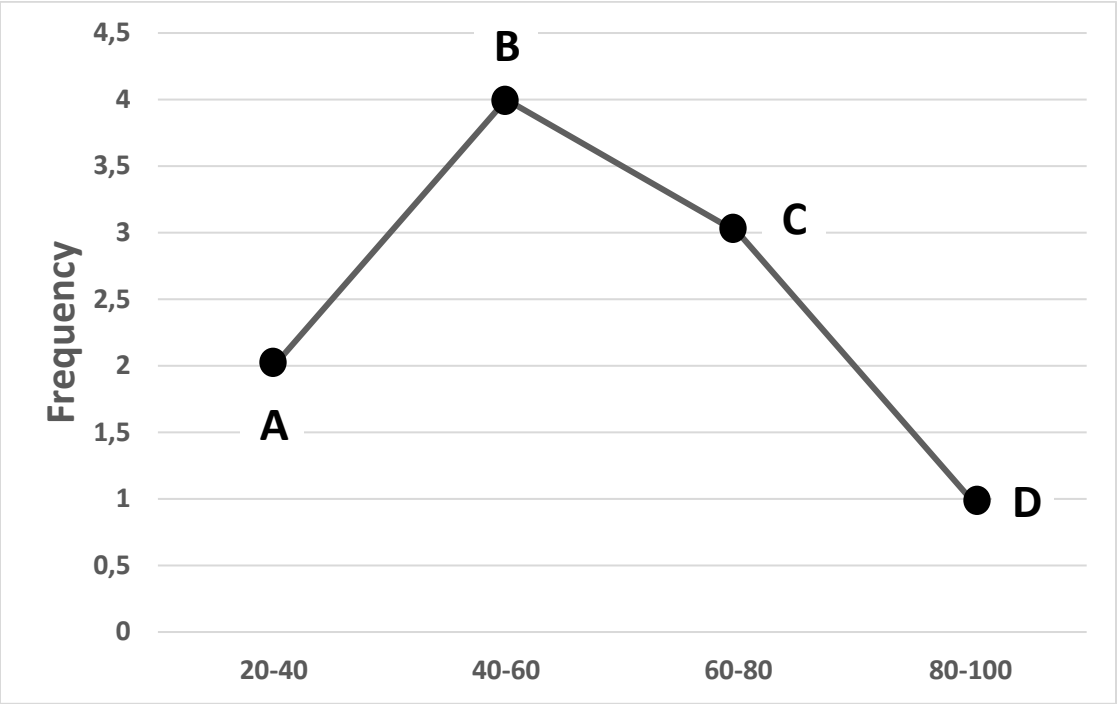


Now we remove the bars so that we only remain with the line.



Mathematicians call such a line a frequency polygon.

At the moment, you have a rough idea of what a frequency polygon is. Now let's do a quick breakdown of the frequency polygon.



What are the coordinates of point A? Well, the *y-coordinate* is clearly 2. As for the x-coordinate, it is literally in the middle of 20 and 40. 30 is the number that is literally in the middle of 20 and 40. So the *x-coordinate* of point A is 30. Therefore, the coordinates of point A are (30;2).

Another way saying that 30 is in the middle of 20 and 40; "30 is the midpoint of 20 and 40". We calculate the midpoint of 20 and 40 by: $\frac{20+40}{2} = 30$.

What are the coordinates of point B?

Well, the y-coordinate is clearly 4. As for the x-coordinate, it is literally in the middle of 40 and 60. 50 is the number that is literally in the middle of 40 and 60. Meaning it is the midpoint of 40 and 60: $\frac{40+60}{2} = 50$

So the x-coordinate of B is 50. Therefore, the coordinates of B are (50;4)

What are the coordinates of point C?

The y-coordinate is clearly 3. As for the x-coordinate, it is in the middle of 60 and 80. 70 is the number that is in the middle of 60 and 80. Meaning it is the midpoint of 60 and 80: $\frac{60+80}{2} = 70$

So the x-coordinate of C is 70. Therefore, the coordinates of C are (70;3)

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What are the coordinates of point D?

The y-coordinate is 1. As for the x-coordinate, it is in the middle of 80 and 100. 90 is the number that is in the middle of 80 and 100. Meaning it is the midpoint of 80 and 100: $\frac{80+100}{2} = 90$

So the x-coordinate of D is 90. Therefore, the coordinates of D are (90;1)

You now have a decent understanding of the **histogram** and the **frequency polygon**. We are going to make a deeper analysis into the concepts.

We redraw the table.

Interval	**Number of students in the interval**
(20 ; 40]	**2**
(40 ; 60]	**4**
(60 ; 80]	**3**
(80 ; 100]	**1**

From the table how many students got a mark of 40 or less?

The answer is 2. We see this from the first interval: (20;40]. These are the only students who got a mark of 40 or less, they fall another the first interval.

From the table, how many students got a mark of 60 or less?

The 4 students in the second interval: (40;60], all got a mark of 60 or less. In addition to that, each of the 2 students in the first interval: (20;40] got a mark of 60 or less. Therefore, there's a total of $4 + 2 = 6$ who got a mark of 60 or less.

From the table, how many students got a mark of 80 or less?

The 3 students in the third interval: (60;80], all got a mark of 80 or less. In addition to that, each of the 4 students in the second interval: (40;60] got a mark of 80 or less. Again, each of the 2 students in the first interval: (20;40] got a mark of 80 or less. Therefore, there's a total of $3 + 4 + 2 = 9$ who got a mark of 80 or less.

From the table, how many students got a mark of 100 or less?

The 1 student in the fourth interval: (80;100], all got a mark of 100 or less. In addition to that, each of the 3 students in the third interval: (60;80] got a mark of 100 or less. Each of the 4 students in the second interval: (40;60] got a mark of 100 or less. Again, each of the 2 students

in the first interval: (20;40] got a mark of 100 or less. Therefore, there's a total of $1 + 3 + 4 + 2 = 10$ who got a mark of 100 or less. That's all the students.

Let's add all of this information into our table. We call this new column "**Cumulative frequency**".

Interval	Number of students in the interval	Cumulative Frequency
(20 ; 40]	**2**	**2**
(40 ; 60]	**4**	**6**
(60 ; 80]	**3**	**9**
(80 ; 100]	**1**	**10**

2 is the number of students who got **40 or less**.

6 is the number of students who got **60 or less**. We add the **2** from the (20;40] interval to the **4**. **2** + **4** = **6**

9 is the number of students who got **80 or less**. We add the **2** and **4** from the previous intervals. **2** + **4** + **3** = **9**

10 is the number of students who got **100 or less**. **2** + **4** + **3** + **1** = **10**

Note that 10 is the total number of the students. The cumulative frequency column above helps us answer questions like the ones we just did.

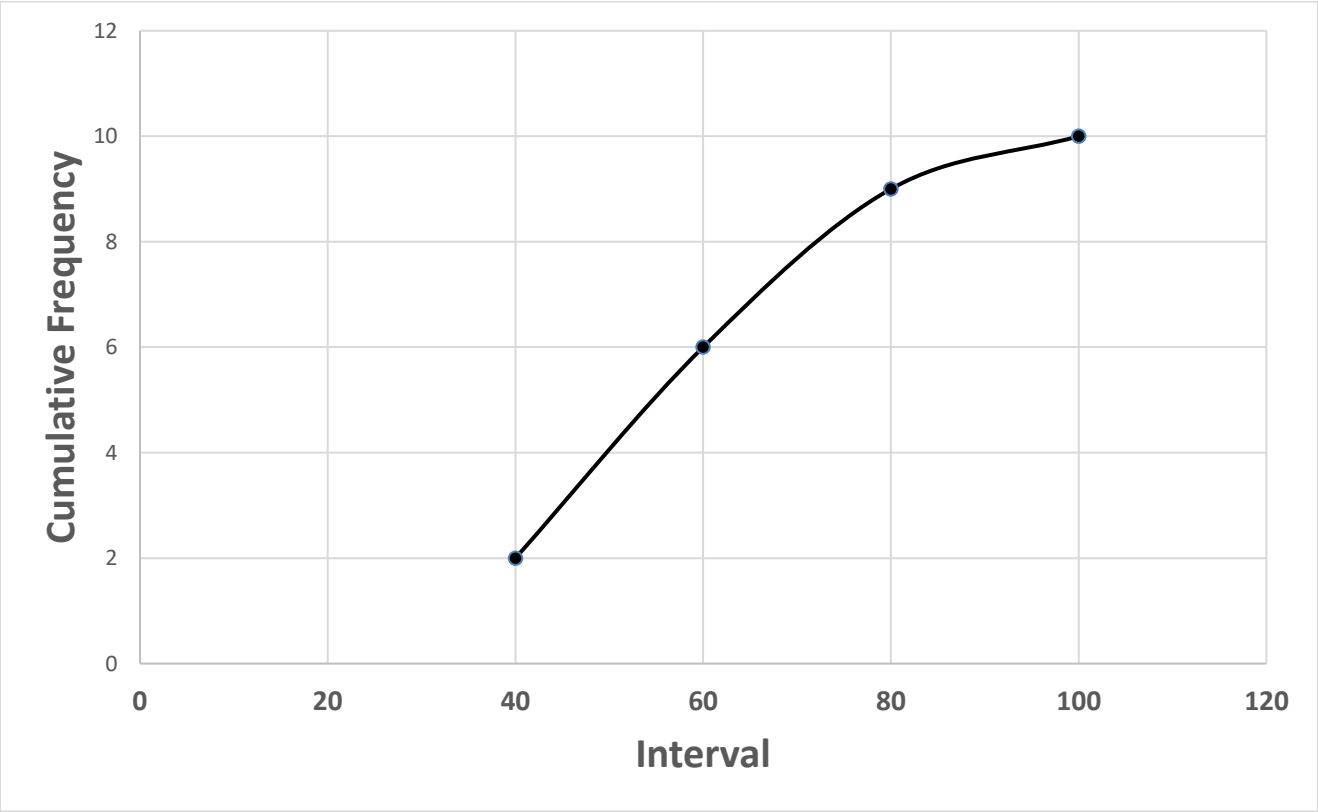
For example, when you want to know how many students got a mark of 80 or less, you just look for the interval with an upper limit of 80.

The interval with an upper limit of 80 is (60;80], then check the value of this interval under the **cumulative frequency** column. The value is 9, so we can safely say that 9 students got a mark of 80 or less. Note that the 3 under the **frequency** column is the number of students who got a mark in the interval (60;80].

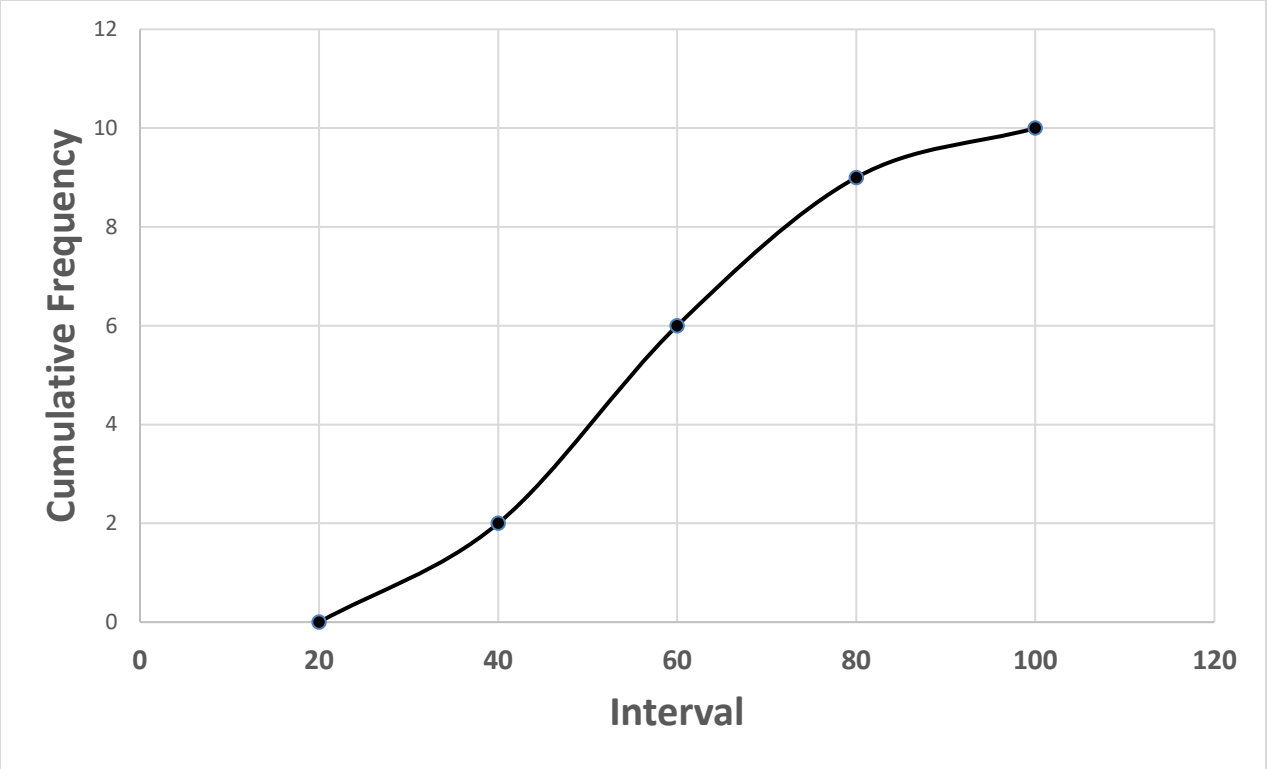
We can also draw another graph that helps us to answer such questions. This graph uses the information on cumulative frequency and the upper bound of each interval. For example, for the first interval (20;40], we plot the upper bound, which is 40, and the value on the **cumulative frequency** which is 2. For the second interval (40;60], we plot the upper bound, which is 60, and the value on the **cumulative frequency** which is 6. We do the same for the other intervals. Let's add this information into our table. We label the column "**coordinates** ".

Interval	Number of students in the interval	Cumulative Frequency	"Coordinates"
(20 ; 40]	**2**	**2**	(40;2)
(40 ; 60]	**4**	**6**	(60;6)
(60 ; 80]	**3**	**9**	(80;9)
(80 ; 100]	**1**	**10**	(100;10)

We now plot these coordinates on a graph. We put "interval" on the x-axis and "**cumulative frequency**" on the y-axis.



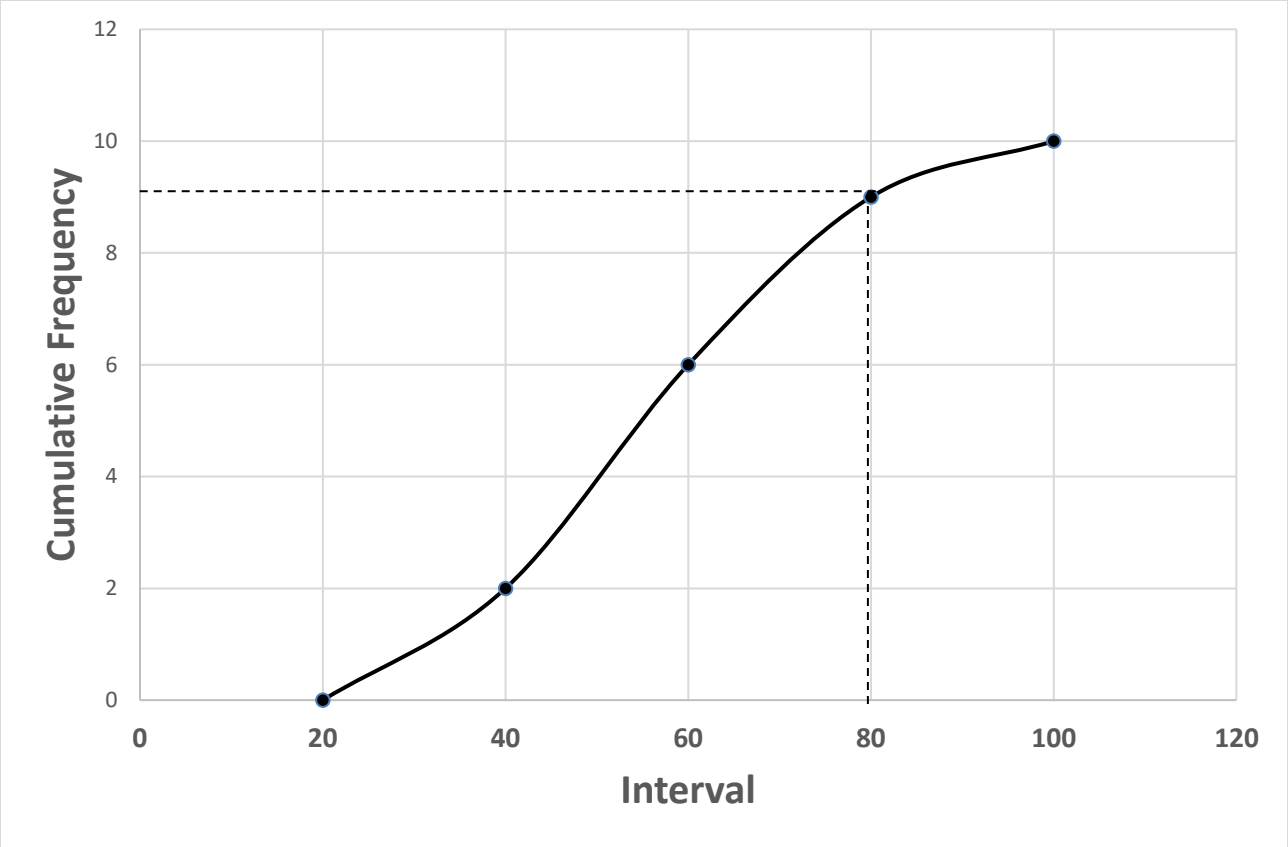
Put an additional point at the lower limit of the first interval. The first interval is (20;40], so put the point "20" on the graph. We include that in the graph below.



This kind of graph is called an **Ogive** or the **Cumulative Frequency Polygon**.

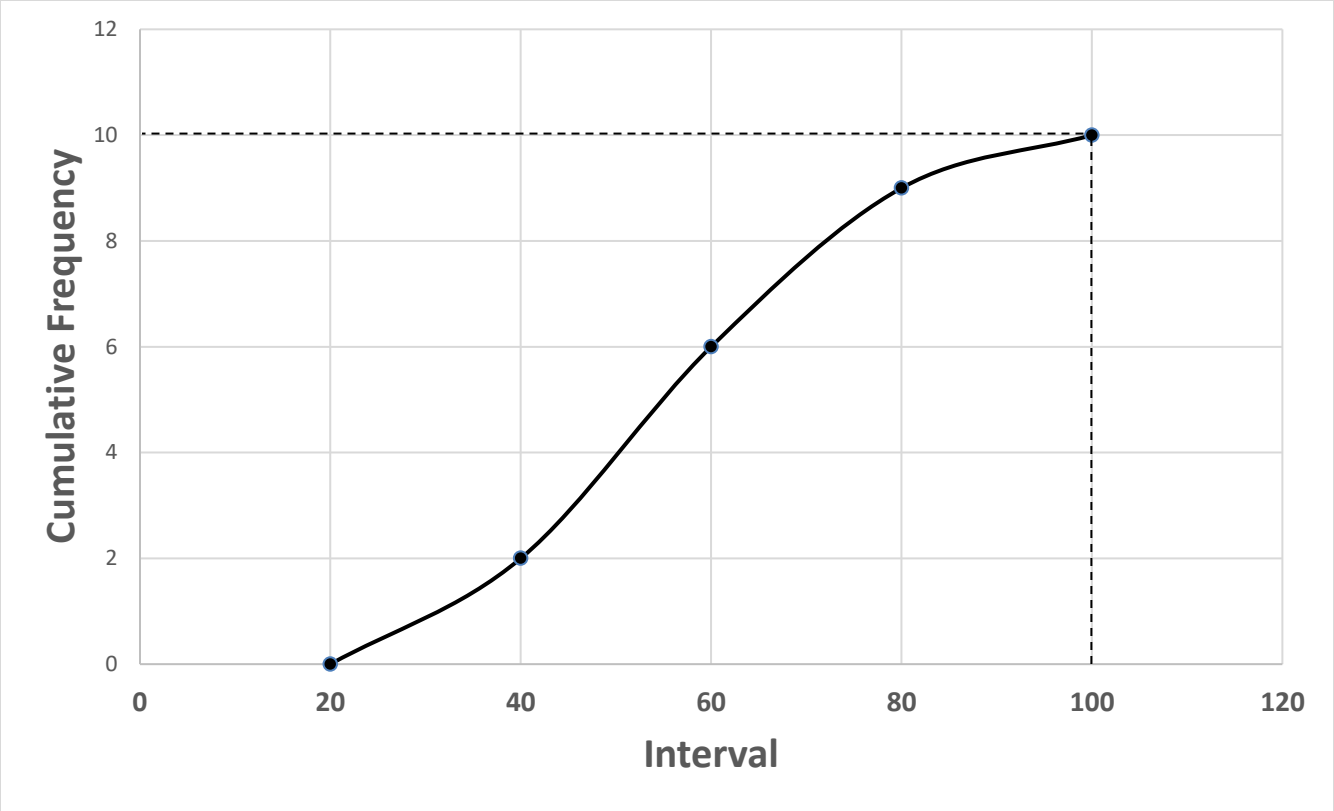
The Ogive is used to answer questions like; How many students got a mark of 80 or less?

To find the answer, you draw a dotted line from the 80 on the x-axis, until the line touches the ogive. You then extend the dotted line straight to the y-axis. Your answer will be the point at which the dotted line stops on the y-axis. Below we follow this procedure to answer our question.



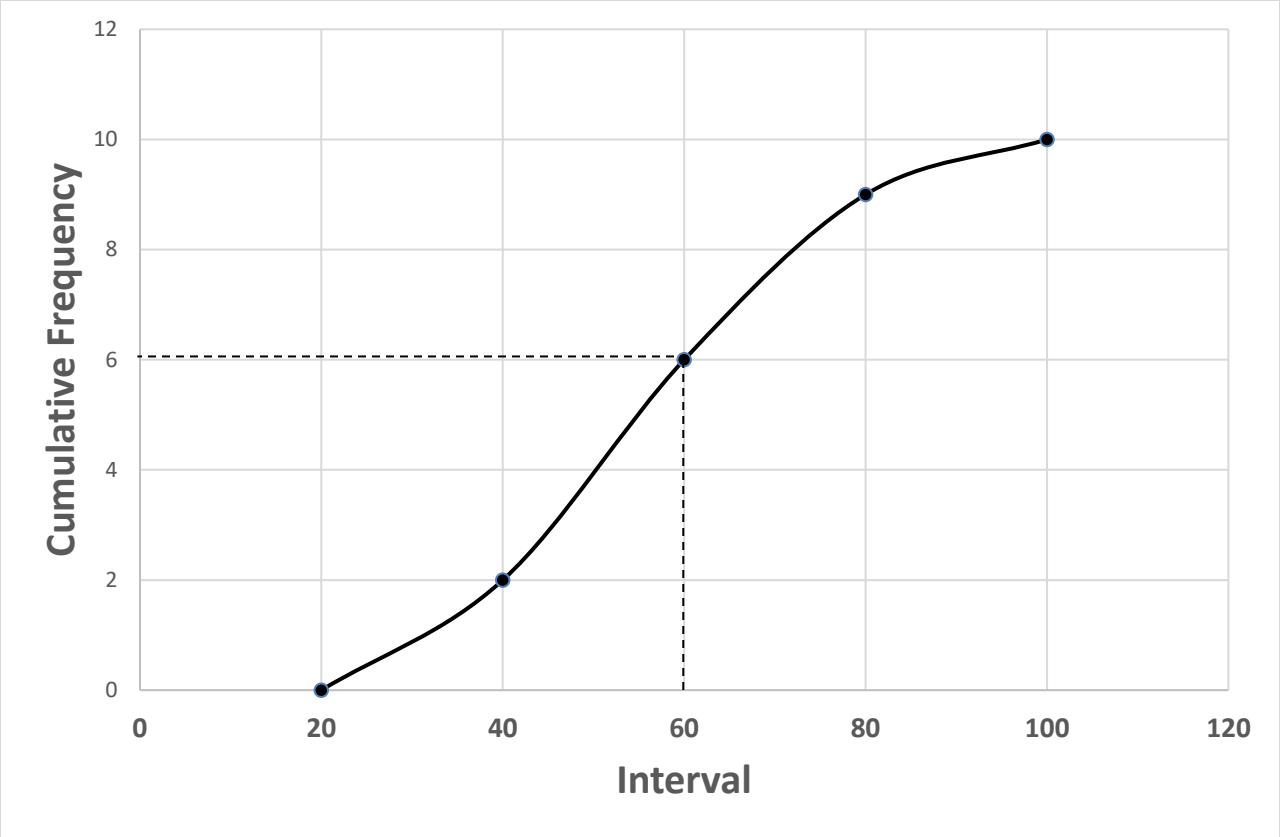
The dotted line stops at point 9 on the y-axis, meaning our answer is 9. Therefore, 9 students got a mark of 80 or less? This gives the same answer as the cumulative frequency polygon gave.

How many students got a mark of 100 or less?



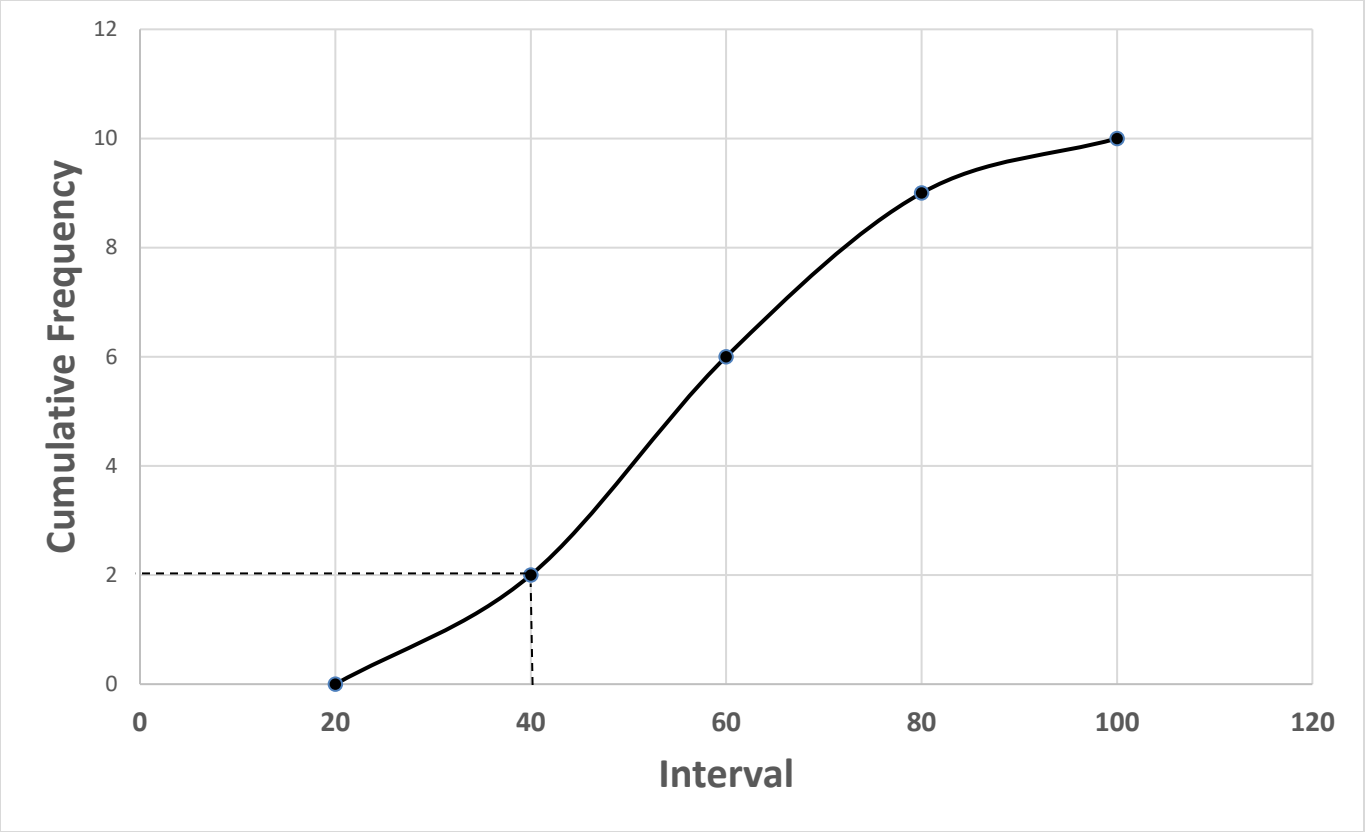
The dotted line stops at 10 on the y-axis, meaning that our answer is 10. Therefore, 10 students got a mark of 100 or less.

How many students got a mark of 60 or less?



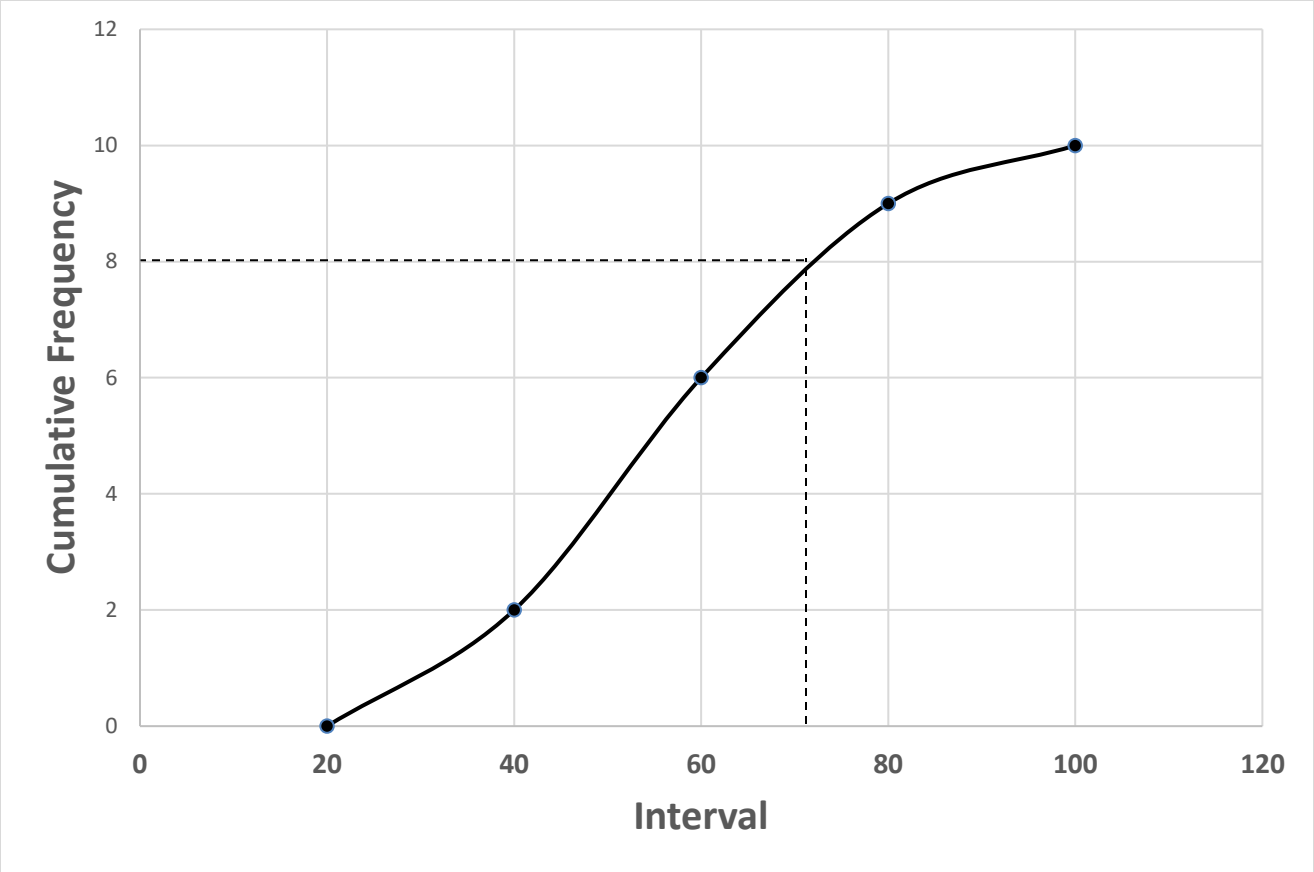
The dotted line stops at 6 on the y-axis. Therefore, 6 students got a mark of 60 or less.

How many students got a mark of 40 or less?



The dotted line stops at 2 on the y-axis. Therefore, 2 students got a mark of 40 or less.

How many students got a mark of 70 or less?



The dotted line stops at 8 on the y-axis. Therefore, 8 students got a mark of 70 or less.

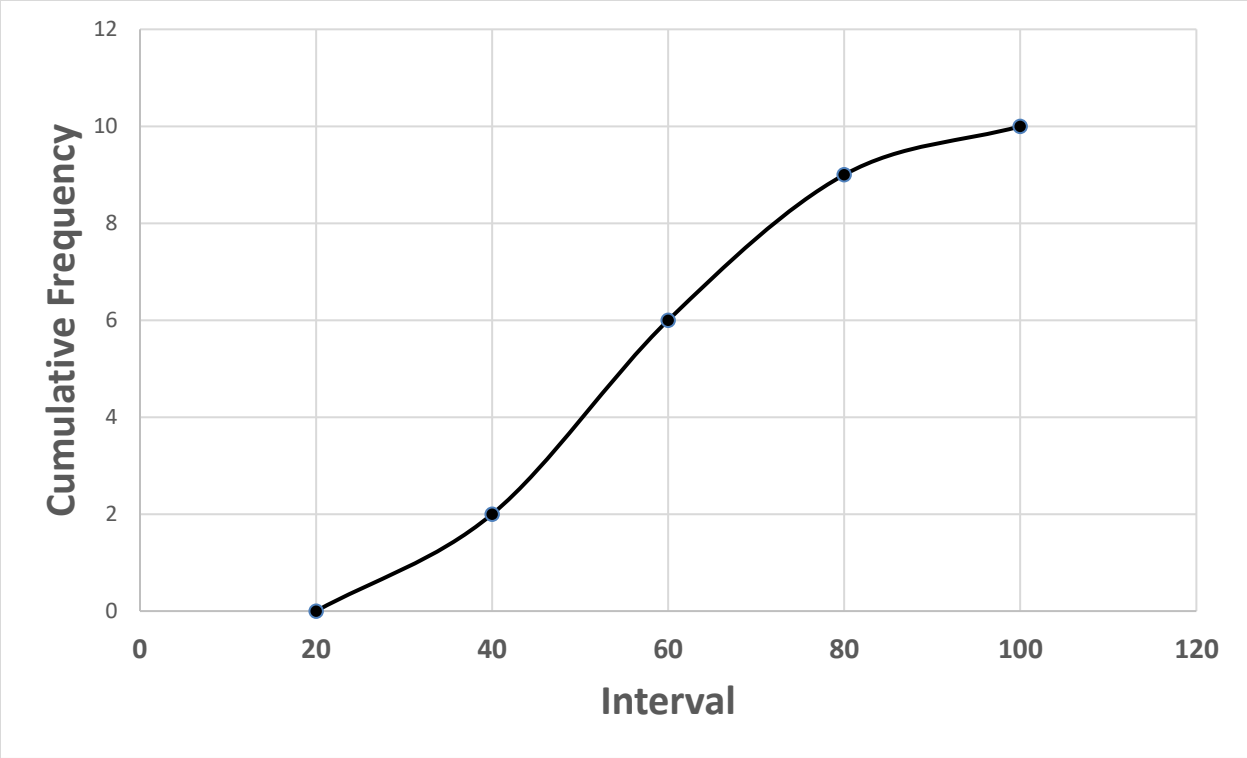
From recent information, we know that there are 6 students who got a mark of 60 or less. We also know that 2 students got a mark of 40 or less. Based on this information, how many students got a mark between 40 and 60? 6 students got a mark of 60 or less. Of the 6 students, 2 got a mark of of 40 or less, so the number of those who got between 40 and 60 is $6 - 2 = 4$. We see this on the tables at the beginning of the chapter, (40 ; 60] has a frequency of 4.

How many students got a mark between 60 and 80?

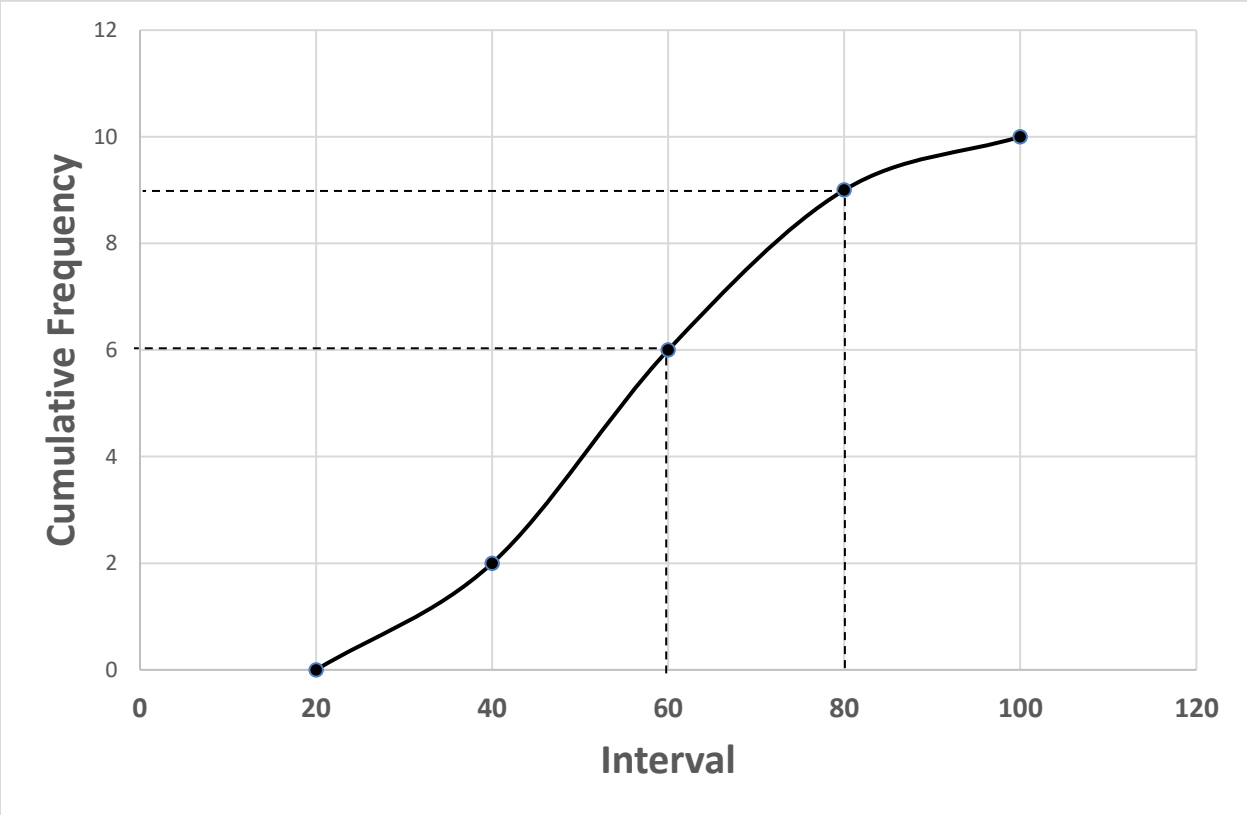
9 students got a mark of 80 or less. 6 students got a mark of 60 or less.

Of the 9 students, 6 got a mark of 60 or less, so the number of those who got between 60 and 80 is $9 - 6 = 3$. We see this on the tables at the beginning of the chapter, (60 ; 80] has a frequency of 3.

We can also use the Ogive to answer such questions. Consider our ogive again.

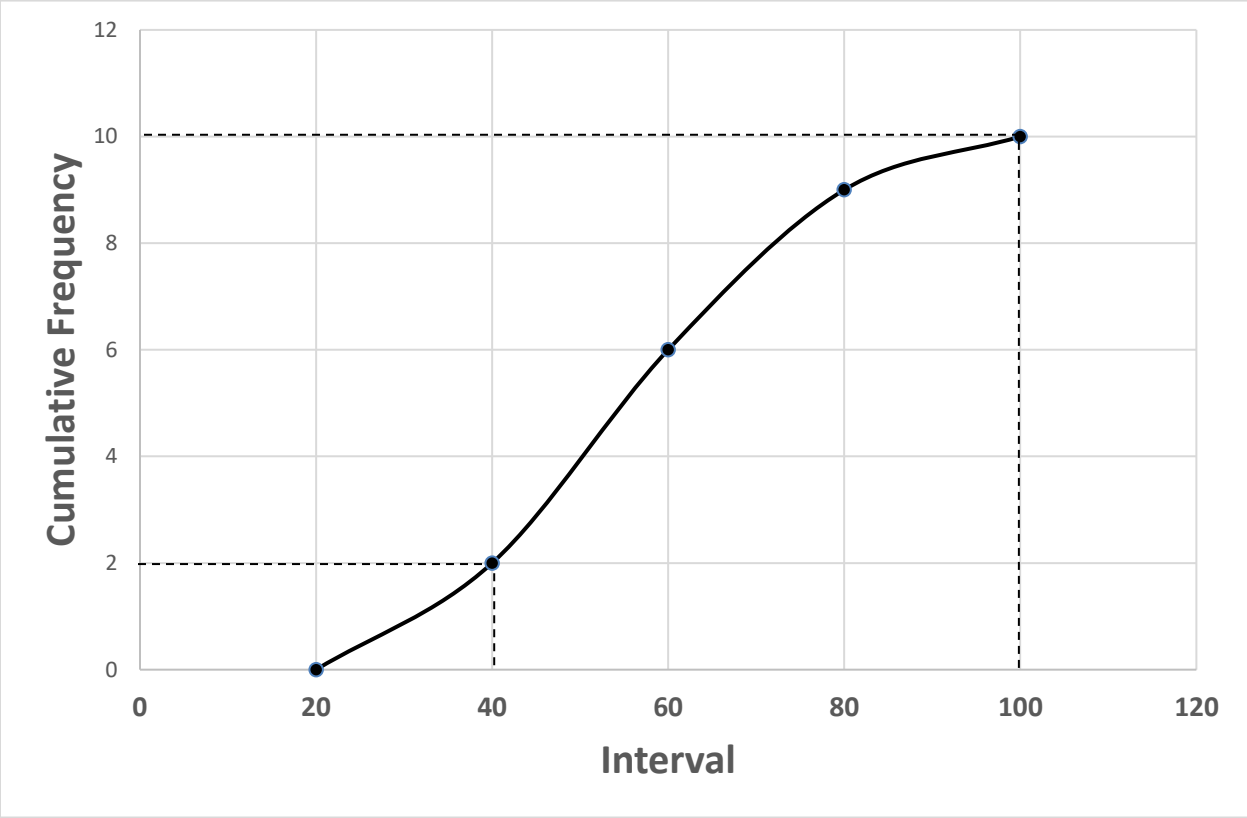


How many students got a mark between 60 and 80? First draw the dotted lines for both 60 and 80.



We then go: $9 - 6 = 3$. Thus 3 students got a mark between 60 and 80. It gives the same answer.

How many students got a mark between 40 and 100? First draw the dotted lines for both 40 and 100.



We then go: $10 - 2 = 8$. Thus 8 students got a mark between 40 and 100.

How many students got a mark more than 60? As in the previous questions, we can either use the **frequency table** or **ogive** to answer such questions. Try to use the frequency table first.

Interval	Frequency
(20 ; 40]	**2**
(40 ; 60]	**4**
(60 ; 80]	**3**
(80 ; 100]	**1**

The students who got a mark of more than 60 are in the intervals (60 ; 80] and (80 ; 100]. There are 3 students in the (60 ; 80] interval and 1 student in the (80 ; 100] interval. Thus they are $3 + 1 = 4$ students who got a mark of more than 60.

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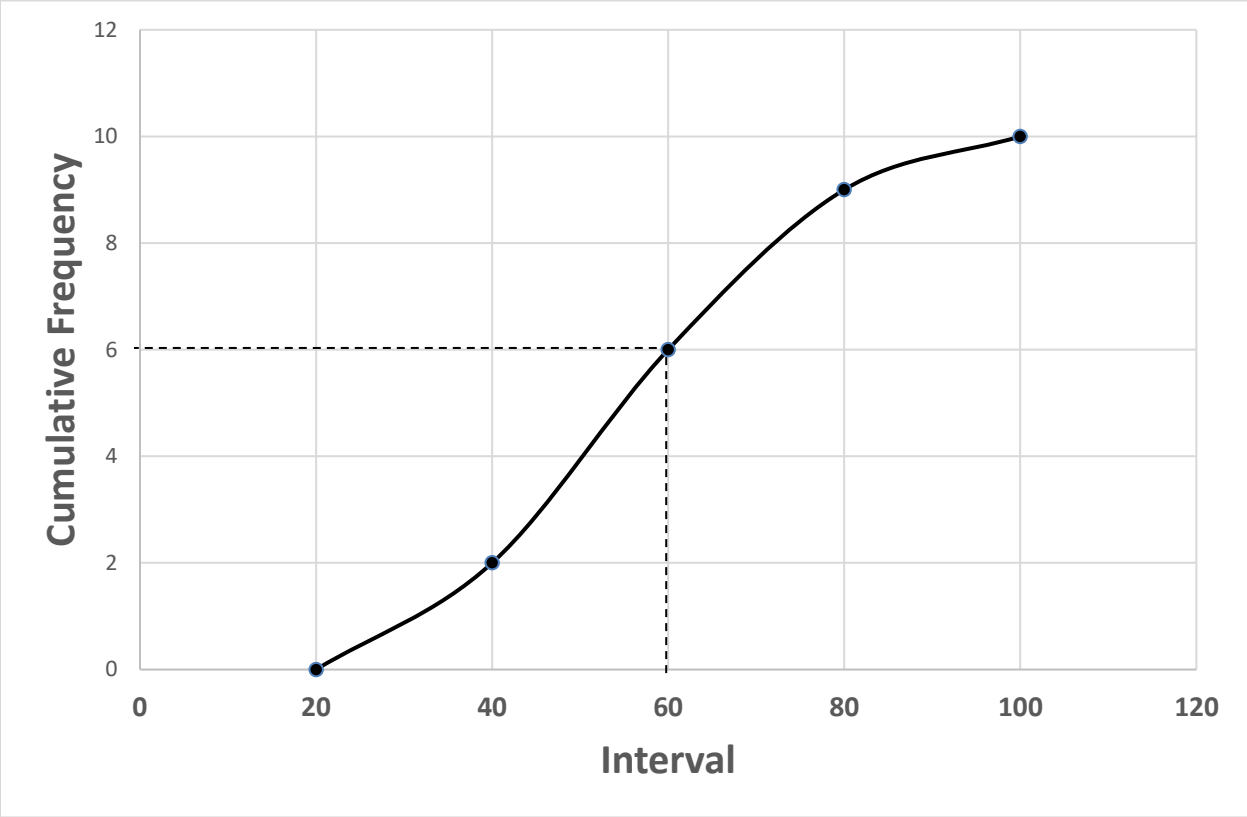
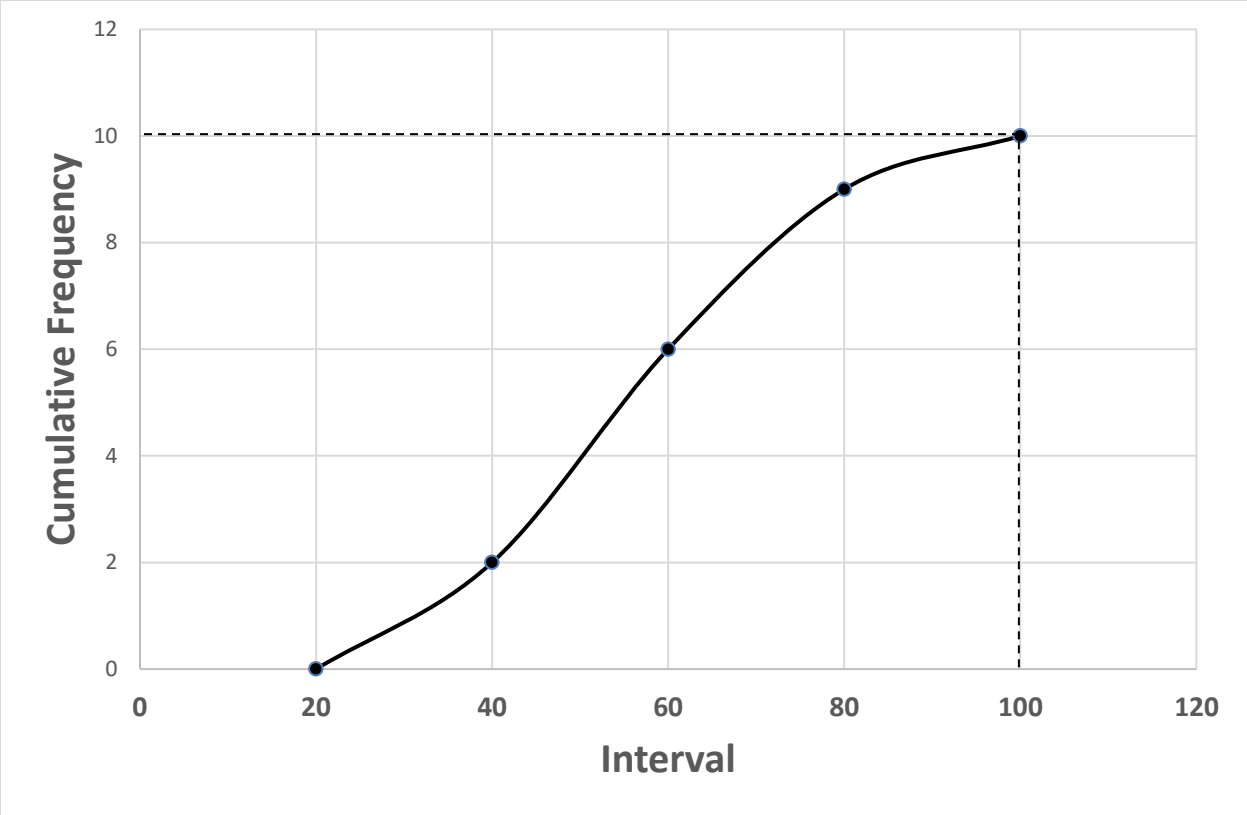
Now try to use the ogive.

We draw our ogive and the dotted line from 60.



We have 6 on the y-axis, those are the 6 students who got a mark of 60 or less. We then "minus" this number from the total number of students. The total number of students is 10. Thus $10 - 6 = 4$ is the total number of students who got a mark of more than 60.

If by any chance you are not given the total number of students (anything), you just read it off from the ogive; it is the y-coordinate of the last point on the Ogive.



The 10 on the last point above is the total number of students (items).

How many students got a mark of more than 40?

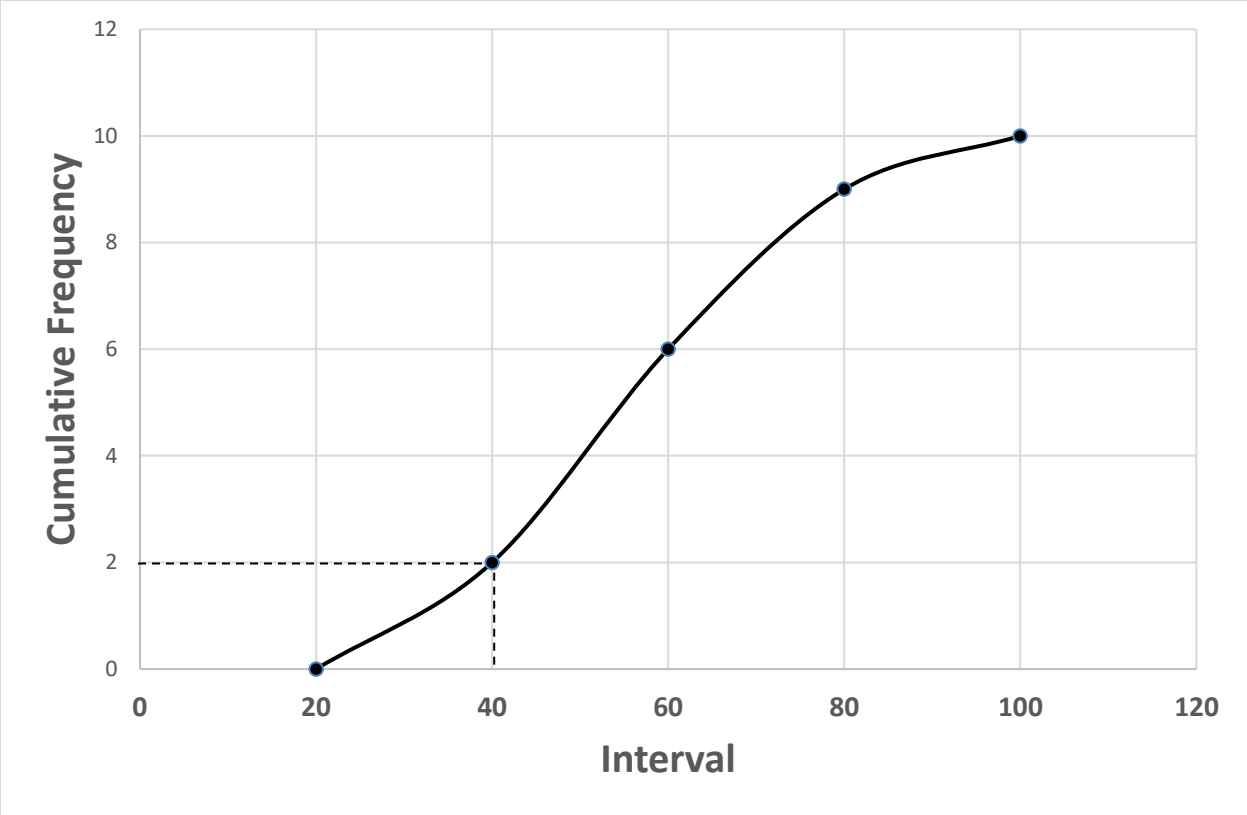
Try to use the frequency table first.

Interval	Frequency
(20 ; 40]	**2**
(40 ; 60]	**4**
(60 ; 80]	**3**
(80 ; 100]	**1**

The students who got a mark of more than 40 are in the intervals (40 ; 60], (60 ; 80] and (80 ; 100]. There are 4 students in the (40 ; 60] interval , 3 students in the (60 ; 80] interval and 1 student in the (80 ; 100] interval. Thus they are $4 + 3 + 1 = 8$ students who got a mark of more than 40.

Now try to use the ogive.

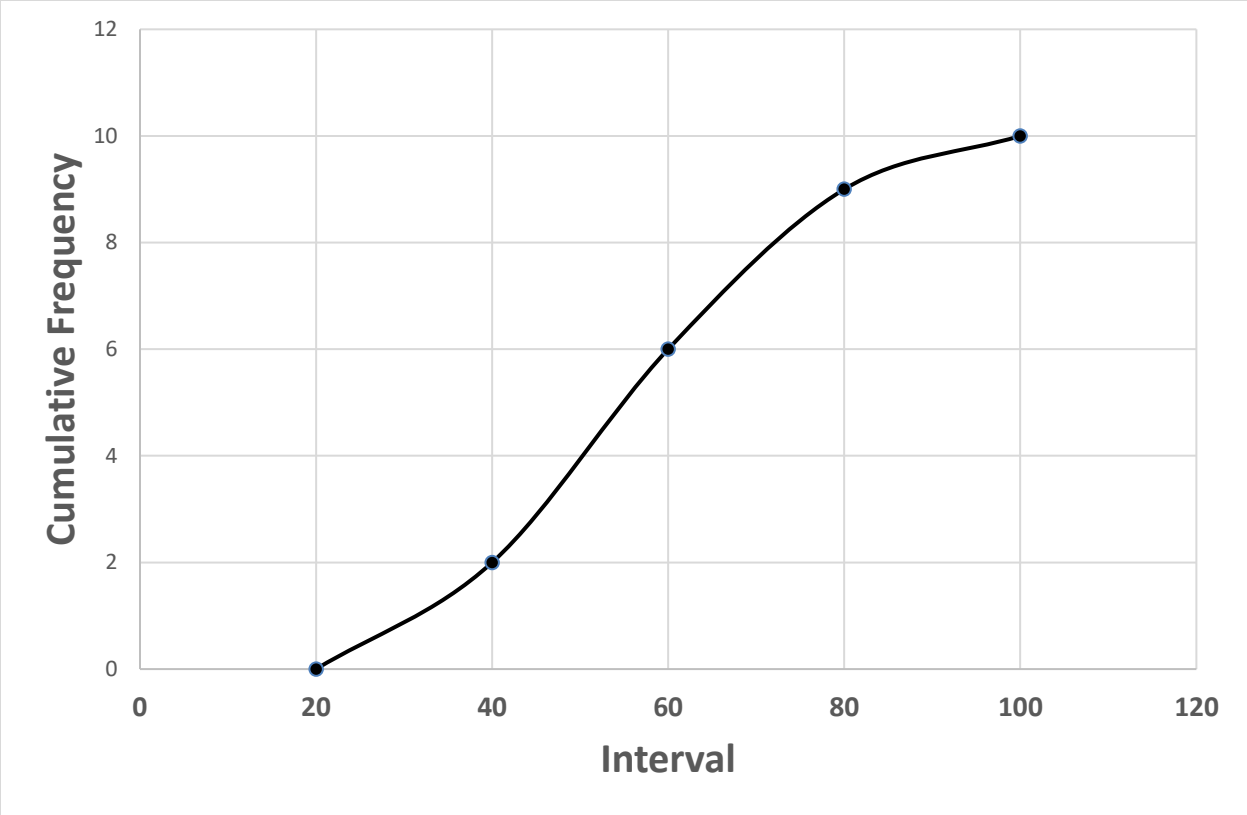
We draw our ogive and the dotted line from 40.



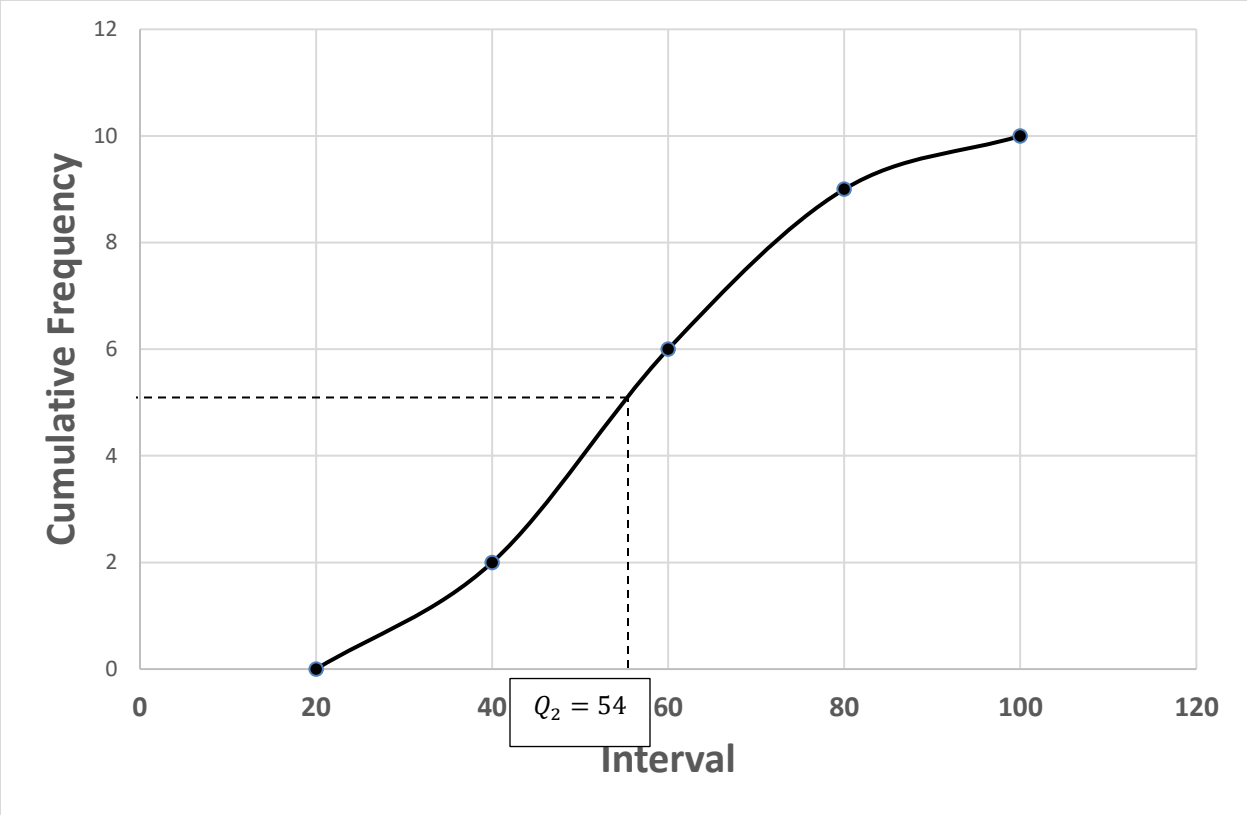
We have 2 on the y-axis, those are the 2 students who got a mark of 40 or less. We then "minus" this number from the total number of students. The total number of students is 10. Thus $10 - 2 = 8$ is the total number of students who got a mark of more than 40. It gives the same answer.

Estimating the median, the 1st quartile and 3rd quartile using the Ogive

We can also use the ogive to find the median, 1st quartile and 3rd quartile of data. We begin with the median on our example of the 10 students.



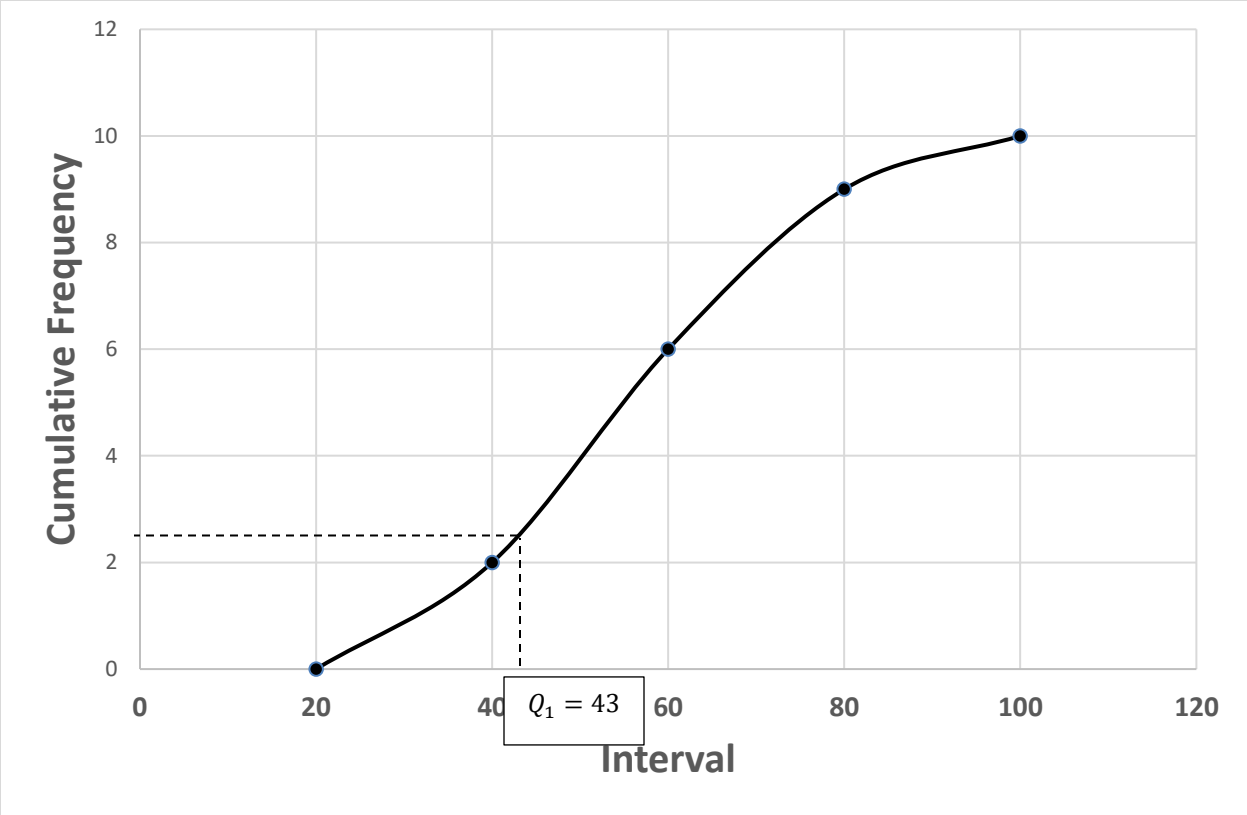
As you know, the median is the middle value. In our example, we have 10 students. If we put the marks of the students in order, from lowest to biggest, the median will be the mark in the middle. We go; $\frac{10}{2} = 5$. We see that the median is more or less the 5th number on the list. To find it from the ogive, we draw a dotted line from 5 on the y-axis until it touches the ogive. We then extend it down to the x-axis. The point at which the dotted line stops on the x-axis is the median.



Thus, the median mark is 54, that is, the estimated middle value is 54; $Q_2 = 54$.

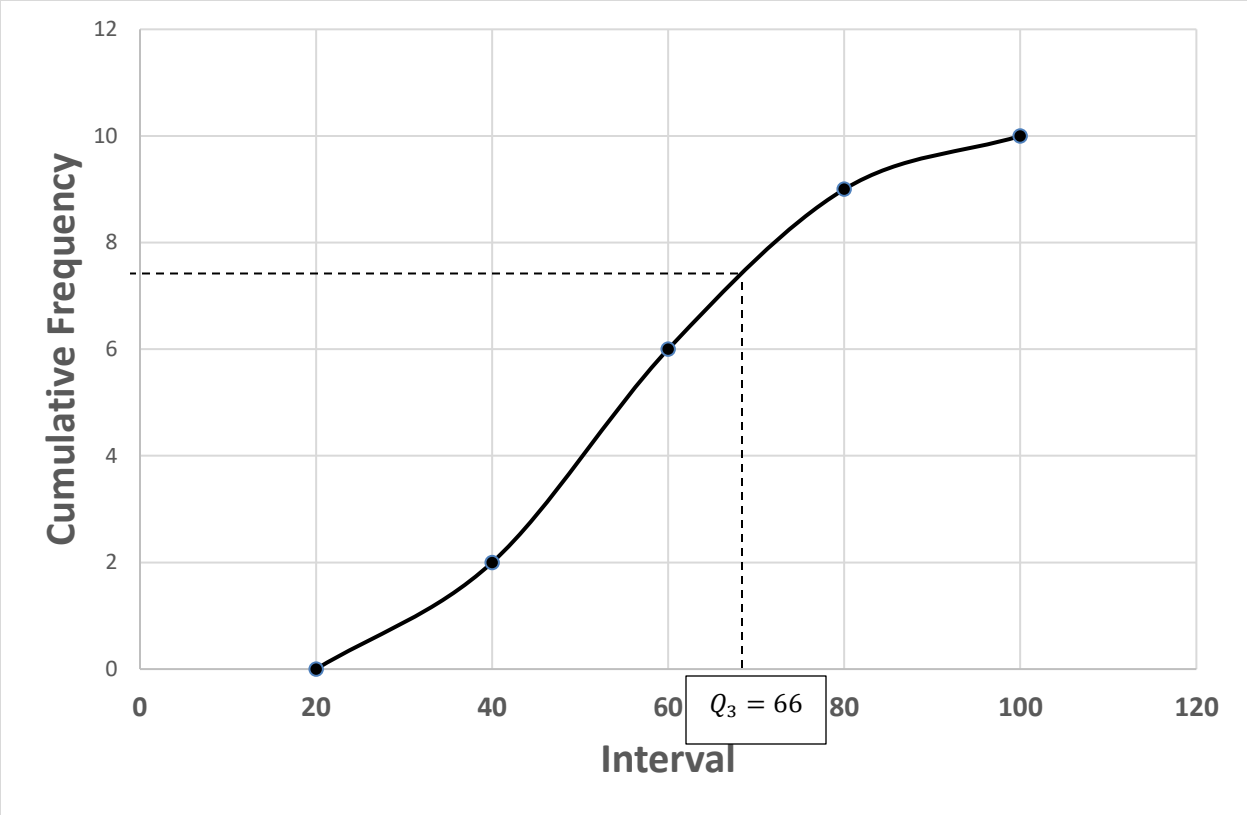
The 1st quartile (Q_1) and 3rd quartile (Q_3) are found in a similar way.

For the 1st quartile, we **divide** the total number of students by 4 to get its position; $\frac{10}{4} = 2.5$. We follow the same steps as in with the median.



Thus the 1st quartile is 43; $Q_1 = 43$.

For the 3rd quartile, we **multiply** the total number of students by $\frac{3}{4}$ to get its position; $10 \times \frac{3}{4} = 7.5$. We follow the same steps as in with the median.



Thus the 3rd quartile is 66; $Q_3 = 66$.

Variance

I am going to explain variance using some examples.

Class A has 4 students.

Class B has 4 students too.

Students from both classes write a math test. The results are shown below.

Class A				**Class B**			
50	53	60	63	19	22	92	95

What is the mean mark (average) of each of the classes?

Class A

The letter used to represent the mean is $\bar{x}$.

$$mean = \overline{x_A} = \frac{50 + 55 + 60 + 63}{4} = 57$$

Class B

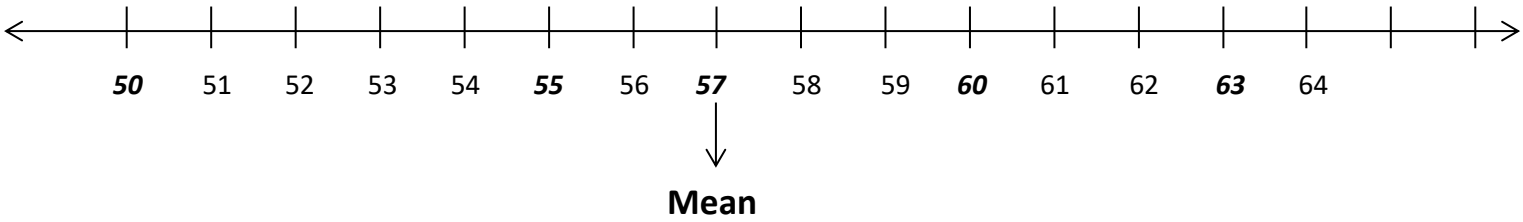
The letter used to represent the mean is $\bar{x}$.

$$mean = \overline{x_B} = \frac{19 + 22 + 92 + 95}{4} = 57$$

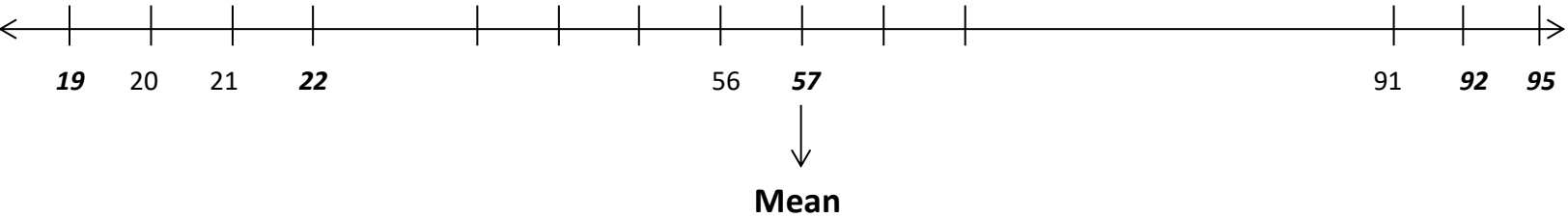
So the mean of class A is $\overline{x_A} = 57$ and the mean of class B is $\overline{x_B} = 57$

Let's draw a number line for each class and indicate their respective means.

Class A



Class B



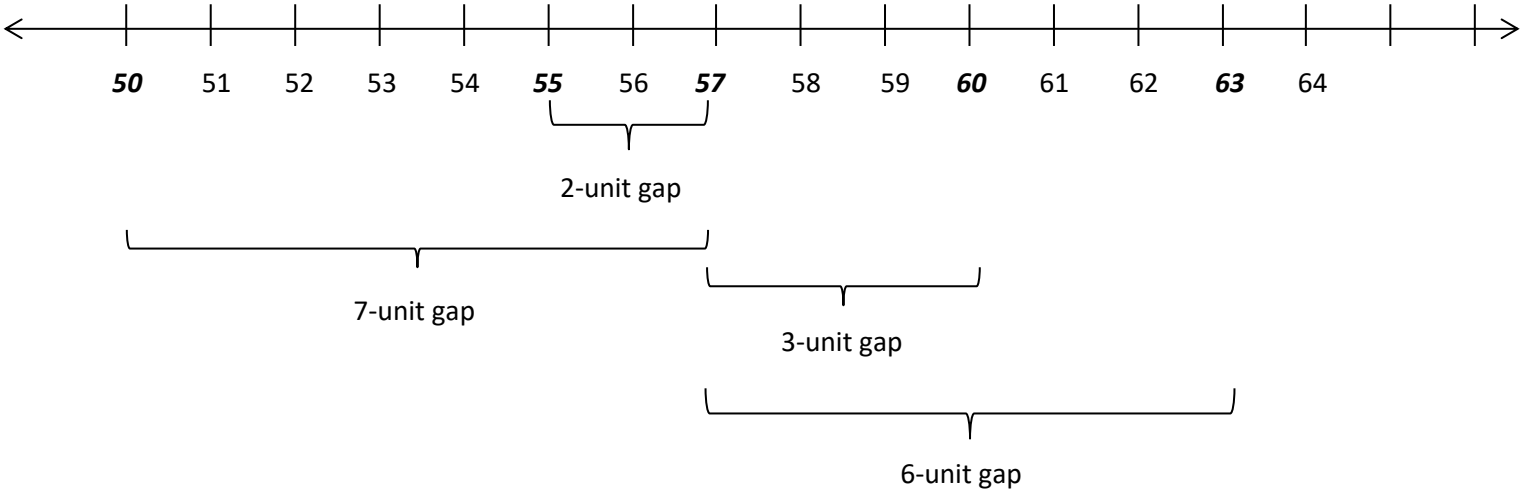
The marks of class A seem to be closer to their mean of 57, the marks of class B are further away from their mean of 57.

The gaps between the marks of class A and their mean of 57 is smaller compared to those of class B with their mean.

Mathematicians say the variance ("gaps") of class A is smaller than the variance ("gaps") of class B.

Let's calculate how big these gaps are.

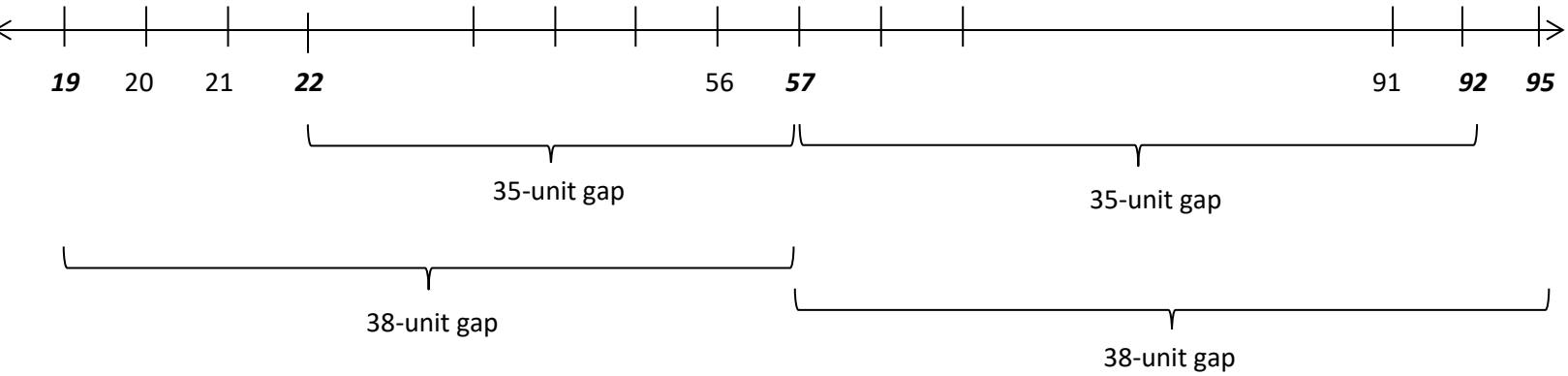
Class A



We add of these gaps: $7 + 2 + 3 + 6 = 18$

The total gap is 18; thus the "variance" is 18.

Class B



We add of these gaps: $35 + 38 + 35 + 38 = 146$

The total gap is 146; thus the "variance" is 146.

Clearly, class B has a bigger variance than class A, meaning the gaps between the mean and the marks is bigger. Students in class B got marks that are generally far away from the mean, while those from class A got marks closer to their mean.

The above was just a rough idea of what variance is. Below is the build-up of the official formula and calculation of variance.

For class A: Master the table below. Remember that $\bar{x}$ is the mean, $\bar{x} = 57$

Marks	$x - \bar{x}$	$(x - \bar{x})^2$
50	$50 - 57 = -7$	$(-7)^2 = 49$
55	$55 - 57 = -2$	$(-2)^2 = 4$
60	$60 - 57 = 3$	$(3)^2 = 9$
63	$63 - 57 = 6$	$(6)^2 = 36$
	$total =$	98

The total is 98. Another way of representing this total is: $\sum(x - \bar{x})^2 = 98$.

The formula for variance is:

$$variance = \frac{\sum(x - \bar{x})^2}{n}$$

, where *n* is the number of items in the data, in this case, we have 4 students in the class. So $n = 4$ and $\sum(x - \bar{x})^2 = 98$. Substitute these values into the formula.

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$$variance = \frac{\sum(x-\bar{x})^2}{n} = \frac{98}{4} = 24.5$$

For class B. Remember that $\bar{x}$ is the mean, $\bar{x} = 57$

Marks	$x - \bar{x}$	$(x-\bar{x})^2$
19	$19 - 57 = -38$	$(-38)^2 = 1444$
22	$22 - 57 = -35$	$(-35)^2 = 1225$
92	$92 - 57 = 35$	$(35)^2 = 1225$
95	$95 - 57 = 38$	$(38)^2 = 1444$
	$total =$	$5\ 338$

The total is 5338. Another way of representing this total is: $\sum(x-\bar{x})^2 = 5\ 338$.

The formula for variance is:

$$variance = \frac{\sum(x-\bar{x})^2}{n}$$

, where *n* is the number of items in the data, in this case, we also have 4 students in the class. So $n = 4$ and $\sum(x-\bar{x})^2 = 5\ 338$. Substitute these values into the formula.

$$variance = \frac{\sum(x-\bar{x})^2}{n} = \frac{5\ 338}{4} = 1\ 334.5$$

The variance of class A is 24.5 and the variance of class B is 5 338.

The variance of class B is way higher than that of class A, which means that the marks if class B are generally further away from their mean, and this of class A are closer to the mean.

Memorize the following table format for calculating variance.

Marks	$x - \bar{x}$	$(x-\bar{x})^2$
	$total =$	

Then substitute into the formula:

$$variance = \frac{\sum(x-\bar{x})^2}{n}$$

Depreciation

FINANCIAL MATHEMATICS

Depreciation

Let's say you buy a car today worth *R100 000*. Three months later, you decide to sell the car. Do you think it would be fair of you to sell it at R100 000, the price you bought it with?

Well, it wouldn't be fair. You must sell it at a lower price because it was used, the tires are wearing out, the engine is no longer 'fresh', it's older and so on. This is why you must sell the car at a lower price. This loss of value of assets overtime is called depreciation. Assets such as computers, phones, printers, vehicles and machines are good examples.

Some assets depreciate (lose value) faster than others. For example, a business has two identical cars, car 1 and car 2, worth *R100 000* each. *Car 1* is used 1 day per week, *Car 2* is used 7 days a week, which car would depreciate (lose value) faster?

Car 2 will obviously drop in value faster because it was used more, the parts of the car become "tired", more rusty and less strong. So the car must be valued much less. So if the business decides to sell both cars at the end of the month, *Car 1* may be sold at something like *R90 000* and *Car 2* at *R70 000* (for example).

Therefore in this section, I am going to teach you how to calculate such values of depreciation.

I am going to explain to you the concept using an example.

Example

- At the beginning of the year *2019*, you go to a garage called "Trading Cars" and buy a brand new car at the cost of *R100 000*.
- At the end of the year, you decide to sell the car back to "Trading cars" garage. The owner of the garage says the car now is worth *R90 000* since you have been using it. So they buy it at *R90 000*.
- By how much did your car depreciate in value over the year 2019?
- The answer is *R10 000*. The car's value depreciated by *R10 000*. The formula for the depreciation amount is;

$$\begin{aligned} Depreciation\ Amount &= Old\ value - current\ value \\ &= 100\ 000 - 90\ 000 \\ &= 10\ 000 \end{aligned}$$

So the car decreased in value by *R10 000* as calculated above.

- Therefore depreciation amount is the amount by which the value of an asset decreases over some time.
- We can also describe such a drop in value using percentages
- By what percentage did the car depreciate in value over the year 2019?
- The answer is *10%*. I calculated it as follows. The car depreciated by the amount *R10 000*, so you want to find *R10 000* as a percentage of *R100 000*.
  We use the formula percentage.

$$percentage = \frac{value}{total\ value} \times 100\%$$

$$= \frac{10\ 000}{100\ 000} \times 100\%$$

$$= 10\%$$

So the car depreciated by 10%. In mathematics, we say the depreciation rate of the car was 10% over the year 2019.

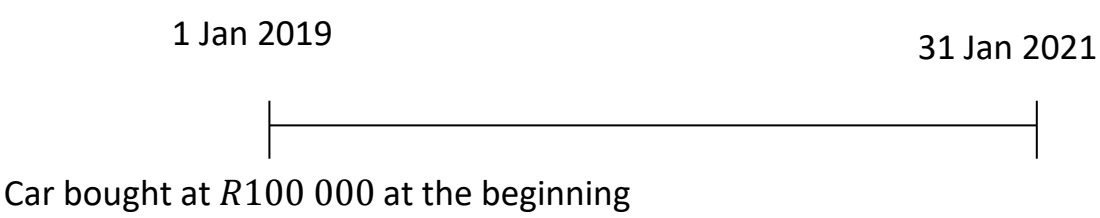
Make sure you don't confuse the depreciation amount with the depreciation rate.

The depreciation amount is R10 000, while the depreciate rate is 10%.

Therefore depreciation rate is the percentage by which the value of an asset decreases over some time. In this section, our prime focus will now be to find the value of the car at the end of some time, given the initial price you bought the car with and percentage by which it depreciates (decreases).

So a typical question would be as follows:

Kano buys a car worth *R100 000* at the beginning of the year 2019.The depreciation rate of the car is 10% per year, what will be the value of the car at the end of the year?



Solution

They say the value of the car decreases by 10% of $R100\ 000$ over the year. And we know that 10% of $R100\ 000$ is $\boldsymbol{R10\ 000}$.

So the value of the car at year end is;

$= current\ amount - depreciation\ amount$
$= 100\ 000 - \mathbf{10\ 000}$

We got the $\mathbf{10\ 000}$ above by finding 10% of $100\ 000$

$= 100\ 000 - 100\ 000 \times 10\%$

$= 100\ 000 - 100\ 000 \times \frac{10}{100}$

10% is basically $\frac{10}{100}$

$= 100\ 000 - 100\ 000 \times 0.1$

The above is in the form;

$$a - a \times b$$

$= 100\ 000 - 100\ 000 \times 0.1$

, where $a = 100\ 000$, $b = 0.1$

Remember that by factorization, $a - a \times b = a(1 - b)$

That means we can also factorize.

$= \mathbf{100\ 000} - \mathbf{100\ 000} \times 0.1$

$\boldsymbol{a} - \boldsymbol{a} \times b$

$= \mathbf{100\ 000}(1 - 0.1)$

$\boldsymbol{a}(1 - b)$

$= 90\ 000$

Thus the value of the car at the end of the year will be $R90\ 000$. So if Kano were to sell it at the end of the year, he would sell it at a price of $R90\ 000$.

Notice we basically used the shortcut formula;

$$Value = 100\ 000(1 - 0.1)$$

, where $100\ 000$ is the value at the beginning, and 0.1 is the percentage by which the car depreciates (decreases). This is also called the ***depreciation rate***.

We can therefore generalize the formula for the value of the car at the end of the year.

$$Value = Price(1 - depreciation\ rate)$$

Let's denote the price as P, the depreciation rate as i, and the value as A. Then new formula becomes;

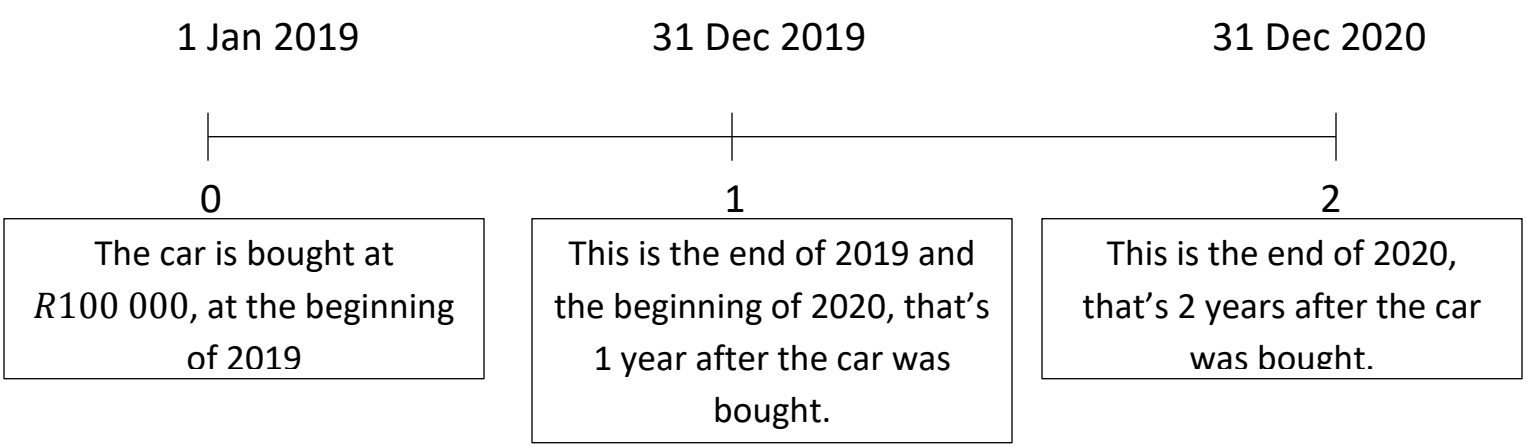
$$A = P(1 - i)$$

External Question

Now let's extend this question to 2 years.

Kano buys a car worth R100 000 at the beginning of 2019. The depreciation rate of the car is 10% per year (meaning the car value drops by 10 000 "per year that passes"). What will be the value of the car at the end of 2 years (at the end of 2020)?

Solution



What we want here is the value of the car after 2 years, which is basically at 31 Dec 2020. They say the value of the car decreases by 10% (R10 000) for every year that passes, so that means we should say R100 000 minus R10 000 depreciation from 2019 and another minus R10 000 from the depreciation in 2020.

$Value\ at\ the\ end\ of\ 2\ years = 100\ 000 - 10\ 000 - 10\ 000$

We simplify this.

$$= 100\ 000 - 100\ 000 \times 10\% - 100\ 000 \times 10\%$$

$$= 100\ 000 - 100\ 000 \times \frac{10}{100} - 100\ 000 \times \frac{10}{100}$$

$$= 100\ 000 - 100\ 000 \times 0.1 - 100\ 000 \times 0.1$$

The above is in the form;

$$a - a \times b - a \times b$$

$$= 100\ 000 - 100\ 000 \times 0.1 - 100\ 000 \times 0.1$$

Remember that $a - ab - a\,b = a - 2ab$,

and when we can also factorize $a - 2ab$ to get $\boldsymbol{a}(1 - 2b)$.

Similarly,

$$100\ 000 - 100\ 000 \times 0.1 - 100\ 000 \times 0.1 = \mathbf{100\ 000}(1 - 2 \times 0.1)$$

$$= 80\ 000$$

Thus the value of the car at the end of the 2 years will be R80 000. So if he were to sell it at the end of 2020, he would have to sell it at a price of R80 000.

We can therefore generalize the value of the car at end of the 2 years;

$Value\ at\ the\ end\ of\ 2\ years = P(1 - 2 \times depreciate\ rate)$

The formula is too wordy, let's replace the words with letters. Let A denote the "value" of the car at the end of years, P is the price and i the depreciation rate per year.

$A = P(1 - 2.i)$

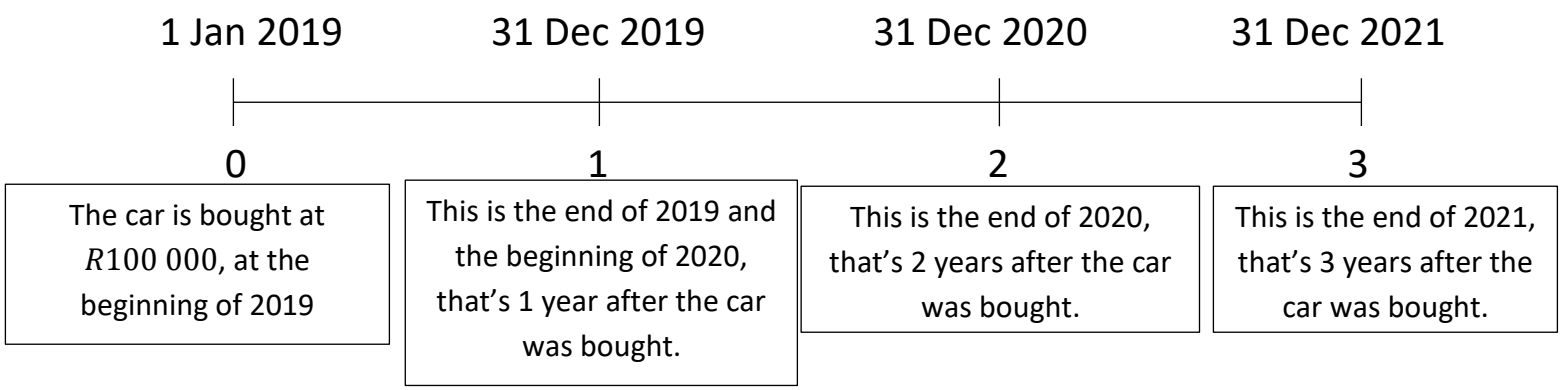
Don't you notice any pattern between the timeline of 2 years, $A = P(1 - 2.i)$, and that of 1 year, $A = P(1 - i)$?

For the depreciation over 1 year, the formula is $A = P(1 - i)$, there is basically a 1 in front of i to represents the 1 year of depreciation. As for the depreciation over 2 years, the formula is $A = P(1 - 2.i)$, the 2 in front of the i represents the 2 years of depreciation. **Keep that in mind**, we are going to do an example with depreciation over 3 years.

Extended example

Now let's extend the question to 3 years. Kano buys a car worth *R100 000* at the beginning of *2019*. The depreciation is *10%* per annum (per year), meaning the value of the car will drop by *10 000* $(that's\ 10\%\ \text{of}\ 100\ 000)$, for every year that passes. What will be the value of the car at the end of *3* years, that is, at the end of *2021*?

Solution



We use the shortcut formula from the previous example. But we use 3 instead of 2 years since we are dealing with 3 years of depreciation.

$Value\ (after\ 3\ years) = Price(1 - 3 \times depreiation\ rate)$

The formula is too wordy, let's replace the words with letters. Let A denote the "value" of the car after 3 years, P the price and i the depreciation rate per year.

$A = P(1 - 3.i)$

Dear student, I'm pretty sure you notice some pattern now.

For 1 year, the calculation formula is $A = P(1 - i)$

For 2 years, the calculation formula is $A = P(1 - 2.i)$

For 3 years, the calculation formula is $A = P(1 - 3.i)$

This pattern will continue for "higher" years.

Thus if a car costs P today and its depreciation rate is i per annum, the value of the car at the end of "n" years will have the formula;

$A = P(1 - n.i)$

Some books write the formula in the form $A = P(1 - i.n)$

They just change the order from $n.i$ to $i.n$, it doesn't make any difference. So the general formula is;

$$A = P(1 - n.i)$$

Mathematicians call this the formula the ***simple interest formula*** for ***depreciation***. We use the words "simple interest" because the value of the asset drops by the same amount every year. On the example we just did, the value of the car dropped by the same amount, R10 000 every year. The formula applies not only for the car but for many other assets. Let's now check if you understand the concept.

Question

Kano buys a car worth R100 000 today. The depreciation rate of the car is 10% per annum. What is the value of the car at the end of 5 years?

Solution

We apply the formula for simple interest depreciation.

$A = P(1 - i.n)$

$P = 100\ 000,\ i = 10\% = \frac{10}{100} = 0.1\ ,\ n = 5$

$A = 100\ 000(1 - 0.1 \times 5)$

$= 50\ 000$

So the car will have the value of *R50 000* in *5 years'* time.

This question is similar to the previous ones. I'm sure you now fully understand the concept. To conclude the simple interest form of depreciation, I am going to draw you a table on the value of the car over the 5 year period. This is based on the example we just did.

Number of years of depreciation	Starting price	i per annum	Substitute into formula	Value of the car at the end
1	100 000	$10\% = 0.1$	$= P(1 - i.n)$	$= 90\ 000$
2	100 000	$10\% = 0.1$	$= P(1 - i.n)$	$= 80\ 000$
3	100 000	$10\% = 0.1$	$= P(1 - i.n)$	$= 70\ 000$
4	100 000	$10\% = 0.1$	$= P(1 - i.n)$	$= 60\ 000$
5	100 000	$10\% = 0.1$	$= P(1 - i.n)$	$= 50\ 000$

We can clearly see from the table that the value of the car drops by *R10 000* for every year. Do the following question to round up the section.

Question

A tractor costs R100 000 today and has a depreciation rate of 5% per annum.

What is the price of the tractor 2 years from now?

Solution

$A = P(1 - i.n)$

$P = 100\ 000,\ i = 5\% = \frac{5}{100} = 0.05\ ,\ n = 7$

$A = 100\ 000(1 - 0.05 \times 7)$

$= 65\ 000$

The tractor will have the value of *R65 000* in *7 years'* time.

Question

Gilbert buys a bus at a cost of R250 000 today. The bus depreciates at 13% per year.

a) What is the value of the bus in 4 years' time?
b) By how much did the value (price) of the bus decrease over the 4 years?

Solution

a)

$A = P(1 - i.n)$

$P = 250\ 000,\ i = 13\% = \frac{13}{100} = 0.13\ ,\ n = 4$

$A = 250\ 000(1 - 0.13 \times 4)$

$= 120\ 000$

b)

$Change\ in\ value = Old\ value - New\ value$

$= 250\ 000 - 120\ 000$

$= 130\ 000$

Question

Jessica bought a car 5 years ago. The car's simple depreciation rate is 7% p.a for the past 5 years. Today, the car's worth R45 000. By how much did Jessica buy the car 5 years ago?

Solution

We use the formula $A = P(1 - i.n)$

However, you are given A as *R45 000* and are asked to find P, the price at which she bought the car.

We are given $A = 45\ 000, i = 7\% = 0.07\ , n = 5$

We then substitute.

$45\ 000 = P(1 - 0.07 \times 5)$

$45\ 000 = P \times 0.65$

We want to remain with P "alone" on one side, so we divide both sides by 0.65

$$\frac{45\ 000}{0.65} = \frac{P \times 0.65}{0.65}$$

$$\frac{45\ 000}{0.65} = \frac{P \times \cancel{0.65}}{\cancel{0.65}}$$

$$\frac{45\ 000}{0.65} = P$$

$P = 69\ 230.7693$

The price (value) of the car was R69 230.7693 *5 years* ago when Jessica bought it.

There is of course a shortcut formula derived from
$A = P(1 - i.n)$

Make P be on its own on one side. You can do this by dividing both sides by $(1 - i.n)$

$$\frac{A}{(1 - i.n)} = \frac{P(1 - i.n)}{(1 - i.n)}$$

$$\frac{A}{(1 - i.n)} = \frac{P\cancel{(1 - i.n)}}{\cancel{(1 - i.n)}}$$

$$\frac{A}{(1 - i.n)} = P$$

$$P = \frac{A}{(1 - i.n)}$$

You could've used this formula from the start.

$$P = \frac{A}{(1 - i.n)}$$

$$P = \frac{45\ 000}{(1 - 0.07 \times 5)}$$

$$P = 69\ 230.7693$$

Question

A car is worth R70 000 today. Ricardo bought the car 11 years ago. The simple depreciation rate of the car is 2% p.a. How much did Ricardo spend on the car 11 years ago?

Solution

Notice you are looking for *P*, ***Not*** *A,* so you should use;

$$P = \frac{A}{(1 - i.n)}$$

$A = 70\ 000,\ i = 2\% = 0.02\ ,\ n = 11$

$$P = \frac{70\ 000}{(1 - 0.02 \times 11)}$$

$= 89\ 743.59$

The car was worth $R89\ 743.59$ 11 years ago.

Question

A car depreciates from R900 000 to R240 000 over *"n"* years. The simple depreciation rate is 8% p.a. How many years did it take the car to depreciate from R900 000 to R240 000?

Solution

$A = P(1 - i.n)$

We are given $A = 240\ 000$, $P = 900\ 000$, $i = 8\% = 0.08$, so we are looking for n

Substitute

$240\ 000 = 900\ 000(1 - 0.08 \times n)$

Remember that the aim is to remain with n on its own on one side. So first divide both sides by 900 000.

$$\frac{240\ 000}{900\ 000} = \frac{900\ 000(1 - 0.08 \times n)}{900\ 000}$$

$$\frac{240\ 000}{900\ 000} = \cancel{\frac{900\ 000(1 - 0.08 \times n)}{900\ 000}}$$

$$\frac{240\ 000}{900\ 000} = (1 - 0.08 \times n)$$

$$\frac{240\ 000}{900\ 000} = (1 - 0.08 \times n)$$

$$\frac{4}{15} = 1 - 0.08 \times n$$

$$\frac{4}{15} - 1 = -0.08 \times n$$

$$-\frac{11}{15} = -0.08 \times n$$

Remember, the aim is to remain with n alone on the right-hand side; so divide both sides by -0.08

$$-\frac{11}{15 \times -\mathbf{0.08}} = \frac{-0.08 \times n}{-\mathbf{0.08}}$$

$$-\frac{11}{15 \times -\mathbf{0.08}} = \cancel{\frac{-0.08 \times n}{-\mathbf{0.08}}}$$

$$-\frac{11}{15 \times -\mathbf{0.08}} = n$$

$$\frac{55}{6} = n$$

$$n = \frac{55}{6}$$

$n = 9.17$

So it took *9.12* years for the car to depreciate from *R900 000* to *R240 000.*

There is a shortcut formula derived from $A = P(1 - i.n)$ to calculate n. See the derivation below.

$A = P(1 - i.n)$, the aim is to make n the subject of the formula. Start by dividing both sides by P

$$\frac{A}{P} = \frac{P(1 - i.n)}{P}$$

$$\frac{A}{P} = \frac{\cancel{P}(1 - i.n)}{\cancel{P}}$$

$$\frac{A}{P} = 1 - i.n$$

$$\frac{A}{P} - 1 = -i.n$$

$$\frac{A}{P} - 1 = -i.n$$

$$\frac{A - P}{P} = -i.n$$

Divide both sides by $-i$

$$\frac{A - P}{P \times -i} = \frac{-i.n}{-i}$$

$$\frac{A - P}{-P.i} = \frac{\cancel{-i}.n}{\cancel{-i}}$$

$$\frac{A - P}{-P.i} = n$$

$$n = \frac{A - P}{-P.i}$$

Move the “minus” on the denominator to the numerator.

$$n = \frac{-(A - P)}{P.i}$$

$$n = \frac{-A + P}{P.i}$$

$$n = \frac{P - A}{P.i}$$

$$n = \frac{P - A}{P.i}$$

Compound Interest formula for Depreciation

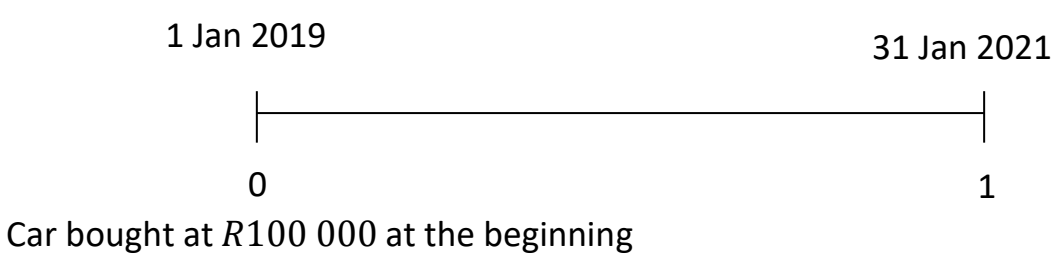
This form of depreciation is slightly different from the ***simple interest*** one. You will see that they differ as we progress with the topic.

Question

Kano buys a car worth R100 000 at the beginning of the year 2019. The depreciation rate of the car is 10% per year, what will be the value of the car at the end of the year?

Solution

Again, let's draw the timeline representation of what just happened on the value of the car over the 1 year.



We use our beloved formula; $A = P(1 - i.n)$

, where $P = 100\ 000,\ n = 1$, and $i = 10\% = 0.1$

Substitute

$A = 100\ 000(1 - 0.1 \times 1)$

$= 90\ 000$

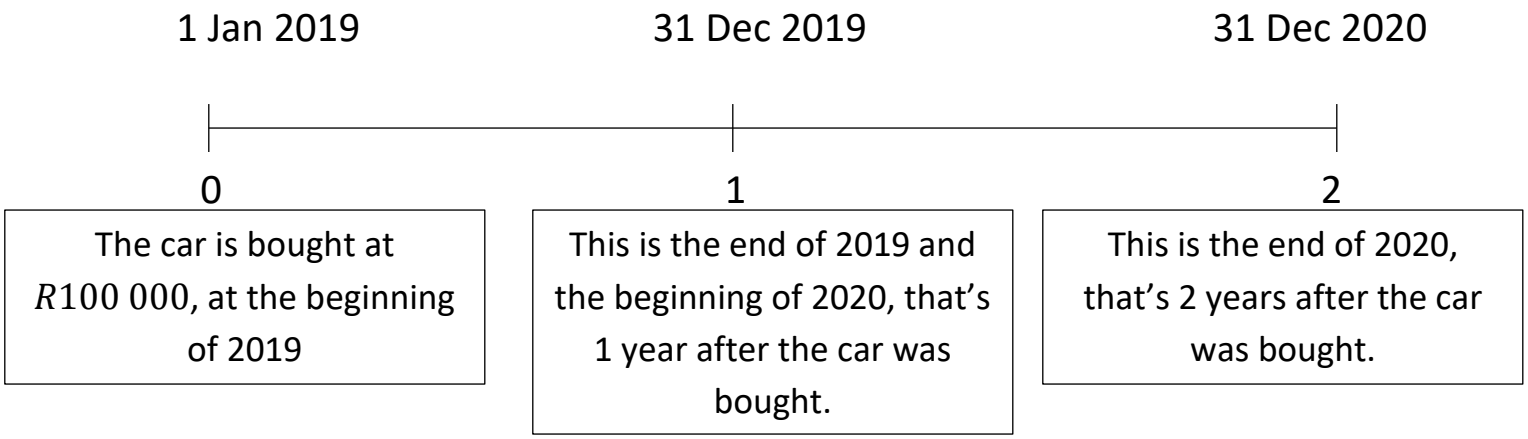
Thus the value of the car will be *R90 000* in a year (at the end of 2019).

Question

Let's extend the question above to 2 years. What will the value of the car be at the end of the 2 years if it depreciates by 10% of the *R90 000* in the second year 2020?

Solution

Extend the timeline above to 2 years and put in the new information.



The value of the car is *R81 000* over *2 years* .This is how we get the answer.

- At the beginning, the car is bought at R100 000.
- The value of the car depreciates by *10%* of *R100 000* (that's *R10 000*) over the first year. The value of the car is *R100 000 – R10 000 = R90 000* at the end of *2019*.
  So $V_1 = 90\ 000$ *(on the 31st of December 2019).*

Over year 2(2020), the car depreciates by 10% of its current value, R90 000. So the value of the car after 2 years is

$V_2 = current\ value - 10\%\ of\ 90\ 000$

$= 90\ 000 - \frac{10}{100} \times 90\ 000$

$= 90\ 000 - 90\ 000 \times \frac{10}{100}$

$= 90\ 000 - \ 90\ 000 \times 0.1$

$= 90\ 000 - \ 9\ 000$

$= 81\ 000$

Take notice that the car depreciates by $10\ 000$in *year 1* and $9\ 000$ in *year 2*.

Let's study the calculation closely to find the shortcut formula for the value of the car after 2 years.

$V_2 = 90\ 000 - 90\ 000 \times 0.1 = 81\ 000$

Remember this $90\ 000$ is the value of the car at the end of *year 1*.

$$V_2 = \underbrace{90\ 000} - 90\ 000 \times 0.1$$

This value was calculated like this:

$$100\ 000 - 100\ 000 \times 0.1 = 90\ 000$$

Now substitute this in the equation of V_2 above.

$V_2 = 100\ 000 - 100\ 000 \times 0.1 - 90\ 000 \times 0.1$

Again, let's replace this $90\ 000$ by $100\ 000 - 100\ 000 \times 0.1$

$V_2 = 100\ 000 - 100\ 000 \times 0.1 - (100\ 000 - 100\ 000 \times 0.1) \times 0.1$

$V_2 = 100\ 000 - 100\ 000 \times 0.1 - 0.1 \times (100\ 000 - 100\ 000 \times 0.1)$

$V_2 = 100\ 000 - 100\ 000 \times 0.1 - 0.1 \times (100\ 000 - 100\ 000 \times 0.1)$

Multiply out this

$V_2 = 100\ 000 - 100\ 000 \times 0.1 - 0.1 \times 100\ 000 + 0.1 \times 100\ 000 \times 0.1$

$V_2 = 100\ 000 - 100\ 000 \times 0.1 - 100\ 000 \times 0.1 + 100\ 000 \times 0.1 \times 0.1$

$V_2 = \mathbf{100\ 000} - \mathbf{100\ 000} \times 0.1 - \mathbf{100\ 000} \times 0.1 + \mathbf{100\ 000} \times 0.1 \times 0.1$

, the above is in the form

$$= \boldsymbol{a} - \boldsymbol{a} \times \boldsymbol{b} - \boldsymbol{a} \times \boldsymbol{b} + \boldsymbol{a} \times \boldsymbol{b} \times \boldsymbol{b}$$

$$= \mathbf{10\ 000} - \mathbf{100\ 000} \times \mathbf{0.1} - \mathbf{100\ 000} \times \mathbf{0.1} + \mathbf{100\ 000} \times \mathbf{0.1} \times \mathbf{0.1}$$

We substituted $\boldsymbol{a} = \mathbf{100\ 000}$ and $\boldsymbol{b} = \mathbf{0.1}$

$= \boldsymbol{a} - \boldsymbol{a} \times \boldsymbol{b} - \boldsymbol{a} \times \boldsymbol{b} + \boldsymbol{a} \times \boldsymbol{b} \times \boldsymbol{b}$

Let's simplify this.

$= \boldsymbol{a} - \boldsymbol{ab} - \boldsymbol{ab} + \boldsymbol{ab^2}$

$= \boldsymbol{a}(1 - \boldsymbol{b}) \underbrace{- \boldsymbol{ab} + \boldsymbol{ab^2}}$

Now factor out $\boldsymbol{ab}$ here.

$= \boldsymbol{a}(1 - \boldsymbol{b}) - \boldsymbol{ab}(1 - \boldsymbol{b})$

Now factor out $1 - \boldsymbol{b}$ here.

$= (1 - \boldsymbol{b})\underbrace{(\boldsymbol{a} - \boldsymbol{ab})}$

Now factor out $\boldsymbol{a}$ here.

$= (1 - \boldsymbol{b})\boldsymbol{a}(1 - \boldsymbol{b})$

Rearrange it.

$= \boldsymbol{a}(1 - \boldsymbol{b})(1 - \boldsymbol{b})$

$= \boldsymbol{a}(1 - \boldsymbol{b})^2$

Now then, substitute $\boldsymbol{a} = \mathbf{100\ 000}$ and $\boldsymbol{b} = \mathbf{0.1}$

$V_2 = 100\ 000(1 - 0.1)^2 = 81\ 000$

The above is the shortcut formula for the question we just did.

Let's break down the question. He bought the car at *R100 000* at the beginning. Over the year *2019*, the car depreciates by *10%* of *R100 000* (which is *10 000*). So the value of the car is *100 000 – 10 000 = 90 000* at the end of *year 1* .Then over *year 2*, the car depreciates by *10%* of *90 000*, which is *90 000*. So the value of the car at the end of *year 2* is *90 000 – 900 = 81 000*.

This question is slightly different from the ***simple interest*** one. The car then depreciated by *10%* of *100 000* in *year 1*, which is *10 000*. In the *year 2*, it depreciated by *10%* of *100 000 too*, which is *10 000*. So every year the car depreciated by the same amount. That's the difference between ***depreciation based on Simple Interest*** and ***depreciation based on Compound Interest***.

Note that the formula follows the same pattern as in simple interest. Use the example above for compound depreciation over 1 year, 2 years and 3 years.

For end year 1, $V_1 = 100\ 000(1 - 0.1)^1 = 90\ 000$

For end year 2, $V_2 = 100\ 000(1 - 0.1)^2 = 81\ 000$

For end year 3, $V_3 = 100\ 000(1 - 0.1)^3 = 79\ 000$

Do you see the pattern?

We can safely establish a general formula for the value of the car if it depreciates by a ***percentage of its previous value***.

$V_n = P(1 - i)^n$

$$A = P(1 - i)^n$$

, where A is the depreciation value of the car

, P is the initial value of the car

, i is the depreciation rate

n is the number of years over which it depreciates.

Mathematicians call this depreciation on a *reducing balance*.

Question

Lee buys a super computer at $R35\ 000$. The computer depreciates at 6% per annum on a reducing balance basis. What is the value of the computer in 5 years' time?

Solution

$A = P(1 - i)^n$

, where $= 35\ 000$, $i = 6\% = 0.06$, $n = 5$

$A = 35\ 000(1 - 0.06)^5$

$A = R25\ 686{,}64$

Question

Kano bought a car 3 years ago. The car's compound interest was 10% per annum over the past 3 years. Today, the car is valued at $R72\ 900$. At what price did Kano buy the car 3 years ago?

Solution

Since the car depreciates at a compound rate, we use the formula $= P(1 - i)^n$. However, you're given that $= 72\ 900$, $i = 10\% = 0.1$, $n = 3$

So you are basically asked to find P, the price at which the car was bought.

$A = P(1 - i)^n$

$72\ 900 = P(1 - 0.1)^3$

Since we are looking for P, we divide both sides by $(1 - 0.1)^3$ so that we may remain with P "alone" on the right-hand side.

$$\frac{72\ 900}{(1 - 0.1)^3} = \frac{P(1 - 0.1)^3}{(1 - 0.1)^3}$$

$$\frac{72\ 900}{(1 - 0.1)^3} = \frac{P\cancel{(1 - 0.1)^3}}{\cancel{(1 - 0.1)^3}}$$

$$\frac{72\ 900}{(1-0.1)^3} = P$$

$$100\ 000 = P$$

Thus Kano bought the car at $R100\ 000$ 3 years ago.

As always, there is a shortcut formula to calculate P from
$A = P(1-i)^n$.

The aim is to make P to be on its own on one side.

Firstly, divide both sides by $(1-i)^n$

$$\frac{A}{(1-i)^n} = \frac{P(1-i)^n}{(1-i)^n}$$

$$\frac{A}{(1-i)^n} = \frac{P\cancel{(1-i)^n}}{\cancel{(1-i)^n}}$$

$$\frac{A}{(1-i)^n} = P$$

$$P = \frac{A}{(1-i)^n}$$

Question

A forklift is valued at $R250\ 000$ today. For the past 7 years, it has been at a depreciation rate of 13% per annum. If depreciation is calculated on a reducing balance basis, what was the value of the forklift 7 years ago?

Solution

We use the formula for P immediately.

$$P = \frac{A}{(1-i)^n}$$

, where $= 250\ 000$, $i = 13\% = 0.13$, $n = 7$

$$P = \frac{250\ 000}{(1 - 0.13)^7}$$

$$P = R662\ 682,1008$$

The forklift was valued at $R662\ 682,1$ 7 years ago.

Question

[Finding $\boldsymbol{i}$ in $A = P(1 - \boldsymbol{i})^n$]

3 years ago, Esther bought a car at $R100\ 000$. The car has depreciated to the value of $R72\ 900$ today. If the car has depreciated on a reducing balance basis, what was the depreciation rate per annum over the past 3 years?

Solution

[Finding $\boldsymbol{i}$ in $A = P(1 - \boldsymbol{i})^n$]

$$A = P(1 - \boldsymbol{i})^n$$

, where $= 72\ 900$, $P = 100\ 000$, $n = 3$

$$72\ 900 = 100\ 000(1 - \boldsymbol{i})^3$$

Since you're working towards having $\boldsymbol{i}$ on its own on one side, we divide both sides by $100\ 000$.

$$\frac{72\ 900}{100\ 000} = \frac{100\ 000(1 - \boldsymbol{i})^3}{100\ 000}$$

$$\frac{72\ 900}{100\ 000} = \frac{\cancel{100\ 000}(1 - \boldsymbol{i})^3}{\cancel{100\ 000}}$$

$$\frac{72\ 900}{100\ 000} = (1 - \boldsymbol{i})^3$$

$$\frac{729}{1\ 000} = (1 - \boldsymbol{i})^3$$

Cube root both sides to get rid of the "exponent 3" on $(1 - \boldsymbol{i})^3$

$$\sqrt[3]{\frac{729}{1000}} = \sqrt[3]{(1-\boldsymbol{i})^3}$$

$$\sqrt[3]{\frac{729}{1000}} = (1-\boldsymbol{i})^{3\times\frac{1}{3}}$$

$$\sqrt[3]{\frac{729}{1000}} = (1-\boldsymbol{i})$$

$$\sqrt[3]{\frac{729}{1000}} = 1-\boldsymbol{i}$$

$$\sqrt[3]{\frac{729}{1000}} - 1 = -\boldsymbol{i}$$

Divide both sides by -1 to get rid of the negative in front of $\boldsymbol{i}$

$$\frac{\sqrt[3]{\frac{729}{1000}} - 1}{-1} = \frac{-\boldsymbol{i}}{-1}$$

$$\frac{\sqrt[3]{\frac{729}{1000}} - 1}{-1} = \boldsymbol{i}$$

$$\frac{\sqrt[3]{\frac{729}{1000}}}{-1} - \frac{1}{-1} = \boldsymbol{i}$$

We do that since $\frac{a-b}{c} = \frac{a}{c} - \frac{b}{c}$

$$-\sqrt[3]{\frac{729}{1000}}+1=\boldsymbol{i}$$

Rearrange the left-hand side.

$$1-\sqrt[3]{\frac{729}{1000}}=\boldsymbol{i}$$

$$\boldsymbol{i}=1-\sqrt[3]{\frac{729}{1000}}$$

$$\boldsymbol{i}=0.1=10\%$$

The shortcut formula for this is derived in the following way;

$$A=\ P(1-\boldsymbol{i}\ )^n$$

$$\frac{A}{P}=\ \frac{P(1-\boldsymbol{i}\ )^n}{P}$$

$$\frac{A}{P}=\ (1-\boldsymbol{i}\ )^n$$

$$\sqrt[n]{\frac{A}{P}}=\sqrt[n]{(1-\boldsymbol{i})^n}$$

$$\sqrt[n]{\frac{A}{P}}=(1-\boldsymbol{i})^{n\times\frac{1}{n}}$$

$$\sqrt[n]{\frac{A}{P}}=(1-\boldsymbol{i})$$

$$\sqrt[n]{\frac{A}{P}}=1-\boldsymbol{i}$$

$$\sqrt[n]{\frac{A}{P}}-1=-\boldsymbol{i}$$

$$\frac{\sqrt[n]{\frac{A}{P}} - 1}{-1} = \frac{-\boldsymbol{i}}{-1}$$

$$\frac{\sqrt[n]{\frac{A}{P}}}{-1} - \frac{1}{-1} = \boldsymbol{i}$$

$$-\sqrt[n]{\frac{A}{P}} + 1 = \boldsymbol{i}$$

$$1 - \sqrt[n]{\frac{A}{P}} = \boldsymbol{i}$$

$$\boldsymbol{i} = 1 - \sqrt[n]{\frac{A}{P}}$$

$$\boldsymbol{i} = 1 - \sqrt[n]{\frac{A}{P}}$$

Question

Sebastian buys a tractor worth $R350\ 000$ today. 7 years later, the tractor will have depreciated to the value of $R170\ 000$. If the depreciation of the tractor is calculated on a reducing balance basis, what was the depreciation rate per annum?

Solution

$$\boldsymbol{i} = 1 - \sqrt[n]{\frac{A}{P}}$$

, where $= 350\ 000$, $P = 170\ 000$, $n = 7$

$$i = 1 - \sqrt[7]{\frac{170\ 000}{350\ 000}}$$

$$i = 0.098$$

$$i = 0.098 \times 100\%$$

$i = 9.8\%$

Question

[Finding n in $A = P(1-i)^n$]

Yassin bought a new car at $R100\ 000$ some years ago. The compound depreciation on the car is at 10% per annum. The value of car is $R72\ 900$ today. How long did it take the car to depreciate from $R100\ 000$ to $R72\ 900$?

Solution

$A = P(1-i)^n$

, where $= 72\ 900$, $P = 100\ 000$, $i = 10\% = 0.1$

$72\ 900 = 100\ 000(1-0.1)^n$

$72\ 900 = 100\ 000(0.9)^n$

The aim is to remain with n "alone" on one side, so divide both sides by $100\ 000$.

$$\frac{72\ 900}{100\ 000} = \frac{100\ 000(0.9)^n}{100\ 000}$$

$$\frac{729}{1000} = (0.9)^n$$

Take logs because you want to bring the n down from the exponent position.

$$log\left(\frac{729}{1000}\right) = log(0.9)^n$$

$$log\left(\frac{729}{1000}\right) = nlog\ (0.9)$$

There's a mathematics rule that goes; $log\ a^b = b\ log\ a$. Based on this;

$log(0.9)^n = nlog\ (0.9)$

$$\frac{log\left(\frac{729}{1000}\right)}{log(0.9)} = \frac{n\ log(0.9)}{log(0.9)}$$

$$\frac{log\left(\frac{729}{1000}\right)}{log(0.9)} = n$$

$$n = \frac{log\left(\frac{729}{1000}\right)}{log(0.9)}$$

$$n = 3$$

There is a shortcut formula for this as well. Check out the derivation below.

$$A = P(1 - i)^n$$

$$\frac{A}{P} = \frac{P(1 - i)^n}{P}$$

$$\frac{A}{P} = (1 - i)^n$$

$$log\left(\frac{A}{P}\right) = log[(1 - i)^n]$$

$$log\left(\frac{A}{P}\right) = nlog(1 - i)$$

There's a mathematics rule that goes; $log\ a^b = b\ log\ a$. Based on this;

$$log(1 - i)^n = nlog\ (1 - i)$$

$$\frac{log\left(\frac{A}{P}\right)}{log(1 - i)} = \frac{n\ log(1 - i)}{log(1 - i)}$$

$$\frac{log\left(\frac{A}{P}\right)}{log(1 - i)} = n$$

$$n = \frac{log\left(\frac{A}{P}\right)}{log(1 - i)}$$

Question

The value of a truck depreciated from $R900\ 000$ to $R500\ 000$ over some years. The depreciation rate of the truck is at 7% per annum, on a reducing balance basis. How long did it take the truck to depreciate from $R900\ 000$ to $R500\ 000$?

Solution

Apply the shortcut formula immediately.

$$n = \frac{log\left(\frac{A}{P}\right)}{log(1-i)}$$

, where $= 900\ 000$, $= 500\ 000$, $i = 7\% = 0.07$

$$n = \frac{log\left(\frac{500\ 000}{900\ 000}\right)}{log(1-0.07)}$$

$n = 8.099504661$

Therefore, it took *8.1 years* for the truck to depreciate from $R900\ 000$ to $R500\ 000$.

Simple interest depreciation vs Reducing balance basis (Compound)

In this section, we are going to study the difference between the two depreciation methods, the simple interest method and reducing balance method. We will do so by using one easy example.

Example

A car is bought at $R100\ 000$ today. It depreciates at a rate of 10% per year. Calculate the value of the car at the end of each year for 5 years. Use both the simple interest method and reducing balance basis. Compare these values using a table.

Solution

Remember the formulas are as follows.

Simple Interest depreciation : $A = P(1 - i.n)$

Compound Interest depreciation : $A = P(1 - i)^n$

End of year	*Value using Simple Interest method*	*Value using Compound Interest method*
1	90 000	90 000
2	80 000	81 000
3	70 000	72 900
4	60 000	65 610
5	50 000	59 049

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Nominal and Effective Interest Rate Calculations

Consider the following statement.

Joseph deposits R100 into his savings account at the beginning of the year. The bank says the interest rate is 18% p.a compounded monthly. How much will Joseph have at the end of the year?

I'm sure this is the first time you come across the phrase "compounded monthly". Before we proceed to the answer, let's first interpret the statement "18% p.a compounded monthly".

"18% p.a" obviously means an interest of 18% per year.

Compounded monthly means the money in the account grows at the end of every month, in the compound sense. Since we have 12 months in the year, the money grows 12 times in a year.

But by what percentage does it grow per month? The statement says 18% per year, which means we would have to divide the 18% by 12 to get the percentage by which the money grows at the end of every month.

$\frac{18\%}{12} = 1.5\%$

So the money will grow by 1.5% at the end of every month in the year, in the compound sense.

Now let's calculate how much Joseph will have at the end of the year. We use the compound interest formula.

$A = P(1 + i)^n$

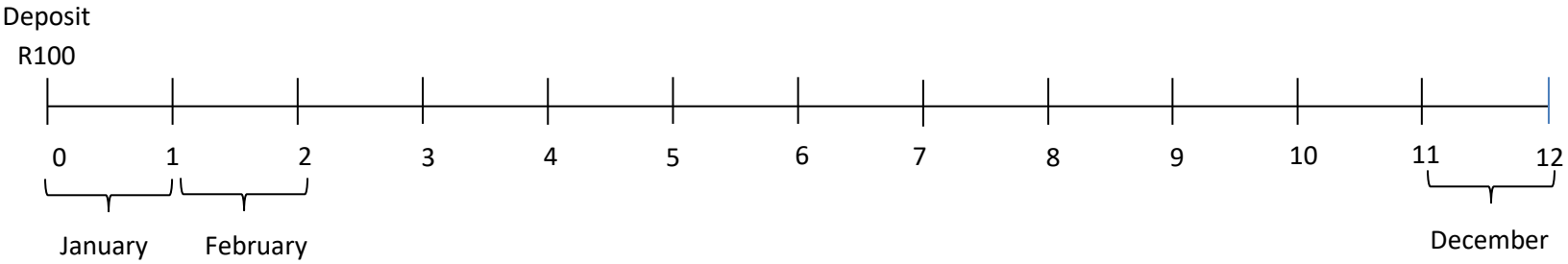
, where $P = 100, \quad i = 1.5\% = \frac{1.5}{100} = 0.015\ , \ \ n = 12$

Note that the "n = 12" tells us that the money gained interest 12 times in the year, that is, at the end of every month.

$A = 100(1 + 0.015)^{12} = 119.56$

Therefore, Joseph will have $R119.56$ in his bank account at the end of the year.

Check out the timeline below. It shows you what actually happens in Joseph's bank account throughout the year.



Below we show the balance of the account each month throughout the year.

End of month 1 (time 1): $100(1 + 0.015) = 101.5$

End of month 2 (time 2): $101.5(1 + 0.015) = 103.02$

End of month 3 (time 3): $103.02(1 + 0.015) = 104.57$

End of month 4 (time 4): $104.57(1 + 0.015) = 106.14$

End of month 5 (time 5): $106.14(1 + 0.015) = 107.73$

End of month 6 (time 6): $107.73(1 + 0.015) = 109.34$

End of month 7 (time 7): $109.34(1 + 0.015) = 110.98$

End of month 8 (time 8): $110.98(1 + 0.015) = 112.65$

End of month 9 (time 9): $112.65(1 + 0.015) = 114.34$

End of month 10(time 10): $114.34(1 + 0.015) = 116.05$

End of month 11(time 11): $116.05(1 + 0.015) = 117.79$

End of month 12(time 12): $117.79(1 + 0.015) = \mathbf{119.56}$

Joseph has a bank balance of R119.56 at the end of the year.

Let's calculate how much he gained over the year (interest amount).

$119.56 - 100 = 19.56$

So he earned an interest of $R19.56$ over the year. Meaning his $R100$ increased by $R19.56$. We now calculate 19.56 as a percentage of the starting amount of $\boldsymbol{R100}$**.**

$\frac{19.56}{\mathbf{100}} \times 100\% = 19.56\%$

So Joseph earned 19.56% of $R100$. This is the real interest rate over the year. Another name for this is the "**Effective Interest Rate**".

Now what about the 18% from the question, one that goes "18% p.a compounded monthly"? Well, this kind of interest rate is called the "**Nominal Interest Rate**" because it is "not" the actual interest rate earned. The Nominal interest rate is just used to do calculations. The actual interest rate is 19.56%, that's the Effective interest rate.

Attempt the examples below.

Example

1) Jessica deposits R200 into her savings account at the beginning of the year. The interest rate is at 17% p.a compounded monthly.

a. How much will Jessica have in her account at the end of the year?
b. Identify the "nominal interest rate".
c. Calculate the "effective interest rate".

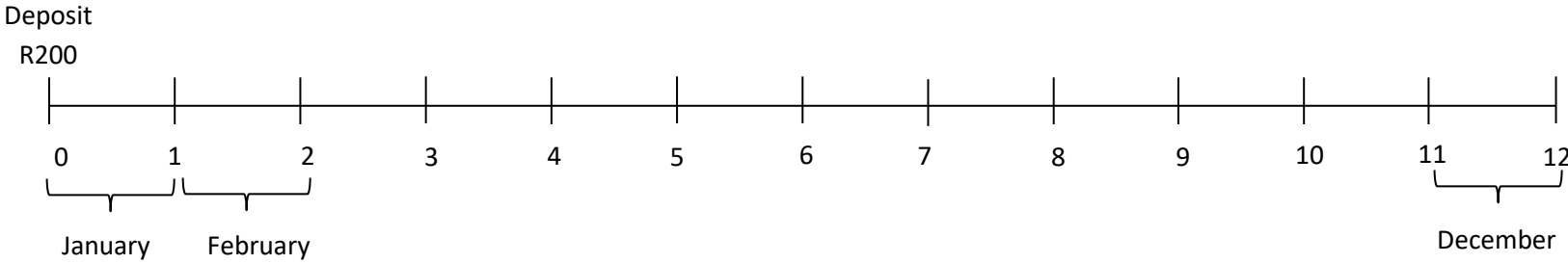
2) Mathew deposits R100 into his savings account at the beginning of the year. The interest rate is at 13% p.a compounded quarterly.

a. How much will Mathew have in his account at the end of the year?
b. Identify the "nominal interest rate".
c. Calculate the "effective interest rate".

Solution

1)

a.



$A = P(1+i)^n$

, where $P = 200, \quad i = \frac{0.17}{12}, \ n = 12$

Note that the "n = 12" tells us that the money gained interest 12 times in the year, that is, at the end of every month.

$$A = 200\left(1 + \frac{0.17}{12}\right)^{12} = 236.78$$

Therefore, Jessica will have $R236.78$ in his bank account at the end of the year.

b. 17% is the nominal interest rate
c. We have to calculate the effective interest rate.
   Firstly, we find the interest earned over the year.

$236.78 - 200 = 36.78$

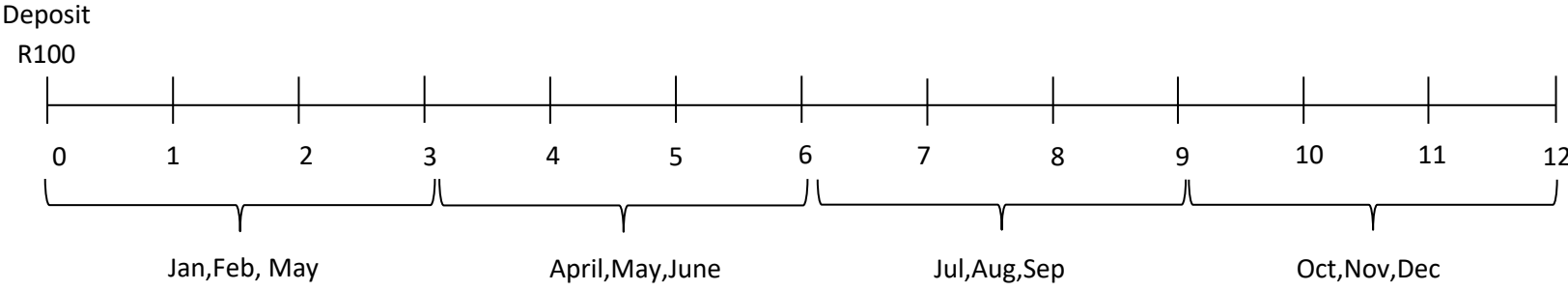
So he earned an interest of $R36.78$ over the year. Meaning her $R200$ increased by $R36.78$. We now calculate 36.78 as a percentage of the starting amount of $R200$.

$\frac{36.78}{\mathbf{200}} \times 100\% = 18.39\%$

Therefore, the effective interest rate is 18.39%

2)

a.



$A = P(1+i)^n$

, where $P = 100, \quad i = \frac{0.18}{4}, \ n = 4$

Note that the "n = 4" tells us that the money gained interest 4 times in the year, that is, at the end of every quarter of the year.

$$A = 100\left(1 + \frac{0.13}{4}\right)^4 = 113.65$$

Therefore, Mathew will have $R113.65$ in his bank account at the end of the year.

b. 13% is the nominal interest rate
c. We have to calculate the effective interest rate.
   Firstly, we find the interest earned over the year.

$$113.65 - 100 = 13.65$$

So he earned an interest of $R13.65$ over the year. Meaning her $R100$ increased by $R13.65$. We now calculate 13.65 as a percentage of the starting amount of $R100$.

$$\frac{13.65}{100} \times 100\% = 13.65\%$$

Therefore, the effective interest rate is 13.65%

There's actually a formula to find the effective interest rate from the nominal rate. Take question 1 above for example, our nominal rate is 17%, and we found the effective rate is 18.39%. Check if the proposed formula below works.

$$1 + i_e = \left(1 + \frac{i_{nom}}{m}\right)^m$$

, where i_e represents the effective interest rate, and i_{nom} represents the nominal interest rate.

Substitute the nominal interest rate and the number of times the money grows ($m = 12$)

$$1 + i_e = \left(1 + \frac{0.17}{12}\right)^{12}$$

$$i_e = \left(1 + \frac{0.17}{12}\right)^{12} - 1$$

$$i_e = 0.1839$$

Multiply this by 100% to put it in the percentage form.

$$0.1839 \times 100 = 18.39\%$$

$i_e = 18.39\%$

Therefore, the **effective interest rate** is indeed 18.39% per year.

We can also get the nominal interest rate using the effective interest rate.

Take question above for example, our **effective** rate is 13.65%, and we found the **nominal** rate is 13%. Check again if the proposed formula below works.

$$1 + i_e = \left(1 + \frac{i_{nom}}{m}\right)^m$$

Substitute the **effective** interest rate and the number of times the money grows ($m = 4$)

$$1 + 0.1365 = \left(1 + \frac{i_{nom}}{4}\right)^4$$

$$1.1365 = \left(1 + \frac{i_{nom}}{4}\right)^4$$

$$\sqrt[4]{1.1365} = \sqrt[4]{\left(1 + \frac{i_{nom}}{4}\right)^4}$$

$$\sqrt[4]{1.1365} = \left(1 + \frac{i_{nom}}{4}\right)$$

$$\sqrt[4]{1.1365} = 1 + \frac{i_{nom}}{4}$$

$$\sqrt[4]{1.1365} - 1 = \frac{i_{nom}}{4}$$

$$4\left(\sqrt[4]{1.1365} - 1\right) = i_{nom}$$

$$0.13 = i_{nom}$$

$$i_{nom} = 0.13$$

Multiply this by 100% to put it in the percentage form.

$$0.13 \times 100 = 13\%$$

$$i_e = 13\%$$

Therefore, the **nominal interest rate** is indeed 13% per year.

Make sure you memorize the formula for effective and nominal interest rate, together with ready-made calculation formulas for the effective and nominal interest rates

$$1 + i_e = \left(1 + \frac{i_{nom}}{m}\right)^m$$

$$i_e = \left(1 + \frac{i_{nom}}{m}\right)^m = -1$$

$$i_{nom} = m\left(\sqrt[m]{1 + i_e} - 1\right)$$

Question

1) Riyaad deposits $R2\ 000$ into his bank account, which pays interest at the rate of 11% p.a compounded monthly. How much will Riyaad have in his bank account 2 years after depositing the $R2\ 000$?

2) Wilburt deposits $R2\ 600$ into his bank account, which pays interest at the rate of 14%p.a compounded monthly. How much will Wilburt have in his bank account 3 years after depositing the $R2\ 600$?

3) Moussa deposits $R50\ 000$ into his bank account, which pays interest at the rate of 14%p.a compounded quarterly. How much will Moussa have in his bank account 5 years after depositing the $R50\ 000$?

4) Calculate the effective interest rate per annum of an investment that earns a nominal interest rate of 13.7% p.a compounded half-yearly.

5) The effective interest rate per annum of an investment is 15%. Calculate the nominal interest p.a if the investment is compounded quarterly.

Solution

1) $A = P(1+i)^n$

, where $P = 2\ 000$, $i = \frac{0.11}{12}$, the 0.11 comes from $11\% = \frac{11}{100} = 0.11$ and $n = 24$.

Remember that "n" is the number of times the money grows. Riyaad left money in the account for 2 years, meaning he left it for 24 months,

$(12\ months \times 2\ years = 24\ months)$.

$$A = 2\ 000\left(1+\frac{0.11}{12}\right)^{24} = 2\ 489.66$$

Notice that we can convert the formula $A = P(1+i)^n$ to $A = P\left(1+\frac{i_{nom}}{m}\right)^{m\times n}$, where i_{nom} is the nominal rate, "m" is the number of times the money grows per year, "n" is the number of years the money is in the account.

Let's redo the question using this formula.

$P = 2\ 000,\ i_{nom} = 0.11, m = 12, n = 2$, substitute into the formula.

$$A = P\left(1+\frac{i_{nom}}{m}\right)^{m\times n} = 2\ 000\left(1+\frac{0.11}{12}\right)^{12\times 2}$$

$= 2\ 489.66$

"It works like a charm". From now on, we apply this formula in such situations.

2) $A = P\left(1+\frac{i_{nom}}{m}\right)^{m\times n}$

$P = 2\ 600,\ i_{nom} = 0.14, m = 12, n = 3$, substitute into the formula.

$$A = P\left(1+\frac{i_{nom}}{m}\right)^{m\times n} = 2\ 600\left(1+\frac{0.14}{12}\right)^{12\times 3}$$

$= 3\ 947.49$

3) $A = P\left(1+\frac{i_{nom}}{m}\right)^{m\times n}$

$P = 50\ 000,\ i_{nom} = 0.14, m = 4, n = 5$, substitute into the formula.

$$A = P\left(1 + \frac{i_{nom}}{m}\right)^{m \times n} = 50\ 000\left(1 + \frac{0.14}{4}\right)^{4 \times 5}$$

$= 99\ 489.44$

4) We use the formula for nominal and effective interest rates.

$1 + i_e = \left(1 + \frac{i_{nom}}{m}\right)^m$, we then leave the effective rate alone on the left-hand side.

$$i_e = \left(1 + \frac{i_{nom}}{m}\right)^m - 1$$

, $i_{nom} = 15\% = 0.15, m = 2$, because "half-yearly" means the money grows at the end of every 6 months in a year. There's two "6-month" periods in a year. Another way of saying "half-yearly" is "semi-annually" and "bi-annually".

$$i_e = \left(1 + \frac{i_{nom}}{m}\right)^m - 1$$

$$i_e = \left(1 + \frac{0.15}{2}\right)^2 - 1$$

$i_e = 0.155625 = 0.155625 \times 100\% = 15.56\%$

5) We use the formula for nominal and effective interest rates.

$1 + i_e = \left(1 + \frac{i_{nom}}{m}\right)^m$, rearrange the formula to have the nominal rate alone on one side. Save yourself time and use the one given.

$$i_{nom} = m\left(\sqrt[m]{1 + i_e} - 1\right)$$

, $i_e = 15\% = 0.15, m = 4$, substitute.

$$i_{nom} = 4\left(\sqrt[4]{1 + 0.15} - 1\right)$$

$i_{nom} = 0.1422 = 0.1422 \times 100\% = 14.22\%$